



DALLAS
FORT WORTH
INTERNATIONAL
AIRPORT

DESIGN CRITERIA MANUAL

Revised 2025

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Adopted Standards

The DFW International Airport currently adopted versions of the International Building Code, International Fire Code, International Plumbing Code, International Mechanical Code, International Fuel Gas Code, International Energy Conservation Code, and National Electrical Code/National Fire Protection Association 70.

Preface

Introduction

The Dallas Fort Worth International Airport (hereinafter referred to as “DFW Airport”) Design Criteria Manual has been developed to ensure a unified approach to the design of the renovation of existing facilities or the construction of new facilities throughout the DFW Airport complex.

Scope and Purpose

This Manual establishes specific design criteria for all public infrastructure, terminal buildings and other public facilities owned, operated, or maintained by the Dallas Fort Worth International Airport Board (hereinafter referred to as "Board"). The Design Criteria manual is not intended to limit or dismiss the experience, knowledge, or talent of the Designer. The Board encourages Designers to recommend alternatives when deviations would be beneficial. However, adherence to these criteria should result in project development that conforms to the goals and objectives of the Board.

Maintenance

Proposed changes to this manual should be submitted to the Building Official for consideration. Proposed changes or additions to this Manual should be submitted to DFW Airport Code Enforcement for consideration. Requests for such change or addition shall include a complete description of the change proposed and shall be accompanied by sufficient technical analyses to support the change or addition and identified as a design improvement, public safety improvement, construction cost/schedule improvement, operation or maintenance improvements, life cycle cost benefit, or other improvement.

Request for Variances and/or Interpretation Statement

It is recognized that this Manual may not identify criteria for all types of Projects or all technologies possible for the design.

Chapter 1 General Requirements

1.0 Purpose

Design of new construction, renovation, remodel, and demolition activities requires various environmental regulations and DFW Airport specific requirements. Please incorporate, address, and abide by the following environmental and wildlife requirements when designing specifications and drawings.

Not incorporating these items in specifications and drawings prior to building permit application phase may delay construction or unexpected cost increases during issue resolution.

1.1 Prohibited Items

1.1.1 Asbestos

No Asbestos Containing Material (ACM) shall be installed on Dallas Fort Worth International Airport (DFW) property. As required by 25 TAC 296 Texas Asbestos Health Protection Rules, an asbestos survey is required prior to the disturbance of any existing building materials. DCC PM will coordinate with EAD to conduct an asbestos survey.

1.1.2 Polychlorinated Biphenyls

No Polychlorinated Biphenyls (PCB)-containing products shall be installed on DFW property.

1.2 Substitution Procedures

DFW Code approval is required for any variance requests to this manual. Discussion of variations before submissions will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. Variations to this manual must be called out in each submission. Failure to point out a variation may cause DFW to require rejection and removal of such work.

1.3 Design Variance (DV) Request Submittals

Design variance and interpretation requests must meet at least one of the following criteria:

1. Improvement to design
2. Improvement to public safety
3. Improvement to construction schedule
4. Construction cost savings
5. Operation and maintenance (O&M) cost savings
6. Resolution of conflict of established savings
7. Design condition not covered by this manual
8. Lifecycle cost benefit

1.4 Request Format

Design variance requests shall be submitted to DFW Code. Variance Form is located on the DFW Code and Construction website.

1.5 Customer Support Services

1.5.1 Contractor Substitutions

The construction contractor may ask for substitution of specified material or equipment if:

1. The contractor provides evidence that a material is not available or,
2. The contractor provides evidence that a material has an unreasonable delivery time through no fault of the contractor or,
3. If the special conditions allow the use of equal.
4. Must be accepted and approved by appropriate stakeholders.

1.5.2 Periodic Work Observation

The consultant shall conduct site visits with DFW Code and/or DFW QA discuss any observed defects and/or deficiencies. The consultant shall not be required to make exhaustive or continuous site visits to check the quality or progress of the work.

1.5.3 Temporary Construction Conditions

Need to define temporary exterior and interior performance criteria based Terminal finishings and requirements. Exterior barricades and visual barriers to be constructed of durable approved materials that use graphics and information as approved by appropriate DFW stakeholders. Any temporary walls for tenant buildout should be constructed of drywall or approved sheathing and separate construction from occupied terminals. Temporary contractor construction trailers and laydown yards shall be detailed.

1.5.1.1 Exterior

Scaffolding, equipment, temporary fencing, safety barricades and structures, caution tape, cones, or similar should not be visible to any traveling public. All exterior construction shall be coordinated with DFW Communications, Marketing, and Planning to determine graphics and materials to be used.

1.5.1.2 Interior

Scaffolding, equipment, temporary fencing, safety barricades and structures, caution tape, cones, or similar should not be visible to any traveling public. All interior construction or temporary facility work shall be blocked from public view. For interior work, wall finishes shall be coordinated with DFW Communications, Marketing, and Planning to determine graphics and materials to be used.

1.5.1.2.1 Temporary Construction Walls

1.5.1.2.1.0 SIDA Wall

1. Temporary construction walls utilized in the SIDA (sterile/secure) shall meet the following minimum standards:
2. Temporary construction walls shall utilize 5/8 inch sheetrock, on both sides, as a hardened surface.
3. Temporary construction walls shall utilize metal studs, on 16 inch centers, as wall supports.
4. Temporary construction walls shall be fastened to each stud, top plate, and bottom plate.
5. Temporary construction walls shall extend from floor to deck with no openings.
6. All temporary construction walls require DPS inspection upon completion, and prior to any continued construction.

7. Temporary construction walls shall not interfere with the operation of CCTV cameras and/or AACS. Prior to the deactivation of CCTV/AACS, ITS and DPS approval is required. If approved, only contractors holding a class B security license will conduct servicing, deactivation or removal of CCTV cameras, alarms, and/or wiring required to be moved during construction.
8. Temporary construction walls shall not interfere with emergency fire door operation. DPS Police, DPS Fire, and DFW ITS will be notified prior to removing access to any fire door or deactivation of alarms. Any emergency fire exit door in a SIDA or Secure temporary wall shall be equipped with CCTV monitoring on both sides of door and contacts on the door to initiate an alarm in the module control relay (MCR). Each emergency fire exit door shall be equipped with a local alarm to sound a horn (minimum of 94db) and strobe (minimum dual cd @115) in the immediate area of the door. Any locks/cores involving SIDA wall construction shall be coordinated with DPS.

1.5.1.2.1.1 Tenant Build Out

1. Shall be on drawings by a signed and sealed design professional.
2. Door shall be 36 inches wide minimum and have a coded lock.
3. Approved means of dust control shall encapsulate the entire project.
4. Masonite shall be installed between floor and bottom sill plate. Any damage to floor shall be repaired to like new status.
5. Bracing at top and mid-level of wall shall be installed to prevent wall from falling.
6. Construction barricade walls shall not encroach into means of egress corridors. Access doors shall be open into the construction area.
7. Construction barricade walls shall be secured, inspected, and repaired on a regular basis.
8. Drawings shall indicate wall attachment methods.
9. Drawings shall indicate location of the barricade boundaries.

1.5.1.2.1.2 Construction Trailer and Modular Building Guidelines

General Requirements

1. Trailers shall follow the currently adopted Building code for relocatable buildings.
2. Provide a detailed location of the trailers on a site map.
3. Provide the manufacturer's permanent data plate and its location on the trailer.
4. Provide a safety plan for the trailers.
5. Provide details how a multiple unit trailer will be connected and protected.
6. Provide the ramp and stairs details.
7. The service life for a temporary structure shall not exceed 180 days.
8. Structures or uses that are temporary but are anticipated to be in existence for more than 180 days are required to conform to code requirements for permanent structures and uses.

9. Trailers and Modular Building that are anticipated to be in existence for more than 180 days shall have a monitored fire alarm system installed.
10. Occupation of trailer(s) is prohibited without an approved TCO inspection and/or CO approval by DFW Code.

Building

1. Temporary Contractor Construction Trailers and Temporary Modular Buildings
 - a. Materials used for exterior horizontal floor assemblies such as stairs, walkways, ramps, and the associated supporting construction members shall consist of:
 - i. Non-combustible construction materials or
 - ii. Fire-resistant-rated construction materials
 - b. Materials used for exterior awnings, canopies, and the associated supporting construction members shall consist of:
 - i. Non-combustible construction materials or
 - ii. Fire-resistant-rated construction materials
2. Permanent Contractor Construction Trailers and Permanent Modular Buildings
 - a. Materials used for exterior horizontal floor assemblies such as stairs, walkways, ramps, and the associated supporting construction members shall consist of:
 - i. Non-combustible construction materials
 - b. Materials used for exterior awnings, canopies, and the associated supporting construction members shall consist of:
 - i. Non-combustible construction materials
3. Occupation of trailer(s) is prohibited without an approved TCO and/or CO inspection by DFW Code.

Electrical

1. Utility Power
 - a. Plans shall include all information for electrical scope of work required to power trailer(s).
 - b. Plans shall include all required Concrete Encased Electrical Duct Banks, both primary and secondary, as required by DFW DCM.
 - c. Plans shall include DFW Concrete Encased Electrical Duct Bank detail per DFW DCM.
 - d. Details shall be provided for installation of electric rack.
 - e. Details shall be provided for all electrical equipment to be installed on the electrical rack.
 - f. All electrical equipment sizes shall be provided on plans in voltage and amperage as per NEC 110.3(A).
 - g. All conductor sizes shall be provided on plans as per NEC 310.16
 - h. All grounding electrode and equipment sizes shall be provided on plans as per NEC 250.66 and 250.122.
 - i. All conduit from equipment rack to trailer(s) shall be buried to prevent damage and tripping hazards as per NEC 110.27(B) and 90.1.

- j. Grounding electrode system is required for both electrical service equipment and the trailer as per NEC 250.32 and 250.66. Each section of the trailer(s) shall be connected to the grounding electrode system.
 - k. Trailers requiring a fire alarm system shall be provided and shown on the submitted plans.
 - l. Electrical for (other) equipment, such as ice machines or chargers, shall be included on the submitted plans with the above required size requirements.
 - m. Energizing electrical energy and trailer are prohibited without an approved electrical inspection from DFW Code Electrical division.
 - n. Occupation of trailer(s) are prohibited without a TCO issued by DFW Code.
2. Generators
- a. Plans shall show that the trailer is powered by a generator.
 - b. The total square foot of the trailer shall be included.
 - c. The generator shall be located as close as possible to the electrical attachment point of the trailer to place the cables from the generator to the trailer, so they are not in an area subjected to physical damage or become a trip hazard, as per NEC 110.27 and 90.1.
 - d. Ground Rod- Grounding electrode is required to bond both the frame of the generator and the frame of the trailer, as per NEC 250.32.
 - e. Cable sizes shall be provided from the generator to the trailer, as per NEC 400.5.
 - f. The grounding Electrode conductor size shall be provided, as per NEC 250.66.
 - g. Trailers requiring fire alarm systems are not permitted to be powered by the generators. The power source for trailers requiring a fire alarm system shall be powered by a reliable source, provided by the utility company (Oncor).
 - h. Energizing the generator and trailer(s) are prohibited without an approved electrical inspection from DFW Code electrical division.
 - i. Occupation of trailer(s) is prohibited without an approved TCO and/or CO inspection by DFW Code.

Plumbing & Mechanical

- 1. An approved water sample shall be obtained prior to scheduling a plumbing or civil final inspection. Additionally, these samples are required to be scheduled through the CSS Portal. Water samples are only taken in the morning, Monday through Thursday. Water samples shall be scheduled by 6am. The water shall be flushing/running at the time a sample is taken.
- 2. All piping shall be heat traced and insulated when subject to freezing. Piping shall also be protected from weather where exposed with a metal jacket.
- 3. All backflows shall be certified prior to receiving a plumbing final.
- 4. An accessible isolation valve, for the domestic water, shall be installed within five feet of the trailer.
- 5. Double cleanouts shall be provided within five feet of the trailer.

6. Provide the method of condensate disposal. Condensate drains shall not directly connect to any plumbing drain, waste, or vent pipe.
7. Indicate where and how the domestic water is supplied.
8. Indicate the method of sanitary sewer disposal.
9. Occupation of trailer(s) is prohibited without an approved TCO and/or CO inspection by DFW Code.

Accessibility

1. Construction trailers on construction sites are exempt from complying with the Texas Accessibility Standards (TAS).
2. TDLR is not required if the estimated construction cost is less than \$50,000.00.
3. DFW Airport DCC Code Accessibility inspection is required for all trailers and modular building:
 - a. All operable parts (light switches, thermostats, card readers, dispensers, hardware.) shall comply with TAS sections 308 and 309.
 - b. Toilets, urinals, drinking fountains, showers, and lavatories shall comply with the requirements under TAS chapter 6.
 - c. Accessible routes shall comply with TAS chapter 4.
 - d. Ramps shall comply with TAS sections 405 and 406.
4. Occupation of trailer(s) is prohibited without an approved TCO and/or CO inspection by DFW Code.

1.5.4 Punch Lists

The consultant shall prepare punch lists and verify that the contractor has addressed all punch list items.

1.5.5 Final Inspection

The consultant shall participate in a final inspection to determine contractor has completed all contract requirements.

1.6 Design Submittal Requirements

As part of the design submittal procedures, the consultant shall ensure each submittal has the minimum requirements listed in the following sections.

Refer to Division 01 on the Procurement website under Solicitations at www.dfwairport.com regarding design specifications for project. Refer any questions to DFW project management team.

1.7 Design Development Drawings

Based upon the approved schematic design documents, the consultant shall prepare design development documents that include the following as a minimum:

1.7.1 Interactive Rendering

A 3D interactive rendering of the project is required at delivery date prior to the consultant proceeding to the construction document phase. This rendering shall allow a 360-degree view of the project.

1.7.2 Architectural

1. Drawings should be organized for subsequent use as contract document drawings. Include title sheets with zoning, building, fire, life safety, plumbing, mechanical, electrical code summaries, and calculations; area and location maps; and a drawing index.
2. Architectural drawings should include abbreviations, symbols, legends, room material code index, and general notes.
3. Architectural site plan with cross-reference notes
4. Key floor plans
5. Demolition drawings shall be prepared which clearly indicate altered, discontinued, and removed work where extensive removal and/or demolition operations are required. Demolition drawings shall indicate, but not be limited to, the following pertinent information:
 - a. Location and size of structural members
 - b. Methods of closing openings
 - c. Other information to clearly indicate the extent of known materials and conditions to be removed
 - d. Symbols that are used for the demolition work shall be the same as those used on the drawings for new construction
 - e. Complete design and detailing of systems interruptions and protection of adjacent or affected systems and operations from damage due to interruptions and/or construction activities
6. Life Safety Plan:
 - a. The Life Safety Floor Plan shall show the complete egress travel path, from the furthest point in each room, to the exit discharge into public way. Each travel path shall show the calculated distance, including the distance to the proper exit stairway used. This shall also include the common pathway to the point of divergence and include the dimensions/ depth of any dead end.
 - b. Each room the egress travel path passes through shall show the calculated occupant load, including the occupant load factor used from the current adopted building code cycle. Provide Life Safety Tags to include Function of Space, Occupancy Group, Area and Occupant Load.
 - c. The floor plan shall show the egress pathway around all fixed and moveable obstructions that the occupants would be required to navigate around to safely exit the tenant space or building and shall follow the 2012 Texas Accessibility Standards.
 - d. Show all fire rated assemblies, walls, shafts, etc. and their ratings.
 - e. Show all existing or future exit signs, directional signs, and emergency light locations within the assigned egress path.
 - f. Show the locations of all portable fire extinguishers, pull stations, knock boxes, and any other fire related devices used for safe exiting as per the current adopted code cycles.
 - g. Show the door schedule with the hardware used for the exiting door(s). The floor plan shall also show the swing direction of each door associated with the egress travel path. The egress travel path shall not pass through any doorways that require special skills or tools, as per the current adopted code. Location of all fire rated partitions and fire rated doors
7. Accessible route plans
8. Floor plans with room names, room numbers, and wall/partition types
9. Reflected ceiling plan
10. Roof plan showing slops, equipment locations, drainage, and penetrations.

11. Interior elevations
12. Interior / Exterior building sections
13. Exterior elevations
14. Preliminary opening schedules
15. Preliminary architectural finish schedule
16. Wall and partition schedules

1.7.3 Civil

1. Site development plan
2. Site grading plan
3. Pavements, parking, and roadways plan
4. Storm Water Management Plan (SWMP)
5. Site utilities plan
6. Landscape plans

1.7.4 Structural

1. Floor plans showing structural foundations systems and sub-slab construction, horizontal and vertical framing systems, suspended slab openings, depressed slab locations, lateral load cross bracing, and typical construction details; final structural design criteria; foundation design criteria; preliminary sizing of major structural components; critical coordination clearances; outline specifications.

1.7.5 Mechanical

1. Mechanical drawings shall incorporate dimensions, symbols, reference to codes, conventions, schedules, diagrams in describing the size and routing of pipes, material to be used, equipment criteria, duct sizes and shapes, amount of flow and the temperature of material in pipes and ducts, valve types and location, floor and wall penetrations, tank construction, equipment, piping insulation, and other facets of mechanical design as required.
2. Provide a legend sheet indicating symbols, line types, and abbreviations used in the contract drawings.

1.7.6 Plumbing

1. Plumbing drawings shall incorporate requirements for construction of the facility design, including pertinent services, equipment, and other features required for the performance of the mechanical equipment. These drawings incorporate dimensions, symbols, reference to codes, conventions, schedules, diagrams, etc. in describing the size and routing of pipes, the kind of material to be used, equipment criteria, duct sizes and shapes, amount of flow and temperature of material in pipes and ducts, valve types and location, floor and wall penetrations, tank construction, equipment, piping insulation, and other facets of mechanical design as required. Drawings shall be completely coordinated with other disciplines.
2. Provide a legend sheet indicating all symbols, line types, and abbreviations used in the contract drawings.

1.7.7 Fire Protection

1. Fire protection drawings delineate the components required to provide fire protection to the facility. This includes all methods of fire protection including, but not limited to wet-pipe, dry-pipe, deluge systems, clean agent systems, foam systems, pre-action systems, fire extinguishers, fire hoses, and standpipes. These drawings shall establish procedures for construction of the fire protection system design, including water connections to the fire main, sprinkler heads, piping, alarms, valves, etc. The delineation for these drawings shall incorporate dimensions, codes, conventions, schedules, diagrams, etc., in describing the fire protection system design.

1.7.8 Electrical

1. Electrical drawings shall generally include, as required, the following types of delineations:
 - a. Block Diagrams
 - b. One Line Diagrams
 - c. Schematic Diagrams
 - d. Connection Diagrams
 - e. Facility electrical power distribution plans
 - f. Facility lighting plans
 - g. Facility communications plans and risers
 - h. Facility lightning protection plans
 - i. Fire alarm plans and risers
 - j. Panel, circuit, and other schedules
 - k. Detail drawings
2. The number of delineation drawings for a project shall be kept to a minimum, consistent with clarity for constructability.

1.8 Construction Contract Documents

Based upon design development documents that have been reviewed and accepted by DFW, the consultant shall prepare complete construction documents.

The following shall be submitted as part of the construction document phase:

1.8.1 35% Submittal

The percentage does not mean 35% of the drawings. EACH drawing required for the construction document package shall be 35% complete, including but not limited to title block, background annotations, object placements, annotations, keynote development, key plan development, schedules, etc. This submittal is a step beyond the typical Schematic drawing of a project. All CADD layering, annotation, and line systems shall be in place. This submittal shall include all Division 1 requirements and but not limited to:

1.8.1.1 Drawings

International Code Council and DFW Amendments to the code shall be referenced to the permitted work.

1. Cover sheet, list of drawings, legends, and general notes
2. Provide statement of Special Inspections and Geotechnical Report(s)
3. Site surveys and boring location plans
4. Civil, landscaping, and utility plans and details, and schedules
5. Measured drawings of existing conditions and demolition plans
6. Building Code Analysis and Diagrams
7. Life Safety Plan
8. Architectural plans
9. Architectural sections
10. Typical wall sections
11. Reflected ceiling plans
12. Interior elevations
13. Typical architectural details
14. Architectural schedules
15. Structural plans
16. Structural sections
17. Structural details
18. Structural schedules

19. Mechanical hydronic piping, duct and control plans
20. Riser diagrams
21. Mechanical isometrics
22. Mechanical details
23. Mechanical schedules
24. Building management and automation systems control diagrams
25. Plumbing plans
26. Plumbing riser diagrams
27. Plumbing details
28. Plumbing schedules
29. Electrical site plans
30. Electrical floor plans
31. Electrical Schedules
32. Fire alarm and fire protection drawings
33. Public address system
34. Baggage handling systems
35. Special Systems including, but not limited to:
 - a. Master clock system
 - b. Security system
 - c. Automated external defibrillator (AED) cabinet monitoring system
 - d. Multi-User Flight Information Systems (MUFIDS)
 - e. Common use terminal equipment
 - f. Premise distribution system
 - g. Information network infrastructure design
 - h. Telecommunication room design
 - i. Uninterruptible Power Supply (UPS) requirements to support special systems
 - j. Blue Light Emergency Telephones (BL-ETELs)
 - k. Fire Fighter Emergency Telephones (FFETs)
 - l. 800 MHz emergency radio supplemental radiating system (SRS)
 - m. Cable Television (CATV)
 - n. Furniture, fixture, and equipment layout and design
 - o. Color selections
 - p. other systems as required

1.8.1.2 Specifications

An outline specification of all applicable sections shall be developed using the most recent edition of the MasterSpec format.

1.8.1.3 Basis of Designs – Calculations

The consultant shall furnish initial calculations for all disciplines that are necessary to determine the requirements and configuration of all parts of all systems required for the execution of all following design work.

1.8.2 70% Submittal

An in-progress submittal of the contract documents shall be scheduled by the consultant when the contract documents are approximately 70% complete. The 70% submittal shall represent the continuation of the 35% submittal, incorporating 35% review comments, and shall be in the final format of the construction documents. The 70% submittal, at a minimum, shall comprise the following documents:

1.8.2.1 Drawings

The drawings shall include all required sheets of the final construction working drawings defined by the drawing list, each at least to the 70% level of completion, with sufficient information included for the preparation of a detailed cost estimate.

1.8.2.2 Specifications

The specifications shall be in the final format in the most recent edition of the MasterSpec format and shall address all applicable subdivisions and shall contain adequate technical information to supplement the drawings to quantify sizes, capacities, and qualify grade, strength, workmanship finishes, and other characteristics of applicable materials and equipment. Provide a list of sole-source items included in the design document and provide sole-source justifications for all items not on the pre-approved list.

1.8.2.3 Calculations

The consultant shall furnish calculations for all disciplines that are necessary to determine the final requirements and configuration of all parts of all systems required for the execution of all construction work.

1.8.2.4 Design Analysis Report

The design report shall address how the information gained from the design development review has been carefully coordinated, cross-referenced, and incorporated into the construction documents, how the construction documents address further developments in design characteristics, code compliance issues, and description of how the design meets or differs from the requirements of the SOW. The design report shall be in its final format and shall include, at a minimum, the following data:

1. Executive summary
2. Existing site conditions
3. Utilities
4. Contractor access and facilities
5. Potential coordination conflicts/phasing issues
6. Overall design philosophy and criteria by disciplines
7. Floor area calculation and allocations
8. Material descriptions and properties
9. Equipment description and properties
10. Coordination of maintenance and operational issues
11. Code report and compliance issues

1.8.3 90% Submittal

An In-Progress Submittal of the contract documents shall be scheduled by the consultant when the contract documents are approximately 90% complete or substantially complete. The 90% submittal shall incorporate the comments and information gained from the 70% submittal and is a comprehensive and complete pre-final construction contract document, suitable for public procurement and construction. The documents shall essentially be 100% complete, pending any work for only minor corrections to resolve discrepancies discovered during the final review and for the incorporation of final client comments.

1.8.3.1 Drawings

The drawing set shall include all required construction working-drawing sheets completed to practically 90% level of completion.

1.8.3.2 Specifications

Specifications shall be complete, comprehensive, and fully coordinated with the working drawings between disciplines and with the contract and special provisions.

1.8.3.4 Calculations

All calculations shall be finalized, incorporating all resolved comments and corrections of the 70% submittal.

1.8.3.5 Design Analysis Report

The Design Report shall be finalized by carefully coordinating, cross-referencing, and incorporating all resolved comments and corrections of the 70% submittal and the contents of all previously submitted design reports and revised to reflect the final design.

Safety Risk Assessment for airside projects must be completed sometime between 70% and 90%. Project team must contact the Airside Operations Safety Manager for scheduling.

1.8.4 100% Final Submittal

The final submittal of the 100% construction contract documents shall incorporate all client comments developed by the 90% review, with all outstanding actions resolved. Deliverables shall include the following documents:

1. Consultant annotated responses to the design review comments from the 90% submittal
2. Final drawings
3. Final specifications
4. Final cost estimate
5. Final Calculations
6. Final DAR

1.8.4.1 Certification

The consultant shall certify in writing that all the RFP or contract documents are in conformance with the DFW DCM, all code agency requirements, and that the documents are complete.

1.8.4.2 Documents

Comply with requirements of the DFW DCM. All contract and RFP documents shall be complete. All design, drafting, coordination, dimensioning and cross references shall be complete, all agency plan comments shall be resolved, and DFW's and approved tenant's comments from all previous reviews must be incorporated into the documents or otherwise be addressed to the satisfaction of DFW. It is necessary that drawings be signed and stamped for this submittal.

1.8.4.3 Specifications

Part I and Part II Specifications shall be complete. They will be in final draft format and shall be submitted in loose-leaf binders.

1.8.4.4 Soils Report

The consultant shall bind separately soils investigations performed by the consultant to establish design criteria for the project. This report shall be issued as an information attachment to the BID documents.

1.9 File Naming Conventions

These conventions standardize the naming and numbering of all project documents for tracking purposes. All CAD documents shall contain a numbering system according to the criteria outlined in the following sections.

1.9.1 DFW File Naming Components - Project Plan Files

The components used to identify a file depend on the project’s complexity and requirements. DFW has a file naming system that consists of four standard components and two optional components.

1.9.1.1 Introduction

This chapter deals with the file setup for C3D model files, naming, and layer standards. Each design drawing (model file) shall be in model space (with nothing in paper space) and shall have no other drawing files externally referenced into it.

1.9.1.2 DFW Model Template

The most current DFW C3D model file template must be used on all projects. The file is available in the DFW DFI Package with the most recent template being ACAD-DFW_C3D_TEMPLATE.dwg.

1.9.1.3 DFW File Naming Components - Model Files and External Reference Files

The DFW file naming system for model files and external reference file conforms to the NCS Uniform Drawing System (UDS) and DFW legacy standards. The DFW external reference file naming system consists of three standard components and two optional components, as follows.

LLNNNNN	-	L	LLL	LL	-	LNLN
(1)		(2)	(3)	(4)		(5,6)

1.9.1.4 File Naming Conventions

These conventions standardize the naming and numbering of all project documents. All CAD documents shall contain a numbering system according to the criteria outlined below.

(1) Project Identifier

This is the DFW contract number provided by the DFW Project Manager. If the project has multiple tasks, denote which task with a decimal point and task number e.g., ce95003.01

ce95003	-	C	GRA	SC	-	AAVP
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(2) Level 1 Discipline Designator

This component identifies the discipline. For example, a civil model file is denoted with a C after the hyphen.

ce95003	-	C	GRA	SC	-	AAVP
---------	---	---	-----	----	---	------

(3) Level 2 Discipline Designator - Optional

This optional, user-defined component further specifies the type of model discipline, as needed. For example, in the civil grading model file a **GRA** is added to the **C** designation. Use **-** if this option is not needed.

ce95003	-	C	GRA	SC	-	AAVP
---------	---	---	-----	----	---	------

(4) Model Type Designator

The example of the 2-character model type designator shows an **SC**, denoting a Section. The example file name denotes the Civil Grading Section for contract ce95003.

ce95003	-	C	GRA	SC	-	AAVP
---------	---	---	-----	----	---	------

(5) User-Defined Model Type Modifiers - Optional

This optional, user-defined component further identifies the model type for aspects such as R1 for Revision 1, A1 for Phase 1. User defined modifiers shall be applied consistently and with a key.

In this example, the file name denotes the first revision to the Civil Grading Section, Phase 1 development for Contract ce95003.

ce95003	-	C	GRA		SC	-	AAVP
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(6) DFW File Naming Components - Identifiers

The following tables provide the identifiers and their descriptions for use in naming project plan files and external reference files.

Table 1 - Level 1 Discipline Designators

Discipline	Description	Discipline	Description
A	Architectural	M	Mechanical
B	Geotechnical	O	Operations
C	Civil	P	Plumbing
D	Process	Q	Equipment
E	Electrical	R	Resource
F	Fire Protection	S	Structural

G	General	T	Telecommunications
H	Hazardous Materials	V	Survey/Mapping
I	Interiors	X	Other Disciplines
L	Landscape	Z	Shop Drawings

Table 2 - Level 2 Discipline Designators

Identifier	Description	Identifier	Description
AER	Aerial	LID	Lighting Detail
AVD	AVI Details	LIN	Lighting Notes
AVN	AVI Notes	LIP	Lighting Plan
AVP	AVI Plans	LOP	Sheet Layout Plan
BLD	Boring Log Data	MSC	Miscellaneous Sheets
BLM	Boring Log Map	MSL	Overall Milestone Layout Plan
BMP	Benchmark Plan	PEN	Pavement Elevations Notes
CFG	Conceptual Foundation Grading	PHM	Potholing Map
COV	Cover/Title Sheet	PMN	Pavement Marking Notes
CRS	Cross Sections	PVE	Pavement Elevations Plan/Det
CST	CDOT Standard Plans List	PVM	Pavement Marking Plans/Pvmt Details
DET	Detail Sheets	QTB	Quantity Tabulations
DMN	Demolition Notes	QTY	Summary of Approximate Quantities

DMP	Demolition Plan	RDT	Roadway Details
DPP	Drainage Plan and Profile	RPL	Roadway Plan
DRD	Drainage Details	RPP	Roadway Plan and Profile
DRL	Drainage Profiles	SCD	Structural Caisson Details
DRN	Drainage Notes	SCL	Structural Caisson Layout
DRP	Drainage Plan	SCN	Structural Construction Layout
ECD	Erosion Control Details	SED	Structural Excavation/Backfill Details
ECN	Erosion Control Notes	SEG	Structural Engineering Geology
ECP	Erosion Control Plan	SGL	Structural General Layout
EES	Excavation Earthwork Summary	SGN	Structural General Notes
ELD	Electrical Details	SLD	Structural Slab Reinforcing Details
ELN	Electrical Notes	SMD	Structural Miscellaneous Details
ELP	Electrical Layout Plan	SPD	Structural Panel Details
FCD	Fence Details	SSD	Structural Details

FCP	Fence Plan	STY	Structural Typical Section
FIO	For Information Only	SUR	Survey Control Plan
FSM	Fill Settlement Markers	TDT	Track/Platform Details

GED	Geometry Data	TOP	Topography
GEP	Geometry Plan	TPL	Track/Platform Plans
GNN	General Notes Sheet	TPP	Track/Platform Plan and Profile
GRA	Grading Plan	TTB	Title Block
GRN	Grading Notes	TYP	Typical Sections
GUT	General Utilities	UTD	Utility Details
HRT	Haul Route	UTN	Utility Notes
IMG	Images	UTP	Utility Plans
IND	Index of Drawings	UTX	Utility Text (Base)
INF	Infrastructure/Site	VMS	Vicinity Map Sheet
JTD	Joint Layout Details	WDT	Retaining Wall Details
JTN	Joint Notes	WGL	Retaining Wall General Layout
JTP	Joint Layout Plan	WTY	Retaining Wall Typical Section
KEY	Key Map		

Table 3 - Model Type Designators

Identifier	Description	Identifier	Description
3D	Isometric/3D	GP	Grading Plan
CP	Construction Plan	PP	Phasing Plan
DG	Diagrams	QP	Equipment Plan
DP	Demolition Plan	RP	Roof Plan
DT	Detail	SC	Section
EL	Elevation	SH	Schedules
EP	Enlarged Plan	SP	Site Plan
ER	Erosion and Sediment Control Plan	XP	Existing Plan
FP	Floor Plan	01, 02, 03	Sequence

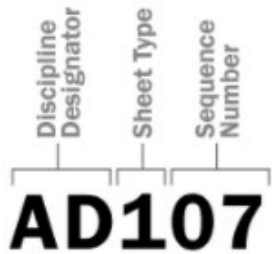
(7) Model File Breakout

All modeled content shall be delivered to DFW in a single model file representing each design discipline.

1.10 Sheet Naming Conventions

DFW follows the United States National CAD Standard United States National CAD Standard and the National BIM Standard - United States National BIM Standard - United States. The information below is adapted from and meets the U.S. National CAD Standard.

Components of the Sheet Number



The Sheet Types designator takes the drawings of a single discipline and organizes them. Drawings are always organized from most general to most specific or specialized. Since plans are most helpful to see the overall design of the project, they come first. Elevations and sections are a bit more specific and come next followed by details and schedules, which are the most specific type of drawings. The following table aligns with the *U.S. National CAD Standard*.

AD107

AD107

Chapter 2 Digital Infrastructure Platform Integration

2.0 General

2.0.1 Intent

The purpose of this chapter is to institute a series of requirements for integration to a Digital Infrastructure Platform. Integrations shall supply the Digital Infrastructure Platform with rich data to facilitate key functionalities in line with both DFW's "Digital Airport Ecosystem" vision and business Key Performance Indicators (KPIs).

2.0.2 Summary

This section details the responsibilities for the Contractor, the Owner, and any other responsible parties in delivering integrations to the Digital Infrastructure platform.

1. Requirements set forth in Chapter 2 shall apply to all subsections in Chapter 2.
2. The Contractor shall be responsible for furnishing and installation of one or more complete integrations to the Digital Infrastructure Platform as outlined in this specification. All Contractor work in this specification shall be included in the base contract.
3. The Owner shall have responsibilities as outlined in this specification.
4. The integrations to the Digital Infrastructure Platform and its components shall support communications between all devices and/or servers using industry standard protocols.
5. All labor, material, equipment, and software not specifically referred to herein or on other project documents that is required to meet the functional intent of all sections of this specification shall be provided at no additional cost to the Owner.
6. Where the drawings and specifications differ, the more stringent requirement shall be adhered to unless specifically communicated otherwise.
7. It is the goal of the Owner to fully realize a Digital Infrastructure platform that accommodates flexibility for assorted manufacturer devices and systems to be completely integrated into a consolidated platform and offers fluid growth and expansion to additional facilities.

2.0.3 Related Documents

Drawings and documents of the Contract, including but not limited to General and Supplementary Conditions as well as Division 01 Specification Sections, apply to, and should be read in conjunction with this section.

2.0.4 Abbreviations

A/C – Air Conditioning
ADT – Azure Digital Twin
AHU – Air Handling Unit
API – Application Programming Interface
ANSI – American National Standards Institute
ASHRAE – American Society of Heating, Refrigeration, and A/C Engineers
BBMD – BACnet Broadcast Management Device
BTL – BACnet Testing Laboratory
BMS – Building Management System
CPU – Central Processing Unit
CSV – Comma Separated Values

DDNS – Dynamic DNS
DNS – Domain Name Systems
DT – Digital Twin
DTD – Digital Twin Definition Language
FAS – Fire Alarm System
FDD – Fault Detection and Diagnostics
GUI – Graphical User Interface
HMI – Human Machine Interface
KPI – Key Performance Indicators
IAQ – Indoor Air Quality
IEEE – Institute of Electrical and Electronics Engineers
IP – Internet Protocol
ISO – International Standardization Organization
ISA – Instrument Society of America
LAN – Local Area Network
LED – Light Emitting Diode
MFA – Multi-factor Authentication
MSTP – Master Slave Token Passing
MQTT – Message Queuing Telemetry Transport
NEMA – National Electric Manufacturer’s Association
NFPA – National Fire Protection Association
oBIX – Open Building Information Xchange
ODBC – Open Data Base Connectivity
OEM – Original Equipment Manufacturer
OIW – Operator Interface Workstation
OLE – Object Linking and Embedding
OPC – Open Platform Communications
OSHA – Occupational Safety and Health Administration
OSI – Open Systems Interconnection
OT – Operational Technology
PC – Personal Computer
PDU – Power Distribution Unit
PICS – BACnet Protocol Implementation Conformance Statement
POE – Power over Ethernet
PPM – Portable Operator’s Terminal
QoS – Quality of Service
RAID – Redundant Array of Inexpensive Discs
RAM – Random Access Memory
REC – Real Estate Core
REST – Representative State Transfer
RFI – Radio Frequency Interference
RH – Relative Humidity
SCADA – Supervisory Control and Data Acquisition
SMS – Short Messaging Service
SMTP – Simple Mail Transfer Protocol
SNMP – Simple Network Management Protocol
SQL – Structured Query Language
SSO – Single Sign-On
SSPC – Standing Standard Project Committee
TLS – Transport Layer Security
TT – Temperature Transmitter
UC – Unitary Controller
UPS – Uninterruptable Power Supply
UL – Underwriter’s Laboratory

VAV – Variable Air Volume Terminal Device
VPN – Virtual Private Network
VFD – Variable Frequency Drive
VLAN – Virtual Local Area Network
WAN – Wide Area Network

2.0.5 Definitions

Algorithm - A procedure, typically in software, for solving a problem or performing a routing.

Analog - A continuously modulating value or signal.

BACnet - The Building Management and Control Network open protocol communications standard developed by ASHRAE, defined in ASHRAE SSPC/135. BACnet can operate in a variety of media including Ethernet and Serial networks.

BACnet Object - A physical or virtual point with a series of associated properties, such as a temperature sensor, including but not limited to Name, Current Value, Maximum/Minimum Values, and High/Low Alarms.

BACnet/IP - The BACnet standard using the Internet Protocol (IP), in compliance with Annex J of ASHRAE SSPC/135.

Binary - A two-state system where an “ON” condition is represented by a high signal level and an “OFF” condition is represented by a low signal level.

Control Process - The software required to perform a complete control loop, including but not limited to the input signal, interlock logic, control parameters, process calculation, and the output signal.

Component - Any individual element of the Digital Infrastructure platform furnished under this subcontract including hardware, software, and other materials.

Fault Detection and Diagnostics - A system, or subsystem, that measures and provides diagnostics for the improvement of the operating performance of buildings by identifying abnormalities in the operation of integrated systems. FDD software employs Artificial Intelligence, deductive modeling, and statistical methods to automatically detect and diagnose deviations between measured and projected performance.

Furnish - Purchase and deliver to the appropriate installing subcontractor, complete with every appurtenance, document, commissioning activity, and warranty.

Integration - The ability of control systems, including devices and components therein, from different manufacturers to connect to the Digital Infrastructure platform, providing coordinated monitoring and control via data exchange through common communications protocols and media. Methods of integration include industry standard protocols such as BACnet and OPC as well as connectors provided by this Contractor.

Modulate - Movement of a control device or monitoring sensor through a range of values.

Ontology - A framework that facilitates the organizing of concepts, information, and ideas into an intuitive framework, capturing parameters for those concepts, information, and ideas such as names, descriptions, locations, and relationships to other portions of the ontology.

Software - Programs that are executed by the Digital Infrastructure platform or any device to which the Digital Infrastructure platform is integrated, encompassing any programs such as operating systems, application programs, operating sequences, FDD, GUI, and databases.

2.0.6 Scope of Work

1. Scope of work for this contractor
 - a. This Contractor shall furnish all labor and materials to construct, install, test, verify, and commission one or more complete integrations to the Digital Infrastructure platform, as outlined in this specification.
 - b. Digital Infrastructure platform shall integrate to facility systems including but not limited to the following:
 - i. Building Management System
 - ii. Lighting Control System
 - iii. Electrical Energy Metering and Submetering
 - iv. Utility Energy Metering and Submetering
 - v. Computerized Maintenance Management Systems
 - vi. Elevator Control System
 - vii. Occupancy Monitoring and Analytics System
 - viii. Physical Security System
 - ix. Fire Alarm System
2. Scope of work for Owner
 - a. The Owner shall be responsible for the following, at a minimum:
 - i. The IP addressing schema to which any new devices shall adhere.
 - ii. Coordination and availability of internal IT resources for setup and troubleshooting activities.
 - iii. If required, the Owner shall permit the installation of hardware connectors, such as an edge gateway, onto the OT network to facilitate the ongoing communication of data from devices and/or servers to the Digital Infrastructure Platform
3. Scope of work for Installation Contractor
 - a. Installing Contractors shall be responsible for the following, at a minimum:
 - i. Coordination of information for all devices and systems to be integrated into the Digital Infrastructure Platform including but not limited to:
 1. IP addresses
 2. Communications protocols
 3. Queries or data aggregation in adherence with topic structures and data payload structures
 4. Points and/or register lists of all points available for integration. A subset of the available points shall be selected by the Owner and/or other Contractors to be integrated
 5. Point naming in adherence with Client and Willow standards

6. Metadata tagging in adherence with Brick, REC, and/or Willow ontology standards
7. Physical installation including but not limited to:
 - a. Wiring, both power and communication
 - b. Mounting of hardware
 - c. Installation, licensing, testing, and verification of software
8. Authentication
 - a. If required, installing contractors shall permit the installation of software connectors on their devices and/or servers to facilitate the ongoing communication of data from the devices and/or servers to the Digital Infrastructure Platform.
 - b. Upgrade of the firmware and software, immediately prior to project closeout, to the latest stable version, on every system and device to which the Digital Infrastructure platform is integrated.

2.0.7 Project Submittals

1. All submittals, as with all project documentation, must be unique and must be generated only for this project. Reuse of prewritten documentation shall not be allowed.
2. Shop Drawings
 - a. 3 copies of shop drawings of the entire integration to the Digital Infrastructure Platform shall be submitted and consisting of the following at a minimum:
 - i. A comprehensive network topology detailing the relevant network information for each integration to the Digital Infrastructure Platform.
 - ii. The proposed integration protocol to be used for each.
 - iii. Asset/register lists for all points available to be integrated.
 - iv. Proposed maintenance or service agreements.

2.0.8 Closeout Submittals

1. Before project closeout can conclude, this Contractor shall upgrade the firmware on every device integrated to the Digital Infrastructure Platform, including virtual machine instances, such that the latest stable version of all firmware is installed at the time of turnover to the Owner.
2. All submittals, as with all project documentation, must be unique and must be generated only for this project. Reuse of prewritten documentation shall not be allowed.
3. Project Documentation including but not limited to:
 - a. As-built network topology
 - b. Complete asset/register list of integrated points, including metadata.
 - c. Finalized maintenance or service agreements.
4. Operation and maintenance data for all integrations to the Digital Infrastructure Platform
5. Test data for all devices for which specification-requested field testing and verification took place.
 - a. Where testing is required for certification, all obtained certifications shall be supplied to Owner.

2.0.9 Quality Assurance

1. Manufacturer Qualifications
 - a. The manufacturer of devices proposed to be integrated to the Digital Infrastructure platform shall have a proven history of successful Digital integrations in projects of like size and complexity.
 - b. The manufacturer of devices proposed to be integrated to the Digital Infrastructure Platform shall make reasonable assurances that the underlying technology/protocol upon which the integration to the Digital Infrastructure Platform is constructed has a projected lifetime far exceeding the proposed uses of the Digital Infrastructure Platform.
2. Installer Qualifications
 - a. This Contractor shall be the manufacturer themselves or an independent installer certified by the manufacturer.
 - b. This Contractor shall be available for virtual meetings with the Client during normal business hours in the time zone in which the project resides.
 - c. This Contractor shall be staffed with knowledgeable engineers and application personnel.
 - d. This Contractor shall be responsible for fit, finish, and operation of the installed integrations to the Digital Infrastructure Platform.
3. Quality Assurance Program
 - a. This Contractor shall implement a quality assurance program.
 - b. This Contractor shall assign a single individual to serve as the quality assurance manager.
 - i. All installation issues and concerns shall be brought to the attention of the quality assurance manager.
 - ii. It shall be the responsibility of the quality assurance manager to keep all interested parties apprised of progress with regards to issues and concerns that have been identified.
4. Governing Code Compliance
 - a. This Contractor shall be responsible to research, understand, and comply with all current relevant governing codes, ordinances, and regulations that apply to the installation of the Digital Infrastructure Platform, potentially including UL, NFPA, and any applicable local, state, and national codes.
5. FCC Regulation
 - a. All electronic equipment shall conform to the requirements of FCC Regulation, Part 15, Section 15, Governing Radio Frequency Electromagnetic Interference and Subpart J, Governing Class A Computing Devices and be so labeled.

2.0.10 Delivery, Storage, and Handling

1. Provide factory-shipping containers for each piece of equipment.
2. Maintain containers throughout shipping, storage, and handling as required to prevent equipment damage.
3. Deliver, store and handle products to project site under provisions of the contract documents.
4. Coordinate all project site deliveries with construction project manager.
5. Accept equipment on-site and verify shipment condition.
6. Damaged equipment shall be reordered and/or replaced immediately and without additional cost to the Owner.
7. Protect equipment from construction conditions such as dust and debris by storing materials in a protected manner.

2.0.11 Field Conditions

1. This Contractor shall perform, at no cost to the Owner, a site survey prior to bidding.
2. This Contractor shall coordinate with the Construction Manager and installing contractor of the Digital Infrastructure Platform to fully assess current and proposed field conditions and limitations prior to the start of work.
3. This Contractor shall coordinate with all relevant installing contractors on an ongoing basis to facilitate the work to be performed.
4. During construction, the Construction Manager shall have the final say on all disputes related to field conditions.

2.0.12 Warranty

1. This Contractor shall agree to repair or replace, without additional cost to the Owner, any controller, device, wiring, or any other component furnished and installed by this Contractor as a part of an integration to the Digital Infrastructure Platform that fails within the specified warranty period.
 - a. Warranty period: 1 year from date of substantial completion.
2. Any componentry furnished and installed by this Contractor that fails during the period prior to the date of substantial completion shall be repaired or replaced without additional cost to the owner.
3. Repair work shall only be undertaken at times approved by the Owner.
4. Repair work shall not include routine maintenance.

2.0.13 Codes, Permits, Standards, and Guidelines

1. All work shall conform to the following codes and standards, at a minimum, where applicable:
 - a. Local Electrical Codes
 - b. Local Building Codes
 - c. National Fire Protection Association
 - d. National Electrical Code
 - e. Local, State and National Energy Codes
 - f. Underwriters Laboratories
 - g. American National Standards Institute
 - h. National Electric Manufacturer's Association
 - i. American Society of Heating, Refrigeration and Air Conditioning Engineers
 - j. Building Automation and Control Network protocol standard as developed by ASHRAE.
 - k. Institute of Electrical and Electronics Engineers
 - l. American Standard Code for Information Interchange
 - m. Electronics Industries Association
 - n. Occupational Safety and Health Administration
 - o. American Society for Testing and Materials
 - p. ANSI/TIA/EIA-862, Building Automation Systems Cabling Standards for Commercial Buildings

2.1 Products**2.1.1 General**

1. The integrations to the Digital Infrastructure Platform shall be comprised of a network of integrated facility systems and their respective network controllers and server(s)
2. When a specific reference to a manufacturer of a product is made and the term "approved equal" is used, substitutions of a product by another manufacturer will be considered only if the substituted product is demonstrated to conform to all specified requirements.

3. All equipment and materials shall be new and without defect.
4. Asbestos and PCB Certification:
 - a. After completion of installation but prior to the date of substantial completion, this Contractor shall certify in writing that products and materials installed do not contain asbestos or polychlorinated biphenyls.

2.1.2 Manufacturers

Subject to compliance with these specifications, manufacturers offering products that may be incorporated into the project shall include the following:

1. TBC

2.1.3 Digital Infrastructure Platform Architecture

1. The integrations to the Digital Infrastructure Platform shall make use of a modular IP network utilizing standard protocols.
2. The Digital Infrastructure Platform shall make use of the Azure Digital Twin framework as well as Snowflake for storage of relevant data. Cloud integrations may be made through these services.
3. All integrations to the Digital Infrastructure Platform shall make use of one or more established ontologies for categorization and navigation of integrated devices and components.
4. Local integrations to the Digital Infrastructure Platform shall have bidirectional communication (read and write) capabilities and shall utilize one of the following protocols:
 - a. BACnet/IP
 - b. Modbus TCP
 - c. OPC-UA
 - d. oBIX
 - e. RESTful API
 - i. Data can be pushed from integrated systems to the Digital Infrastructure Platform using the public API for the Digital Infrastructure Platform.

2.1.4 Design Files

Design Files are Model Files where most design activity takes place.

2.1.5 Data Reference Files

Data Reference Files (DREFs) are Model Files containing the original AEC objects produced by Civil 3D for the purpose of sharing their data across multiple Design Files. Data Shortcuts are the Civil 3D mechanisms utilized to share this data. Creating a Reference is the action taken to establish the link between the host and the source files for a particular AEC object.

2.1.6 Reference Files

Reference Files are Model Files. Reference Files can be DWG, DGN, PDF, DWF, and all image file formats supported by Civil 3D. Reference files usually provide background information for a model file or design information for a sheet file. DWG files that are referenced into other DWG files are often referred to as XREF's. Model files can be referenced (XREF'd) into Sheet files. Sheet files are never used as XREF. The XREF type should always be "Overlay".

2.1.7 Sheet Files

A sheet file is synonymous with a plotted CAD drawing file. A sheet file is a selected view or portion of referenced Model File(s) within a border sheet introduction.

2.2 Building Automation System (BAS)

The Building Automation System (BAS) is a central connection point for systems and devices with automated control or monitoring capabilities, particularly those without their own centralized system. For example, the Baggage Handling System has a central monitoring system of its own and is not connected to the BAS. Other systems such as the Fire Alarm system and some Computer room equipment systems have both their own centralized system and some level of connection to the BAS.

All systems that use automated controls will need to be integrated into the Airports BAS system unless stated otherwise by the BAS Administrator. This is a Tridium Niagara based system referred to as Enterprise Integrated Operating System (EIOS).

Examples of the systems to be integrated include but are not limited to:

1. HVAC (all aspects)
2. Automated plumbing devices (valves, flow meters, etc.)
3. Conveyance systems (elevators, escalators, moving sidewalks)
4. PBB systems (RTU's, PCA's Bridge controls, APU's, etc.)
5. Sump Systems such as grease/storm/sanitary sewer (pumps, level control, heaters/heat trace, etc.)
6. Lighting Controls
7. Electrical gear (switchgear breakers, power meters, ATS, UPS, etc.)

The definition of BAS controls should include but by no means be limited to HVAC control. If any questions, contact the BAS Administrator.

Chapter 3 Off-Site Construction

3.0 Application and Administration

3.0.1 Administrative Provisions

3.0.1.1 Purpose

The purpose of this standard is to provide minimum requirements to safeguard public health, safety, and general welfare and to address societal and industry challenges for the inspection and regulatory compliance of off-site construction. This standard is intended for adoption by government agencies and organizations for use in conjunction with model codes to achieve uniformity in the inspection and regulatory compliance of off-site construction.

3.0.1.2 Scope

The scope of this standard is to provide minimum requirements to safeguard the public health, safety and general welfare and address societal and industry challenges in multiple facets of the off-site and modular construction process, including planning, designing, fabricating, transporting, and assembling commercial building elements, and for the inspection and regulatory compliance of off-site and modular construction.

3.0.1.3 Provision for Compliance

Where requirements are not provided by this standard, the applicable provisions of the construction codes adopted by the authority having jurisdiction (AHJ) shall apply to the off-site construction. Where these provisions are not applicable, methods and practices in accordance with the appropriate sections of the International Building Code (IBC) or applicable adopted codes as identified in the construction documents as applicable for the intended use of the structure shall be used.

3.0.1.4 Compliance Alternative

Nothing in this standard is intended to prevent the use of designs, technologies, or products, including listed and labeled modular components, as alternatives to any prescriptions in this standard, provided that equivalence is demonstrated and approved by the authority having jurisdiction. Listed and labeled modular components shall be installed in accordance with their listing, the manufacturer's installation instructions, and the applicable requirements of the construction codes adopted by the authority having jurisdiction.

3.0.1.5 Referenced Standards

The specific year, date, and editions of the code listed below serve as a reference; however, for accuracy and completeness, it is essential to always use the most current version of the code.

1. ASCE 7 - with Supplement 1 Minimum Design Loads and Associated Criteria for Buildings and Other Structures
2. IBC - International Building Code 101.3, 103.1, 202, 301.2, 310.4
3. ICC/MBI 1205 - Standard for Off-Site Construction: Inspection and Regulatory Compliance 104.1, 302.4, 503.1
4. ISO 6707 - Buildings and Civil Engineering Works – Vocabulary 803.1
5. NFPA 221 - Standard for High Challenge Fire Walls, Fire Walls, and Fire Barrier Walls 302.3

3.0.2 General Requirements

3.0.2.1 General

Off-site construction shall comply with the general requirements of this Design Criteria Manual and all project contract documents. The project requirements vary, thus precluding absolute adherence to the items identified in all cases. However, unless there is adequate written justification, these guidelines will govern design and specifications.

3.0.2.2 Planning Considerations

Projects that include off-site construction shall identify and consider the following:

1. Entities of the AHJ at the manufacturing facility and the project's location.
2. The AHJ's specific requirements on review and approval of construction documents of off-site components.
3. The AHJ's specific requirements on inspections of off-site components.
4. Potential restrictions on a project due to manufacturing process of off-site components.
5. Restrictions due to transportation limitations of off-site components.
6. Impacts of sequencing of installation of off-site components at project site.
7. Restrictions on-site on the use of cranes.
8. Responsibilities of entities involved, including registered design professionals, manufacturers, installers of off-site components, and general contractors.

3.0.2.3 Specific Requirements of Authorities Having Jurisdiction (AHJ) Over Off-Site Construction

The project shall comply with the requirements of the AHJ.

3.0.2.3.1 Statewide Modular Programs

Where there is an existing statewide Modular (or Industrialized) Buildings Program, the project shall comply with both state and local jurisdiction requirements for plan approval and inspections of off-site components in manufacturing plants and at project sites. Where there is no existing statewide Modular (or Industrialized) Buildings Program, the AHJ shall determine requirements for a plan approval and inspections of off-site components in manufacturing plants and at project sites.

3.0.2.3.2 Duties and Responsibilities of Building Officials

Building officials in the installation or erection of a modular building or building component.

1. Verify through inspection that the modular building or component displays the required certification label and the label of the third-party inspection agency or complies with the labeling requirements of the AHJ.
2. Verify through inspection that the modular building or component has not been damaged in transit to a degree that would render it unsafe. If the building or component has been damaged, the building official is authorized to require evaluation by a registered design professional or test on structural elements for strength.
3. Verify that all off-site components are installed on-site in accordance with the approved plans.
4. Prevent the use or occupancy of a modular building that, in the opinion of the building official, contains a serious defect or imminent safety hazard and notify the AHJ immediately.
5. Notify the state program administrator, if applicable, of any violations of this standard.

3.0.2.4 Responsible Parties

The AHJ shall request identification of the responsible parties for the following:

1. Texas Licensed Professional Engineer/Architect for on-site and off-site elements.
2. Manufacturer of off-site construction.
3. On-site assembly of off-site elements, including volumetric modular units and panelized constructions.
4. Construction of site-built elements, including any items shipped loose to the site by the off-site manufacturer.

3.0.2.5 Qualifications

References to each chapter and section that has qualification requirements shall pertain to designers and constructors.

3.0.2.5.1 Manufacturers Experience

Off-site manufacturers must have a minimum of five years' experience as a qualified manufacturer and a member of the Modular Building Institute. Provide engineering services by an authorized, licenses engineer in the State of Texas, having a minimum of four years of experience as an engineer knowledgeable in building design analysis, protocols, and procedures for the ICC/MBI Standards for Off-Site Construction.

3.0.3 Submittal Document

3.0.3.1 General Requirements

The submittal documents shall be in accordance with Section 107 of the International Building Code, and the construction documents shall meet the following requirements:

1. The construction documents shall comply with the requirements of the AHJ regarding off-site components. The AHJ responsible for approval of submittal documents shall determine where separation of off-site components from site-built components in a set of construction documents is required.
2. The construction documents shall include accommodation, clearances, chase ways, and equipment to facilitate the completion of the systems onsite.
3. Appropriate delineation of off-site and site-built components shall be achieved. Graphical representations on drawing sheets shall clearly distinguish the off-site components from the site-built components.
4. Where required by the AHJ for projects having multiple off-site volumetric units and/or panels, a Key Plan for off-site components shall be provided, and all such off-site components delivered to the site shall be identified with unique numbers.
5. Where required, mate lines shall be identified.
6. Specifications for off-site components shall be provided.
7. A construction sequencing plan for off-site components shall be provided. The sequencing plan shall include a key plan that identifies all components manufactured off-site. Each component shall be identified by a unique number.
8. Shop drawings shall be approved by the registered design professional.
9. Documents shall be reviewed by any third party or special inspector against submitted permit drawings.
10. Engineering Analysis states, "An engineering analysis for design loads to support the selection of all structural members and connections in compliance with applicable codes shall be provided."

11. Design Calculations

a. Design calculations shall:

- i. Identify reference standards and/or code table and present design methodology in a step-by-step reviewable format.
- ii. Include applicable design loads and load paths.
- iii. Demonstrate compliance with the loads required by the building code adopted by the AHJ.
- iv. If the design is using a prescriptive method of design, clearly identify the source utilized.

3.0.3.2 Additional Requirements for Nonsite-Specific Buildings

A nonsite-specific building shall be identified as not having an available address on construction documents. The construction documents for nonsite-specific buildings shall meet the requirements of the AHJ.

3.0.3.3 Additional Requirements for Construction Documents for Panelized Systems

The construction documents for panel systems shall include:

1. Systems and components requiring special inspections.
2. Statement on the sequence of construction.
3. Details of connections for panels to the building structure.
4. Details of connections for panel-to-panel attachments.

3.0.4 Inspections

3.0.4.1 General

Inspections shall include in-plant inspections and on-site inspections. Inspections shall be in accordance with the applicable building code.

3.0.4.2 In-plant Inspections

In-plant inspections shall verify that constructions are compliant with the approved construction documents.

3.0.4.3 On-site Inspections

On-site inspections of off-site components shall verify installation is compliant with approved manufacturer's installation instructions and connections performed on-site are compliant with approved construction documents. When on-site inspections are conducted by AHJ, inspection procedures prescribed by the AHJ shall be followed. When on-site inspections are conducted by other than AHJ.

On-site connections that require inspections shall include:

1. Interconnections between off-site components installed at site.
2. Connections between off-site components and adjoining site-built components, including the building structure.
3. Other connections involving off-site components that require inspections by the AHJ.
4. Connections of "shipped loose off-site items" installed at site.

3.0.5 Warranty

Submit all manufacturer's warranties prior to final commissioning and acceptance.

3.0.5.1 Modular Building System Warranty

Furnish manufacturer's no-dollar-limit warranty for the off-site construction building system. The warranty period is to be no less than 15 years from the date of acceptance of the work and be issued directly to the Airport. The warranty must provide that if, within the warranty period, the off-site construction building system shows evidence of deterioration resulting from defective materials or workmanship, correcting any defects is the responsibility of the building system manufacturer. Repairs that become necessary because of defective materials and workmanship while metal building system is under warranty are to be performed within 32 hours after notification. Failure to perform repairs within 32 hours will constitute grounds for having emergency repairs performed by others and will not void the warranty.

3.0.5.2 Roof System Weather-Tightness Warranty

Furnish manufacturer's no-dollar-limit warranty for the metal panel system. The warranty period is to be no less than 20 years from the date of acceptance and is to be issues directly to the Airport. The warranty is to provide that if, within the warranty period, the roof panel system shows evidence of corrosion, perforation, rupture, loss of weather-tightness or excess weathering due to deterioration of the panel system resulting from defective materials and correction of the defective workmanship is to be the responsibility of the building system manufacturer. Repairs that become necessary because of defective materials and workmanship while roof panel system is under warranty are to be performed within 24 hours after notification. Failure to perform repairs within 24-hour after notification will constitute grounds for having emergency repairs performed by others and not void the warranty. Immediate follow-up and completion of permanent repairs must be performed within 10 days of notification.

3.0.5.3 Modules and Modular Components Warranty

Furnish manufacturer's no-dollar-limit warranty for the off-site construction building system. The warranty period is to be no less than 1 year from the date of acceptance of the work and be issued directly to the Airport. The warranty must provide that if, within the warranty period, the modules or modular components manufactured off-site shows evidence of deterioration resulting from defective materials or workmanship, correcting any defects is the responsibility of the building system manufacturer. Repairs that become necessary because of defective materials and workmanship while under warranty are to be performed within 32 hours after notification. Failure to perform repairs within 32 hours will constitute grounds for having emergency repairs performed by others and will not void the warranty.

3.0.5.4 Modular Wall Panel Warranty

Furnish manufacturer's no-dollar-limit warranty for the off-site construction building system. The warranty period is to be no less than 10-year from the date of acceptance of the work and be issued directly to the Airport. The warranty must provide that if, within the warranty period, the wall panels manufactured off-site show evidence of deterioration resulting from defective materials or workmanship, correcting any defects is the responsibility of the building system manufacturer. Repairs that become necessary because of defective materials and workmanship while under warranty are to be performed within 32 hours after notification. Failure to perform repairs within 32 hours will constitute grounds for having emergency repairs performed by others and will not void the warranty.

3.1 Definitions

3.1.0 General

For the purposes of this standard, the terms listed in this section shall have the indicated meaning.

3.1.0.2 Undefined Terms

The terms not specifically defined in this standard or in standards referenced herein shall have ordinarily accepted meanings such as the context implies.

3.1.1 Definitions

Abbreviations - The following abbreviations, when used in this standard, shall have the following meanings, unless the context clearly indicates otherwise.

AOR - Architect of Record
ASCE - American Society of Civil Engineers
ASTM - American Society for Testing and Materials
CFR - Code of Federal Regulations
EOR - Engineer of Record
IBC - International Building Code.
ICC - International Code Council, Inc.
IEC - International Electrotechnical Commission
ISO - International Organization for Standardization
LEED - Leadership in Energy and Environmental Design
MDF - Medium-density Fiberboard
NFPA - National Fire Protection Association
OSHA - Occupational Safety and Health Administration
POAC - Plan of corrective action
QAQC - Quality Assurance Quality Control
STC - Solid Transmission Class

Alteration - Any construction, other than ordinary repairs of the house or building, to an existing modular building, modular component or panelized system after the manufacturer has attached the certification label. Industrialized housing or buildings that have not been maintained shall be considered altered.

Applicable Building Code - The versions of the building code that have been adopted by the state or jurisdiction in which an industrialized house or building is to be constructed.

Assembly - A collection of modular components assembled into a whole or partial module or modular building.

Authority Having Jurisdiction (AHJ) - Organization, political subchapter, office, or individual charged with the responsibility of administering and enforcing the provisions of the applicable building code. The authority having jurisdiction shall include a state agency or local DFW Code.

Building Shell - The structural framework, exterior walls and cladding that make up the building envelope, excluding the electrical, mechanical, or plumbing systems.

Building Thermal Envelope - The basement walls, exterior walls, floor, roof, and any other building elements that enclose conditioned space or provide a boundary between conditioned space and exempt or unconditioned space.

Certification Label - A decal, insignia, or alteration decal.

Compliance Assurance Program - Procedures that state the guiding principles and define the framework for ensuring that construction documents approved by a design review agency, or that modular buildings inspected by a third-party inspection agency, comply with the applicable building codes.

Construction Documents - Designs, plans and specifications, including written, graphic, and pictorial documents, prepared, or assembled for describing the design, location, and physical characteristics of the off-site construction necessary to show compliance with the applicable building codes.

Data Plate - A plate attached by the manufacturer or installer to a modular building or modular component that contains identifying information allowing code officials or end users to determine if the structure is suitable for installation in their jurisdiction, location, or project.

Decal - The approved form of certification issued by the authority having jurisdiction, to be permanently attached to the modular building, modular component or panelized system indicating that it has been constructed to meet or exceed the applicable building code requirements.

Design Package - The aggregate of all construction documents, including on-site documentation, and the compliance control program, to be submitted by the manufacturer to the design review agency, or required by the design review agency for compliance review. A design package shall include model- or project-specific plans and calculations, typical system packages and calculations, or any combination thereof. Unique on-site construction details and site-specific foundation drawings prepared for specific projects are not a part of the design package.

Industrialized Building - A commercial structure that is constructed in one or more modules, or constructed using one or more modular components, built at a location other than the commercial site and is designed to be used as a commercial building when the module or modular component is transported to the commercial site and erected or installed.

Industrialized Housing - A residential structure that is designed for the occupancy of one or more families; is constructed in one or more modules, or constructed using one or more modular components, built at a location other than the permanent site; and is designed to be used as a permanent residential structure when the module or modular component is transported to the permanent site and erected or installed on a permanent foundation system.

Industrialized Installer - An entity who:

1. Is engaged in or responsible for the assembly, connection, on-site construction and erection of modular buildings or modular components at the building site.
2. Is engaged in, or responsible for, the alteration or recertification of modular buildings.

Industrialized Retailer - An entity who sells, leases, or offers to sell or lease to the public modular buildings, panelized systems, or modular components.

Insignia - The approved form of certification issued by the authority having jurisdiction to the manufacturer to be attached to the modular building, modular component or panelized system indicating that it has been constructed to meet or exceed the applicable building code requirements.

Installation - The assembly of a modular building, modular component, or panelized system on-site and the process of affixing the modular building, modular component or panelized system to land, a foundation, footings, or an existing building.

Integration - Act or process of coordinating the independent work of two or more manufacturers to merge a building shell for a modular building with the electrical, mechanical, and plumbing systems and equipment prior to delivery to the installation site.

Key Plan - A building perimeter of the project, which shows the layout of the off-site volumetric units or panels with a unique identifier assigned to each of such units or panels.

Manufacturer - The entity responsible for the manufacturing of assemblies, panelized systems, modular buildings, or modular components.

Manufacturing Plant - The location other than the building site at which modular buildings, modular components, modules, or panels are assembled or manufactured prior to transport to the final construction site.

Manufacturing Plant (Primary) - A facility that completes the construction or fabrication of a modular building, modular component or module that was begun in the subsidiary manufacturing plant.

Manufacturing Plant (Subsidiary) - A facility that constructs or fabricates portions of a modular building, modular component, or module before it is moved to the primary manufacturing plant for completion.

Modular Building - Industrialized housing and buildings.

Modular Component - A subassembly, subsystem, or combination of elements, including panelized systems, building shells or bathroom pods, for use as a part of a modular building that is not structurally independent, but is a part of structural, plumbing, mechanical, electrical, fire protection or other systems affecting life safety.

Module - A three-dimensional, volumetric section of a modular building designed and approved to be transported as a single section, independent of other sections, to a site for on-site construction.

Nonsite-Specific Building - A building for which the permanent site location is unknown at the time of construction.

Off-Site Construction - A modular building, modular component or panelized system that is designed and constructed in compliance with this standard and is wholly or in substantial part fabricated or assembled in manufacturing plants for installation—or assembly and installation—on a separate building site and has been manufactured in such a manner that all parts or processes cannot be inspected at the installation site without disassembly, damage to, or destruction thereof.

On-Site Construction - Preparation of the site, foundation construction, construction of the supporting structure, assembly, and connection of off-site or open construction elements and completion of site-related construction in accordance with the construction documents and details.

Open Construction - A modular building, modular component or panelized system manufactured in such a manner that all portions can be readily inspected at the building site without disassembly, damage, or destruction thereof.

Ordinary Repairs - The removal and replacement of existing materials, elements, equipment, or fixtures using like or the same new materials, elements, equipment, or fixtures that serve the same purpose.

Panel - A distinct, sectional element in a panelized system.

Panelized System - Wall, roof, or floor components that are constructed at a location other than the building site in a manner that prevents the construction from being inspected at the building site without disassembly, damage, or destruction thereof.

Permanent Foundation System - A foundation system for modular buildings designed to meet the applicable building code. In a permanent foundation system, anchorage of the building to the foundation is provided to resist the uplift and sliding forces that result from the application of the prescribed loads. The use of ground anchors or earth augers to resist uplift or sliding forces is not considered a permanent foundation.

Person - An individual, partnership, company, corporation, association, or any other legal entity, however organized.

Public - The people of the state, including individuals, companies, corporations, associations or other groups, however organized, and governmental agencies.

Quality Control - Controls and inspections implemented by the manufacturer, as applicable, to ensure the material provided and work performed meet the requirements of the approved construction documents and applicable building codes.

Registered Design Professional - An individual who is registered or licensed to practice their design profession as defined by the statutory requirements of the professional registration laws of the state or jurisdiction in which the project is to be constructed. Must be in good standing in the state of Texas.

Registrant - A person who is registered in accordance with this standard or the applicable states' statutory requirements as a manufacturer, an industrialized installer, industrialized retailer, a design review agency, a third-party inspection agency, a third-party inspector, or a third-party site inspector.

Station - A defined area within a manufacturing plant in which one or more work activities and inspections have been assigned to take place.

Third-Party Inspection Agency - An approved person or entity determined by this standard or applicable states statutory requirements to be qualified by reason of facilities, personnel, experience, demonstrated reliability and independence of judgment to inspect industrialized housing, buildings, and portions thereof for compliance with the construction documents, compliance control program and applicable codes.

Third-Party Inspector - An approved person determined by applicable statutory requirements to be qualified by reason of experience, demonstrated reliability and independence of judgment to inspect modular buildings, and portions thereof, for compliance with the construction documents, compliance control program and applicable building code. A third-party inspector works under the direction of a third-party inspection agency.

Third-Party Site Inspector - An approved person determined by applicable statutory requirements to be qualified by reason of experience, demonstrated reliability and independence of judgment to inspect construction of the foundation and installation of modular buildings, and portions thereof, for compliance with the construction documents and the applicable code.

Unique On-Site Construction Details - Construction details that are not part of, or that differ from, the manufacturer's approved on-site construction details. Unique on-site construction details include additions that affect the code compliance of the house or building such as car ports, garages, porches, decks, and stairs.

3.2 Design (Planning, Design, Fabrication and Assembly)

3.2.1 General

3.2.1.1 Application

Design of off-site construction shall be in accordance with the provisions of the applicable codes and standards adopted by the authority having jurisdiction. The requirements in this standard shall be in addition to the requirements of the applicable codes and standards and all standards, performance requirements and design in the contract documents.

3.2.1.1.1 Seal & Signature of Architect or Engineer

Design/data shall bear the seal and signature of a professional architect or engineer, registered in Texas, attesting that the structures meet the requirements of the specification.

3.2.1.1.2 Design Parameters

The design and manufacture of off-site modules and modular components shall be coordinated with site utility services, accessibility requirements, vehicular and pedestrian access, mechanical, electrical, plumbing, fire protection requirements, interior construction and finishes, and such other items as may be necessary for a complete, functional building.

3.2.1.1.3 Design of Products for Transportation

The design of modules and manufactured products shall be adequate for transport and shall meet the shipping standards for both the federal highway administration (FHWA) shipping standards and the Texas Department of Transportation.

3.2.1.2 Alternative Materials, Design and Methods of Construction and Equipment

The provisions of this standard are not intended to prevent the use of alternate materials and methods per IBC Section 104.11.

3.2.1.3 Transportation

Transportation considerations shall be in accordance with Transportation and Storage standards.

3.2.2 Fire and Smoke Protection Features

3.2.2.1 Scope

The provisions of this chapter shall govern the materials, systems, and assemblies used for structural fire resistance and fire-resistance-rated construction separation of adjacent spaces to safeguard against the spread of fire and smoke within a building and the spread of fire to or from buildings for off-site construction.

3.2.2.2 Application

Design and installation of fire and smoke protection features shall be in accordance with the provisions of the applicable codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.2.3 Construction Documents

Construction documents shall be submitted to the authority having jurisdiction and shall include:

1. How vertical continuity of shaft enclosures and other vertical openings are being provided, where required by applicable codes and standards.
2. A schedule or drawings of fire-resistant joint systems, where required by applicable codes and standards.
3. How fireblocking is provided, where required by applicable codes and standards.
4. How draftstopping is provided, where required by applicable codes and standards.
5. Delineation of off-site and on-site portions of structural fire resistance and fire-resistance-rated construction.
6. Where load-bearing walls are provided, the construction documents shall include the structural load paths and must demonstrate continuous load paths for the full height of each load-bearing wall.

3.2.2.4 Inspections

Inspections shall be in accordance with applicable codes and standards, including ICC/MBI 1205.

3.2.2.4.1 Certified Firestop Inspection

1. A 3rd- party certified Firestop inspection is required for all firestop system applications, including, but not limited to, assemblies, membrane penetrations, annular spaces, and rated openings. Third party inspection documentation is needed to be uploaded to DFW permit portal prior to final inspection. Cementitious annular fill shall be field verified. Collapsible pipe collars shall have their UL listed documentation uploaded to the permit portal. All existing penetrations, in the project area, shall be sealed and inspected.
2. All located existing assembly penetrations (fire, water, etc.), that fall in the footprint of the current project, shall be the responsibility of the current contractor to have sealed and inspected prior to the final inspections.
3. Provide material documentation for spray on fire resistant material (SFRM) of the floor/deck/ceiling/structural members to DFW Code permit portal prior to Building Final.

4. Documentation for “Third Party” certifications for SFRM and/or spray in insulation shall be uploaded to DFW Code permit portal prior to Building Final.

3.2.2.5 Secondary Attachments to Structural Members

Where primary and secondary members require fire protection, secondary attachments to those structural members shall be protected with the same fire-resistive material and thickness as required for the structural member. The protection shall extend away from the structural member a distance of not less than 12 inches or shall be applied to the entire length where the attachment is less than 12 inches long. Where an attachment is hollow and the ends are open, the fire-resistive material and thickness shall be applied to both the exterior and interior of the hollow attachment.

Exception: Secondary attachments that are protected by different materials or thicknesses to the structural member shall be approved by the authority having jurisdiction.

3.2.3 Mechanical

3.2.3.1 Scope

The provisions of this section shall govern the design and installation of mechanical and fuel gas systems for off-site construction.

3.2.3.2 Application

Design and installation of mechanical and fuel gas systems shall be in accordance with the provisions of the mechanical and fuel gas codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.3.2.1 Rack & Riser Systems

Rack and riser systems manufactured offsite:

1. Shall be designed to support the required loads, including the operational weight of the racked materials.
2. Shall be designed to meet seismic design requirements.
3. Shall be designed for adequate clearance for installation, access for maintenance, and access for inspection.
4. Shall be designed to isolate components from vibration and to prevent the transmission of noise and vibration.
5. Must be able to be supported by the structure.
6. Must be designed with the required clearance from other equipment and structures to meet fire and safety codes.

3.2.3.2.2 Skid-Mounted Equipment

Skid-mounted equipment manufactured offsite:

1. Shall be designed to meet relevant safety regulations, such as OSHA and NFPA.
2. Shall be designed in a manner to provide easy access to all parts of the equipment for maintenance and repair.
3. Skid shall be capable of supporting the weight of the equipment, plus any additional loads such as personnel or materials.

3.2.3.3 Construction Documents

The design documentation shall describe and delineate the portions of the systems that are to be constructed off-site from those that are to be constructed on-site and identify the registered design professional responsible for off-site and on-site elements.

3.2.4 Electrical

3.2.4.1 Scope

The provisions of this section shall govern the design and installation of electrical systems for off-site construction.

3.2.4.2 Application

Design and installation of electrical systems shall be in accordance with the provisions of the electrical codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.4.2.1 Rack & Riser Systems

Rack and riser systems manufactured offsite:

1. Shall be designed to support the required loads, including the operational weight of the racked materials.
2. Shall be designed to meet seismic design requirements.
3. Shall be designed for adequate clearance for installation, access for maintenance, and access for inspection.
4. Shall be designed to isolate components from vibration and to prevent the transmission of noise and vibration.
5. Must be able to be supported by the structure.
6. Must be designed with the required clearances from other equipment and structures to meet fire and safety codes.

3.2.4.2.2 Skid-Mounted Equipment

Skid-mounted equipment manufactured offsite:

1. Shall be designed to meet relevant safety regulations, such as OSHA and NFPA.
2. Shall be designed in a manner to provide easy access to all parts of the equipment for maintenance and repair.
3. Shall be capable of supporting the weight of the equipment, plus any additional loads such as personnel or materials.

3.2.4.3 Construction Documents

The design documentation shall describe and delineate the portions of the systems that are to be constructed off-site from those that are to be constructed on-site and identify the registered design professional responsible for off-site and on-site elements.

3.2.5 Plumbing

3.2.5.1 Scope

The provisions of this section shall govern the design and installation of plumbing systems for off-site construction.

3.2.5.2 Application

Design and installation of plumbing systems shall be in accordance with the provisions of the International Plumbing Code and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.5.2.1 Rack & Riser Systems

Rack and riser systems manufactured offsite:

1. Shall be designed to support the required loads, including the operational weight of the racked materials.
2. Shall be designed to meet seismic design requirements.
3. Shall be designed for adequate clearance for installation, access for maintenance, and access for inspection.
4. Shall be designed to isolate components from vibration and to prevent the transmission of noise and vibration.
5. Must be able to be supported by the structure.
6. Must be designed with the required clearance from other equipment and structures to meet fire and safety codes.

3.2.5.2.2 Skid-Mounted Equipment

Skid-mounted equipment manufactured offsite:

1. Shall be designed to meet relevant safety regulations, such as OSHA and NFPA.
2. Shall be designed in a manner to provide easy access to all parts of the equipment for maintenance and repair.
3. Shall be capable of supporting the weight of the equipment, plus any additional loads such as personnel or materials.

3.2.5.3 Construction Documents

The design documentation shall describe and delineate the portions of the systems that are to be constructed off-site from those that are to be constructed on-site and identify the registered design professional responsible for off-site and on-site elements.

3.2.6 Fire Protection and Life Safety Systems

3.2.6.1 Scope

The provisions of this section shall govern the design, installation, and operation of fire protection systems for off-site construction.

3.2.6.2 Application

Design and installation of fire protection systems shall be in accordance with the provisions of the fire protection codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.6.2.1 Skid-Mounted Equipment

Skid-mounted equipment manufactured offsite:

1. Shall be designed to meet relevant safety regulations, such as OSHA and NFPA.
2. Shall be designed in a manner to provide easy access to all parts of the equipment for maintenance and repair.

3. Shall be capable of supporting the weight of the equipment, plus any additional loads such as personnel or materials.

3.2.6.3 Construction Documents

The design documentation shall describe and delineate the portions of the systems that are to be constructed off-site from those that are to be constructed on-site and identify the registered design professional responsible for off-site and on-site elements.

3.2.7 Exterior Walls

3.2.7.1 Scope

The provisions of this chapter shall govern the minimum requirements for exterior walls, exterior wall coverings, exterior wall openings, exterior windows and doors, and architectural trim.

3.2.7.2 Application

The requirements for exterior walls shall be in accordance with the provisions of the codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.7.3 Exterior Walls

Exterior walls shall comply with requirements for system continuity.

3.2.7.3.1 Fire Resistance of Systems

Walls shall be properly constructed or sealed to ensure continuity of fire-resistive construction.

3.2.7.3.2 Control Layers

Where applicable, air, vapor and weather barrier systems shall be designed and installed to provide continuity of the barrier.

3.2.7.3.3 Continuity of Exterior Insulation

Exterior insulation systems shall be designed and installed to provide continuity of the insulation.

3.2.8 Wall Assemblies

3.2.8.1 Scope

The provisions of this chapter shall govern the minimum requirements for wall assemblies, prefabricated or stick-built that are constructed off-site and assembled on-site.

3.2.8.2 Application

The requirements for exterior walls shall be in accordance with the provisions of the codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.8.3 Design Criteria

1. Shall be free-standing, floor supported with perimeter fascia support.
2. Shall have the capability to incorporate ceiling cross beams to support ceiling grid, tiles, lighting, and mechanical systems.
3. Shall be able to adjust/level height up to 3 inches.

4. Shall have the capability to adjust decrease/increase width dimensions in 1/16 inch increments as required by the facility.
5. Shall have the capability to add a stud stacker for future height adjustment as required by the facility.
6. Shall be comprised of super studs and components which can be field cut (if required).
7. Shall provide a pre-engineered electrical system equipped with a 25 foot power whip, factory- mounted rough-in boxes, face plates, and cladding cutouts.
8. Shall provide faceplates and cladding cutouts for rough in boxes containing combination power and data.
9. Shall have a limited 10 year warranty.
10. Shall accommodate and provide provisions for additional embedded elements (wayfinding, security systems, touchless entry mechanism, millwork, plumbing fixtures, etc..) within the wall cavity.
11. Shall provide a seamless installation designed to install up to 3 times faster than conventional construction.
12. Shall provide a clean installation process omitting the need to seal off areas from dust and debris.
13. Shall be able to contribute to LEED v4 – Building Design and Construction.
14. Shall have a 4 week manufacturer lead time after receipt of approved drawings.
15. Shall be designed utilizing industry-standard software, Revit / cloud base software for key stakeholder digital design collaboration, convenience, and ease.
16. Coordinate shipping to optimize the ease of staging and ensure the best cost options. Knock-down costs.
17. Shall be designed to meet the performance criteria as per the contract documents – including but not limited to fire rating, acoustical, deflection, uniformity, surface burning, seismic, etc. Coordinate openings and required loading criteria.

3.2.8.4 Execution

3.2.8.4.1 Installers

Manufacturer of off-site construction building system, or manufacturer approved and trained installer.

3.2.8.4.2 Examination

1. Verify locations of conceal construction for support and anchorage.
2. Verify that openings are plumb, level, and square.
3. Verify that floor and ceiling surfaces are in place.

3.2.8.4.3 Installation

3.2.8.4.3.1 General

1. Comply with manufacturer's installation instructions.
2. Do not cut metal components except where trimming is indicated on Shop Drawings.
3. Install system without gaps at joints with other construction.

3.2.8.4.3.2 Framing

1. Install framing plumb, accurately aligned, and free of warp or twist.
2. Install components with securely fastened full-contact joints.
3. Anchor framing system rigidly and securely to adjacent construction without damaging surfaces.

4. Install perimeter gaskets without gaps to provide continuous light and acoustical seals.

3.2.8.4.3.3 Opaque Panels

1. Install panels per factory-numbered sequence.

3.2.8.4.3.4 Glazing

1. Install glass panels per factory-numbered sequence.
2. Install glass on resilient setting blocks in glazing channels.
3. Install glazing gaskets with joints only at corners and to provide continuous barrier to air and sound.
4. Install glass panels with open vertical joints of uniform width.
5. Double Glazing: Clean surfaces that will be inaccessible after installation in framing.

3.2.8.4.3.5 Swinging Doors

1. Install doors with uniform jamb and head clearance.
2. Adjust doors for smooth, accurate operation and secure latching.
3. Adjust door closers to comply with ADA Standards and Texas Accessibility Standards.

3.2.8.4.3.6 Sliding Doors

1. Align track for smooth, quiet operation.
1. Adjust end stops for accurate closed and fully open positions.

3.2.8.4.3.7 Systems Integration

1. Coordinate wiring connections.

3.2.9 Structural Design

3.2.9.1 Scope

The provisions of this chapter shall govern the minimum requirements for structural design. All structural designs require a Texas registered PE stamp with a signature.

3.2.9.2 Application

Structural design shall be in accordance with the provisions of the codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.2.9.3 Special Requirements for Modules

Modules shall comply with the requirements of Section 3.4.2.

3.2.9.4 Tolerances for Panelized System Construction

Tolerances for the attachment of panelized systems to other systems and interfacing construction shall be as established by the manufacturer of the attachment device.

3.2.10 Referenced Standards

The specific year, date, and editions of the code listed below serve as a reference; however, for accuracy and completeness, it's essential always to use the most current version of the code.

1. ASCE 7 - with Supplement 1 Minimum Design Loads and Associated Criteria for Buildings and Other Structures

2. IBC - International Building Code 101.3, 103.1, 202, 301.2
3. ICC/MBI - 1205 Standard for Off-Site Construction: Inspection and Regulatory Compliance 104.1, 302.4, 503.1
4. ISO 6707-1 - Buildings and Civil Engineering Works – Vocabulary 803.1
5. NFPA 221- Standard for High Challenge Fire Walls, Fire Walls, and Fire Barrier Walls 302.3

3.3 Special Requirements Based on Product Type (Planning, Design, Fabrication and Assembly)

3.3.1 General

3.3.1.1 Scope

The provisions of this chapter shall govern special building construction.

3.3.2 Modules

3.3.2.1 Special Requirements for Structural Design

The following information shall be applicable to construction for the structural design.

3.3.2.1.1 Stiffness

Modules shall incorporate stiffness to limit deflections, lateral drift where applicable, vibration or any other deformations that adversely affect the intended use and performance of the module or building. Modules shall have the strength and stiffness for temporary conditions such as transportation, hoisting and erection and permanent conditions.

3.3.2.2 Unit to Unit Interconnection

The interconnection points of modules to one another shall have the strength to transfer the necessary design forces and stiffness to enable the collection of modules to behave as a complete building structure.

3.3.2.3 Unit to Site-Built Connection

The connection of modules to site-built components shall have the strength to transfer the necessary design forces.

3.3.2.4 Diaphragm and Shear Walls

Analysis and design of the completed building shall account for the effect of discontinuities in the diaphragm and shear walls between modules.

3.3.2.5 Performance Criteria

1. Provide tolerance requirements by system. Tolerances must be coordinated with tolerances of field construction and must meet industry standards.
Contractor/manufacturer shall design the modules with construction and installation tolerances for fitting each module to each other and to the support structure.
2. All deflection requirements of all materials and systems within prefabricated module shall be identified and not exceeded during off-site fabrication during manufacturing, transportation, storage, and installation.
3. Provide access and maintenance requirements.
4. Define requirements of Construction Documents and Submittals - i.e., minimum drawings, calculations, sequence, and temporary conditions

5. Temporary Supports and Shoring - Manufacturer to provide details where required to retain integrity of off-site fabrication during manufacturing, transportation, storage, and installation.
6. Building systems shall meet requirements, applicable codes and standards and Airport system specification and requirements.

3.4 Manufacturing Plant (Planning, Design, Fabrication and Assembly)

3.4.1 General

Off-site construction and manufacturing plants shall comply with the provisions of this chapter.

3.4.2 Location

Where facilities used to manufacture or construct modules or modular components in accordance with this standard are located directly on the work site where the proposed modules, panelized systems or modular components are to be installed, they shall not be considered off-site construction, and this standard shall not apply.

3.4.2.1 On-site Panelized System Fabrication

Where an operation specifically intended for the fabrication of panelized systems is not located separately from the structure under construction, the operation shall not be considered off-site construction, and this standard shall not apply.

Exception: Where the proposed modules, panelized system or modular components are utilized as a portion of the manufacturing plant.

3.4.3 QA/QC Plan

3.4.3.1 QA/QC Plan Requirement

Each manufacturing plant shall have an approved QA/QC plan in accordance with Chapter 5 of ICC 1205 prior to commencing fabrication or construction activities. The quality control plan shall include both inspections and testing to ensure that the manufactured product meets the specification and design requirements of the project and complies with relevant building codes. Quality Control program shall include shop verification and pre-installation verification of compliance to all tolerances and deflection criteria.

3.4.3.2 QA/QC Plan Availability

Each manufacturing plant shall have a printed copy of the approved QA/QC plan available for inspection at reasonable times without prior announcement.

3.4.4 QA/QC Process

3.4.4.1 QA/QC Personnel

Each manufacturing plant shall identify a responsible party that shall implement the QA/QC processes within the facility. The person responsible for the quality program shall have the authority necessary to ensure compliance with this standard. Quality program personnel shall demonstrate to the inspection agency that they have adequate knowledge of the product, factory operations, and the codes and standards to which the product is being manufactured and shall also demonstrate the ability to perform their required duties. QA/QC personnel must also demonstrate that the manufactured product meets the design specifications and quality requirements and is free from defects.

3.4.4.2 QA/QC Frequency

Observation of the manufacturing and construction activities shall be performed as often as is necessary to ensure compliance with the QA/QC plan.

3.4.5 QA/QC Records

The QA/QC records shall comply with the requirements of this section.

3.4.5.1 Documentation Preparation

The party responsible for implementing the QA/QC plan shall prepare verification documents in accordance with the approved QA/QC plan.

3.4.5.2 Documentation Availability

All documentation required by the QA/QC plan shall be made available to the AHJ upon request.

3.4.5.3 Documentation Retention

All documentation required by the QA/QC plan shall be retained by the manufacturer for not less than 12 months after the delivery of the module, panel, or component.

Exception: Where the approved QA/QC plan indicates a document retention duration more than 12 months.

3.4.5.4 Availability

The manufacturing facility shall not restrict access by the AHJ, or their authorized representative, at any time when manufacturing or construction activities are occurring.

3.4.5.5 Notification

The manufacturing facility shall notify the AHJ, or their authorized representative, prior to the commencement of fabrication or construction projects in accordance with the approved QA/QC plan.

3.4.6 Sustainability**3.4.6.1 Materials Protection**

All building materials shall be protected from weather, moisture, and contaminated sources in accordance with the material manufacturer's recommendations.

**3.5 Fabrication
(Planning, Design, Fabrication and Assembly)****3.5.1 General**

Off-site construction and manufacturing of modules, panelized systems or modular components shall be in accordance with the approved plans and provisions of the applicable codes and standards adopted by the authority having jurisdiction. The requirements in this section shall be in addition to the requirements of the applicable codes and standards.

3.5.2 Volumetric Construction

Each modular building or module shall be identified where it individually or in combination forms structural components. Attention to the connections and erection of the unit or sections that form the structure as well as the lateral force-resistance system shall be identified in addition to the connections to the foundation.

3.5.3 Deformation Compatibility

Materials expected to be subjected to deformation during the fabrication process shall comply with the requirements of this section.

3.5.3.1 Design

Buildings and components shall be designed and detailed for deformation compatibility where required. These shall include interior and exterior non-bearing, non-shear walls; structural separation walls; and interconnected modules – diaphragms, chords, and collectors.

3.5.4 Temporary Supports and Shoring

Where required by the AHJ, the designer shall provide the details where required to retain the same vertical support arrangement throughout and preserve the integrity of the structure during manufacture, transportation, and on-site storage.

3.5.5 Off-Site Storage

3.5.5.1 Stacking

The manufacturer shall provide details to consider additional vertical or horizontal loading on modules while stacked in temporary storage. These details shall include measures prescribing maximum stacking of modules.

3.5.5.2 Weather Protection

The manufacturer shall take precautions to protect stacked and stored modules from weather events.

3.6 Transportation and Storage (Planning, Design, Fabrication and Assembly)

3.6.1 General

Transportation of units that have been manufactured or constructed off-site shall comply with the provisions of this chapter. With each set of units that is to be installed in the same manner, the manufacturer shall provide instructions that include transport, lifting and placement procedures where applicable.

3.6.1.1 Packaging

Package and deliver modular components, sheets, metal panels, and other manufactured items so as not to be damaged or deformed and protected during transportation and handling.

3.6.1.2 Staging

Units shall be shipped for ease of project staging.

3.6.1.3 Cost

Units shall be shipped to ensure the most economical shipping cost.

3.6.2 Transportation Permitting

Transportation of modules, panelized systems or modular components that have been manufactured or constructed off-site shall be in accordance with the applicable transportation permitting requirements for each jurisdiction that the shipment will enter.

3.6.2.1 Transportation Route

The route used to transport modules, panelized systems or modular components that have been manufactured or constructed off-site shall be of a width and height to accommodate the load and the transportation vehicle.

3.6.2.2 Transportation Weight

The gross weight of the modules, panelized systems, or modular components to be transported and the transportation vehicle shall not exceed the permitted requirements for each jurisdiction that the shipment will enter.

3.6.3 Methods of Transport

Modules, panelized systems, or modular components that have been manufactured or constructed off-site shall be transported using one or more of the following methods:

1. An integrated chassis with axles and wheels suitable to support the weight and size of the object being transported.
2. A separate trailer suitable to support the weight and size of the object being transported.
3. Other transportation systems that are accepted by the applicable jurisdictions and are suitable to support the weight and size of the object being transported.

3.6.4 Loading and Unloading

The manufacturer shall create with the erector a schedule that limits the need for storage and a loading and unloading manifest that protects the modules from stress and mechanical damage. The logistics for unloading the staging processes and the type of crane required for lifting the elements or modules are to be detailed by the erecting contractor.

3.6.4.1 Erection Procedure

Provide a written plan detailing the erection/lifting procedure with required plans and calculations clearly showing the intended sequence and method of erection in accordance with CFR 1926.752 - Site Layout, Site-Specific Erection Plan, and Construction Sequence.

3.6.5 On-Site Storage of Modules**3.6.5.1 On-site Storage**

To minimize the number of modules to be stored, the erection contractor shall designate a staging area on-site.

3.6.5.2 Stacking

The erection contractor shall utilize stacking details provided by the manufacturer to counter additional vertical or horizontal loading on the modules while stacked in temporary storage. These details shall limit the maximum stacking of modules.

3.6.5.3 Weather and Mechanical Protection

The erection contractor shall take all steps to protect stacked and stored modules from weather events as well as mechanical damage.

3.6.5.3.1 Protection of Materials

Cover with suitable weather-tight and ventilated covering to ensure dryness, with positive slope for drainage of water and controlled humidity. Store in a manner to prevent bending, warping, twisting, and surface damage. Do not store wall panels in contact with other materials that might cause denting, staining, or other surface damage. Retain strippable protective covering on panels for entire period up to panel installations.

3.6.5.3.2 Sun Exposure

Do not expose to sunlight, except as necessary for installation and concealment. Always protect against ignition. Do not deliver foam-plastic insulation materials to the project site before installation time.

3.6.5.3.3 Ignition

Always protect against ignition. Do not deliver foam-plastic insulation materials to the project site before installation time.

3.6.5.4 Staging

The length of time any modules are staged shall be scheduled to minimize the length of time they are on-site.

**3.7 On-Site Installation
(Planning, Design, Fabrication and Assembly)****3.7.1 General**

The manufacturer shall provide instructions that include installation procedures and layout information.

3.7.2 Foundation

A separate plan, prepared where required by a registered design professional, shall be submitted for permit.

3.7.3 Loads

The foundation shall be designed to support the building, all live and dead loads, and all construction loads. The foundation shall be designed to consider all geotechnical limits placed on the building and foundation at the site.

3.7.4 Installation Tolerances

The registered design professional shall detail in the construction documents the required construction tolerances for fitting each module to each module and to the foundation. Terms shall be as defined in ISO 6707-1.

3.7.5 Connections

Structural details shall be generated in anticipation of the unique connections and sequences required for modular construction, including locations for temporary connections and transportation loading.

3.8 Referenced Standards (Planning, Design, Fabrication and Assembly)

3.8.1 Referenced Standards

This chapter lists the standards that are referenced in various sections of this document. The standards are listed herein by the promulgating agency of the standard, the standard identification, the effective date and title, and the section or sections of this document that reference the standard. The specific year, date, and editions of the code listed below serve as a reference; however, for accuracy and completeness, it's essential always to use the most current version of the code.

1. ASCE 7- with Supplement 1: Minimum Design Loads and Associated Criteria for Buildings and other Structures 603.2
2. IBC- International Building Code: 101.3, 103.1 202, 301.2
3. ICC/MBI 1205- Standard for Off-Site Construction: Inspection and Regulatory Compliance
4. ISO 6707-1- Buildings and Civil Engineering Works – Vocabulary 803.1
5. NFPA 221- Standard for High Challenge Fire Walls, and Fire Barrier Walls 302.3

3.9 Plan Approvals and Inspection Procedures (Inspection and Regulatory Compliance)

3.9.1 General

3.9.1.1 Scope

The provisions of this chapter shall apply to off-site construction and shall include any on-site connection details and instructions for items installed on site. In order to obtain plan approval, the manufacturer shall be required to provide documentation to the AHJ. Documents shall consist of plans, specifications, calculations, test results, product evaluation reports and critical manufacturer's instruction, and other documentation that describes in detail the product and manufacturing processes employed to produce off-site buildings or components. The documents shall also include plans and details for site-completed portions of the project that require direct interface with the modular completed portions.

3.9.1.2 Remote Virtual Inspections.

The AHJ shall have the option to conduct remote virtual inspections (RVI). Except for unmanned aerial vehicles (drones), the camera shall be operated by the permit holder at the direction of the inspector. The inspector shall have the ability to communicate directly with the camera operator to perform a thorough inspection of the project. The response of the camera operator to the directions of the inspector shall be verified by live video feed. State and local requirements shall be followed as if in-person inspections are being conducted. Except for unmanned aerial vehicles (drones), the camera shall be operated by the permit holder at the direction of the inspector. The inspector shall have the ability to communicate directly with the camera operator to perform a thorough inspection of the project. The response of the camera operator to the directions of the inspector shall be verified by live video feed. State and local requirements shall be followed as if in-person inspections are being conducted.

3.9.2 General Requirement for Plan Approval

3.9.2.1 Plans, Specifications, Calculations, and Other Documentation

Where required by the AHJ, construction documents and other documentation shall bear the signature and seal of a registered design professional. If the registered design professional is the same firm as the third-party inspection or design agency, that relationship must be disclosed to the AHJ.

3.9.2.2 Structural Calculations Cover Page

The structural calculations cover page shall be dated, be identified with the project information, include an index, and be signed by a registered design professional.

3.9.2.3 Name and Location

Documents shall indicate the name and location of the manufacturer, registered design professional or other entity submitting the plans.

3.9.2.4 Drawing Cover Sheet

The following shall be included on the drawing cover sheet:

1. List of design firms and professionals used for the project, including the business address and phone number with the listing.
2. An index for the drawing set.
3. The use and occupancy classification.
4. The applicable codes.

3.9.2.5 Insignia of Approval

Drawings shall indicate the location of the insignia of approval.

3.9.2.6 Details, Lists, or Tables

Drawings shall provide details, lists or tables delineating the aspects of the project that are constructed and installed in the factory and in the field, with a clear delineation of scope on the AHJ inspection responsibilities.

3.9.2.7 Note of Revisions

Drawings shall provide a note that any revisions made to the certified unit on site shall be submitted to and inspected by the AHJ to verify code compliance.

3.9.2.8 Space for Stamp of Approval

Each sheet shall include a blank space in the lower right corner for the stamp of approval. The blank space shall not be less than 3 inches wide by 3 inches high.

3.9.2.9 Modular Building Layout and Sequencing Plan

Construction documents shall include a layout plan of modular building and a sequencing plan. The location of cranes shall also be identified.

3.9.3 General Building and Architectural

3.9.3.1 Table, Schedule, or Schematic Identifier

A combination of one or more tables, schedules or schematics shall be provided to identify the following requirements:

1. Use and occupancy classifications
2. Type of construction

3. Building area and number of occupants
4. Building height in feet and stories
5. Methods used to determine height and area increases
6. Fire-rating requirements for the building elements and any minimum separation distance requirements
7. Structural design loads
 - a. Roof live and dead load
 - b. Floor live and dead load
 - c. Design wind speed and exposure
 - d. Seismic design criteria and risk category
 - e. Ground and roof snow load
 - f. Climatic design zones

3.9.3.2 Floor Plan and Elevation

Floor plans and typical elevations shall be provided with dimensions and notations showing compliance with space requirements including room areas, horizontal dimensions, location of space relative to finished grade level, ceiling height and overall building height.

3.9.3.3 Cross Sections

Cross sections and interior elevations shall be provided to identify major building components.

3.9.3.4 Details of All Finished Materials

Details of finished materials shall be provided; for example, flashing material and gauge to be used such as at openings and at penetrations through roofs.

3.9.3.5 Attic Access and Attic Ventilation

Attic access and attic ventilation shall be provided. Where attic access is provided, attic floor loading criteria shall be indicated.

3.9.3.6 Exterior Envelope

Information on exterior envelope, including water-resistive barriers and required fire-resistance-rated assemblies, shall be provided in accordance with the applicable building code.

3.9.3.7 Interior Wall

Information on interior wall and floor/ceiling material, including any required fire-resistance-rated assemblies, shall be provided.

3.9.3.8 Door and Window Schedule

A door and window schedule shall be provided that identifies the following:

1. Dimensions. Example: emergency escape and rescue requirements. Required and provided light and ventilation where applicable.
2. Data satisfying the applicable energy conservation construction code. Required and provided design pressure where applicable.

3.9.4 Building Accessibility

3.9.4.1 General

If accessibility provisions are not a requirement for the project, a statement identifying the code qualifying exception shall be provided.

3.9.4.2 Detail Sheets

Typical detail sheets for required accessible features and facilities for the modular portions of the project shall be provided.

3.9.4.3 Scaled Clearances

Typical scaled clearances visible on plans, in the areas where accessibility is required by the applicable building code, shall be provided.

3.9.5 Fire Safety

3.9.5.1 Fire and Smoke Protection Features

The construction documents shall provide the:

1. Type of construction.
2. Fire-resistance rating of building elements.
3. Fire-resistance-rated assemblies.
4. Fire-resistant joint systems.
5. Through-penetration firestop systems.
6. Methods of determining fire resistance, such as listed assemblies, test reports and engineering analysis.
7. Information on how to meet the requirements for the concealed spaces created between off-site components.

3.9.5.2 Fire-Resistance-Rated Assemblies

A separate sheet that shows detail for fire-resistance-rated assemblies, including reference to listing, test report, engineering analysis or methodology used to determine the fire-resistance rating, shall be provided, and shall include the following:

1. Applicable rated assemblies detailed including separation from unit to unit, unit to common spaces, shafts, plumbing, HVAC, stair, and elevator.
2. Plans shall identify applicable fire-resistance-rated locations.

3.9.5.3 Life Safety Plan

Where required by the AHJ, a life safety plan to describe the means of egress, including details that describe aisles, exits, corridors, passageways, travel distance and stairway enclosures, shall be provided.

3.9.5.4 Flame Spread and Smoke-Developed Classifications

Flame spread and smoke-developed classifications for interior finish materials shall be provided.

3.9.5.5 Draft Stops and Fire Blocking

Locations of required draft stops and fire blocking, including the identification of how provisions are satisfied, shall be provided.

3.9.5.6 Opening Protectives in Fire-Resistance-Rated Systems and Assemblies

Details for opening protectives in fire-resistance-rated systems and assemblies, including reference to listings for required doors, windows, frames, and hardware to complete the opening protective requirements, shall be provided.

3.9.5.7 Fire Protection Systems

Drawings for fire protection systems, including sprinkler systems, standpipes, fire alarms, and detection systems where required in modular portions of the project, shall be provided.

3.9.6 Mechanical Systems**3.9.6.1 Equipment Location, Size and Material Specifications.**

The location, size and material specifications for equipment and components provided by the manufacturer as part of the factory-installed portion of the project shall be provided, including electric heating systems; hydronic heating systems; heating, ventilating and air-conditioning systems; water reclamation systems; and appliances.

3.9.6.2 Heat Loss Calculation

Where required by the applicable building code, a heat loss calculation shall be provided.

3.9.6.3 Ratings and Manufacturer's Listings

Input/output ratings and manufacturer's listing requirements shall be provided for equipment and appliances provided by the manufacturer as part of the factory-installed portion of the project.

3.9.6.4 Method to Supply Combustion Air

The method to supply combustion air shall be provided where applicable.

3.9.6.5 Ventilation Air

The method for providing ventilation air shall be provided.

3.9.6.6 Makeup Air

The method for providing makeup air shall be provided where applicable.

3.9.6.7 Flues, Vents and Chimneys

The location of flues, vents and chimneys shall be provided, including the clearances from air intakes, combustible materials, and other vents and flues, where applicable.

3.9.6.8 Manufacturer's Data

Manufacturer's data, instructions, and listing information for the installation of fuel-burning equipment shall be provided including specific details on the drawings, where applicable.

3.9.7 Plumbing and Gas Systems**3.9.7.1 Drawings of the Plumbing System**

Drawings of the plumbing system, Drain-Waste-Vent System (DWV) and water lines shall be provided, including the size of piping, fitting locations, traps, vents, cleanouts, and valves.

3.9.7.2 Site-Installed Plumbing

All plumbing on drawings to be site installed shall be identified.

3.9.7.3 Plumbing Materials

Plumbing materials shall be identified.

3.9.7.4 Locations of all Equipment

The locations of all equipment, appliances, and safety controls to be installed shall be shown.

3.9.7.5 Gas Line Sizing

Gas line sizing information, including developed length and pressure rating, shall be shown.

3.9.8 Electrical System**3.9.8.1 General**

The equipment manufacturer's instructions shall be provided as part of the factory-installed components of the project.

3.9.8.2 Load Calculations for Service and Feeders

Load calculations for service and feeders shall be provided.

3.9.8.3 Main Disconnect and Overcurrent Protective Devices

The size, rating, and location of the main disconnect and overcurrent protective devices shall be shown and identified.

3.9.8.4 Locations of Outlets, Junction Boxes, Fixtures and Appliances

The locations of outlets, junction boxes, fixtures, and appliances, including required locations of GFCI protection, shall be shown, and identified.

3.9.8.5 Lighting Plan

A lighting plan shall be provided. Exterior and interior lighting locations shall be shown. Required smoke detecting alarm device locations and circuitry shall be identified.

3.9.8.6 Wire Type and Size

The wire type and size shall be provided.

3.9.9 Energy Conversation System**3.9.9.1 Compliance with Mandatory Energy Code Requirements**

Details and calculations to show compliance with mandatory energy code requirements shall be provided.

3.9.9.2 Building Thermal Envelope

Drawings and details of materials and assemblies that describe the building envelope shall be provided.

3.9.10 Structure**3.9.10.1 Load Listings**

1. Construction documents shall include the information required by IBC Section 1603. Provide certified engineering calculations for the following products: Roof and wall wind loads with basic wind speed, exposure category, co-efficient, importance factor, designate type of facility, negative pressures for each zone, method, and requirements of attachment: Roof Dead and Live Loads.

2. Collateral Loads.
3. Foundation Loads
4. Roof Snow and Ground Snow Loads
5. Seismic Loads and Assumed Soil Site Classification
6. Risk Category
7. Floor Live Load
8. Earthquake Design Data
9. Geotechnical Information
10. Flood Design Data
11. Special Loads

3.9.10.2 Engineering Analysis

An engineering analysis for design loads to support the selection of all structural members and connections in compliance with applicable codes shall be provided. The contractor shall supply information related to the design requirements in the construction phase, including temporary loadings and considerations for incomplete structures.

3.9.10.3 Design Calculations

Design calculations shall:

1. Identify reference standards and/or code tables, and present design methodology in a step-by-step review-able format.
2. Include applicable design loads and load paths.
3. Demonstrate compliance with the loads required by the building code adopted by the AHJ.
4. If the design is using a prescriptive method of design, clearly identify the source utilized.

3.9.10.4 Specifications of Materials

Materials shall comply with the material requirements of the IBC, IRC, or adopted codes as applicable, and shall provide information to enable determination of design strengths.

3.9.10.4.1 Details of Structural Elements and Subsystems

Details of structural elements and subsystems, including framing details, spacing, size, connections, and fasteners, shall be provided. This includes:

1. Engineered roof details.
2. Truss specifications data sheet.
3. Truss or rafter uplift connections.
4. Attachment of roof coverings.
5. Braced wall lines identified on the drawings.
6. Specifications for shear walls and diaphragms.
7. Structural adequacy of shear walls and diaphragms confirmed.
8. Locations and specifications for hold-down devices.
9. Column and post schedule.
10. Header and beam schedule.
11. Continuity of uplift resistance confirmed.
12. Details of module interconnection.
13. Windborne debris protection, as applicable.
14. Floor details.
15. Connections of all framing members.

3.9.10.4.2 Blocking Plan and Foundation Loads Imposed by Modules

The resulting uniform and concentrated load vectors for magnitude and direction imposed by the modules on the suggested foundation plans shall be provided. All loads, both gravity and lateral loads due to wind and seismic, shall be identified.

3.9.10.4.3 Foundation Plans

Where required by the AHJ, foundation plans and details of connections of modules to foundations shall be provided by a registered design professional. This information includes:

1. Allowable soil bearing pressure.
2. Minimum concrete strength required for specified anchoring devices.
3. Locations and specifications for anchoring devices.
4. The actual gravity, lateral and uplift loads.

3.9.11 Connections**3.9.11.1 On-site Connections**

On-site connections, including interconnection details and instructions and connections from site-built elements to modules or panels, shall be provided.

3.9.11.2 Information about On-Site Connections

Construction documents shall include details of the following types of connections:

1. Interconnections between off-site components at site.
2. Connections and attachments between off-site components and site-built building assemblies.
3. Connections between all construction components connected and completed at the site, including fire-rated architectural elements such as walls; floors; ceiling; roofs; and structural/nonstructural, mechanical, plumbing, electrical, communications, fire protection, technology and HVAC systems and assemblies.

3.9.11.3 Manufacturer's Instructions

On-site connections shall follow approved manufacturer's instructions.

3.9.12 Inspection Procedures**3.9.12.1 General**

Typical inspections for projects with off-site components include in-plant and on-site.

3.9.12.2 Responsible Parties

Documentation identifying individuals or parties responsible for inspection of components shall be identified and provided prior to an inspection process.

3.9.12.3 In-Plant Inspections

In-plant inspections that are conducted by employees of a third-party inspection agency shall verify that constructions comply with the approved construction documents. The third-party inspection agency shall inspect each modular or panelized unit in a phase of construction or inspect modular components to verify that construction is in compliance with the approved construction documents and the approved quality documents.

3.9.12.3.1 Remote Virtual Inspections

The AHJ shall have the option to conduct remote virtual inspections (RVI).

3.10 Third Party Review and Inspection Agencies (Inspection and Regulatory Compliance)

3.10.1 General

3.10.1.1 Third-Party Review and Inspection Agency

Third-party agencies, other than the state, that perform in-plant inspections, evaluate quality control procedures, approve engineering manuals, and approve plans shall be approved by the AHJ. Approval requests shall contain the following requirements as applicable:

1. Name and address of the entity making application.
2. Disciplines for which the third-party agency seeks approval.
3. A list of key personnel indicating their primary functions or duties.
4. The number of years the entity has actively engaged in the business for which it seeks approval.
5. A statement by the evaluation agency that it shall adhere to the requirements of the regulations of the AHJ.
6. A copy of the accreditation certificate, issued by an ISO/IEC 17011-compliant accreditation body, that establishes the ability to provide services in accordance with the ISO/IEC 17065 standard for product evaluation or the ISO/IEC 17020 standard for inspection service, as appropriate.
7. A copy of the registered design professional certification for any employee that shall be responsible for plan review and approval work.
8. A copy of the professional license or plan review certification for all employees that shall be performing plan review. The plan reviewer must only review plan and specific disciplines in the area in which they are certified or licensed.

Exception: Unlicensed plan reviewers shall be permitted to review plans, provided they work directly under the supervision of a professional engineer, architect or certified plan reviewer.

9. A copy of the professional license or certifications for personnel that shall be performing in-plant inspections. The inspector shall only inspect in areas in which they are licensed or certified.
10. A statement that the third-party agency is independent and does not have any actual or potential conflict of interest and is not affiliated with, influenced by, or controlled by any present or potential client manufacturer in any manner that might affect its capacity to render service or reports of findings objectively and without bias.
11. A statement that the third-party agency shall review each design as required for plan approval and identify deficiencies on a corrective action report item by item with citations to the standards. The third-party agency shall verify that deficiencies have been corrected prior to issuing acceptance. The third-party agency shall signify approval of a design by placing its stamp of approval or authorized signature on each page and submit a copy to the AHJ and manufacturer.

12. A statement that the third-party agency shall perform an initial plant certification and document a written certification report to the manufacturer and the AHJ. The initial plant certification shall require 100 percent inspection of the aspects of construction on not fewer than one module or modular component and verify that the manufacturer has a functioning quality assurance process. This inspection shall be made by one or more qualified engineers who have reviewed the approved design and by a qualified inspector. If the initial module or modular component fails to conform with the design or standard, additional modules or modular components shall be inspected until the third-party agency is satisfied that the manufacturer complies to the approved design, the standards, and the quality assurance manual.
13. A statement that for ongoing inspections, the third-party agency shall inspect each module in a phase of construction to verify that the quality assurance process is in compliance with the initial manufacturing plant certification. If the third-party agency finds deficiencies, additional action shall be taken to verify the quality assurance process is in compliance with the initial manufacturing plant certification.
14. A statement that for ongoing inspections, the third-party agency shall determine the frequency of inspection of a modular component or panelized system as necessary to verify the quality assurance process is functioning in compliance with the initial manufacturing plant certification. If the third-party agency finds deficiencies, additional action shall be taken to verify the quality assurance process is in compliance with the initial manufacturing plant certification.
15. The third-party agency shall provide to the manufacturer a written or electronic inspection report after each inspection identifying each deficiency to the standard or approved designs. The third-party agency shall only issue labels after all deficiencies have been corrected.
16. A procedure for maintaining records to control the agency's labels.
17. A procedure to identify all modules or modular components produced and inspections performed.
18. The third-party agency shall have insurance to cover liabilities arising from possible errors and omissions from its operations.
19. The request for approval shall contain the signature of a responsible officer, owner, or partner of the submitting third-party agency.

3.10.1.2 Unlicensed Inspector

An unlicensed inspector shall be considered an inspector in training and must receive a professional license or certification within two years of the date of employment. The third-party inspection agency shall have an inspector training procedure that is to be monitored by the state or the accreditation agency. The training shall be done by a licensed professional or a certified inspector.

3.10.2 Performance Evaluation of Evaluation Agencies

3.10.2.1 General

The state or accreditation agency shall monitor the reliability of each evaluation agency at any reasonable time, with or without prior announcement. Each examination shall investigate the application of evaluation procedures, including construction documents, test results and analysis of compliance assurance programs. Copies of each report shall be sent to the evaluation agency. Evaluation agencies shall be notified of any deficiencies and of the manner in and time by which such deficiencies must be corrected. Not more than 20 days after receiving such notification, the evaluation agency shall provide a plan of corrective action (POCA) to the state or accreditation agency outlining the steps to be taken to correct the deficiency. The state or accreditation agency shall evaluate the POCA, through monitoring, and determine if the POCA corrected the deficiency or if further action shall be necessary.

3.10.3 Quality Assurance Personnel

3.10.3.1 Quality Assurance Administrator

The quality assurance administrator shall be a registered design professional with not less than two years of actual or supervisory experience in modular or component products, and not less than two years of experience in the quality assurance function.

3.10.3.2 Quality Assurance Supervisor

The quality assurance supervisor shall have not less than two years of experience as an inspector in the modular or component industry.

3.11 Requirements for Compliance Assurance Programs (Inspection and Regulatory Compliance)

3.11.1 General

The quality assurance manual shall be submitted to the AHJ on 8 ½ inch × 11 inch paper bound into an indexed booklet or electronic equivalent. The cover shall have the words “Compliance Assurance Program” and identify the manufacturer, inspection, and evaluation agencies.

3.11.1.1 Revision to the Manual

A procedure for revision to the manual shall be outlined.

3.11.1.2 Permission for Inspection

The manufacturer shall provide the state or third-party inspection agency with a signed document that allows the state or third-party inspection agency to inspect the manufacturing plants, products and building sites at a reasonable time without prior announcement.

3.11.1.3 Organizational Chart and Key Personnel

A graphic organizational chart of the company indicating the various departments, including management and production, shall be provided, including a list of key personnel by title in charge of the Compliance Assurance Program and management personnel by title and job description.

3.11.1.4 Manufacturing Plant Layout

A schematic plan of the manufacturing operation showing location and testing stations for mandatory inspections shall be provided.

3.11.1.5 Station-by-Station Description

A detailed list of work performed at each station shall be provided.

3.11.1.6 Inspection Procedure

An outline of inspection and test procedures, including accept/reject criteria and mandatory inspection characteristics, shall be provided. The procedure shall identify the frequency of inspection and how nonconformances are remedied.

3.11.1.7 Inspection Checklist

A detailed list of inspections performed at each station and signoffs shall be provided.

3.11.1.8 Training

A procedure for training the management individuals involved in quality control and employees shall be provided. The procedure should include training to the approved documents.

3.11.1.9 Internal Auditing

A procedure on auditing the quality assurance process and how nonconformances are evaluated and remedied, and a plan of corrective action (POCA) procedure, shall be provided.

3.11.1.10 Construction Documents

A procedure on how documents including revisions are approved shall be provided. A procedure to verify that all construction documents conform to the approved building system shall be provided.

3.11.1.11 Manufacturing Records

A method of keeping manufacturing records, site operations, storage preservation and access to these records shall be outlined identifying length of time records are kept.

3.11.1.12 Serial Numbering

A system to serial-number the modules and modular components shall be outlined.

3.11.1.13 Testing and Inspection Equipment

Procedures for testing and use of inspection equipment shall be provided. This shall include the schedule of maintenance inspection and calibration of the equipment.

3.11.1.14 Storage of Module

A procedure for storage of modules at manufacturing plant, at other storage points and after delivery to site shall be provided.

3.11.1.15 Packing for Shipping

Procedures for packing, packaging, storage, and transportation and related inspections shall be provided.

3.11.1.16 Forms

Forms to be used as part of the quality assurance program and a written procedure on how the forms are used shall be provided.

3.11.2 Material Control**3.11.2.1 Inspection of Materials**

The procedure for the inspection of materials at point of receipt and at the point of use on the production line shall be provided.

3.11.2.2 Protection of Materials

An explanation of how materials are stored for protection from damage and deterioration shall be provided. Materials shall be stored in accordance with the manufacturer's instructions.

3.11.2.3 Rejected Materials

A procedure for the segregation and disposal of rejected materials shall be outlined.

3.11.3 Certification Label**3.11.3.1 General**

A procedure for certification label control and month end reporting shall be provided. The manufacturer shall purchase and apply the certification label in accordance with the requirements of the AHJ.

3.11.3.2 Final Inspection and Labeling

Procedures for final inspection, identification and labeling shall be provided.

**3.12 Authorities Having Jurisdiction
(Inspection and Regulatory Compliance)****3.12.1 General****3.12.1.1 Registration of Manufacturers**

Manufacturers shall not engage in business activity relating to the construction of modules, modular components, or modular buildings, without being approved by the AHJ.

3.12.1.2 Requirements for Manufacturers

Requirements for manufacturers are as follows:

1. The approval of a manufacturer shall be in compliance with the AHJ.
2. Each separate manufacturing facility shall be approved. A manufacturing facility is separate if it is not on property that is contiguous to a registered manufacturing facility.
3. An approved manufacturer shall notify the AHJ in writing if:
 - a. The corporate or firm name is changed.
 - b. The main address of the registrant is changed.
 - c. There is a change in not less than 25 percent of the ownership interest of the company in not more than a 12-month period. A change in ownership shall require a new registration if the new owners do not accept responsibility for units constructed under the previous owners.
 - d. A new manufacturing facility is established.
 - e. There are changes in principal officers of the firm.
4. An application for original registration or renewal shall be rejected if any information contained on, or submitted with, the application is incorrect or incomplete. The certificate of registration shall be subject to revocation or suspension by the AHJ for violation of this standard.

3.12.1.3 Inspection and Enforcement

The AHJ is designated as the representative for the enforcement of this standard.

3.12.1.4 List of Approved Third-Party Inspection Agencies

The AHJ shall maintain a list of approved third-party inspection agencies. Each manufacturer shall contract with one or more inspection agencies for evaluation, monitoring and inspection services. The contract shall delineate the services to be provided by the inspection agency. The inspection agency shall notify the AHJ not more than 30 days after signing a new contract or terminating an existing contract with any manufacturer.

3.12.1.5 Right of Entry and Examination by AHJ

The AHJ shall have access, during reasonable hours, to enter a manufacturing facility and the building sites upon permission of any person who has authority or shares the use, access, or control over the building, or upon request from local officials having jurisdiction, for examination as to compliance with this standard.

**3.13 Third-Party Insignia, Identification and Data Plates
(Inspection and Regulatory Compliance)****3.13.1 General****3.13.1.1 Third-Party Agency Certification Insignia and Identification**

Modular components or modules of an approved modular building shall be marked with an insignia or other identification supplied by the third-party inspection agency that includes the name and address of the third-party inspection agency and the certification label number.

3.13.1.2 Mounting of Insignia

The insignia or identification shall be affixed so that it cannot be removed without destroying it. The insignia shall be affixed or applied in the vicinity of the electrical distribution panel or in another location that is readily accessible for inspection. Where a building comprises more than one modular component or module, the required insignia shall have the option of being furnished as a single label for the entire building, provided each module is marked by the third-party inspection agency in a clearly identifiable manner provided with or on the insignia.

3.13.1.3 Insignia Control

The insignia shall be under direct control of the third-party inspection agency until applied by the manufacturer to the modules or modular components that comply with this chapter. The manufacturer shall not acquire insignia from other sources. Each inspection agency shall keep a list of the serial numbers of insignia issued to each manufacturer's plant such that a copy of the record can be submitted to the administrator upon request.

3.13.1.4 Manufacturer's Data Plate

A manufacturer's data plate shall be provided for each dwelling unit or building. The following information shall be placed on a permanent manufacturer's data plate as identified on the cover sheet of the plans and in a location that is readily accessible for inspection. The compliance assurance agency shall approve the form and location of the data plate and shall verify that the data plate is complete with the following information:

1. Manufacturer's name and address
2. Compliance assurance agency certification number
3. Serial number of each module of the building
4. Serial number of the state registration seal
5. Date of manufacture of the building

6. List of codes and standards under which the building was evaluated and constructed and the type of construction and occupancy classification under those codes and standards.
7. Design live roof load; design floor live load; design wind speed; design ground snow load; and seismic design and risk category.
8. Thermal resistance (“R”) values
9. Special conditions or limitations concerning the use of the building under the codes and standards applicable to the building. A list of such conditions or limitations that is furnished separately with the building shall comply with this requirement.
10. Special instructions for handling, installation, and erection of the building. A list of such instructions that is furnished separately with the building shall comply with this requirement.
11. Designation of electrical service ratings, directions for water and drain connections and, where applicable, identification of permissible type of gas for appliances.

3.13.1.5 Record Retention

The manufacturer shall maintain copies of the data plate and reports of inspection, tests and any corrective action taken for not less than a period of 10 years from the date of manufacture of the building.

3.14 Referenced Standards (Inspection and Regulatory Compliance)

3.14.1 Referenced Standards

This chapter lists the standards that are referenced in various sections of this document. The standards are listed herein by the promulgating agency of the standard, the standard identification, the effective date and title, and the section or sections of this document that reference the standard. The specific year, date, and editions of the codes listed below serve as a reference, however; for accuracy and completeness, it's essential always to use the most current version of the code.

1. IBC: 101.3, 103.1, 202, 310.1, 310.4
2. IRC: 101.3, 103.1, 202, 310.4
3. ISO/IEC 17011 Conformity assessment – General requirements for accreditation bodies accrediting conformity assessment bodies 401.1
4. ISO/IEC 17020 Conformity assessment – General requirements for accreditation bodies accrediting conformity assessment bodies 401.1
5. ISO/IEC 17065 Conformity assessment – General requirements for accreditation bodies accrediting conformity assessment bodies 401.1

Chapter 4 Existing Conditions

4.0 Plan Assessment

4.0.1 Airport Layout Plan

The Airport Layout Plan (ALP) is a scaled graphic presentation of existing and ultimate airport facilities, as approved by the Federal Aviation Administration (FAA). Reference is made to FAA Advisory Circular (AC) 150/5070-6 (current edition) for ALP requirements. The ALP is a public document, which serves as a record of land and facility requirements, both present and future, as well as a source document for land use proposals. All proposed capital improvements must comply with the ALP. Any such project must be submitted through the Building Official to the FAA to determine if the project is in compliance with the ALP or if the ALP can be changed to accommodate the proposed project. Must submit airspace study application for approval with the FAA. For detailed information concerning the most current ALP, consult with DFW Code.

Radio/Wireless/Radar or similar installations are classified as a “Permanent” change to the ALP and require an FAA Air Space Application and Study Determination for possible interference with FAA or DFW Airport Board operations.

4.0.2 Site Plans

A site plan shall be prepared for all new and site projects. The site plan shall delineate all existing and proposed facilities and features. The site plan shall provide a clear schematic of the intended land use, project or building layout, site and project dimensions, access points, proximity to existing structures, etc. This plan will be used to initiate coordination among the Airport departments, the FAA, and tenants adjacent to the site. The site plan is also used to initiate changes to the ALP and to address potential line-of-sight issues.

The maximum building or equipment heights shall be indicated on the site plan prior to determining line-of-sight acceptability. A Traffic Impact Analysis (TIA) will be required for all landside development projects that will generate and distribute vehicular traffic along the Airport’s roadway network. A Level of Service “C” or better is required for all the airport’s roadways and intersections.

A landscape plan for each site shall be submitted with the site plan.

4.1 Tree Conservation Plan

1. The DFW Tree Conservation Plan establishes tree conservation requirements for all tree-disturbing activities. Refer to the most recent version of the Tree Conservation Plan for the detailed requirements. All tree-disturbing activities shall comply with the DFW Tree Conservation Plan.
2. A Tree Survey shall be completed for the entire project area. Additions to the project area may require additional Tree Surveys to be completed. The Tree Survey will be used to determine the Preservation Requirement and Tree Requirement. Refer to the Tree Conservation Plan for a complete description of Tree Survey requirements.
3. A Plan Sheet shall be developed to demonstrate compliance with the Preservation Requirement and Tree Requirement. The Plan Sheet shall identify trees to be protected, removed, and planted. Refer to the Tree Conservation Plan for a complete description of the Plan Sheet Requirements.

4. Tree Conservation Plan submittals shall be submitted for review and approval prior to issuance of a building permit. A project tree conservation plan shall include at a minimum a tree survey report, design plan sheets, and TCP worksheet. Construction shall adhere to the project's approved tree conservation plan. Protective measures shall be put in place prior to disturbing vegetation at the site and maintained throughout the project.
5. Tree Review shall be conducted prior to closing the building permit. The Tree Review will verify trees preserved, removed, and planted based on the as-built drawings. The Tree Review will identify any required mitigation planting or additional penalties required prior to building permit.

**REFER TO DFW TREE CONSERVATION PLAN MANUAL FOR MORE INFORMATION AND
DETAILED SPECIFICATIONS**

Chapter 5 Civil

5.0 Utilities

5.0.1 General Design Criteria for Water, Sanitary Sewer, and Reclaimed Water

5.0.1.1 Purpose and Scope

Sections 5.0.1, 5.0.2, 5.0.3, 5.0.4 and 5.0.5 (Utility Sections) of this Manual are to be used by engineering professionals for use in design and construction of water, wastewater and reclaimed water systems owned and operated by DFW.

The Utility Sections apply to all distributed assets designed for all contracts where the asset(s) will be dedicated back to DFW for Operations and Maintenance (O&M). Additionally, any distributed assets constructed by a private party but intended for dedication back to DFW must follow the provisions of these sections.

This technical resource is not intended to substitute for any professional engineering judgment by the designer who will assume ultimate responsibility for selection, reference, and appropriate application of the Utility Sections.

If a conflict is found between this DCM and other local, state, or federal regulations, the most stringent criteria shall be used during design. The Engineer of Record shall notify ETAM in writing about any such conflicts in these sections.

The term ROW is utilized throughout this manual, and as the DFW Airport does not have Rights of Way for its roadways, this term is meant to include the property between the individual lease lines on the opposite sides of the roadways, and between the opposite sides of the AOA on the roadways where control of access exists.

5.0.1.2 General Information and References

Material and construction for potable water sanitary sewer, and reclaimed water main additions or extensions shall be accomplished in accordance with the "Standard Specifications for Public Works Construction", North Central Texas Council of Governments, latest edition, unless otherwise revised or altered by requirements of these sections and/or DFW.

Aircraft loading shall be considered while designing potable water, sanitary sewer, reclaimed water main and their appurtenances inside the Air Operation Area (AOA). All applicable Federal Aviation Administration (FAA) rules, regulations, standards, and specifications shall be implemented during design.

Design Criteria for Potable Water Distribution Systems are as follows:

1. TCEQ Title 30, Chapter 290, "Public Drinking Water", along with all applicable laws, regulations, codes, and standards
2. AWWA Standards
3. Design Criteria for Wastewater Collection and Distribution Systems are as follows. TCEQ Chapter 217, "Design Criteria for Domestic Wastewater Systems"
4. Design Criteria for Reclaimed Water Distribution Systems are as follows: TCWQ Chapter 210: "Use of Reclaimed Water", and TCEQ Chapter 210.7: Reclaimed Water Facilities

5.0.1.3 Internal Record Search

The designer shall research all pertinent DFW records including, but not limited to, the following:

5.0.1.3.1 Design Records

These are water main, sanitary sewer lines, and reclaimed water line construction plans and do not necessarily represent “as built” information or have not yet been constructed.

5.0.1.3.2 As Built/Record Drawings

As-Built records that show the actual construction alignment of existing mains. These records sometimes vary from the original design due to unforeseen construction problems.

5.0.1.3.3 Water, Wastewater, and Reclaimed Water Master Plans

The latest water, wastewater, and reclaimed water master plans must be reviewed to be in compliance with any recommendations.

The EOR shall notify ETAM. This notification must be in writing with any explanations with design deviation from the utility master plan.

5.0.1.3.4 DFW Geographical Information System (GIS)

DFW GIS integrates various geographically referenced utility infrastructure records of the DFW Airport.

5.0.1.4 Utility Location Request

The designer must conduct a comprehensive investigation of all nearby existing and proposed utilities in order to avoid possible conflicts. This shall include calling DFW 811 to schedule a surface marking of known utilities. This shall include, but not limited to, the following utilities:

1. Gas (G)
2. Telephone (T)
3. Underground Electric (UE) and Overhead Electric (OE)
4. Cable (C)
5. Fiber Optic (FO)
6. Storm Drain (SD)
7. Petroleum (P)
8. First Flush Storm Water System
9. Potable Water
10. Reclaimed Water
11. Sanitary Sewer
12. Irrigation Systems
13. Spent Aircraft Deicing Systems

Non franchise utilities, DFW owned, will require coordination with DFW staff and is required in the design phase where a Level A SUE is required.

5.0.1.5 Field Investigation

Field investigations including Geotechnical Investigation, Subsurface Utility Engineering (SUE) or Environmental Investigation shall be conducted for all water, wastewater, and reclaim water main designs.

5.0.1.5.1 Geotechnical Investigation

The design and construction of water, wastewater, and reclaimed water systems must account for the variability of the uncertain subsurface conditions, and the potential project cost associated with the variability. This is especially critical on large projects or projects containing complex or difficult geotechnical problems where alignment and/or grade changes may be appropriate based on geotechnical recommendations. The general criteria for geotechnical investigation, is described as follows.

5.0.1.5.1.1 Investigation Requirements and Criteria

A geotechnical investigation may be conducted prior to design and/or construction of a project. However, data from earlier project design activity can be incorporated if sufficient and reliable data is available for the current project.

1. If required, the geotechnical report shall be prepared by a professional engineer with considerable geotechnical, design and construction experience relevant to the anticipated project.
2. Investigate subsurface materials and conditions according to these requirements, which represent the minimum acceptable level of care. Higher levels of care, which would involve more extensive sampling, testing, analyses, and reporting, may be required for certain projects.
3. Boring locations shall be within an offset distance of no more than 20 feet from the centerline alignment of the water/wastewater/reclaimed water main or at the location of the proposed structure. If this cannot be achieved, Geotechnical Engineer shall provide the exhibit with proposed bore location with explanation for exception from this requirement to DFW Code.
4. Investigate subsurface materials and conditions on all pipeline projects except those involving "small" repairs, pipe replacement along the exact same alignment as the existing pipe, in-place lining of existing pipe, or pipe bursting where the proposed pipe will not be more than one to two standard pipe sizes larger than the existing pipe. The projects to which these requirements apply include tunneling, guided boring, directional drilling, pipe bursting, pipe jacking, and auger boring. These methods are defined herein as "tunneling and trenchless methods."
5. Perform investigations that are appropriate for the project. The requirements given herein address a broad category of projects. It is not possible in these requirements to identify all possible geotechnical issues that may arise or that may be unique to a particular project.
6. Tailor the investigation to the type of construction, the anticipated geology, the landforms and topography, and the project schedule and budget.
7. Involve professionals who are experts in the particular type of underground construction. For pipelines constructed by tunneling and trenchless methods, the site exploration, laboratory testing, geotechnical analyses, and reporting shall be planned and implemented in conjunction with engineers whose expertise is in tunneling and trenchless construction methods and with geologists or engineering geologists experienced in civil engineering construction.

8. Locate geotechnical borings by taking into account topography, and landforms, expected subsurface materials and conditions, and proposed type of construction. Borings for tunnels and trenchless methods must be located in conjunction with engineers and engineering geologists who are experts in those types of construction.
9. Where truck-mounted drill rigs cannot access critical boring locations, obtain the necessary specialized drilling equipment.
10. Locate geotechnical borings, including piezometers, which are part of investigations for tunnels and trenchless methods far enough off of the proposed pipe alignment so that they do not impact construction.
11. Advance borings in soil or soil-like materials using continuous flight auger, hollow stem auger with drag bit, or thin-walled tube. Obtain samples using thin-walled tube or split spoon. In general, reserve split-spoon sampling for cohesion less materials.
12. Advance borings and obtain samples in rock or rock-like material using double-tube core barrel.
13. Conduct in situ tests, such as Standard Penetration Tests and packer tests, as needed to characterize the subsurface materials and conditions.
14. Conduct geophysical tests (resistivity, ground penetrating radar, seismic, and very low frequency) as needed to locate the soil/rock interface, cavities, porous rock, and faults.
15. Locate geotechnical borings at all proposed work and access shafts and pits for tunnels and trenchless methods.
16. Install and abandon piezometers and ground water monitor wells in compliance with State law.
17. Backfill boreholes outside pavement using non-shrink grout from the bottom of the borehole to within three feet of the ground surface. Plug the upper three feet with cuttings from the borehole.
18. Reference the location of each boring to the Texas State Plane Coordinate System and the ground surface elevation to USGS MSL with horizontal and vertical positional tolerances of +/- one foot. The Engineer of Record (EOR) should contact DFW survey group to get the latest requirements.

5.0.1.5.1.2 Depth

Minimum-boring depths shall be as follow:

Table 4 - Minimum Boring Depths

Incoming Pipe Trench Depth (TD)	Minimum Bore Depth
0' To 10'	TD+5'
10' to 25'	TD+10'
25' and deeper	1.5*TD

1.0.1.5.1.3 Laboratory Testing for Open Cut Construction

1. Conduct Unconfined Compressive Strength, Atterberg limits, Moisture content, Grain size analyses test for soil and soft rock, including shale, mudstone, clay shale and claystone.
2. Special testing as required to characterize collapsible soils, soils susceptible to particle migration, etc.
3. Unconfined compressive strength for rock.

5.0.1.5.1.4 Laboratory Testing for Tunneling and Trenchless Methods

1. In addition to the testing described in Section 1.0.1.5.1.3, conduct slake durability, cell pressure and as required for soil and soft rock, including some shale, mudstone, clay shale, and claystone.
2. In addition to the testing described in Section 1.0.1.5.1.3, conduct cerchar abrasivity, point load, Brazilian tensile and punch penetration test for rock.

5.0.1.5.1.5 Laboratory Testing for Corrosion studies

1. Conduct tests for resistivity, pH, chlorides, and sulfates, as needed.

5.0.1.5.1.6 Geotechnical Analyses

1. Check for, identify, and reconcile inconsistencies in subsurface information or subsurface materials and conditions that may impact design or construction. Compare the boring logs to one another, compare the logs to mapped stratigraphy as contained in Environmental Geology of the DFW area and compare the logs to personal experience in the same area. Assess the potential for "differing site condition" claims based on these comparisons and undertake additional investigations, as needed, to resolve such inconsistencies.
2. Provide geotechnical parameters and construction recommendations appropriate for the type of construction and covering the topics requested. In all cases, provide project-specific recommendations. Avoid generalized discussions and detailed explanations of theory or phenomena.
3. If the project is open cut construction, provide bearing capacity for thrust blocking, provide maximum lift thickness for backfilling, identify need for excavation retaining system to protect surrounding infrastructure, identify aspects of the behavior of the in situ subsurface materials that could affect design (for example, highly plastic soils and extremely flat pipeline grades are incompatible for gravity line construction if water can enter the trench during construction), provide lateral earth pressures for underground structures (do not provide lateral earth pressures for excavation safety systems).
4. If the project is utilizing Trenchless construction methods, provide soil and rock properties, characteristics, and geotechnical parameters required by the tunneling and trenchless construction engineers.

5.0.1.5.1.7 Geotechnical Report

1. Use only 8½ inch by 11 inch and 11 inch by 17 inch sheets. If needed, prepare color illustrations, maps, drawings, photographs, and other graphics to adequately convey the information.
2. Characterize soil by reporting, at a minimum, the engineering classification (group name and symbol), color, relative stiffness, shrink / swell potential, unconfined compressive strength, unit weight, moisture content, liquid limit, particle size distribution, and plasticity index. Where appropriate, report SPT results and collapse potential.
3. Characterize rock by reporting, at a minimum, the geologic material, color, degree of weathering, relative hardness, RQD, percent recovery, unconfined compressive strength, unit weight, and relative frequency and infilling of discontinuities. Where appropriate, report SPT results.

4. Characterize groundwater by reporting observations and expected occurrence. Provide mitigation methods in case groundwater is encountered.
5. Submit preliminary reports for review with the preliminary design documents. Submit final reports with the 100 percent submittal of design documents.
6. If the project is open cut construction, the final report shall include two separate reports, Geotechnical Data Report (GDR) and Geotechnical Design Memorandum (GDM). The geotechnical data report shall include the boring location map, boring logs, text describing the investigation and the subsurface materials and conditions that were encountered. It shall also include the State Plane coordinate and USGS ground surface elevation for each boring on the boring log. Geotechnical Design Memorandum shall provide geotechnical recommendations specific to the project and only for items or issues requested.
7. If the project is utilizing tunneling and trenchless methods, the final report shall include three separate reports, GDR, Geotechnical Baseline Report (GBR), and GDM. Geotechnical data report and Geotechnical memorandum shall be per above list. Geotechnical Baseline Report shall provide the geotechnical information to establish what constitutes "anticipated site conditions for the project." The experts in underground construction must author or co- author the GBR because it is a contractual document that allocates risk associated with construction. Use clear, concise prose organized in parallel structure. Provide specific, quantifiable, and measurable baselines, not subjective, vague descriptors. Describe baseline conditions reach by reach, sequentially along the project alignment. (A reach is defined as a segment having consistent ground conditions and excavation methods.) Address all baseline conditions and design and construction considerations for each reach in a single section of the GBR. Write the GBR using parallel structure to present information reach by reach in the same repetitive manner. Discuss distinctly different elements of work separately. For instance, discuss tunnels separately from shafts. Minimize the presentation of information not related to baseline conditions.

5.0.1.5.2 Subsurface Utility Engineering (SUE)

Subsurface Utility Engineering (SUE) process may be conducted in project planning, design, or construction phase(s) to obtain reliable subsurface utility information. Using this technology, it will be possible to avoid many utilities relocation before construction and many unexpected encounters during construction, thereby eliminating many costly, time-consuming project delays. In addition, all existing utilities shall be located and marked prior to initiation of survey for design.

5.0.1.5.2.1 SUE Provider requirements

SUE shall be conducted by well-trained, experienced, and capable individuals using state-of-the-art designating equipment, vacuum excavation, or comparable non-destructive locating equipment, state-of-the-art surveying and other data recording equipment and software systems, as necessary.

1. Quality Level D is simply the process of gathering record data (which is not necessarily so simple). This data could be digital records, paper records, internet search, site inspection or just talking to people who are familiar with the site. The available data could be limited and not produce a complete picture of what is onsite. In addition, if the source of some data is unknown or questionable, the completeness and accuracy of the investigation could be compromised. In any event, once the data is collected, it is compiled into a drawing that states the quality level.
2. Quality Level C involves the process of surveying the visible utility features, or if they have already been surveyed, checking the surveyed locations for their accuracy. This data is compiled with the Quality Level D data and both levels are labeled accordingly. Quality Level C data could, and often will, conflict with the Quality Level D data and raise questions as to the locations of some underground utility lines.
3. Quality Level B involves designating the underground utilities, for example, by markings provided through an 811 call or by contacting an individual utility company. This data is added to the data collected from Quality Levels D and C. The designated utilities are then surveyed and added to the drawing which can then be identified as representing the results of a Quality Level B investigation.
4. Quality Level A involves physically locating the actual utility often by 'potholing.' Once the utility is potholed it is located horizontally and vertically by survey measurements. This location is typically to the top of pipe, which should be noted on the drawing.

The SUE provider shall call 811 and others for utility locates for exposing utilities.

5.0.1.5.3 Environmental Investigation

The designer shall evaluate all available resources to identify any potential environmental issues, including possible soil or groundwater contamination, during the preliminary design phase of water, wastewater, or reclaimed water projects. This process shall be coordinated with DFW Environmental Affairs Department (EAD).

5.0.1.6 Crossings

Throughout the airport, utility crossings exist, this section is meant to determine how specific attributes of the airport are to be crossed with the utilities.

5.0.1.6.1 Roadway, Runway, Taxiway, and Ramps

Encasement pipe shall be used to encase the carrier pipe. Carrier pipe shall have restrained joints by internal restraints system. The crossing should be perpendicular to the roadway, runway and taxiway's alignment based on existing conditions.

Encasement pipe for roadway crossing shall be minimum of 10 feet away from edge of pavement. The ends of the casing pipe must be sealed with rubber boot or an equivalent sealing method.

Encasement pipe for runway and taxiway crossing shall be minimum of 50 feet away from runway and taxiway safety areas or obstacle free zone whichever is greater.

Raci-spacers shall be set every 4 feet center on all pipes in casing.

5.0.1.6.2 Utility Crossing

Foreign mains are typically defined as water, wastewater, telephone, electric, cable, fiber optic, storm drain, petroleum and any other franchise or public utility own and/or operated by outside entities (non-DFW).

Each utility line is governed by a separate license agreement. Please contact the Commercial Development team to obtain permit information and crossing requirements for each specific utility line.

DFW Airport Franchised utilities:

1. Menzies (Jet Fuel)
2. Flint Hills (Jet Fuel)
3. Oncor (Electrical)
4. Atmos (Gas)
5. TRA (80% of Sanitary Sewer)
6. City of Irving (20% of Sanitary Sewer)
7. Explorer (Fuel, Gasoline)
8. Frontier (Fiber, communications)
9. Water Line (City of Euless)

5.0.1.6.2.1 Foreign Mains Crossing DFW Water/Wastewater/Reclaimed Water Mains

1. All foreign lines crossing DFW Utility mains shall be at 90 degree (preferable) or 60 degrees (minimum).
2. The vertical separation between proposed foreign main and existing DFW Utility mains shall be in accordance with Table 5 while meeting all applicable DFW, state and federal requirements.
3. All foreign wastewater lines crossing over DFW mains shall be encased with 150 psi pressure-rated pipe. The length of encasement shall be minimum 18 feet centered at the DFW water main.

Table 5 - DFW Main (in) versus Foreign Pipe (in)

		Foreign Pipe (in)			
		8-12	16-24	30-66	72 & Larger
DFW Main (in)	8-12	2 ft	2.5 ft	3 ft	3.5 ft
	16-24	2.5 ft	3 ft	3.5 ft	4 ft
	30-66	3 ft	3.5 ft	4 ft	4.5 ft
	72 & Larger	3.5 ft	4 ft	4.5 ft	5+ ft

Note: All dimensions in (ft.) shall be measured from outside diameter (O.D.) to outside diameter (O.D.)

5.0.1.6.3 Railroad Crossing

All mains crossing under railway tracks shall be placed in an encasement pipe, from the railroad ROW-to-ROW. Casing pipe and joints shall be made of metal capable of withstanding the railroad loadings and other loads superimposed upon them.

1. The engineer of record shall contact the railroad authority and the appropriate regulatory agency for any additional design and/or construction requirement.
2. The engineer of record shall copy ETAM on all correspondence with each regulatory agency.

5.0.1.6.4 State Highway Crossing

The engineer of record shall contact the Texas Department of Transportation (TxDOT) and/or the appropriate regulatory agency to determine any special design, construction requirement and/or permitting requirements. The EOR shall notify ETAM of all correspondence with each regulatory agency.

5.0.1.6.5 Culvert Crossing

Encasement pipe shall be used to encase the carrier pipe. Carrier pipe shall have restrained joints by internal restraints system. The minimum vertical clearance between encasement pipe and the bottom or top of culvert shall be 2 feet. The encasement pipe shall be extended a minimum of 15 feet from the outside edge of culvert. The Engineer of record shall coordinate with DFW Energy Transportation and Asset Management (ETAM) System Performance/Watershed Management. All correspondence shall be copied to ETAM.

5.0.1.7 Tunneling and Trenchless Technology

These technologies shall be considered in order to minimize the disturbance of the environment, traffic, congested living or working areas while utilizing more efficient methods of installation, inspection, repair, rehabilitation, and replacement of underground utilities.

5.0.1.7.1 Technical References

The following references, as applicable, may be reviewed in conjunction with this manual: "Trenchless Technology Pipeline and Utility Design, Construction, and Renewal", by Mohammad Najafi and Sanjiv Gokhale, 2004.

5.0.1.7.2 General Consideration

1. Location of Excavation Pits: The location, size and depth of the boring pits shall be evaluated during design. The location of the entry pit is preferred at the lower elevation end of the tunnel when there is a slope. This will allow any groundwater or any boring slurry to flow by gravity from the tunnel into the bore pit where it can be pumped out during construction.
2. Size of Encasement Pipe: The encasement pipe size shall be determined by the designer, with the knowledge that DFW requires spacers in the annular space instead of grout.
3. Loading: Encasement pipe and joints shall be designed to withstand the imposed loading.
4. Annular Space: The annular space between the encasement and carrier pipe shall have spacers around the entire pipe with a spacing designed by the designer.
5. Corrosion Protection: Encasement pipe shall be protected against corrosion. Where both carrier pipe and casing pipe are both metallic. They shall be separated electrically by using insulated spacers.

5.0.1.8 Casing and Carrier Pipes

Water/sanitary sewer/reclaimed water mains shall be considered for encasement under the following condition:

1. Highway/railroad crossing
2. Runway and Taxiway
3. Creek/water body crossing
4. As required by the design

The carrier pipe shall be restrained or welded all around joints or be a monolithic pipe between span sections.

5.0.1.8.1 Steel Encasement Pipe

Designer shall be responsible for determining the minimum wall thickness of steel encasement pipe as required for any specific application with following minimum conditions:

1. Steel pipe with minimum yield strength of 36,000 psi, or as required by FAA for airfield applications.
2. Minimum thicknesses of steel encasement pipe for different diameters are shown under Table 6 which must be verified by the designer. These wall thicknesses shall be increased by at least 0.063 inch for uncoated casing and by at least 0.063 inch for coated or uncoated steel casing to be installed by jacking or boring.
3. If the casing pipe is placed by open cut, the casing pipe shall be coated.

Table 6 - Minimum Thickness of Steel Encasement Pipe

Encasement Pipe I.D. (in)	Coated or Cathodically Protected
12 or under	0.188
14	0.188
15	0.250
16	0.219
18	0.250
20	0.281
21	0.312
22	0.281
24	0.312
26	0.344
27	0.438
28	0.375
30	0.406
32	0.438
34- 36	0.500
38	0.500
40	0.531
42	0.562
44- 46	0.594
48	0.625
50	0.656
52	0.688
54	0.719
56- 58	0.750
60	0.781
62	0.812
64	0.844
66- 68	0.875
70	0.906
72	0.938
21	0.312

5.0.1.9 Permits

All the required permit approval notice(s) must be included in the final construction documents. These includes, but not limited to, the followings:

1. TXDOT Right-of-Way Access Permit
2. TXDOT Utility Permit
3. USACE permits (example 404, NW-12. Etc.)
4. Approval letter from other entities or agencies having jurisdiction.

5.0.1.10 Instrumentation and Control

All instrumental and control systems shall comply with the existing Supervisory Control and Data Acquisition (SCADA) systems and control systems that are currently being used by DFW. Plans and specifications shall be provided by the Engineer of Record with each plan submittal for the project.

5.0.1.11 Pipe Color

All mains will follow the following pipe color scheme.

Table 7 - Utility Pipe Color Scheme

Utility	Pipe Color
Potable Water	Blue
Sanitary Sewer	Green
Storm Drainage	Green (if PVC)
Reclaimed Water	Purple

5.0.1.12 Markers and Detector wires

All pipes shall be placed with detector wire over the pipe, and a utility marking tape placed one foot above the pipe. The color of the tape and detector wire shall match the color of the pipe material and shall identify what the specific utility is. Tracer wire must be rerouted to a test station. When in landscaped area, a 2'x2' concrete pad shall be installed.

5.0.1.13 As-Built Drawing

All projects shall be constructed in conformance with approved sealed plans and specifications as possible. Any field change shall be documented in As-Built drawings for all water, wastewater, and reclaimed water pipeline projects. Refer to Chapter 1 Specifications for these requirements.

5.0.1.14 Record Drawing

All as-built changes shall be incorporated in Record Drawings by the Engineer of Record. Refer to Chapter 1 Specifications for these requirements.

5.0.1.15 Asset Identification Number

The Design Engineer shall co-ordinate with DFW ETAM System Performance Group and DFW ETAM Utilities Group for proposed asset identification number. Asset Identification number shall be called out on design plan. At minimum follows asset shall have asset identification number in the drawing.

1. Isolation Valve
2. Pumps and Motor
3. Fire Hydrant
4. Cathodic Protection Test Station
5. Water Meter
6. Air Release, Air/Vacuum, and Combination Air Valves
7. Flush Points
8. Sampling Stations
9. Backflow Preventer
10. Manhole, Junction Structure
11. Lift Station

5.0.1.16 Aircraft Deicing Facilities

Standards and guidance for designing aircraft deicing facilities shall be in accordance with FAA's Advisory Circular (AC) number: 150/5300-14 (current edition). In an event of revisions to the referenced AC, it will be the design engineer's responsibility to follow latest FAA guidelines for design of aircraft deicing facilities.

5.0.2 Water Utility Distribution

5.0.2.1 DFW Airport Water System

This Section includes various aspects of water main design including replacement criteria, sizing, depth, embedment, and location. Also included are design criteria for various water appurtenances including different types of valves, fire hydrants, flush points, meters along with corrosion protection system.

Water distribution mains shall be designed in conformance with "30 TAC §290: Public/Drinking Water", as enforced by Texas Commission on Environmental Quality (TCEQ), Latest Edition, and with all applicable laws, regulations, codes, and standards.

5.0.2.2 Water Main Sizing

5.0.2.2.1 Water Pipeline Network

The DFW potable water pipeline network is classified according to

Table 8 - DFW Airport Water Main Classification

Type	Typical Size Range (in)	Direct Service Connection
Distribution Main	16" and Smaller	Permitted
Transmission Main	Larger than 16"	Not Permitted

5.0.2.2.2 Looped Mains

Where practical, new water main installations shall be looped to improve the system's water flow.

5.0.2.2.3 Water Demand

DFW Airport water system must be able to supply water at rates which fluctuate over a wide range during different times of year and hours of the day. Per capita usage can vary greatly depending on the area's zoning and the efforts made by the owners for water conservation.

5.0.2.2.4 Size/Capacity Determination Criteria

Pipe shall be sized for the combined maximum day demand and fire flow demand.

A Hazen William Friction Coefficient $C=80$ shall be used for all service line flow calculations. A higher C coefficient may be used for new mains only upon approval from DFW Utility Group.

The Engineer of Record shall submit water use calculations showing average day demand, maximum day demand, peak hourly demand, building square footage and total acres for the proposed use. The average day demand, peak day demand, and peak hour demand used in design shall be approved by DFW Utility Group.

Pressure reducing valves (PRV), as required by the International Plumbing Code, shall be installed inside the building, and must be illustrated and identified on a site utility plan.

Minimum operating pressure is 50 psi at the highest elevation meter location using average day demand.

The maximum allowable velocity at peak hour demand shall not exceed 5 feet per second (fps).

The minimum pressure at any point must not be less than 35 psi residual.

5.0.2.2.4.1 Fire Flow

Service lines shall be adequately sized to provide the minimum required fire flow and duration at a residual pressure of 20-psig or greater.

Required Fire flow shall be as specified by the International Fire Code (IFC), latest edition, or by the National Fire Protection Association (NFPA) Standard, whichever is more stringent. Design Engineer shall coordinate with DFW Fire Protection Engineer for fire flow requirement.

Emergency demands are considered to be fire flow requirement plus peak day demand.

Maximum allowable velocity during emergency demand shall not exceed 10 feet per second (fps).

The minimum residual pressure at any point during emergency demand must not be less than 20 psi.

5.0.2.2.5 Sizing Criteria

The water mains must be sized in accordance with any approved master plan established for that area.

If a master plan is not available, the sizing of the water main must be based on engineering analysis of initial and future demand of the area to be served. Water transmission and distribution mains must be sized to meet peak daily water demand plus required fire flow plus any additional criteria as needed. When site-specific data is unavailable, designer shall use the most conservative data while meeting or exceeding the following minimum criteria for sizing distribution mains.

5.0.2.2.5.1 Minimum Pipe Size

5.0.2.2.5.1.1 General Area

Minimum 8 inch main shall be used for all general areas.

5.0.2.2.5.1.2 Industrial Area

A minimum 12 inch main for industrial areas shall be used.

5.0.2.2.5.2 Non-Standard Pipe Sizes

14 inch and 18 inch water pipes are considered nonstandard for the DFW Potable Water System (PWS) and shall not be used.

5.0.2.3 Depth of Cover

The depth of cover is measured from the top of the pipe to the natural or finished ground surface above the pipe. Minimum depth of cover shall be 6 feet – 0 inches.

5.0.2.3.1 Minimum Cover

The following Table 9 applies to water main installation in public right-of-way.

Table 9 - Minimum Depth of Cover for Water Mains

Size of Main	Min. Depth of Cover (ft)		
	Standard Pipeline	Highway Roadway	Runway Taxiway
12" and smaller	6	6	Engineering Evaluation & FAA Regulations
12" – 16"	6	6	
20" and larger	7	7	

**Mains shallower than 6 feet will require special engineering evaluation and engineering controls.*

***Water mains under highway and railroad right- of-way must meet all additional criteria as required.*

****Water mains under aircraft taxiways and runways must also meet all additional FAA standards as required.*

5.0.2.4 Location**5.0.2.4.1 New Main Installation**

New water mains shall be placed outside paved areas and as dictated by the design requirements.

The main should be located, where maintenance can be accomplished with the least interference with traffic, structure, and other utilities.

5.0.2.5 Horizontal Alignment**5.0.2.5.1 Change in Direction**

Change in Direction Changes in horizontal alignment shall be achieved by deflection of joints or by use of fittings. Deflection of pipe joints at fittings is prohibited.

Longitudinal bending of pipe is not allowed. The maximum bend for waterlines is 45 degrees.

5.0.2.5.2 Pipe Laying

All water mains shall be laid as straight as possible between intersections and follow right-of-way or centerline alignment curves at a uniform distance from the right-of-way or centerline, as appropriate. Due to any instances of design limitations, if the pipe is laid near the curb and gutter, the designer shall ensure that pipe trench alignment is at least 3 feet away from the edge of curb and gutter to avoid any future maintenance issues.

5.0.2.5.3 Joint Deflection

The maximum deflection angles of pipe joint are typically restricted to 50% of the manufacturer's recommendation. Otherwise, horizontal bends will be required.

5.0.2.5.4 Bends

Horizontal bends shall be restrained type fittings/joints and shall also be blocked with concrete as necessary. Horizontal bends shall also be placed such that the concrete blocking can be poured against undisturbed earth and will not bear against the backfill or bedding of another utility.

5.0.2.5.5 Stationing

Stations must be to the tenth of foot (Ex: STA. 1+90.5). If necessary, station equations can be used at a point along the alignment where the stationing changes. The station equation generally represents the meeting of two stationing systems or the change in authority over the centerline.

5.0.2.6 Vertical Alignment**5.0.2.6.1 Change in Direction**

Changes in vertical alignment shall be achieved by deflection of joints or by use of fittings. Deflection of pipe joints at fittings is prohibited. Longitudinal bending of pipe is not allowed.

The maximum bend for waterlines is 45 degrees.

5.0.2.6.2 Pipe Laying

Mains are to be installed as straight as possible, but excessive depths shall be avoided. For reference, excessive depths are pipes designed to be installed in depths over 20 feet. This is due to the limited ability of standard equipment by operations to reach these mains.

5.0.2.6.3 High Points

Excessive high points that trap air and restrict water flow must be avoided. High points should be designed to coincide with the location of proposed fire hydrants, where possible. Where high points are unavoidable, air valves should be considered.

5.0.2.6.4 Bends

Vertical bends shall be restrained type fittings/joints and blocked with concrete as necessary. All pipes will be restrained with retainer glands and concrete blocking.

5.0.2.6.5 Combined Angle Bend

A combination angle bend can be used where horizontal point of intersection (PI) and point of vertical intersection (PVI) are located at the same point. A combined angle can be shown in Figure 1.

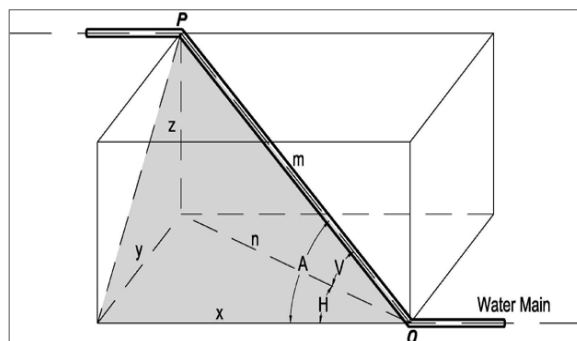


Figure 1 - Combined Horizontal and Vertical Angle

$$\cos A = (\cos H) (\cos V)$$

Where,

$\cos A = x/m$	$A = \text{Combined Angle}$
$\cos H = x/n$	$H = \text{Horizontal Angle}$
$\cos V = z/m$	$V = \text{Vertical Angle}$

5.0.2.6.6 Slope

All water mains shall be designed to have a minimum 0.1% slope to allow draining and flushing, if necessary. The vertical change in slopes is restricted to 50% of the manufacturer's recommended deflection. Otherwise, vertical bends will be required. Vertical bevels can be considered to accommodate long vertical curves of bevel or deflected joints in lieu of a bend for large concrete main in accordance with manufacturer's recommendations. Vertical curves are not to be less than 100 feet in length. The Polyvinyl Chloride (PVC), PVI, and PVT should be at quarter, half, or full stations (Ex: PC Sta. 0+00, PI Sta. 0+50 and PT Sta. 1+00).

5.0.2.6.7 Slope Designation

Design slopes shall be to the nearest hundredth of a percent (Ex: Slope 5.20%).

5.0.2.6.8 Elevation Designation

Elevations shall be shown to the nearest hundredth of a foot (Ex. El. 495.95).

5.0.2.7 Pipe Material & Embedment

5.0.2.7.1 Pipe Material Selection

Pipe Material Selection Recommended pipe material is summarized in Table 8.

Table 10 - Recommended Water Pipe Material & Embedment

Pipe & Material Specification	Pipe Size (in)	Joint Specification	Embedment
PE	¾ - 3	Fusible C901	Engineering Analysis Required (See 5.0.2.7.3)
PVC AWWA C900 (DR-14)	4 - 12	Bell & Spigot Joints: ASTM D3139	
PVC AWWA C905 (DR-14)	16	Fusible Fusible C900 Certa-Lok Certa-Lok C900/RJ	
RCCP Bar Wrapped ANSI/AWWA C303 Class 150	20 – 42	Bell & Spigot ANSI/AWWA C303	

Pipe material shall be selected based on the lowest life cycle cost. PVC is the preferred material for smaller pipe sizes (≤16 inch diameter). Reinforced concrete cylinder pipe (RCCP) is preferred for larger pipe sizes (>16 inches). Other materials such as ductile iron and steel may be specified upon approval by the DFW Airport Utility.

The Engineer of Record shall prepare a Technical Memorandum that includes recommendations for preferred pipe materials, with a case study that includes pros and cons on pipe type.

When a metal pipe including concrete cylinder, ductile iron or steel is specified, the pipe must be protected from corrosion. Corrosion protection measures should be part of any pipeline design using these materials. For ductile iron pipes a minimum single layer of 8-mil liner low density polyethylene (LLDPE) wrapping is required. When steel pipe is utilized, the pipe interior lining shall be cement-mortar and exterior coating shall be either cement-mortar, tape or polyurethane as approved by the Utilities Group. All joints on metal pipe shall be bonded, and in locations with reactive soils or induced currents a cathodic protection system may be necessary.

5.0.2.7.2 Fittings

All PVC and ductile iron pipe shall use full body ductile iron fittings. Compact fittings are not allowed. All 90-degree bends shall be avoided in the system, if possible. All the fittings used in water pipe installations shall be in accordance with the latest “American National Standards Institute” (ANSI) section A21.10 AWWA C 110 for pipes.

5.0.2.7.3 Embedment Requirements

Designer shall specify class of embedment and its detail on design drawings. The type of embedment to be used is determined by pipe material and depth of cover.

For flexible pipe, the Engineer of Record shall prepare a technical memorandum that includes the maximum pipe deflection anticipated by the pipe. Pipe deflection shall not be more than 50% of manufacturer's recommendation.

5.0.2.8 Separation Distance Between Water & Wastewater Mains

When a water main is installed near an existing wastewater facility, conveyance, or appurtenance, separation requirements of 30 TAC §290.44(e), as enforced by TCEQ, governs the minimum separation distances.

5.0.2.8.1 Cross Connection

No physical connection shall be made to a drinking water supply system where a potential or actual contamination hazard exists unless the public water system is protected from contamination. Any appurtenance shall be designed and constructed to prevent any possibility of sewage entering the drinking water system.

5.0.2.8.2 Backflow and Siphonage

The rules of 30 TAC §290.44(h) apply to backflow and siphonage control.

5.0.2.9 Connection to Existing Mains

Testing and Chlorination: A new valve shall be installed at the point of connection for water main extensions to facilitate testing and chlorination of the new main prior to its placement into service.

Tapping Water Mains: Where services require the tapping of any existing reinforced concrete cylinder water distribution main, the Designer shall specify that the Contractor employ a qualified specialty contractor to perform this service. Tapping of all water mains shall be done in accordance with AWWA standards and coordinated with DFW Utilities Services. The tap shall be performed only at the horizontal tangent of the main in conjunction with a tapping gate valve with a dielectric insulating gasketed flange.

Tapping Fee: All new or enlarged, permanent water meter installations, served by DFW Airport's potable water distribution system, will be subject to an impact tapping/connection fee. Tapping impact fee will be charged with the water bill after the permanent meter and account is established.

Wet Connection: At all points where wet connections are made to existing lines, the tapping connection fittings shall be supported by blocking up to the spring line with minimum 2,000 psi concrete.

5.0.2.9.1 Tapping Sleeve and Valve

Tapping sleeve and valve shall be used whenever possible for connections to existing mains to avoid interruption of water services. Taps are restricted to at least one standard pipe size smaller than the tapped pipe. (e.g. If the existing main is 16 inches, the largest pipe that can be tapped will be 12 inches.)

Full body tapping sleeves shall be used. A tapping sleeve will not be allowed if the materials and conditions of the existing main preclude tapping. "Size-on-size" taps will not be permitted, unless made by use of an approved full bodied mechanical joint tapping sleeve.

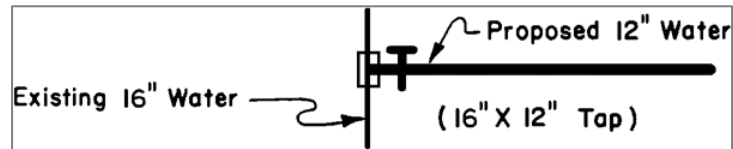


Figure 2 - Tapping Criteria for Water Mains

5.0.2.9.2 Cut in Connections

Connections 4 inches and larger of new mains to existing mains shall be made by cutting in a tee. Tapping sleeves may be allowed in lieu of cutting in a tee on a case-by-case basis. If the system needs an additional valve, then a cut-in connection with a valve and tee should be used. If the system needs an additional valve, then a cut-in connection with a valve and tee should be used. 5 feet of existing pipe in each direction shall be removed and replaced at the Cut-In Connection.

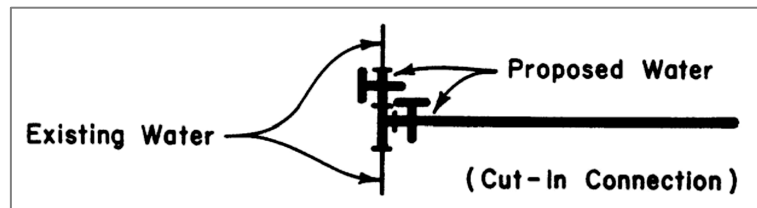


Figure 3 - Cut-In Connection

5.0.2.9.3 Standard Connection

When connecting to a crossing water main, two tees with intervening valve(s) are required.

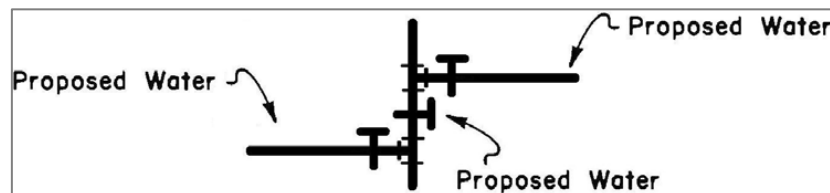


Figure 4 - Standard Connection

5.0.2.9.4 Four Way Connection

Crosses are not permitted. However, Type-D connections may be allowed when other types of connection are not feasible.

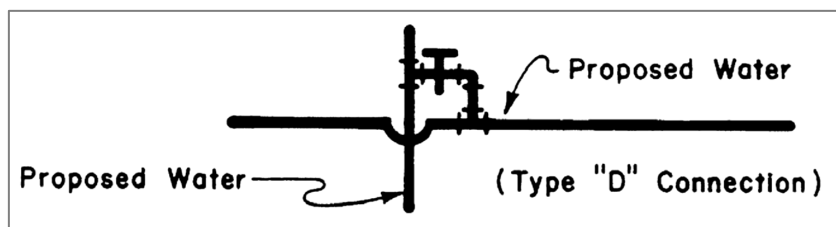


Figure 5 - Type D Connection

5.0.2.10 Water Services and Connections

5.0.2.10.1 General Requirements

Water services shall be in accordance with DFW Utility Group. The design engineer shall co-ordinate with DFW Utility Group for service connection location and detail.

Meter box should be located inside the Terminal building for CTA. Except for CTA, meters may not be located inside the building or fence line and must be vehicular accessible from roadways.

Water services are not to cross railroad, interstate, or state highways.

Water meter boxes and its appurtenances are not allowed in sidewalks, driveways, or load bearing pavement.

Service taps, regardless of type, shall not be made in vaults.

Permanent domestic water services shall not be supplied from fire hydrant leads. Domestic water service and fire flow line shall be isolated and tapped separately.

New facilities shall have one water service tap from the existing or new watermain. Fire and Domestic water supplied from the one water service tap shall be separated outside the facility or within the building's valve room or fire riser room. The Tenant/Lessee shall be responsible for the maintenance of the water service from the demarcation line (normally the Watermain tap isolation valve flange facing the facility). DFW Utility Group shall be responsible for the water meter and water meter box.

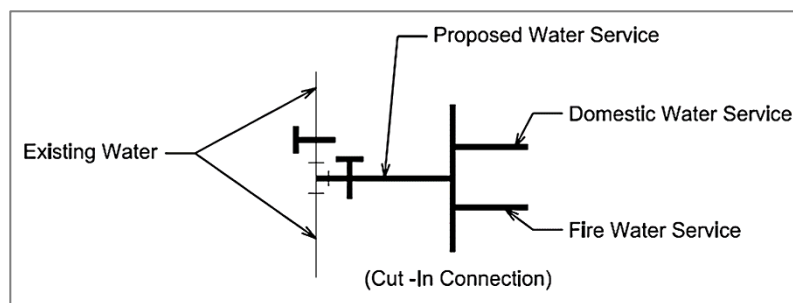


Figure 6 - Domestic Water Tap to Lead-in or Fire Loop

Domestic water service may be tapped into fire loop or lead-in, provided that the domestic tap is located prior to the backflow preventer.

Service taps on fire loop after the backflow preventer are prohibited.

Each site (tenant) must have its own water service which must not cross any existing or projected lot (or lease) line(s).

The water service shall be at least one size smaller than the proposed or existing main. Size on size connections are generally not allowed unless special permission is granted from DFW Utility Group.

5.0.2.10.2 Small Water Service

Small water services range from 1 inch to 2 inch in diameter. Water service over 30 feet shall be a minimum of 1 inch diameter. AWWA C901 poly- pipe with Ford couplings and fittings shall be used for small water service.

Designer shall show all DFW service meters on drawings.

5.0.2.10.2.1 Large Water Service

Large water services are greater than 2 inches in diameter. A service line of minimum 4 inches will be required for a 3 inch meter.

PVC C900 (DR 14) pipe shall be used for large water services.

Large service meter vaults shall be located at the lease-line (except for Terminal buildings) with easy access and with protection from vehicular traffic.

Designer shall work with DFW's Utilities Group on the evaluation of all large meter vaults.

5.0.2.11 Dead-End Mains

5.0.2.11.1 General Requirements

Dead-end main situations are to be avoided whenever possible. "Dead-end main" means a water main over fifty feet long and not being fed from both ends at the time of installation.

5.0.2.11.2 Special Consideration

If the dead-end main is 8 inch or larger, a fire hydrant shall be installed 5 feet from the end.

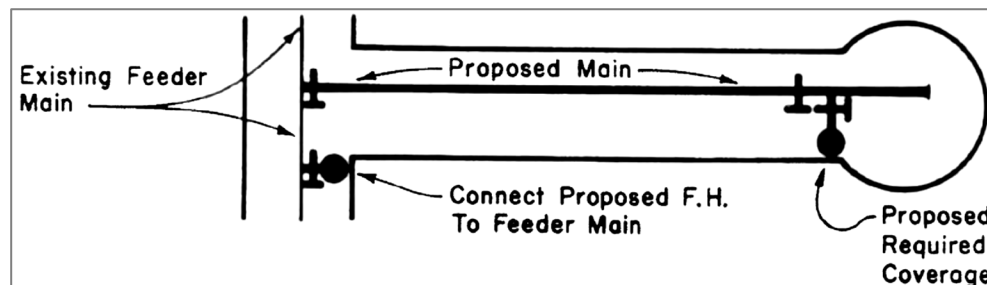


Figure 7 - Dead-End Main Eight Inch or Larger

All dead-end mains shall include flusher with timer.

5.0.2.12 Abandonment of Water Mains

Where possible abandoned water mains shall be removed. Water mains shall be abandoned by cutting and plugging where it is not possible to remove the existing pipe.

The cut and plug shall be as close to the main left in service as practical unless there is other impending utility work planned that could disturb the plug. If the new main is to be constructed to connect to the existing main at the point of cut, a cut and plug is not required. If the main to be abandoned at a tap and valve, the abandoned tapping sleeve and valve shall be removed.

The contractor shall verify with DFW International Airport's Environmental Affairs Department that the pipe to be removed has no asbestos containing materials.

5.0.2.12.1 Abandonment of Water Appurtenances Fire Hydrants

Removed/Salvaged fire hydrants shall be handed over to DFW Utilities Group.

5.0.2.12.2 Valves

Small Valve: Valves smaller than 16 inches are not to be salvaged. Upon removal of the valve cover, stack, and stem extension, the valve body must be abandoned by filling with 2 sacks per cubic yard mix of sand to a point at least 12 inches below the pavement.

Large Valve: Valves 16 inch and larger may be salvaged if requested by DFW.

5.0.2.12.2.1 Valve Boxes

All remote-control valves, isolation valves, pressure valves, altitude valves, and reducing valves for onsite reclaimed water systems shall be installed below grade in a valve box. Valve boxes located in streets or any areas subject to vehicular traffic shall be provided with concrete collars. Engineer shall coordinate with DFW regarding the standards of concrete collars. Collars around such valve boxes shall be formed and finished off neatly in an acceptable manner as approved by DFW.

5.0.2.12.3 Abandonment of Vaults

Vaults shall be abandoned by filling with sand and/or gravel compacted to 90% (95% in pavement) of maximum standard proctor dry density. The bottom of the vault shall be filled with class B concrete up to the top of the abandoned pipe opening.

5.0.2.13 Corrosion Protection System (Corrosion Control)

This section is applicable to all metal water pipes and fittings where corrosive environments, or soil may potentially damage the pipes and appurtenances. Typically, soil resistivity less than 1000 ohms-cm can be considered as extremely corrosive soil.

Within the Airport boundary, all metal piping and fittings shall be protected by a cathodic protection system that is designed by a NACE certified professional and installed by a licensed contractor.

5.0.2.13.1 Preventive Requirement

The following requirements shall be used as a guide for the Engineer of Record for the design of a system to reduce corrosion as much as possible.

5.0.2.13.1.1 Material Selection

PVC pipe shall be used for all mains 16 inches and smaller.

All RCCP shall be used with required dielectric coatings, as necessary.

5.0.2.13.1.2 Dielectric Coating

External and internal protective coating shall be considered for all metallic pipes as necessary to limit the rate of cathodic reaction.

5.0.2.13.1.3 Electrical Isolation

Isolating Joints by using insulating kits or other means are required to prevent galvanic corrosion for all metallic pipes in the following locations:

1. Changes in pipeline materials
2. Connections to existing piping (i.e. old and new piping)
3. Inlet and outlet piping of plant facilities
4. Laterals from transmission mains
5. Taps to existing RCCP
6. Valve to RCCP
7. Metallic casing spacer to RCCP

5.0.2.13.2 Mastic Coating and Plastic Encasement

All fittings shall be coated with mastic. No wrap.

5.0.2.13.3 Monitoring Requirements

The system shall include test stations and electrical continuity as follows.

5.0.2.13.3.1 Corrosion Test Station (CTS)

All RCCP and PCCP shall be designed with CTS at least at every 1000 feet to measure any potential current or resistance.

Designer shall provide GPS coordinates for all CTS locations.

5.0.2.13.3.2 Electrical Continuity

All pipelines shall be electrically continuous between CTSs through joint bonding wires, or welded joints, as necessary.

5.0.2.13.4 Corrosion Protection System (CPS)

Corrosion Protection System (CPS) is to be designed to introduce an external DC current which makes the structure a cathode. The CPS must be designed by a NACE certified professional engineer with considerable experience in corrosion engineering.

5.0.2.13.5 Corrosion Survey

A detailed corrosion survey shall be conducted along the proposed or existing water main alignments. This investigation shall include, but not be limited to the following:

1. Field soil resistivity measurements
2. Soil and groundwater sample analysis
3. Stray DC earth current and foreign line cathodic protection system activity
4. Identification of potential corrosion problems

5.0.2.13.6 Data Evaluation

All field and laboratory data obtained from the corrosion survey shall be used to develop corrosion prevention and monitor design recommendations.

5.0.2.13.7 Final Design

Based on the corrosion survey and subsequent data evaluation and approval by DFW, a corrosion protection system shall be designed by a NACE professional engineer. This system may include, but not be limited to, the following methods:

Galvanic Protection (GP) System: Current generated from metal at higher energy level.

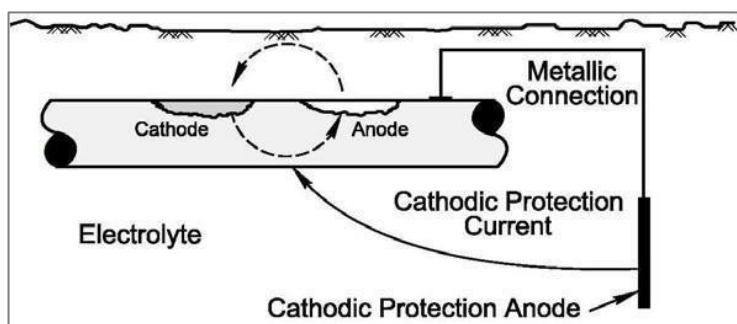


Figure 8 - Galvanic Protection (GP) System

Impressed Current Cathodic Protection (ICCP) System: Current generated from transformer-rectifier energizing a relatively inert anode.

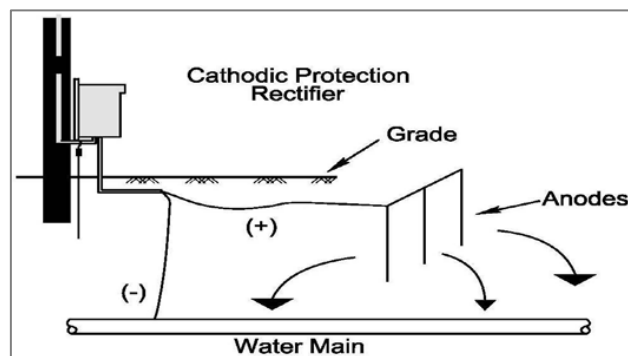


Figure 9 - Impressed Current Cathodic Protection (ICCP) System

5.0.2.14 Testing Procedures

All tests shall be coordinated with and verified by DFW Utilities Group.

5.0.2.14.1 Hydrostatic Testing and Chlorination

All water mains shall be hydrostatically tested and chlorinated before being put in service. The Contractor shall be responsible for conducting hydrostatic tests and chlorination.

5.0.2.14.1.1 Hydrostatic Testing

Perform Hydrostatic pressure and leakage test using methods and per performance requirements of AWWA M23 for PVC Pipe and M9 for Concrete Pressure Pipe or 150 psi for 2 hours for water main and 200 psi for fire main whichever is more stringent.

Prior to establishing the required test pressure, the designer must consider the operating condition of the system including but not limited to Normal working pressure, maximum sustained operating pressure, and Maximum transient pressure along the pipeline.

5.0.2.14.2 Disposal

The chlorinated water can be hauled off in water trucks or discharged into wastewater manholes as approved by DFW Utilities Group and DFW Environmental Affairs Department. Flushed water shall be metered.

The chlorinated water can be discharged to a storm drain or sheet flow in open field with proper de-chlorination method as approved by DFW DCC and DFW EAD.

5.0.2.14.3 Disinfection and Sterilization

Disinfection and sterilization shall be performed in accordance with AWWA C651, latest edition, and will be performed by the installation contractor.

All Bac-T samples will be collected by a DFW Utility employee and tested by the third-party National Environmental Laboratory Accreditation Program (NELAP) certified laboratories.

5.0.2.15 Water Utility Distribution System Isolation Valves

5.0.2.15.1 Gate Valves

Gate valves shall be non-rising stem, solid- wedge gates with cast iron body and bronze mountings. Valve bodies shall be flange-by-flange when tapping into DFW water distribution mains. If possible, all gate valve stems should be vertical.

Unless otherwise specified, valves 3 inches to 12 inches in size with working pressures of 200 psi or less shall be in strict accordance with AWWA Standard Specification for "Gate Valves for Ordinary Water Works Service," designation C509, latest revision. Gate valves with resilient seated gates in accordance with AWWA C509, latest edition.

All new gate valves with resilient wedge: gate valves 16 inches and larger shall be installed with an integral bypass valve in a vault. The valve shall be non-rising stem unless otherwise indicated, shall have a gear reduction actuator, and shall turn counterclockwise to open. Valves shall be provided with a 2 inch square nut operator unless otherwise designated.

Valves or corporation stops smaller than 16 inches for buried service shall be provided with 2 inch square nut operator. The operating nut or extension of any valve shall be between 18 inches and 24 inches below finished grade.

There shall be a valve on each fire hydrant lead restrained to the main. These, and all valves 24 inches and smaller, shall be resilient seated gate valves.

Valves shall be located at the intersection of two or more mains and shall be spaced so that no more than 3 customers will be without water during a shutout. For lines smaller than 24 inches, typical spacing should be 1500 feet. Mains 24 inches and larger shall be valved at intervals not to exceed 2,000 feet.

Branch piping (both new and future branches) shall be separated from the main with gate valves.

Valves shall be located so that isolating any segment of water main requires closing of no more than 3 valves.

The operating nut or extension of any valve shall be between 18 inches and 24 inches below finished grade.

If possible, all gate valves should be vertical. All gate valves 16 inches and larger shall have the bonnet located in a vault or manhole.

All valves 16 inches and larger shall have a gear reduction actuator.

Valves having "push on" joints are not permitted for fire hydrant leads and laterals.

Butterfly valves shall not be allowed.

Water mains shall be designed so that valves can be installed vertically unless conditions dictate otherwise.

5.0.2.15.2 Valve Stacks and Vault

Valves for 18 inch and up or corporation stops buried in the ground shall be provided with cast iron valve stacks of proper dimensions to fit over the valve bonnets, and to extend to the finished ground line in paved areas or slightly above finished grade in other areas. Tops shall be complete with covers and shall be adjustable. Valve stacks shall be set vertical and concentric with the valve stem. A concrete pad shall be poured around all valve stacks when not in paved areas. Design engineer shall provide the detail for the concrete pad.

5.0.2.15.3 Valve Flanges

Flanges for valves shall be drilled to match connecting flanges. All flanges shall conform to the standard specification of the American National Standards Institute. Flanges shall be Class 125 for all pipe, fittings, and valves 3 to 12 inches in diameter with a working pressure of 200 psi or less, and flange bolts shall be coated for corrosion control.

5.0.2.15.4 Blocking Under Valves

All gate valves 8 inches and larger, which are buried shall rest on a concrete pad. The pad shall extend for the full width of the trench to the back of the bell (or flange). Concrete shall be 2000 psi for blocking valves.

5.0.2.16 Tapping Sleeve

Tapping sleeve and valve shall be used whenever possible for connections to existing mains to avoid interruption of water services. Taps are restricted to one standard pipe size smaller than the tapped pipe. If the existing main is 16 inches, the largest pipe that can be tapped will be 12 inches.

All taps on RCCP shall be in accordance with the manufacture's recommendations.

5.0.2.17 Bends

All bends shall be DI with factory Mastic Coating and Poly wrapped. All fittings shall be flanged by MJ fittings.

5.0.2.18 Water Utility System Fire Hydrants

Fire hydrants shall be 5¼-inch. Hydrants shall be a breakaway traffic model with 6 inch mechanical joint shoe to be buried at 5 feet, except where different depth is shown in the hydrant schedule. Fire hydrants that are not installed in pavement shall be surrounded by a 6-foot by 6-foot concrete pad.

Subsurface hydrants are not allowed for use on the airport except when approved by the Fire Marshal. Subsurface (or flush-mount) fire hydrants shall be flush type models with box and cover and shall be back-filled as approved by DCC Code Enforcement and the Fire Marshal's Office.

Hydrants shall have one 5¼ inch outlet and 2 - 2½ inch outlets with National Standard Threading. All hydrants will be fitted with (supplied and installed by the contractor) a 5 inch quick coupler adapter.

Hydrant operation shall be 1½ inch flat to point, open to left, by not more than eight or nine full turns of the operating nut. Hydrants shall meet the requirements of AWWA C502.

Hydrants shall be installed on both sides of all divided road/highways to provide adequate firefighting coverage. Roads/highways where opposing lanes of traffic are separated by a vehicle obstruction shall be considered a divided road/highway.

The entire fire hydrant assembly shall have restrained joints.

Fire hydrants shall not be located within 9 feet in any direction of any sanitary sewer main, lateral, or service regardless of material of construction.

Fire hydrants shall be designed so as not to interfere with sidewalk ramps, trash receptacles, and streetlight and signal pole foundations.

To avoid sidewalks, ramps, and other features, fire hydrants placed near a street corner should be located outside the curve radius and a minimum of 4 feet from ramps. Exceptions may apply in the existing development or long (i.e. >5 feet) radius curb return.

Placement of fire hydrants should take into consideration above ground improvements, landscaping, critical root zones, grades, and other utilities.

When replacing existing water lines along with new fire hydrant leads, the drawings shall indicate that existing fire hydrants in service more than 10 years are to be replaced with new.

5.0.2.18.1 Fire Hydrant Spacing

Fire hydrants shall be spaced at a maximum interval of 500 feet. A fire hydrant shall be located so to reach each building with no more than 250 feet of hose along the most direct route considering all physical impediments.

At minimum, one fire hydrant shall be located within 100 feet of the Fire Department Connection (FDC)

5.0.2.18.2 Fire Hydrant Locations

The location of all public fire hydrants shall meet the following criteria:

1. Fire hydrants shall be located as near to the street intersections as possible but out of the radius of curb turnouts, within 2.5 to 7.5 feet behind curb or projected future curb.
2. No more than one fire hydrant will be allowed on a dead end main.
3. Fire hydrant locations between street intersections shall be at the projection of a property line between lease holders.
4. New fire hydrants shall be placed as close the location of the existing fire hydrant to be replaced as possible.
5. Rail lines, controlled access highways, divided roadways, fences, and walls will inhibit laying the fire hose in the most direct route and must be considered as barriers.
6. Add TxDOT blue surface reflectors along centerline of roads or in the close vicinity of the hydrants.

5.0.2.18.3 Fire Hydrant Leads

Water mains serving a private fire hydrant must be 8 inches or larger and the minimum acceptable fire hydrant lead is 6 inches.

Maximum length of fire hydrant lead shall be 100 feet unless otherwise approved by DFW. Any lead over 50 feet in length requires an 8 inch lead.

A double check backflow preventer is required on fire hydrant leads over 100 feet in length. Backflow preventer must be at the main tap

5.0.2.18.4 Fire Flow Tests

The following items shall be addressed when performing a fire flow test:

1. The pressure hydrant should be closer to a feed main than the flow hydrant.
2. The number of flow hydrants should be determined. The following data need must be recorded during a fire flow test:
 - a. Static Pressure: This refers to the pressure reading before water flows. It is taken from the pressure hydrant just prior to time of the fire flow test.
 - b. Residual Pressure: This refers to the pressure reading while water is flowing. It is taken from the pressure hydrant while the flow hydrants are flowing full.
 - c. This pressure is taken to determine the fire flow for sufficient fire coverage.
 - d. Pitot Pressure: This reading is taken by a pitot gauge from the flow hydrants.
 - e. The pitot gauge should be inserted into the center of the flowing outlet at approximately half of the diameter away from the nozzle.

A validation fire hydrant flow test is required to close-out all applicable permitted construction, as proscribed by the Fire Marshal.

5.0.2.18.5 Backflow Preventer and Detector Check Valve

See Section 5.0.2.24 (Backflow Preventer and Detector Check Valves) for detailed requirements.

Backflow preventer assemblies that are installed in private plumbing systems, fire protection systems, and process water systems that are directly or indirectly connected to or on properties serviced by the DFW Airport water distribution system shall obtain laboratory and field-testing approval as required by Federal, State Rules and Regulations.

To prevent contamination of the PWS from stagnant water in dead end potable water service lines (e.g., private fire systems, private fire main, section for future use, etc.), the installation of an approved backflow prevention assembly is required downstream of the DFW meter, and an approved detector-check backflow preventer assembly is required on fire loop at the lease line.

5.0.2.18.6 Blocking and Drainage

Concrete blocking shall be poured behind hydrants against undisturbed earth. Washed pea gravel shall be placed appropriately around the shoe of the hydrant to effectively drain the hydrant barrel.

5.0.2.18.7 Isolation for Maintenance

All fire hydrants shall be isolated from the main waterline and from other water services by installation of a gate valve between the main waterline and the fire

hydrant in order to facilitate repair of the fire hydrant without having to shut off the main waterline.

5.0.2.18.8 Fire Hydrant Extensions

Water Mains must be brought to 5 feet below grade at fire hydrant locations. A maximum 1 foot extension is allowed for adjustment.

5.0.2.18.9 Cathodic Protection

All fire hydrants shall be cathodically protected.

5.0.2.18.10 Subsurface Fire Hydrants

All subsurface fire hydrants shall have a 7 foot solid red circle painted around the access cover helping to identify its location and use as a fire hydrant. The marking shall be reduced in size when appropriate to keep from encroaching on any airfield operations pavement marking.

5.0.2.19 Water Meters

Water meters shall be required at all service points and shall conform to the following.

5.0.2.19.1 Water Meters

Water meters shall be line sized in accordance with good design practice for the service intended. Meters specified shall be compatible with electronic data collection equipment to enable on-site electronic collection of meter readings. Water usage shall be recorded on both a visual odometer and in electronic memory. Also note the following:

1. All new applications shall utilize Badger Beacon with HR-E LCD encoders, LTE endpoints.
2. Meters up to 2 inches in size shall be positive displacement disc type meters. These water meters shall meet or exceed the requirements of AWWA C700 for Cold Water Meters - Displacement Type.
3. Meters greater than 2 inches, and all irrigation meters regardless of size, shall be turbine type. All turbine meters shall have a strainer. These water meters shall meet or exceed AWWA C701 Class 2 standards.
4. If both high flow and low flows are anticipated, a compound type meter shall be used. These meters shall meet or exceed the requirements of AWWA C702 for Cold Water Meters-Compound Type.
5. Registers for all meters shall read in straight U.S. Gallons.
6. Meters shall be constructed of compatible metals throughout to prevent any corrosive reaction between component metals.

5.0.2.19.2 Water Meter Sizing

Water meters shall be sized in accordance with the International Plumbing Code and with AWWA M22: Sizing Water Service Lines and Meters, latest edition. Accordingly, all applicable water usage including domestic, irrigation, mechanical and fire flow demand shall be considered while sizing a water meter.

5.0.2.19.3 Meter Boxes

Where possible, meters and meter boxes shall be located at the lease line. Meter boxes or vaults shall be specified for each meter. Provide clay dam and link seal at pipe penetrations into vault.

Meter locations shall be identified on the plans and shall be outside the facility served and at a location that is always accessible to utility personnel and service equipment.

Construction details shall be shown on the plans including the details of all internal piping. Piping details shall include a minimum requirement of two isolating valves, the meter, backflow device downstream of the meter and any necessary additional fittings required to remove or service the water meter without closing valves at any location.

All installations requiring 3 inch or larger meters shall be provided with a bypass line with gate valve and arranged such that the bypass can provide unmetered service to the facility during periods of time when the meter is being serviced or replaced. All meter boxes or vaults containing 3 inch or larger meters shall be free draining or be provided with a sump pump which discharges into the nearest storm drain. All drainage design for meter vaults shall be coordinated and approved by DFW.

Meter boxes and vaults shall have cast iron or steel deck plate covers as required to provide adequate service access to the meter location. Deck plate covers shall have an opening for the transmitter mount. Where the weight of the cover or cover sections exceeds 25 pounds, all meters shall be provided with sensor extension cable for mounting the sensor on the box or vault, which shall be easily readable through a small access door in the cover.

A concrete pad shall be placed around all meter boxes, extending 12 inches minimum around perimeter of the box.

A 12 inch-thick gravel bed shall be placed below each meter box.

5.0.2.19.4 Meter Depth

Meters shall be positioned at 18 inches below the top of the Meter Box.

5.0.2.19.5 Positive Displacement Meters with Electronics

5.0.2.19.5.1 Pit or Vault Application

1. 1-inch meter with HR-E LCD encoder and LTE Cellular endpoint with pit-mount fittings with 3 foot lead (specify wire length if more than 3 feet needed)
2. 1½-inch meter with test plug and HR-E LCD and pit electronics with 3 foot lead (specify wire length if more than 3 feet needed)
3. 2 inch meter with test plug and HR-E LCD, and pit electronics with 3 foot lead (specify wire length if more than 3 feet needed)

5.0.2.19.5.2 Mechanical Room or Interior Room Application

1. 1 inch meter with HR-E LCD encoder and LTE Cellular endpoint with wall-mount adapter (specify lead wire length if more than 25 feet needed)
2. 1½ inch meter with test plug HR-E LCD and outdoor remote NOT PREWIRED (specify lead wire length if more than 25 feet needed)
3. 2 inch meter with test plug HR-E LCD and outdoor remote NOT PREWIRED (specify lead wire length if more than 25 feet needed)

5.0.2.19.6 Turbine Meters with Electronics

5.0.2.19.6.1 Pit or Vault Application

1. 3 inch turbo meter with strainer, HR-E LCD encoder and LTE Cellular endpoint with pit-mount fittings (specify lead wire length if more than 3 feet needed)
2. 4 inch turbo meter with strainer, HR-E LCD and pit electronics (specify lead wire length if more than 3 feet needed)
3. Compound meters, HR-E LCD and pit electronics (specify lead wire length if more than 3 feet needed)

5.0.2.19.7 Compound Meters

A compound meter shall be used where high flow rates are necessary, but also where smaller rates of flow are present and require accurate measurement.

Compound meters shall have two measuring elements and a check valve to regulate flow between them.

5.0.2.20 Outlets

Blind flanges or plugs, as applicable, shall be furnished and installed on all valves located at outlet points or terminal points where the water main does not continue. Dead-end main structures shall be avoided whenever possible. If unavoidable, dead-end mains shall be designed to accommodate periodic flushing. The following two design alternatives shall be considered:

1. Locate a fire hydrant less than 50 feet from the main's end.
2. Install a smart flush at the main's end.

5.0.2.21 Air Release Valves, Air/Vacuum Valves and Combination Air Valves

Automatic air release (and vacuum release) valves shall be installed at high points on water transmission mains 16 inches and above to exhaust and admit air to prevent vacuum conditions and air related surges.

Air valves are not required on water distribution mains 12 inch and smaller where fire hydrants and service connections provide a means for venting trapped air.

5.0.2.21.1 Air Release Valve Design

The Engineer is responsible for determining the size and type of air release valves necessary to assure the water system operates properly based upon the water system characteristics and shall provide calculations determining the size and type of valves for review by DFW Utility Group when requested. Air release valves may be necessary on any size of main.

At a minimum, on water mains 16 inches in diameter and larger, and on smaller mains where appropriate, combination air valves will be placed at all high points and air/vacuum valves shall be placed at the down-slope side of all gate valve locations. Air/vacuum and vacuum release valves shall be approved on a case-by-case basis.

All mains 24 inches and larger will include an 18 inch outlet with flange including a 1 inch corporation (minimum) for installation at high points where the installation of an air release valve (ARV) would be necessary. In the absence of an ARV requirement, an 18 inch outlet with flange including a 1 inch corporation shall be placed every 2,500 feet. Proposed waterline connections to air release valve piping are prohibited.

Air release valves shall conform to the requirements of AWWA C512.

5.0.2.21.2 Surface Boxes

Surface box for manual air release where required shall be cast iron box and lid, 12 inches deep and 24 inches in diameter. Raised letters on the cover shall read "WATER".

The top of the air release valve shall be no less than 24 inches below the surface box lid.

In unimproved areas provide a 24 inch marker, 4 feet high, with a 4 foot square concrete pad.

5.0.2.22 Purging and Flush Points

5.0.2.22.1 Flush Points

Flush points are to be installed primarily to flush water mains as needed.

5.0.2.22.1.1 General Requirements

1. Fire hydrants shall not be used in lieu of flush points.
2. Flush points shall not be designed to flush water to storm drains in order to prevent migration of chlorine residual in storm water system.
3. The developer may need to extend a lateral to the end of the cul-de-sac so that it can be used to flush the water through a flush point.
4. Hyper chlorinated water must be dechlorinated during any flushing activity

5.0.2.22.1.2 Type

Smart flush is recommended as flush unit. At minimum the flush unit should have the following probes/sensors:

1. Combined Chlorine
2. Temperature
3. Total Chlorine
4. PH

Dechlorination option should be included in the flush system.

5.0.2.22.1.3 Size

Flush points shall be 2 inches or larger based on design calculations.

5.0.2.22.1.4 Location

1. Dead-end mains
2. High/Low Valve Assembly

5.0.2.23 Sampling Stations

The designer shall coordinate the need for and the location of Sampling Stations with DFW Utilities Group.

5.0.2.24 Backflow Preventer and Detector Check Valves

Backflow prevention device shall be considered at the following locations to protect public water system from cross contamination:

1. Commercial property water service line
2. Dedicated irrigation lines
3. Fire Lines

5.0.2.24.1 Water Service Connections

All temporary construction water services shall be provided with a line sized reduced pressure zone-double check backflow preventer valve assembly and water meter. Services shall not be initiated until backflow prevention devices have been tested and approved for operation by a TCEQ certified tester. All original test forms shall be returned to the Plumbing Inspector at DFW Airport Design, Code, & Construction Department, P.O. Box 612008, DFW Airport, TX 75261- 2008.

Temporary water service for construction purposes may be provided by a Fire Hydrant Construction Meter. These meter/backflow assemblies shall be obtained from the DFW Airport Water Services Department. An application and a monetary deposit will be required for each assembly needed. Fire Hydrant Construction Meters shall not remain connected to a fire hydrant for more than 180 consecutive days nor be allowed to provide water service to construction office trailers. When temporary water service to construction office trailers or the need for water service exceeds 180 consecutive days, a physical connection to an existing water main shall be required by either making a tap or by cutting in a tee fitting, depending on size of service line needed. These connections shall also require the installation of a water meter and a backflow assembly.

5.0.2.24.2 Backflow Preventers

Where the service line provides potable water for a domestic service and connects with other closed or chemical treated systems that could foreseeably contaminate the potable water line, a line sized backflow preventer shall be installed. Drains off the backflow preventer assembly shall be drained to the sanitary sewer. Taps to mains, to provide water for fire protection or other closed pipe systems, shall have a double check valve assembly at the fire line tap. An alternate method of backflow prevention consisting of a 12 inch air gap between an unrestricted overflow of an atmospheric makeup tank and the source of water is also acceptable. All double check and reduced pressure backflow preventers must be certified for operation after installation by a TCEQ certified tester. All original test forms shall be returned to the Utility Inspector at DFW Airport Design, Code, & Construction Department, P.O. Box 612008, DFW Airport, TX 75261-2008.

5.0.2.24.3 Detector Check Valves

A single Detector Check (DC) on closed fire line shall typically be used to measure fire flow for approved automatic fire sprinkler only.

1. Typical size of a Detector Check (DC) is 4 inches (min.), 6 inches, 8 inches and 10 inches as necessary. A 5/8 inch – 1 inch Nutating Disc Positive Displacement (PD) meter is also typically used in a bypass line conjunction with Detector Check (DC).
2. Bypass line in a dedicated fire line should not be used for domestic use. On looped private fire mains, a Detector Check (DC) shall be installed on both taps to the public main.

5.0.2.24.4 Reduced Pressure Backflow Preventer

All pressure reducing backflow preventers that are installed to protect high-hazard services from back-flowing must be tested annually from the date they are installed and certified by a TCEQ certified tester. All original test forms shall be returned to the Plumbing Inspector at DFW Airport Design, Code, & Construction Department, P.O. Box 612008, DFW Airport, TX 75261-2008. Provide weather protection and automatic heated enclosure (i.e. hotbox) if located outside.

Examples of high-hazard services are:

1. Aspirators
2. Autoclaves
3. Sterilizers
4. Lab bench equipment
5. Sewage pumps
6. Sewage ejectors
7. Potable water handling and treatment equipment

5.0.2.25 Thrust Blocking**5.0.2.25.1 Thrust Blocks**

Concrete with a minimum strength of 3,000 psi in 28 days shall be placed for blocking at each change in direction of the pipeline, in such manner as will substantially brace the pipe against undisturbed trench walls. All fittings shall be mastic coated and wrapped in poly-plastic prior to thrust block placement to allow future removal of concrete without damage to fittings. Concrete blocking shall have been in place 4 days prior to testing the pipeline. Thrust blocks shall be used at:

1. Changes in direction, as at tees and bends
2. Changes in size, as at reducers
3. Changes in elevation, as at tees and bends (concrete block anchors preferred)
4. Stops, as at a dead end
5. Valves, where thrusts may be expected

5.0.2.25.2 General Requirements

All pressurized water mains shall be restrained against unbalanced thrust forces due to change in pipeline diameter or alignment in order to prevent joint separation or movement.

5.0.2.25.3 Thrust Restraint System

Any or a combination of the following thrust restraint systems are acceptable for potable water mains:

1. Horizontal Thrust Block
2. Vertical Thrust Block/ Anchor Block
3. Joint Restraint System

5.0.2.25.4 Horizontal Thrust Block

Horizontal thrust block shall be designed to transfer horizontal thrust force to larger bearing area, as needed.

5.0.2.25.4.1 Design Considerations**5.0.2.25.4.1.1 Design Pressure (P)**

The design pressure for horizontal thrust restraint system shall be the maximum anticipated water pressure. Hydrostatic test pressure can be used as the thrust restrained design pressure for most applications.

5.0.2.25.4.1.2 Thrust Force (T)

1. Unbalanced thrust force occurs at the following configurations shall be restraint as needed (Figure 10).
2. Bend
3. Wye
4. Deadend
5. Bifurcation
6. Tee
7. Reducer
8. Closed Valve

5.0.2.25.4.1.3 Bearing Area (Ab)

Thrust bearing area can be calculated as follows:

$$A_b = S_f T / S_b$$

Where,

A_b = Bearing Area

S_f = Factor of Safety, typically 1.5

S_b = Horizontal Bearing Strength

T = Hydrostatic Thrust

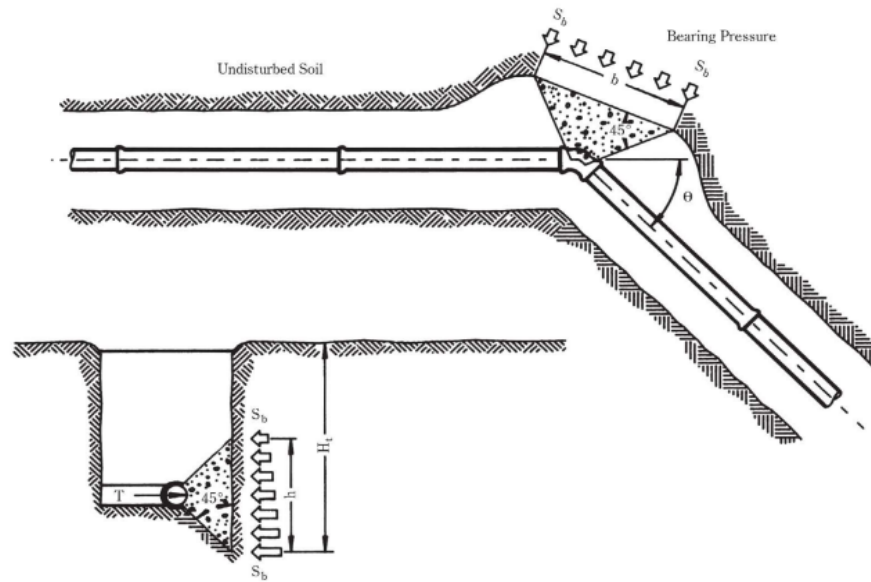


Figure 10 - Horizontal Thrust Bearing Area

Horizontal Hydrostatic Thrust: $T_y = PA \sin \Delta$

Horizontal Hydrostatic Thrust: $T_x = PA \sin \Delta$

Where,

P = Internal Pressure (psi)

A = Bearing Area

Δ = Deflection Angle (degree)

In the absence of actual Horizontal Bearing Strength, Table 10 can be used to estimate soil bearing strength:

Table 11 - Soil Bearing Strength

Soil	Bearing Strength (lb/ft ²)
Muck	0
Soft Clay	1,000
Silt	1,500
Sandy Silt	3,000
Sand	4,000
Sandy Clay	6,000
Hard Clay	9,000

Source: Thrust Restraint Design for Ductile Iron Pipe, by DIPRA, 2000

5.0.2.25.4.2.4 Block Dimension

Block dimension shall be determined based on the required bearing area with the following criteria:

$$b = A_b/h$$

Where,

A_b = Bearing Area

b = Block Width

h = Block Height

Block height (h) shall be equal to or less than one-half the total depth to the bottom of the block, (H_t), but not less than the pipe diameter (D).

$$h \leq 0.5 H_t \quad h > D$$

Block height (h) should be chosen such that the calculated block width (b) varies between one and two times the height.

$$h = b \sim 2b$$

Installation

Bearing surface shall be placed against undisturbed soil extending beyond joints. If not, the fill between the bearing surface and undisturbed soil must be compacted to at least 90% Standard Proctor density.

Thrust block shall not be used in swamps or marshes.

Horizontal thrust block may not be practical for large water main due to size restriction.

5.0.2.25.5 Vertical Thrust Block/Anchor Block

Vertical thrust blocks shall be designed to counter vertical thrust force by weight and to transfer horizontal thrust force to a larger bearing area, as needed. Vertical thrust blocks will require all- thread rods between bends for added restraint.

5.0.2.25.5.1 Design Consideration**5.0.2.25.5.1.1 Design Pressure (P)**

The design pressure for horizontal thrust restraint system shall be the maximum anticipated water pressure. Hydrostatic test pressure can be used as the thrust restrained design pressure for most application.

5.0.2.25.5.1.2 Thrust Force (T)

Unbalanced thrust force occurs at the configuration shown in **Figure 10**.

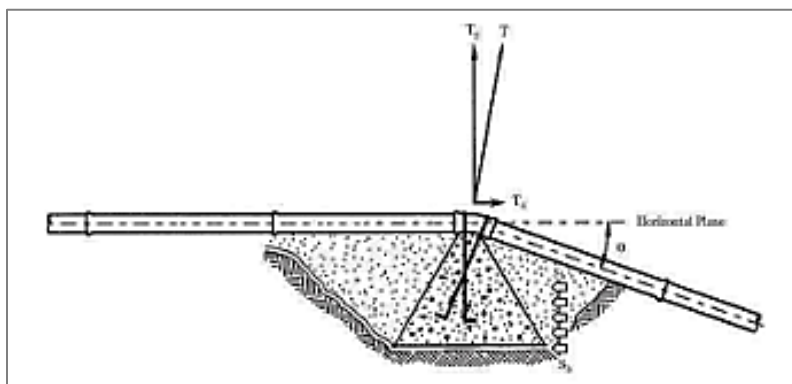


Figure 11 - Vertical Thrust Block

5.0.2.25.5.1.3 Volume of Vertical Thrust Block (V)

Vertical thrust block shall be designed to provide equilibrium by the weight of the block.

Required Volume,

$$V = S_f T_y / W_m$$

$$V = S_f P A \sin \Delta / W_m$$

Where,

S_f = Factor of Safety, typically 1.5

T_y = Vertical Component of Thrust

S_b = Horizontal Bearing Strength

W_m = Density of Block Material

5.0.2.25.6 Restraint Pipe Joint

Joint restraint for pipes larger than 16 inches shall utilize integral, factory joint restraint systems.

Joint restraint shall be provided for all pipe bends and where necessary when joint deflection is utilized. A minimum safety factor of 1.5 shall be used when calculating restrained water pipe length. When joint restraint is required in intersections, extend the joint restraint, at a minimum, to the point of curvature (PC) of the curb line. Notes shall be placed in both plan and profile views and shall include at a minimum the type of restraint to be utilized and the beginning and ending stations of the restraint. Concrete thrust blocking may be approved on a case-by-case basis. In cases where concrete thrust blocks are utilized, at a minimum the Engineer shall include block dimensions and locations on the plans. The proximity of other utilities and structures must be considered when specifying the use of thrust blocking.

5.0.2.25.6.1 Applicability

Restraint Pipe Joint shall be designed to transfer horizontal and vertical thrust force, as needed.

5.0.2.25.6.2 Design Consideration

The design pressure for horizontal thrust restraint system shall be the maximum anticipated water pressure. Hydrostatic test pressure can be used as the thrust restrained design pressure for most applications.

5.0.2.25.6.3 Reference

The following technical references shall be used for calculating thrust restraint system, as required:

1. AWWA M9: Concrete Pressure Pipe by AWWA, Latest Edition
2. AWWA M11: Steel Pipe- A Guide for Design and Installation, Latest Edition
3. Thrust Restraint Design for Ductile Iron Pipe, 2002, or Latest Edition.

5.0.2.26 Manhole Frame and Cover

All manhole frames and covers must meet the minimum requirements as set in 30 TAC 217.55, and for manholes located within the AOA all FAA requirements must be met.

All manhole frames/covers, valve castings are standard steel castings with a special heavy-duty cover.

1. Manhole covers for potable water mains shall be 36 inches diameter.
2. Valve boxes for light traffic are the standard concrete box with a special cover.
3. Valve boxes for traffic areas are of heavy- duty traffic design.

Table 12 - Water System Approved Product List

Basic Product Category	Product Sub-Category	Manufacturer	Model, Type, or Style Approved	Reference Specifications (NCTCOG)	National Specifications
Automatic Flushers		Manufacturer			
Automatic Flusher	Smart Flush	The Kupferle Foundry Company	9800i-WC Eclipse		
Water Meters		Manufacturer			
Water Meters	All Meters	Badger	Badger, Becon with HR-E encoders, LTE Endpoints		AWWA C700 & C701
Water Pipes		Manufacturer			
PVC Pressure Pipe	PVC 8"-12" DR14	Pipelife Jet Stream	DR14	501.9	AWWA C900
PVC Pressure Pipe	PVC 8"-12" DR18	Pipelife Jet Stream	DR18	501.9	AWWA C900
PVC Pressure Pipe	PVC 8"-12" DR14	Diamond Plastics Corporation	DR14	501.9	AWWA C900
Corporation Stop		Manufacturer			
Corporation Stop*	Quick Joint Ball Valve	Ford	FB-1000-x-Q-NL	502.5	AWWA C800
MUST HAVE STAINLESS STEEL COMPRESSION GRIP RING					
Fire Hydrants		Manufacturer			
Fire Hydrants		Muller or M&H	Medallion F2545	502.6	AWWA C502
Fire Hydrants		Mueller	Super Centurio N 250	502.6	AWWA C502
Fire Hydrants		M & H	M & H 129	502.6	AWWA C502
MUST HAVE FULL BODY GLANDS AND RED PRIMER					
Fittings		Manufacturer			
Fittings, Full-Bodied	Ductile Iron	American Cast Iron Pipe		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Star Pipe		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Tyler/Union/UFC O		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Sigma		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	US Pipe & Foundry		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	SIP		502.6	AWWA C110
COMPACT FITTINGS AND GLANDS ARE NOT ALLOWED					

Manhole Appurt.		Manufacturer			
Manhole Ring & Lid	32-inch	Bass & Hays	V1420		
Manhole Ring & Lid	40-inch	Bass & Hays	#1225-40		
Manhole Ring & Lid	40-inch	Neenah	R-1901		
Service Saddles		Manufacturer			
Service Saddles	Bronze Double Strap	James Jones	J-979	502.10	AWWA C800
Service Saddles	Bronze Double Strap	Ford	202B	502.10	AWWA C800
Service Saddles	Bronze Double Strap	McDonald	3825	502.10	AWWA C800
Service Saddles	Bronze Double Strap	Mueller	BR2B	502.10	AWWA C800
Service Saddles	Bronze Double Strap	Smith – Blair	323	502.10	AWWA C800
Tapping Sleeve	Stainless Steel	Manufacturer			
Tapping Sleeve	Stainless Steel*	Cascade	CST-EX	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Ford	FTSS	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	JCM	432	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Mueller	H304	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Romac	SST III	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Smith – Blair	665	502.10	AWWA C 800
*OUTLET FLANGES MUST ALSO BE STAINLESS STEEL. SHALL BE RESTRICTED TO PIPE SIZES 8" AND SMALLER AND SHALL NOT BE USED FOR TAPS GREATER THAN 75% OF THE MAIN SIZE.					
Air Valves		Manufacturer			
Valves, Air	AIR RELEASE AND VACUUM BREAKER	CLA-VAL		502.6	
Valves, Air	AIR RELEASE AND VACUUM BREAKER	VAL-MATIC		502.6	
Valves, Air	AIR RELEASE AND VACUUM BREAKER	VALVE & PRIMER (APCO)		502.6	

Valves, Air	AIR RELEASE AND VACUUM BREAKER	VENT-O-MAT	RBX SERIES	502.6	
Valve Stack	Covers and Lid	Manufacturer			
Valve Covers	Lid	Bass & Hays	#340-1	806.4	
Valve Covers	Lid	Sigma	VB1651M	806.4	
Valve Covers	Cover	Sigma	VB1652M	806.4	
Valve Covers	Lid	Accucast	115002	806.4	
Valve Covers	Stack	Accucast	115000	806.4	
Valve Covers	Lid	Star Pipe Products	Shorty	806.4	
Valve Covers	Stack	Star Pipe Products	Shorty	806.4	
Valves	Vertical Resilient	Manufacturer			
Valves, Gate*	Vertical, Resilient Wedge	Mueller	2360, 2362	502.6	AWWA C509
Valves, Gate*	Vertical, Resilient Wedge	M & H	4067, 4068	502.6	AWWA C509
* MUST HAVE 316 SS BOLTS. BOTH RED EPOXY OR BITUMINOUS EXTERIOR COATINGS ARE ALSO ACCEPTABLE. AWWA C-515 RSGVs WITH REDUCED WALL THICKNESSES ARE NOT ACCEPTABLE.					
Water Meter Vaults	Large Services	Manufacturer			
Water Meter Vaults	Concrete	American Industrial Precast	MUST HAVE APPROVED LIFTING DEVICES		
Water Meter Vaults	Concrete	Brooks/Oldcastle Precast	MUST HAVE APPROVED LIFTING DEVICES		
Water Meter Vaults	Concrete	Concrete Products (CPI)	MUST HAVE APPROVED LIFTING DEVICES		
Water Meter Vaults	Concrete	Hanson	MUST HAVE APPROVED LIFTING DEVICES		
Water Meter Vaults	Concrete	New Basis (Dalworth Quickset)	MUST HAVE APPROVED LIFTING DEVICES		
Water Meter Vaults	Concrete	Turner	MUST HAVE APPROVED LIFTING DEVICES		

Water Meter Boxes	Small Services	Manufacturer			
Water Meter Boxes for 1" Services	PLASTIC WITH DI RING & LID & AMR Lids	Accucast	846211 COMPLETE, 842100 BOX, 872065 LID, 872061 RING		
Water Meter Boxes for 1" Services	PLASTIC WITH DI RING & LID & AMR Lids	Bass & Hays Mid States	MS18DAL BOX, BH-3PD LID, BH- 4PD RING		
Water Meter Boxes for 1" Services	PLASTIC WITH DI RING & LID & AMR Lids	Sigma	MB151-DFW- 18DAL		
Water Meter Boxes for 1 1/2" & 2" Services	PLASTIC WITH DI RING & LID & AMR Lids	Bass & Hays Mid States	MS30DAL BOX, BH-5PD LID & RING		
Water Meter Boxes for 1 1/2" & 2" Services	PLASTIC WITH DI RING & LID & AMR Lids	Accucast	846215 COMPLETE, 843000 BOX, 872071 RING AND 872070 LID		
* GALVANIZED BOXES ARE NOT ACCEPTABLE.					
•Engineer of record shall conform the applicability of these approved products in AOA and CTA area. This standard product list has been developed only for projects located outside the AOA and CTA area. When technical specification for specific products are included as part of the construction contract documents, the requirements of the technical specifications will override these approved product list. It is engineer of record (EOR)'s responsibility to approve the product submittal. DFW utility group recommend EOR to coordinate with DFW Utility group prior to approval.					

5.0.3 Sanitary Sewage Utilities

5.0.3.1 DFW Airport Sanitary Sewer Systems

Wherever possible, disposal of sewage shall be by gravity to the Sanitary Sewer System (SSS). Airport mains shall be extended as required to establish gravity flow disposal. The DFW Airport (DFW) owns and maintains a complete SSS, and the following information is presented as criteria for expansion of the system, or modifications to the existing system. All new connections and modifications shall be in accordance with 30 TAC §217 (Design Criteria for Domestic Wastewater Systems).

5.0.3.2 Sanitary Sewer Main Sizing

The sanitary sewer mains shall be sized based on 30 TAC §217.53, with the following additional requirements. The Engineer of record shall complete an analysis of the initial and future flows of the area to be served and review the latest utilities master plan. The collection and interceptor main shall be sized for the peak hr. flows, which is based on the estimated average daily flow. When site- specific data is unavailable, designer shall use the most conservative data while meeting or exceeding the following criteria for sizing sanitary sewer mains:

Sanitary sewer lines shall be designed to minimize turbulence to prevent release of sulfide gases and subsequent corrosion.

Velocity: The velocity of gravity sanitary sewer main shall be maintained between 2 and 10 ft/sec. (No Variances are allowed). Velocity shall be determined by the Manning Equation:
Size: For sewer mains, 15 inches in diameter or smaller, use the larger size as determined below:

1. The main shall be designed such that the Peak Dry Weather Flow (PDWF) shall not exceed 65% of the capacity of the pipe flowing full. The main shall be designed such that the Peak Wet Weather Flow (PWWF) shall not exceed 85% of the capacity of the pipe flowing full.
2. For sewer mains, 18 inches in diameter or larger, the main shall be designed such that the PWWF shall not exceed 80% of the capacity of the pipe flowing full.
3. Service: The size of the services must be at least one standard size smaller than the proposed and existing sanitary sewer main.

5.0.3.2.1 Minimum Pipe Sizes

The minimum pipe diameter for any public gravity sanitary sewer collection main shall be 8 inches for maintenance and easy cleaning.

5.0.3.2.2

Minimum and Maximum Slope Refer to 30 TAC §217.53, for selecting sanitary sewer main slopes.

5.0.3.3 Sanitary Sewer Pipe Materials and Embedment

5.0.3.3.1 Pipe Material Selection

1. Designers are responsible for specifying the type of pipe to be used in any sanitary sewer main design. Typically, minimum 150-psi pressured rated sanitary sewer pipes is preferred for small diameter (<15") gravity mains.
2. The Engineer of Record shall present a technical memorandum addressing the pipe material, geological conditions, and embedment requirements to DFW Utilities with the design plans for all sanitary sewer main projects.

Table 13 - Pipe Material and Joints

Pipe & Material Specification	Min. Pipe Stiffness (PS)	Allowable Size (in)	Joint Specification
PVC- Pressure Rated* AWWA C-900 (DR-25)	PS= 165	4" – 15"	Bell & Spigot Joints: ASTM D3139 Gasket: ASTM F477 Fusible Fusible C 900(R) Certa-Lok Certa-Lok C900/RJ(TM) EAGLE LOC
PVC- Pressure Rated* AWWA C 900 (DR 21)	PR= 200	18" – 48"	Bell & Spigot Joints: ASTM D3139 Gasket: ASTM F477 Fusible Fusible C 900(R)
PVC- Pressure Rated* AWWA C 900(DR 18)	PR= 235	18" – 48"	EAGLE LOC Eagle Loc 900 Certa-Lok Certa-Lok C900/RJ(TM)
HDPE- Pressure Rated AWWA C906 (DR-11) PE 4710 or	PS= 200 psi	8" – 36"	Fusible Fusion ASTM F2620
Pressure- Rated RTRP α & Pressure- Rated- RPMP β ASTM D 3754 (Fiberglass)	SN $\geq$ 150 psi	18" – 60"	Bell & Spigot Joints: ASTM D3212 Gasket: ASTM F477

5.0.3.3.2 Embedment Requirements

1. Designers shall specify the class of embedment on the design drawings. The type of embedment to be used is determined by pipe material and the depth of cover. The depth of cover is measured from the top of the pipe to the natural or finished ground surface above the pipe.
2. The Engineer of Record shall prepare a technical memorandum that includes the maximum pipe deflection anticipated by the pipe. Pipe deflection shall not be more than 50% of manufacturer's recommendation.
3. General terminology of a typical pipe trench is as follows:

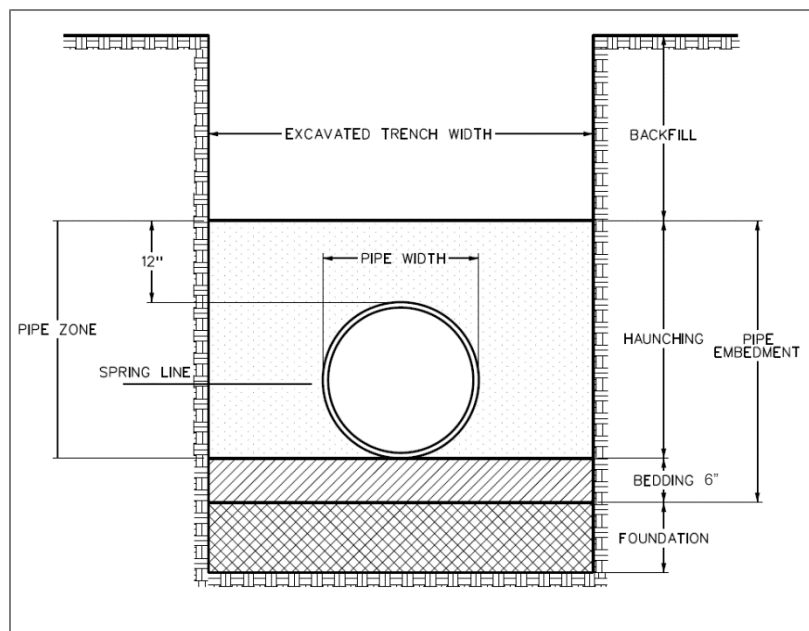


Figure 12 - Typical Pipe Trench

4. The minimum width of the trench where the pipe is being installed shall be wide enough to allow for proper alignment of the piping system.

5.0.3.4 Depth of cover

The depth of cover is measured from the top of the pipe to the natural or finished ground surface above the pipe. The main must be deep enough to serve adjacent properties. Buoyancy of sewers shall be considered, and flotation of the pipe shall be prevented with appropriate construction methods where high groundwater conditions are anticipated.

If fill or embankment placed over existing sanitary sewer mains exceeds 6 feet above the existing ground, DFW Utility Group approval is required. If cuts exceed the minimum depth of cover stated below, DFW Utility Group approval is required. The minimum depth of cover over the upper-most projection of the main shall be as follows table.

Table 14 - Pipe Size and Depth of Cover

Size of Main	Min Depth of Cover (ft)		
	Normal Installation	Highway/Roadway	Runway Taxiway
12" and smaller	6	6	Engineering Evaluation & FAA Regs
12" – 16"	6	6	
20" and larger	7	7	

5.0.3.4.1 Minimum cover

The above guideline applies to sanitary sewer main installation in public right-of-way:

5.0.3.5 Locations

Mains shall be located outside of the roadway footprint within the ROW and should follow the roadway.

1. Sanitary sewer manholes shall not be located in the flowline of an existing creek or drainage area.
2. Install the replacement main in the same trench as the existing main at 6 to 12 inches below the existing grade, if feasible.
3. Install the replacement main 3 feet parallel to the existing main, as measured from the outside edge of both pipes.

5.0.3.6 Horizontal alignment

The horizontal alignment must meet the minimum requirements as set in 30 TAC §217.53, with the following additional requirements.

1. All mains should be laid as straight as possible.
2. Stations shall be at the tenth of a foot (Ex: STA. 10+11.4).

5.0.3.7 Vertical alignment

The vertical alignment must meet the minimum requirements as set in 30 TAC §217.53., with the following additional requirements.

1. Vertical bends are not typically allowed unless otherwise approved by DFW.
2. Design slopes shall be to the nearest hundredth of a percent (Ex: Slope 5.20%).
3. Elevations shall be shown to the nearest hundredth of a foot (Ex. El. 495.95).

5.0.3.8 Separation Distance between Sanitary Sewer and Reclaimed and Water Mains

When a sanitary sewer main is built near an existing water facility, conveyance, or appurtenances, 30 TAC §217.53, governs the minimum separation distances.

5.0.3.9 Connections to Existing Mains

1. The connection, shall always be at a manhole, including all laterals.

5.0.3.10 Sanitary Sewer Service Lateral

1. Each lease property must have its own sanitary sewer service lateral which must not cross any existing or projected lease line(s).
2. Sample and inspection ports are required for service lines when industrial waste monitoring is required. They shall be located at the lease line within the public right-of-way (ROW) to indicate the line of responsibility of the utility.
3. They shall not be located in traffic areas, paved parking areas or sidewalks, these areas shall require manholes.

5.0.3.10.1 Location

1. Sanitary sewer service lateral should be located a minimum of 10 feet downstream of the water service.
2. A sanitary sewer clean-out shall be installed at lease line as a demarcation between public and private line.
3. Sanitary sewer clean-outs are not allowed in sidewalks, paved areas, load bearing pavement, or driveways, these areas will require a manhole.

5.0.3.10.2 Size

1. The minimum size of the sanitary sewer service lateral from the main to the property line shall be 6 inches unless other conditions warrant a larger size. The following criteria may be used as a guide:

Table 15 - Sanitary Sewer Lateral Sizing Criteria

Fixtures	Flow (gpm)	Size (in)
0 – 720	0 – 180	6
720 – 2640	180 – 4000	8
2640 - 4680	4000 - 7000	10
4680 - 8200	7000 - 11600	12

5.0.3.10.3 Depth

1. The sanitary sewer service lateral should be at a depth sufficient to ensure the dwelling to be connected, will be served using a preferred lateral grade of 2 % (min. 1%) and minimum cover of 4 feet.
2. The top of the downstream manhole should be a minimum of 18 inches below the finish floor (FF) elevation of the dwelling to be connected. In cases where this is not achievable approval must come from DFW and will generally require a backwater valve device and damage waiver from the user.

5.0.3.11 Inverted Siphon

The inverted siphons/sag pipes must meet the minimum requirements as set in 30 TAC §217.53, with the following additional requirements.

5.0.3.11.1 General

1. Design flow shall be the hydraulic capacity of the upstream flow or the future projected flow.
2. All siphons shall be completely buried with a minimum of 48 inches of cover when going across a natural or man-made depression,
3. The maximum slope of the downstream (rising) leg of the siphon shall be 15%.
4. The total computed head loss should be increased by 10% as a factor of safety.

5.0.3.11.2 Components

A typical siphon consists of the following components.

5.0.3.11.2.1 Conduits

1. The siphon shall consist of two or more conduits with a minimum pipe diameter of 8 inches.
2. The arrangement of inlet and outlet details must divert the normal flow to one conduit and allow overflow into the second conduit. Usually, the conduit taking the overflow will be right above the point where the approaching conduit is carrying the flow that the first conduit is able to carry at full flow.
3. When multiple conduits are to be used, all conduits should be arranged so that additional conduits can be brought in service as sanitary sewer flows increase.
4. Pressure-rated pipes and fittings shall be used.

5.0.3.11.2.2 Inlet and Outlet Structures or Junction Structures

Inlet and Outlet Structures, two manholes or junction structure must be designed including one upstream and one downstream with adequate clearance for cleaning equipment, inspection, and flushing.

5.0.3.12 Aerial Creek Crossing

Aerial crossing may be used only when all other alternatives have been evaluated, and other options are either not available or not feasible. The engineer of record shall get approval from ETAM. for aerial crossing design. The Engineer of record shall use steel encasement pipe or ductile iron pipe around all aerial carrier pipe.

All aerial crossings when allowed must meet the minimum requirements as set in 30 TAC §217.53. (o), "Bridged Sections".

5.0.3.13 Force Mains

Force mains are required whenever lift stations are required and will be in full compliance with 30 TAC §217, with the following additions.

Lift station/force main systems shall be evaluated for their sulfide generation potential and their ability to achieve scouring velocities during average dry weather flow periods. If the evaluation indicates that sulfide concentration of greater than 2 ppm and solids deposition are likely, the design shall:

1. Define a workable sulfide control technique that will minimize sulfide formation in the force main.
2. Include "pig" launching stations and recovery points to allow cleaning of the force main.
3. Protect the gravity main and manholes downstream of the force main from corrosion. The length of pipe to be protected shall be determined on a case-by-case basis.

The maximum time required to flush the force main shall be calculated on the basis of average dry weather flow. Flush time shall be calculated for average dry weather flow. Design Engineer shall provide the calculated flush time for DFW Utility Group approval.

The force main shall discharge into its own distinct manhole (i.e. multiple force mains shall not discharge into a single manhole.)

Thrust restraint when required shall be shown on the plan view.

5.0.3.13.1 Pipe Materials

All force main piping must meet the minimum requirements as set in 30 TAC §217.64, and the pipe joints shall comply with 30 TAC §217.65, with the following additional requirements.

All force mains shall be approved PVC or an approved High Density Polyethylene Pipe (HDPE) with a minimum diameter of 4 inches. Force main pipe within the station shall be flanged. Flexible fittings shall be provided at the exit wall.

5.0.3.13.2 Identification of Force Main Pipes

All force main piping must meet the identification requirements as set forth in 30 TAC §217.66, and section 5.0.1.11 of this manual.

5.0.3.13.3 Velocities

All pipe velocities shall meet requirements as set in 30 TAC §217.67, with the following additional requirements.

Force mains shall be sized so that the flow velocity is between 3 and 6 feet per second at initial and ultimate development.

5.0.3.13.4 Detention Time

The detention time requirements shall meet the minimum requirements as set in 30 TAC §217.67.

5.0.3.13.5 Connection to Gravity Main

1. Shall meet the minimum requirements as set in 30 TAC §217.67.
2. A force main must terminate in an appropriate structure and either at a manhole.
3. The discharge end of a force main inside a manhole must remain steady and produce non-turbulent flow.
4. A receiving sanitary sewer collection system must accept the maximum pump discharge without surcharging.

5.0.3.13.6 Odor Control

Odor control shall be in accordance with the requirements as set in 30 TAC §217.67.

5.0.3.13.7 Air Release Valves in Force Mains

All force main air release valves must meet the minimum requirements as set in 30 TAC §217.67, with the following additional requirements.

Location and size of all air release valves shall be evaluated for odor or nuisance potential to adjacent property by the design engineer. The use of air release valves shall be restricted to installations where there are no possible alternatives.

5.0.3.13.8 Valves

All force main valves must meet the minimum requirements as set in 30 TAC §217.67.

5.0.3.13.9 Force Main Testing

All force main testing must meet the minimum requirements as set in 30 TAC §217.68,

5.0.3.13.10 Water Meter

A water meter shall be installed along with reduced pressure back-flow preventer, hot box, electric, and signage.

5.0.3.14 Sanitary Sewer Lift Stations

All sanitary sewer lift stations shall be designed strictly in accordance with 30 TAC §217.59-63. Lift stations utilizing dry wells will not be allowed. All concrete utilized in lift stations shall be polymer concrete.

1. Engineer of Record shall submit maximum wet weather flow, maximum dry weather flow, average dry weather flow and minimum dry weather flow to DFW Utility Group for approval. Lift station shall be designed to handle the maximum wet weather flow for its service area.
2. Force main pipe size shall be per maximum dry weather flow. Average dry weather flow shall be used to determine the average detention time in the wet well. Minimum dry weather flow shall be used to determine the maximum detention time in the wet well.
3. The bottom of the wet well shall have a minimum slope to the intake of 2 vertical to 1 horizontal. There shall be no projections in the wet well, which would allow deposition of solids.
4. The wet well volume shall be sized to provide adequate storage volume at peak design flows and a pump cycle time of sufficient duration to prevent pump short cycling and consequential motor damage. Pump cycle time, defined as the sum of "pump off" time plus "pump on" time, shall be as follows:

Table 16 - Motor Size and Cycle Times

Motor (H.P.)	Minimum Cycle Time (Min)
2 to 50	10
51 to 75	15
76 to 250	30
251 to 1500	45

5.0.3.14.1 Preliminary Design Submittal:

A preliminary design submittal will be required for each lift station. The submittal report shall be prepared by a professional engineer register in the State of Texas. The report shall include a brief summary of project scope, plans, a basin map, influent hydraulic calculation, wet well volume calculation, pipe size calculation, velocity, ground water levels in proposed site area, proposed system's effect on existing system's capacity, and opinion of probable construction cost for lift station, force main and annual operating, and maintenances cost.

Engineer of record shall provide the odor control measure. The potential odor determination must include the estimated flows immediately following construction and throughout a system's 50 year expected life cycle.

1. Lift stations will only be considered a viable option if the cost analysis clearly shows that the gravity sanitary sewer mains are not economically feasible.
2. All lift station designs shall consider the potential for future expansion. The design of the lift station shall incorporate a wet well sized for the final capacity of the lift station.

5.0.3.14.2 Lift Stations

Wet Well Detention Time Design Engineer shall calculate the detention time in the wet well for the maximum wet weather flow. The static head shall be calculated for "pump on" and "pump off" elevations in the wet well.

5.0.3.14.3 Site Selection

Lift station sites must meet the minimum requirements as set in 30 TAC §217.59.

5.0.3.14.4 Wet Well and Valve Vaults

The design of all wet wells valve vaults must meet the minimum requirements as set in 30 TAC §217.60.

5.0.3.14.5 Lift Station Pumps

All lift stations pumps must meet the minimum requirements as set in 30 TAC §217.61, with the following additions.

1. A minimum of 2 pumps shall be required for all lift stations. The capacity of the pumps shall be such that the maximum wet weather flow can be handled with the largest pump out of service.
2. All "pump on" levels shall have a minimum separation of 1 foot between levels. All "pump off" levels shall be at least 6 inches above the top of the pump casing. For more than 2 pumps, the "pump off" levels shall be staged with a minimum separation of one foot between levels. At minimum, design engineer shall provide the pump staging sequence for High level alarm, lag pump on, lead pump on, lag pump off, lead pumps off, and low-level alarm. The high-level alarm shall be at least 1 foot above the last (highest) "pump on" level in the wet well and also at least 1 foot below the flow line of the lowest influent line into the wet well.
3. The net positive suction head (NPSH) required by the pump selected shall be compared with the NPSH available in the system at the eye of the impeller. The engineer shall consult the pump manufacturer for the NPSH required values for that pump and compare them with calculated values for the NPSH available. The NPSH available should be greater than the NPSH required for a flooded suction pump. Design engineer shall provide the NPSH calculation to DFW Utility Group for approval.

5.0.3.14.6 Lift Stations Piping

All lift station piping must meet the minimum requirements as set in 30 TAC §217.62, with the following additional requirements.

All suction piping shall be flanged ductile iron and have a minimum diameter of 4 inches. Each pump shall have a separate suction pipe.

1. Suction piping shall have a velocity of 3 to 5 fps.
2. All suction pipes inside the wet well shall be equipped with a flare type, down- turned intake. The distance between the bottom of the flare and the floor of the wet well shall be between $D/3$ and $D/2$ where D is the diameter of the flare inlet.

5.0.3.14.7 Electrical Requirements

All electrical appurtenances and equipment must meet the minimum requirements as set in 30 TAC §217.63, and §217.326, and the latest adopted National Electrical Code (NEC).

5.0.3.14.8 Head Loss Curves

Data points for the system capacity curve shall be provided in tabular form and graphed with pump head capacity curve on the same graph. Two system capacity curves shall be plotted using the Hazen Williams coefficient values of $C = 100$ and $C = 140$.

1. Pump output in gpm at maximum and minimum head shall be clearly shown on the system curve for each pump and combination of pumps.
2. For stations with 2 or more pumps operating in parallel, multiple, and single operation points shall be plotted on the system curve.
3. Pumps with the highest efficiencies at all operating points shall be used.

4. If pumps are equipped with smaller impellers during start up to handle lower than design flows, impellers sized to handle the design flow shall also be provided.

5.0.3.14.9 Buoyancy Calculations

The lift station design shall include a complete analysis of buoyant forces on the entire lift station structure.

5.0.3.14.10 Water Hammer

Design Engineer shall calculate water hammer showing maximum pressures, which would occur upon total power failure while pumping.

5.0.3.14.11 Suction Specific Speed

Suction specific speed should be below 9,000 rpm to ensure that the pump will not cavitate because of internal recirculation.

5.0.3.14.12 Stiffness Ration

In order to ensure that the pump shaft does not bend an excessive amount, the engineer shall calculate the stiffness ratio of the shaft. The stiffness ratio shall not exceed 60.

5.0.3.15 Abandonment of Sanitary Sewer Mains and Appurtenances

Sanitary sewer mains shall be removed from the ground whenever possible, when not possible, the line will be abandoned in place by filling the main with class B concrete.

5.0.3.15.1 Manhole, Cleanout and Sanitary sewer Access Device

1. Manhole, cleanout, and sanitary sewer access devices shall be abandoned by filling with sand and/or gravel compacted to 90% (95% in pavement) of maximum standard proctor dry density.
2. The bottom of manhole, cleanout and sanitary sewer access device shall be filled with class B concrete up to the top of sanitary sewer pipe(s).

5.0.3.16 Corrosion Control

For corrosion control of Sanitary Systems, refer to section 5.0.2.10 of this manual. In addition to the soil reactivity for corrosion control, consideration shall be given to corrosion control for exposures to the gases produced by the sanitary sewer system.

5.0.3.17 Pre and Post CCVT

1. Prior to project close-out, and after installation of the entire Sanitary Sewer system has been completed, all Sanitary Sewer lines and structures shall be cleaned, inspected with a video system, and reviewed by the contractor before copies of the videos are submitted to DCC and ETAM for review.
2. Modifications to existing lines or replacement of existing mains shall be inspected both preconstruction, and post construction.

5.0.3.18 Testing Procedures

All installed SSS pipe must be tested in accordance with the provisions of 30 TAC §217.57.

5.0.3.19 Manholes

All manholes shall meet the minimum requirements as set in 30 TAC §217.55, with the following additional requirements.

Manholes shall be located and spaced so as to facilitate inspection and maintenance of the sanitary sewer main. All manholes must be accessible to maintenance equipment, including 2½ ton straight trucks, dump trucks, vacuum trucks, and standard (not compact) sizes of backhoes and loaders. In isolated cases, construction of all-weather access roads may be necessary for manhole and/or Sanitary sewer line access. Manholes shall be placed at the following locations:

1. Intersections of mains
2. Horizontal alignment changes
3. Vertical grade changes
4. Change of pipe size
5. Change of pipe material
6. The point of discharge of a force main into a gravity sanitary sewer main
7. For commercial developments containing more than 4,000 square feet of air-conditioned space and requiring a water meter greater than 2 inches, a manhole is required on the main at the point of connection to the Sanitary sewer service.
8. At the upstream end of mains

Manholes shall be constructed of a corrosion resistant material. Where new construction ties into an existing manhole, the existing manholes must be lined, coated, or replaced with a corrosion resistant material. The Design Engineer shall provide Manhole Inspection report for sanitary sewer Manhole replacement or rehabilitation for both CIP and non-CIP projects to DFW Utility Group. Where concrete is utilized, all concrete shall be polymer concrete.

Manholes constructed on existing sanitary sewer mains may have a cast-in-place base. All other manholes shall have a pre-cast base.

Manhole and junction box inverts shall have a minimum slope of 2.5% between the inlet and outlet pipe inverts.

5.0.3.19.1 Manhole Spacing

All manhole spacing must meet the minimum requirements as set in 30 TAC §217.55, with the following additional requirements.

Manholes may be spaced no further apart than the distance specified in the following table:

Table 17 - Manhole Spacing Criteria

Pipe Diameter (in)	Max. Manhole Spacing (ft)
8 – 15	500
18 & Larger	800

The maximum allowable manhole spacing for collection systems with horizontal curvature is 300 feet. A manhole must be at point of curvature (PC) and the point of termination of a curve (TC).

5.0.3.19.2 Manhole Frames and Covers

All manhole frames and covers must meet the minimum requirements as set in 30 TAC §217.55, with the following additional requirements.

1. For all manholes located within the AOA, all FAA requirements must be met.
2. All sanitary system manhole covers must be clearly marked with "Sanitary Sewer" visible on the cover.

All manholes must include a concrete pad around the frame and cover with an ID number clearly marked on the upstream side of the manhole pad. The Design Engineer shall co-ordinate with DFW Utilities for Manhole ID number. Design engineer shall provide the detail for manhole ID.

5.0.3.19.3 Manhole Sizes

All manholes shall be sized to meet the minimum requirements as set in 30 TAC §217.55, except that deep manholes shall be required to be a minimum of 60 inches in diameter when over 15 feet deep, and a minimum of 72 inches in diameter when the depth is over 25 feet deep.

Design Engineer shall check following scenario prior to sizing the manhole:

In the event a structure is utilized inside a manhole, the clear space between the structure and the manhole wall shall be a minimum of 48 inches.

If more than two mains connect to a manhole, or if two mains connect to a manhole at an angle other than 180 degrees from each other, a larger diameter manhole may be required in order to accommodate mandrel insertion and hydraulically efficient flow.

New pipe connections to existing manholes shall provide a minimum of 12 inches clearance between the existing pipe ID and the new core hole ID measured on the inside surface of the manhole, regardless of the orientation of the pipes with respect to one another. New precast manholes and manholes with cast-in-place bases shall have holes for pipe penetrations in the manhole wall separated by a minimum of 7 inches, designed by the manhole manufacturer and as measured from the inside diameter of the cored holes on the inside wall of the manhole to ensure the structural integrity of the manhole wall. A Junction structure should be used where a standard 8' diameter pre-cast manhole is not adequate.

Formed concrete collars are required at all existing and precast structure connections. All formed collars will have a minimum of (2) #3 rebar bands around the pipe with #3 rebar transversely at 12 inches OC and located at mid span. The collar shall be 18 inches in width and surrounding the pipe rubber boot a minimum of 6 inches on all sides. The EOR shall provide a detail in the design set for the contractor to use or submit TxDOT Concrete Collar & Pipe Cap Details.

The vertical distance between the highest point of the invert shelf and the bottom of any horizontal or near-horizontal surface protruding into a manhole or junction box, shall be at least 6 feet, when the depth of the main is sufficient.

5.0.3.19.4 Drop Manholes

This can be used where the incoming pipe(s) is 2 feet or higher than the outgoing pipe:

1. Manhole with 2 to 10 foot Drop: External (preferred) or internal drop connection shall be used.
2. Manhole with > 10 foot. Drop: Shall be evaluated on a case-by-case basis.

5.0.3.19.5 Meter Manholes

In all manholes used for metering purposes, all metering devices, electrical equipment, and or electrical wiring shall be “intrinsically safe” to prevent any electrical spark in the presence of potential sewer gas.

5.0.3.19.6 Concrete Pad

When manholes are to be located within a public street, concrete manhole pads shall be used around the manhole frame.

Table 18 - Collar Size and Thickness

MH Size	Collar Size (ft)	Collar Thickness (in)
48	6x6	10
60	8x8	12
>60	Engineering Evaluation	Engineering Evaluation

5.0.3.20 Sanitary Sewer Junction Structure

A sanitary sewer junction structure is generally required where two or more sanitary sewer pipelines cannot be connected at a manhole due to inadequate separation distance between the outer diameters of the mains. The following items, but not limited to, must be considered for designing a sanitary sewer Junction structure.

5.0.3.20.1 General

1. Junction structures are used typically where a standard 8 foot diameter pre-cast manhole is not adequate.
2. The separation distance between pipes must be at a minimum of 1 foot inside junction structure.
3. All pipes entering a junction structure must be at right angle at the junction wall.

5.0.3.20.2 Structural Consideration

1. Roofs of junction structures must be structurally designed for AASHTO loading standard M-306. Except in AOA areas, where FAA requirements shall be used.
2. Non-shrink grout shall be used between the junction structure and sanitary sewer mains along structure wall.

5.0.3.20.3 Corrosion Protection

1. Corrosion protection provisions shall be considered for all junction structures.
2. All moving components of the structure must be Stainless Steel 316 unless otherwise approved by DFW.

5.0.3.20.4 Manhole and Junction Boxes Located Below Ground Water

1. When the interior surface of a concrete manhole or junction box is coated with a urethane, polyurethane, or epoxy liner, the exterior surface of that portion of a manhole or junction box located below ground water level shall be waterproofed using a flexible system applied to the exterior surface. The drawings shall indicate which structures must be waterproofed and the elevation to which waterproofing must be applied (2 feet above ground water level).
2. Manhole joints below the ground water level and/or located in the 100year floodplain shall be sealed by installing a joint wrap material over the joint on the manhole exterior.
3. Construction joints in cast-in-place junction boxes shall be waterproofed using water stops.

5.0.3.21 Sanitary Sewer Access Devices and Cleanouts

Manholes shall be used at all sanitary sewer access points. Cleanouts will not be allowed.

5.0.3.22 All Weather Access Road for Manhole, Junction Structure, and Lift Station

1. All-weather access roads should be at least 14 feet wide. It is intended for emergency use by maintenance equipment. This 14 foot maintenance access road should be outside the toe of any fill slope and the top of any cut slope and shall not have a post construction longitudinal slope greater than 15% nor a post construction transverse slope greater than 5%, shall not have a vertical grade break of greater than 12%, should have an inside turning radius of no less than 35 feet, an outside turning radius of no less than 50 feet, shall be cleared of all vegetation and graded, and should maintain a horizontal and vertical clearance from existing and proposed vegetation and all other objects of no less than 20 feet.
2. The access road shall include a means for equipment to turn around when located more than 200 feet from a paved public roadway. Turn around shall meet the above listed design criteria. Access roads for manhole and junction structure shall be cleared, graded, and stabilized with stones.
3. The Access roadway for lift station shall have a concrete or asphalt concrete pavement as the roadway's surface. Base material and sub-base material as recommended by Geotechnical Report for the site-specific soil and load condition. Roadway shall include a means for equipment to turn around.

Table 19 - Sanitary Sewer Approved Product List

5.0.3.23 Sanitary Sewer Approved Products List							
Mainline-Pipeline (Material)	Manufacturer	Pipe Standard	Joint Specification	Gasket Specification	Notes	Reference Specification (NCTCOG)	National Specification
PVC-Pressure Rated 8"- 15"	JM Eagle or Diamond Plastics	AWWA C-900	BELL AND SPIGOT-ASTM D3139	ASTM F477	PS= 165 psi	501.15	AWWA C 900
PVC-Pressure Rated 18"- 48"	JM Eagle or Diamond Plastics	AWWA C-900 (SDR 21)	BELL AND SPIGOT-ASTM D3139	ASTM F477	PS=200 psi	501.15	AWWA C 900
PVC-Pressure Rated 18"- 48"	JM Eagle or Diamond Plastics	AWWA C-900 (SDR 18)	BELL AND SPIGOT-ASTM D3139	ASTM F477	PS= 235 psi	501.15	AWWA C 900
HDPE-Pressure Rated	Performance Pipe	ASTM F 714 (DR 11)	FUSION ASTM F2620	ASTM F2620	Specify DIPS Only with Min. PS= 200 psi		ASTM F 714
RTRP-Pressure Rated	Hobas or Ameron	ASTM D3754	BELL AND SPIGOT-ASTM 4161	ASTM F477	Min. SN=150 psi	501.24	ASTM D3262
RTRP-Pressure Rated	Hobas or Ameron	ASTM D3754	BELL AND SPIGOT-ASTM 4161	ASTM F477	Min. SN=150 psi	501.24	ASTM D3262
NOTE: DESIGN PLANS DESIGNATE ONLY ACCEPTABLE MATERIALS FOR PROJECT							

Pipe/Manhole Corrosion Protection	Manufacturer	Standard/Model		Notes	Reference Specification (NCTCOG)	National Specification
Protective Liner-Epoxy	Raven Lining Systems	Raven 405		Existing MHs Only	502.9	
Precast Concrete Additive	Conshield Technologies	E.P.A. registration number 75174-2-47000		For use on 4.0' Pre-cast MH only	502.9	
Precast Concrete Additive	BASF Corp	MasterLife AMA 100		For use on 4.0' Pre-cast MH only	502.9	
Protective Liner-Polyurethane	SprayRoq Lining Systems	ASTM D639/D790		N/A	502.9	ASTM D639/D790
Protective Liner-Polyvinyl Chloride	Ameron International	T-Lock		For diameter s of 30" and above. All PVC liners must be white or close to white in color.	502.9	

Manholes	Manufacturer	Standard		Notes	Reference Specification (NCTCOG)	National Specification
Precast	Thompson Pipe Group				502.1	ASTM C478
Precast	Tuner				502.1	ASTM C478
Fiberglass	LF Manufacturing			Non traffic areas only	502.1	ASTM 3753
Precast Steel Reinforced Polymer Concrete Structure	U.S. Composite Pipe			Use only when recommended by engineer	502.1	ASTM D 6783 and ASTM C 478

ALL NEW MANHOLES SHALL HAVE CORROSION PROTECTION

Manhole Lid/Cover/Frame	Manufacturer	Standard/Model		Notes	Reference Specification (NCTCOG)	National Specification
Standard 32"	Bass and Hays	V1420DRING, V1480LID			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 40"	Bass and Hays	122540RING, 122540LID			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Pressure 32"	Bass and Hays	V1420DW T			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Pressure 40"	Bass and Hays	122540W T			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 32"	Accucast	211007			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 32"	Accucast	211009			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating

Pressure 32"	Accucast	210024			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Pressure 32"	Accucast	211023			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 32"	East Jordan Iron Works	V1420/1480Z (Frame) 1480A(Cover)			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 40"	East Jordan Iron Works	BH 1225 40 (Frame & Cover)			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Pressure 32"	East Jordan Iron Works	V1420/1480ZPT (Frame), 1480APT (Cover)			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating

Pressure 40"	East Jordan Iron Works	BH 1225 40PT (Frame & Cover)			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Standard 32"	Star Pipe	MH32DW W CL35			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Pressure 32"	Star Pipe	MH32DW TCL35			806.4	ASTM A48 Class 35B, Cast Iron Only Must have Bituminous Coating
Cleanouts	Manufacturer	Standard/Model		Notes	Reference Specification (NCTCOG)	National Specification
Cleanout Lid	Bass and Hays, Star	339LID			502.2/806.4	
Other	Manufacturer	Standard/Model		Notes	Reference Specification (NCTCOG)	National Specification

Drop Bowl	Reliner/Duran, Inc	Inside Drop Bowl				
Manhole to Pipe Connectors	A-LOK	A-lok Series				ASTM C923
Manhole to Pipe Connectors	NPC Kor-N- Seal	106/406 Series		Precast Manholes Only		ASTM C923
PVC Adapter Coupling	JM Eagle or Diamond Plastics	NA		For PVC Pipe		
Manhole Insert	Doug Meadows	No Flow-Inflow Manhole				ASTM D 1248
NOTES						
UNLESS OTHERWISE IDENTIFIED BY SPECIFICATION OR STANDARD, NCTCOG STANDARDS APPLY TO INSTALLATION						
FOR STANDARD DRAWING AND/OR SPECIFICATIONS NOT LISTED OR LISTED AS NA, REFER TO MANUFACTURER'S SPECIFICATIONS						
ALL ITEMS NOT ON THIS LIST NEED APPROVAL BY OWNER BEFORE PLACEMENT INTO SYSTEM						
Engineer of record shall confirm the applicability of these approved products in AOA and CTA area. This standard product list has been developed only for projects located outside the AOA and CTA area. When technical specification for specific products are included as part of the construction contract documents, the requirements of the technical specifications will override these approved product list. It is engineer of record (EOR)'s responsibility to approve the product submittal. DFW utility group recommend EOR to coordinate with DFW Utility group prior to approval.						

5.0.4 Stormwater Drainage Utilities

5.0.4.1 Design Criteria

The following documents (latest versions) shall be used as a guidance for stormwater drainage design:

1. AC 150/5320-5D Airport Drainage Design2. AC 150/5200-33B Hazardous Wildlife Attractants on or Near Airports
2. ACRP Research Report 174 Green Stormwater Infrastructure
3. Stormwater Drainage Master Plan (SDMP)
4. NCTCOG Integrated Stormwater Management (iSWM) Criteria Manual and associated Technical Manuals
5. Federal Flood Risk Management Standard

Any conflict between these referenced documents and the criteria shall be brought to the attention of DFW for resolution.

5.0.4.2 Drainage Design and Report

A drainage study is required of all projects which involve the change in runoff coefficient for an area greater than one acre and shall include a Preliminary iSWM Plan, completed in accordance with the SDMP, for review and approval by DFW. All drainage designs shall be contained in a Drainage Study which shall be submitted to DFW for review and approval. The Drainage Study shall be in such form as providing the basis for timely and consistent review and will be made a part of the permanent record for future drainage evaluations. The Drainage Study shall, at a minimum, contain the following:

1. Description and plan of existing drainage facilities.
2. Description and plan of proposed drainage facilities with iSWM Criteria Manual Section 3.6.1 compliance.
3. Drainage area map.
4. Description of the analysis conducted for meeting the following four storm event criteria per iSWM:
 - a. Storm Water Quality
 - b. Streambank Protection
 - c. Conveyance
 - d. Flood Mitigation -The analysis shall also include addressing the following questions as outlined in iSWM for acceptable downstream conditions to determine the requirements of any non-structural or structural controls:
 - i. Are the post-development discharges greater than the pre-development discharges?
 - ii. Are the post-development velocities greater than the pre-development discharges?
 - iii. Are the post-development velocities greater than the velocities allowed for the receiving system?
 - iv. Will the floodplain or inundation area water surface elevation increase?
 - v. Does the development in the floodplain increase the base flood elevation more than 0.1 foot for the 100-year or 500-Year design storm event?

5. U.S. Environmental Protection Agency (EPA) Stormwater Management Model (SWMM) with a 2D Component shall be used to develop a drainage design that includes a sufficient hydrologic analysis to determine the existing and proposed drainage conditions such that the post development drainage design meets existing conditions in regard to the four storm event criteria listed above.
6. All calculations associated with the determination of runoff coefficients, volume of runoff, time of concentration, inlet size, culvert or pipe size and elevation of hydraulic gradient and any other items pertinent to the drainage design.
7. Stormwater drainage analysis for the 1-, 10-, 25-, and 100-year design storms.
8. Consideration of drainage alternatives and recommended facilities.
9. Description of measures taken for velocity dissipation to ensure non- erosive velocities at points of discharge.
10. Flood mitigation designs shall meet the iSWM Criteria Manual Section
11. 3.5.2 – Flood Mitigation Design shall meet the SDMP requirements with detention, storage or staging to meet pre- development conditions. Detention ponds shall be designed to detain the 100-Year 24-hour design storm, which will also provide for a duration for capturing the 500-Year storm event that includes the free board for the additional storage.
12. Floodplains or 100-Year inundation zones shall have a 25 foot buffer, which shall not be encroached by any structure. The buffer shall not be landscaped but instead left in its natural state.
13. All pre- and post-development calculations associated with the drainage design to include, in tabular form in the final design plans.
14. A drainage area map with no smaller than 1 inch equals 200 feet scale, and show all streets, building pads and other existing and proposed features. The drainage area map shall show the boundary of the drainage area contributing runoff into the proposed system. The area shall be further divided into numbered sub- areas to determine flow concentration points or inlet location(s). Drainage area maps shall show streets, land- use and land-use boundaries, existing ground elevations on 1 foot contours, and a summary table of peak design flows for sub-areas with acreage, runoff coefficient, and inlet time shown.
15. The quantity and direction of design flow within streets, alleys, natural and manmade drainage ways and at all system intersections clearly shown on the drainage area map. Existing and proposed drainage inlets, storm drainage systems and drainage channels shall be clearly shown and differentiated on the drainage area map.
16. All data used to create the drainage area maps should be submitted including but not limited to existing and proposed changes in contours, proposed impervious surfaces such as buildings, parking, roadways, or other paved surfaces, stormwater and first- flush stormwater pipes, channels, inlets, manholes, outfalls.

The Drainage Study and drainage design drawings shall be sealed and signed by a Licensed Professional Engineer in the State of Texas, which will be reviewed by DFW for compliance with the requirements of these criteria. A Determination on the drainage design will be issued by DFW/ETAM/Systems Performance/Watershed Management (WSM). SWMM files will be provided by WSM for existing baseline conditions.

5.0.4.3 Determination of Design Discharge

In order to properly determine the design storm runoff for a given installation, consideration must be given to the design storm rainfall, the runoff coefficient as affected by the surface condition and by the geometry of the watershed, plus the influence of the time of concentration. The runoff coefficients and minimum inlet times (time of concentration) to be used in determining runoff shall be documented for review and approval by DFW. Adjustments to the runoff coefficients may be required to account for future build-out conditions. Hydrologic methods to be utilized for drainage analysis shall be selected based on the SDMP.

5.0.4.4 Design of Closed Storm Drainage System

In the preparation of hydraulic designs, a thorough investigation shall be made of all existing structures and their performance on the waterway in question. The design frequency for all new closed drainage systems shall be in accordance with the iSWM Criteria Manual Section 3.6.2 – Hydraulic Design Criteria for Streets and Closed Conduits, except as noted below. The total capacity of the drainage facility, including surface flow within limits of available right-of-way or easements, shall be equal to or greater than the runoff of a storm of 100- year design frequency. Shall the 100- year storm runoff exceed the capacity of the above design, then the closed storm system shall be designed based on a minimum 25-year frequency, or larger, to develop a 100-year emergency overflow system. The hydraulic gradient shall be calculated for all storm drain lines, culverts and inlets and shall not be designed above the entrance flowline of any inlet. The permissible difference between the hydraulic gradient and top- of-curb shall not be less than 2 feet.

Storm drains shall be designed to have a minimum mean velocity of 2.5 fps flowing full. Velocities equal to or greater than 10 feet per second shall not be acceptable.

Break-In taps are not allowed. Manufactured factory fittings (wyes) are required for all new connections to existing drainage piping.

5.0.4.5 Design of Open Channels

For each individual project, the iSWM Criteria Manual Section 3.6.3 – Hydraulic Design Criteria for Structures/Open Channels shall be adhered to except as noted below. Backwater analysis is to be developed for major channels to establish water surface elevation and to avoid adverse impacts on adjacent properties. All new channels shall be designed using the design storm events and methods outlined in iSWM or as approved by DFW. Channels shall not be concrete lined, therefore, other means acceptable to DFW to manage velocities over 3 feet per second must be considered. A 2 foot freeboard will be incorporated into the design calculations. Velocities in excess of 3 fps will not be allowed to enter an unlined channel without energy dissipation measures to reduce flows below 4 fps.

To provide positive drainage, a slope of no less than 2 percent for turf areas is required. Slopes shall be designed to ensure non-erosive runoff velocities. Turf banks, where required, shall be graded to permit the use of gang mowers, providing a maximum slope of 4 horizontal to 1 vertical. The tops and bottoms of all slopes shall be gently rounded in a transition curve for optimum appearance and ease of maintenance.

Unlined channels shall use a temporary mat of jute, straw, or curled wood in order to reduce erosion and sediment during the vegetation establishment phase unless sod is used to facilitate vegetation growth.

Channel rehabilitation shall incorporate restoration measures that include low and high flow sections that mimic natural channel dimensions. Refer to the SDMP- Green Stormwater Infrastructure Guidance Manual.

No concrete channels or flumes are allowed.

5.0.4.6 Design of Culverts

Drainage culverts shall pass storm flow from the upstream side of highway, road, or railroad to the downstream side without causing excessive backwater head and without creating excessive downstream velocities. The Designer shall keep the discharge velocities within safe limits (usually 4 fps) while selecting the most economical structure that will provide satisfactory service. Methods outlined by iSWM shall be used with the 100-year design frequency.

5.0.4.7 Flow in Gutters

The permissible spread of water into the street or thoroughfare shall in all cases govern the hydraulic capacity of a given street. On multiple lane roadways, the permissible spread of water will not close more than one travel lane in each direction. For two-lane roadways, the water shall be limited to one-half of each lane width.

5.0.4.8 Design of Storm Drain Inlets

Various charts are available in the Drainage Manual that shall be used to determine the capacity and efficiency of the particular type of inlet chosen. When designing inlets, freedom from clogging or from interference with traffic shall take precedence over hydraulic considerations. Precast units may be used for load bearing applications only with the approval of DFW.

Stormwater inlet grates and frames shall be treated to prevent corrosion and shall have aprons around them.

5.0.4.9 Design and Placement of Manholes and Inlets

Manholes or combination manholes and inlets shall be placed wherever necessary for clean- out and inspection purposes.

Place manholes at changes in direction, junctions of pipe run, and at intervals of 300 to 500 feet in long pipe runs where the size or direction is not changed as well as follow iSWM guidelines. The invert of the manhole section shall be rounded to match the inverts of the pipes entering the manhole in order to reduce eddying and resultant head losses. For manholes that are larger than the incoming or outgoing pipes, expansion losses can sometimes be significant. The use of heavy duty or aircraft rated manholes and covers may be required in some cases depending on location and shall be coordinated with DFW.

5.0.4.10 Drainage of Unpaved Areas Adjacent to Buildings

Unpaved areas adjacent to buildings shall be sloped to direct surface water and roof drainage away from buildings at a minimum slope of 5 percent in the first 10 feet of horizontal distance. Unpaved areas shall be permanently stabilized with vegetative cover to prevent erosion and soil loss. Surfaces paved with concrete or bituminous pavement shall have a slope of not less than 0.5 percent in the direction of drainage, to prevent ponding.

5.0.4.11 Drainage of Unpaved Areas not Occupied by Buildings

Portions of the site not occupied by buildings or pavement shall have adequate continuous slopes to drain toward watercourses, drainage swales, roadways, and storm drainage inlets. Drainage swales or channels shall be sized and sloped to accommodate the design runoff. Sheet flow across sidewalks is allowable. The concentrated runoff shall be carried under walkways in pipes or by suitable sidewalk drains. Swales shall be used to intercept water at the top and bottom of banks where large areas are drained. To provide positive drainage, a slope of not less than 2 percent for turf areas is desirable. Slopes shall be designed to ensure non-erosive runoff velocities. Turf banks, where required, shall be graded to permit the use of gang mowers, providing a maximum slope of 4 horizontal to 1 vertical. The tops and bottoms of all slopes shall be gently rounded in a transition curve for optimum appearance and ease of maintenance.

5.0.4.12 Runoff Reduction

The use of parking pavers and porous pavement/asphalt is encouraged to reduce stormwater runoff volumes in vehicle parking and waiting areas.

5.0.4.13 Design of Detention Structures

Open surface detention structures shall comply with iSWM Criteria Manual Section and shall have a minimum 1-foot freeboard.

5.0.4.13.1 Hydraulic Design Criteria for Structures/Detention Structures.

However, detention structures shall include water quality measures such as an infiltration trench or filter strip or infiltration area at the outfall with a grassy low flow channel as referred to in SDMP - Green Stormwater Infrastructure Guidance Manual.

Underground detention structures will be designed to capture and store the Water Quality Volume (WQv), for release after 24 hours after the storm event, to the First Flush Stormwater System (FFSW). Stormwater capture beyond the WQv will discharge into a Stormwater must incorporate filtration media to filter the WQv, Streambank Protection volume (SPv) criteria set forth in iSWM Section 3.4 Option 3.

5.0.4.13.2 GI/LID Strategies

Green Infrastructure or Low Impact Development (GI/LID) strategies is a selection of stormwater drainage management approaches and technologies that utilize and/or mimic the Dallas Fort Worth Airport natural hydrologic cycle methods of infiltration, evapotranspiration, and reuse. Every development shall use GI/LID strategies as an aid to meet pre- development conditions for the 1-Year storm event as well as treat the FFSW WQv. A few applicable GI/LID strategies are:

1. Infiltration trench
2. Rain garden
3. Bioswale
4. Permeable pavement
5. Filter Strip
6. Rainwater harvesting

Refer to the SDMP GSI Manual for designing GI/LID strategies.

Roadway developments or expansions shall incorporate GI/LID strategies in compliance with the SDMP.

5.0.4.14 Storm Utility Drainage Piping

Pipe Material (latest edition)

1. Reinforced Concrete Pipe: ASTM C 76
2. Reinforced Concrete D-Load Pipe: ASTM C 655
3. Reinforced Concrete Arch Pipe: ASTM C 506
4. Reinforced Concrete Elliptical Pipe: ASTM C 507
5. Precast Reinforced Concrete Box Sections: ASTM C 789 and C 850
6. PVC Pipe minimum pressure rating DR 21: ASTM D 1784, 3139, 3212 & F477, UL 157; solvent weld joints ASTM D2855-90, 2564-90 and/or 2235-90 Corrugated Polyethylene Drainage Tubing: AASHTO M 252
7. Corrugated Polyethylene Pipe 12 to 24 Inches in Diameter: AASHTO M 294
8. Polyethylene Pipe Large Diameter: ASTM F894 Type S
9. Polypropylene Pipe (PP): AASHTO M330 Type S
10. High Performance Polypropylene Pipe (HP): ASTM F2881, AASHTO M330 Type S
11. High Density Polyethylene (HDPE) pipe; AASHTO M294 Type S

5.0.4.15 Pipe Bedding**5.0.4.15.1 Rigid Pipe**

Bedding for rigid pipe shall conform to one of the following classes:

1. Class A: Continuous concrete cradle.
2. Class B: Bed of granular materials having a thickness of at least 6 inches below the bottom of the pipe and extending up around the pipe for a depth of not less than 30 percent of the pipe's vertical outside diameter. The layer of bedding material shall be shaped to fit the pipe for at least 10 percent of the pipe's vertical diameter and shall have recesses shaped to receive the bell of bell-and-spigot pipe. The bedding material shall be sand or selected sandy soil conforming to the following:

Table 20 - Bedding Material Sieve Size & Passing Rate

Sieve	% Passing
2 inches	100
1 inch	90 – 100
½ inch	50 – 80
No. 4	30 – 60
No. 100	0 – 5

3. Class C: Class C bedding shall consist of bedding the pipe in its natural foundation to a depth of not less than 10 percent of the pipe's vertical outside diameter. The bed shall be shaped to fit the pipe and shall have recesses shaped to receive the bell of bell-and-spigot pipe.

5.0.4.15.2 Flexible Pipe

Pipe bedding for flexible pipe, the bed shall be roughly shaped to fit the pipe, and a bedding blanket of sand or fine granular material shall be provided as follows:

Table 21 - Required Pipe Bedding Depth

Pipe Corrugation Depth	Minimum Bedding Depth
½ inch	100
1 inch	90 – 100
2 inches	50 – 80
2 ½ inches	30 – 60

5.0.4.16 PVC and Polyethylene Pipe

PVC and Polyethylene pipe connections to RCP pipe for underdrains must be installed with “Inserta” Tee fittings.

5.0.4.16.1 Bedding Material

1. Shall consist of coarse sands with a maximum particle size of ¾ inch.
2. For pipe installed under paved areas, no more than 12 percent of the material shall pass the No. 200 sieve.
3. For all other areas, no more than 50 percent of the material shall pass the No. 200 sieve.
4. The bedding shall have a thickness of at least 6 inches below the bottom of the pipe and extend up around the pipe for a depth of not less than 12 inches.

5.0.4.16.2 Pipe Joints

1. Rubber Gaskets for rigid pipe shall conform to the requirements of ASTM C 43 Rubber Gaskets.
2. Rubber Gaskets for PVC Pipe shall conform to the requirements of ASTM F 477.

5.0.4.17 RCP Drainage Pipe

1. Shall have filter fabric around every joint, overlapping 12 inches on each side of the pipe joint and box culverts.
2. Plastic gaskets shall conform to the requirements of AASHTO M 198 (Type B).
3. Pipe joint mortar shall consist of one part PCC and two parts sand. The PCC shall conform to the requirements of ASTM C 150, Type I. The sand shall conform to the requirements of ASTM C 144.
4. Poured filler for joints shall conform to the requirements of ASTM D 1190.
5. Oakum for joints in bell and spigot pipe shall be made from hemp (CannabisSativa) line, or Benares Sunn fiber, or from a combination of these fibers. The oakum shall be thoroughly corded and finished.

5.0.4.18 Concrete Collars

Formed Concrete Collars at all new and existing and precast structure connections with new stormwater infrastructure. All formed collars will have a minimum of (2) #3 rebar bands around the pipe with #3 rebar transversely at 12 inch OC and located at midspan. Rebar drilled into box will need epoxy. The collar shall be 18 inch in width and surrounding the pipe a minimum of 6 inch on all sides. Concrete Collars shall also be required where new pipe is connected to existing pipe. These collars shall be 24 inches in width. The EOR shall provide a detail in design set for contractor to use or submit TxDOT Concrete Collar & Pipe Cap Details.

5.0.4.19 Pipe Deflection

Longitudinal deflection at each pipe joint shall not exceed one degree in any direction.

5.0.4.19.1 Pipe Cover

A minimum of 3 feet of cover will be required.

5.0.4.20 Storm Water Pipe Inspections

Prior to project close-out, and after installation of the entire Storm Water Drainage system has been completed, all Storm Water Drainage lines and structures shall be cleaned, inspected with a video system and reviewed by the contractor before copies of the videos are submitted to DCC and ETAM for review.

5.0.4.21 First Flush Stormwater System

The First Flush Stormwater System (FFSW), formerly referred to as the Pretreatment Waste System or Industrial Waste System, shall be used only for the handling of stormwater. The design shall take into consideration the usage charges associated with all discharges to the FFSW and construction of conveyances which only transport authorized discharges. Authorized discharges are defined as storm water discharges, allowable non-storm water discharges, and other individually permitted non-storm water discharges. Individually permitted non-storm water discharges will be determined by Environmental Affairs on a case-by-case basis, whereby pre-approval from specified regulatory bodies may be required.

Process water discharges to the FFSW are explicitly forbidden. Process water is defined in the Clean Water Act, and includes, but is not limited to, apron and ramp wash-down, fuel spills, sanitary sewage, deicing pad runoff, and aircraft and vehicle wash water.

The FFSW may handle discharges that contain excessive pollutant concentrations that would otherwise be directed to the main Stormwater System. Typically, the system is used to transport first flush storm runoff that contains elevated levels of pollutants. The FFSW has occasionally handled non-permissible discharges (e.g., emergency overflow relief line for deicing storage tanks), such uses shall only be allowed by written authorization of the EAD Vice President.

Prior to any connection to the FFSW, an engineering analysis shall verify that there is available incremental capacity in the system to handle the proposed discharges for all nonemergency discharge conditions. The anticipated characteristics of the discharge shall be compliant with the Pretreatment Rule contained in the Airport's Code of Rules and Regulations.

Lift Stations shall be located with due consideration for screening, maintenance access and emergency access.

5.0.4.22 Design Criteria

5.0.4.22.1 Pipe

FFSW pipe shall meet the requirements of AWWA C900 PVC, C905 PVC or designed and manufactured in accordance with the applicable provisions of ASTM D 2680 and D 3212. The pipe will be joined using Elastomeric gasketed joints meeting ASTM D 3212 or solvent weld joints meeting ASTM D 2855-90, ASTM D 2564-90 and/or ASTM D2235- 90 or latest edition. The AWWA C900, C905 or P303 RCCP is required for all pressurized applications. All pressurized pipe shall be installed within casings under roadways, taxiways, and runways. Casing pipe shall be installed with RACI-spacers or equivalent on 4 foot centers. Steel casing pipe can be used with proper cathodic protection. Corrugated steel pipe shall not be used as casing pipe.

5.0.4.22.2 Manholes and Covers

Manholes shall be cast-in-place reinforced concrete without embedded steps. Manholes shall be installed at every 300 feet, at connections to mains and at 45 degree and 90-degree changes in runs. Ring and cover shall be cast iron, 24 inch diameter and adjusted to finish grade or existing ground elevation, as appropriate. Covers shall have "First Flush System" imprinted on them. Joints shall be sealed to prevent infiltration and exfiltration. Existing FFSW system manhole covers may be stamped "Industrial Waste", "Industrial Waste System", Pre-Treatment Waste System (PTWS) or "Waste System." Old covers shall be replaced as encountered with new covers stamped as indicated above. Manhole ring and covers not in pavement will require a concrete pad around them. Engineer of record will need to provide detail.

5.0.4.22.3 Pumps

FFSW Lift Stations shall include a minimum of two submersible electrically operated sewage pumps with low level shutoff, high level alarm and intermediate level sensors required for pump cycling. Pumps shall be manufactured to safely pump wastewater.

Wet pits shall be concrete, cast-in-place with access for inspection and pump maintenance.

5.0.4.22.4 Testing

All FFSW pipe installed will be tested so that the completed pipe system will have a zero-exfiltration loss per 2 hours, where the hydrostatic head at the design hydraulic grade line is not less than 4 feet of test stack with the height of the test stack matching the highest level of the FFSW. A pneumatic test may be substituted for the hydrostatic test to substantiate exfiltration criteria. Maintain 3 pounds per square inch gauge (PSIG) for 5 minutes with zero PSIG drop.

5.0.5 Reclaimed Water System

5.0.5.1 DFW Airport Reclaimed Water System

DFW has decided to use highly treated effluent or reclaimed water to augment other sources of water to meet current and future water demands. A reclaimed water system is currently in use by DFW throughout the airport where reclaimed facilities are available.

1. Reclaimed water main shall be designed in conformance with “30 TAC §210: Use of Reclaimed Water”, “30 TAC §217.70: Reclaimed Water Facilities”, and with all applicable laws, regulations, codes, and standards. Before the Permit is issued the Engineer of Record (EOR) shall submit the letter stating that the plans are designed per TCEQ Chapter 210 and Chapter 344.
2. Engineer will need to submit a cross connection control plan for the contractor to follow to ensure no connections between Potable and Reclaim Water. This will need to be sent to DFW Utility Inspector to upload into impact.
3. Before the Certificate of Occupancy is approved, the Landscape Contractor will need to submit a letter stating that the irrigation was placed and operates per design documents. EOR will need to visually observe the system functioning and submit a letter stating that the system was placed and operates per design documents. This will need to be sent to DFW Utility Inspector to upload into impact.

5.0.5.2 Reclaimed Water Main Sizing

The reclaimed water mains must be sized based on engineering analysis of initial and future demand of the reclaimed water user(s). The designer shall meet or exceed the following minimum criteria for sizing reclaimed water mains:

1. Velocity: The velocity of a reclaimed water main shall be maintained between 2 and 10 feet per second.
2. Service: The size of the services must be at least one standard size smaller than the proposed and existing raw water main.

5.0.5.3 Pressurized Main

Reclaimed water pressure mains shall be designed in accordance with 30 TAC §210.25.

5.0.5.4 Depth of Cover

The depth of cover shall be the same for water mains as covered in Section 5.0.2.3 and Table 9 of this manual.

5.0.5.5 Location

Location of all new reclaimed water mains shall be in accordance with Section 5.0.2.4 of this manual.

5.0.5.6 Horizontal Alignment

Horizontal alignment of reclaimed water mains shall be conducted in accordance with Section 5.0.2.5 of this manual relating to horizontal alignment of water main.

5.0.5.7 Vertical Alignment

Vertical alignment of reclaimed water mains shall be conducted in accordance with Section 5.0.2.6 of this manual relating to vertical alignment of water mains.

5.0.5.8 Reclaimed Water Pipe Material and Embedment

Designers are responsible for specifying the type of pipe to be used in any design.

Typically, all reclaimed water pipes shall be pressure rated with a minimum working pressure rating the same as if the main was being used for potable water. Refer to Table 8 for requirements. Although PVC is the preferred material for smaller mains, there are some restrictions on its use. PVC may not be used within the following areas or circumstances:

1. Elevated crossings or situation where pipe will be permanently exposed.
2. Encased pipes where the method of anchoring is by means of hold-down jacks.

When a metal pipe including concrete cylinder or steel is specified, the pipe must be protected from corrosion. In addition, all reclaimed water mains must be designated with a purple coloration. They may be fabricated from purple base materials, painted purple, or encased in a purple polyethylene sleeve. In addition to the purple color or purple sleeve all pipe materials shall have the words “non-Potable” painted on them in 2 inches high letters every two feet.

5.0.5.8.1 Fittings

All PVC pipe shall use ductile iron fittings with a minimum working pressure rating of 250 pounds per square inch (PSI). Compact fittings are not allowed. All 90 degree bends shall be avoided in the system, if possible.

5.0.5.8.2 Embedment Requirements

Designers shall specify class of embedment on design drawings and shall be in compliance with an engineering analysis of the pipe location and geological conditions.

5.0.5.8.3 Pipe Material for Reclaimed Water Mains

The following table summarizes recommended pipe material for different reclaimed water mains:

Table 22 - Pipe Material and Joints Specification

Pipe & Material Specification	Allowable Size (in)	Joint Specification
Purple PVC AWWA C900 (DR-14)	8 – 16	Bell & Spigot Joints: Certa-Lok C900/R (TM) J
RCCP w/Purple Tape Coating**	24 – 42	Welded Joints

5.0.5.10 Separation Distance between Reclaimed Water and Water/Wastewater Mains

When a reclaimed water main is built near an existing water/wastewater facility, conveyance, or appurtenance, 30TAC §210.25 as enforced by TCEQ, governs the minimum separation distances.

5.0.5.11 Connection to Existing Reclaimed Water Mains

Typically, service lines from individual properties are not allowed to be connected to a reclaimed water main unless otherwise approved by DFW. The following type of connections, however, are permitted:

1. Tapping Sleeve and Valve
2. Cut in Connection
3. Standard Connection
4. Four Way Connection

All connections shall comply with the same requirements as per section 5.0.2.9 of this manual, except that all connections shall include fittings and pipe that comply with reclaimed water criteria, i.e., purple in color.

5.0.5.12 Corrosion Control

This section shall be governed by section 5.2 of this manual.

5.0.5.13 Cross Connections

No physical connection shall be made between a reclaimed water line and potable water or wastewater line. All appurtenances shall be designed and constructed so as to prevent any possibility of cross contamination. The engineer of record shall submit a cross connection testing plan with all reclaimed water system design documents.

5.0.5.13.1 Testing Procedures

1. All reclaimed water mains shall be hydrostatically tested in accordance with §332.14 of this manual before being put in service, in addition all reclaimed water lines will be dye tested. However, no chlorination will be required unless otherwise requested by DFW.
2. Hydrostatic testing of all reclaimed water mains must also be in accordance with the NCTCOG Standards, Latest Edition.

5.0.5.14 Reclaimed Water Isolation Valves

Isolation valves on reclaimed water lines shall meet the requirements of section 5.0.2.15 in this manual.

5.0.5.15 Reclaim Water Meters

All meters in the distribution system for the reclaimed water will meet the requirements for potable water as required in section 5.0.2.19, except that each meter will have purple covering and be clearly marked as non-potable water. Water meters shall be installed at all irrigation service taps, and at all flushing taps.

5.0.5.16 Outlets

Blind flanges or plugs, as applicable, shall be furnished and installed on all valves located at outlet points or terminal points where the reclaimed water main does not continue. Dead-end main structures shall be avoided whenever possible. If unavoidable, dead-end mains shall be designed to include a flush point and the end of the main. Each flush point shall be clearly marked as non-potable water, and painted purple.

5.0.5.17 Air Release, Air/Vacuum, and Combination Air Valves

Air and vacuum valves must be provided in high points according to section 5.0.2.21 of this manual.

All below ground appurtenances for blow-off and air and vacuum valve assemblies must be consistently color-coded purple and clearly marked to differentiate recycled water facilities from potable water and/or wastewater facilities.

5.0.5.18 Thrust Blocking

Thrust blocking for reclaimed water mains must be provided in accordance with section 5.0.2.25 of this manual.

5.0.5.19 Backflow Prevention Device

Backflow prevention Device shall be installed at each delivery tap.

5.0.5.20 Reclaimed Water Blow off Assembly

Either an in-line or end-of-line blow-off drain assembly must be installed to remove water or sediment from the reclaimed water pipeline. Blow-off assemblies must be installed in a low point of the pipeline. The pipeline tap for the assembly must be no closer than 18 inches from a valve, coupling, joint, or fitting unless it is at the end of the pipeline.

1. The reclaimed water blow-off assembly will require 2 vaults.
2. First vault from reclaimed water main will house valve, water meter and second valve with bypass.
3. Second vault will house gate valve, back flow preventer, and second gate valve.

The discharge from blow-offs should be designed to drain into the sanitary sewer system. Discharge of recycled water to storm drains is restricted. If there is no sanitary sewer that can receive the discharge from a blow-off, DFW Utilities must be consulted regarding acceptable alternatives.

5.0.5.21 Manhole Frame and Cover

All manhole frames/covers, valve castings shall comply with section 5.0.2.26 of this manual with the following addition, covers must be painted purple and be clearly denoted as non-potable.

5.0.5.22 Reclaimed Water System Approved Product List

Table 22 lists the approved products for reclaimed water.

Table 23 - Reclaimed Water System Approved Product List

Reclaimed Water System Approved Product List					
Basic Product Category	Product Sub-Category	Manufacturer	Model, Type, or Style Approved	Reference Specifications (NCTCOG)	National Specifications
Water Meters		Manufacturer			
Water Meters	All Meters	Badger	Badger Becon with HR-E encoders, LTE Endpoints with Purple covers for Reclaimed water		AWWA C700 & C701
Water Pipes		Manufacturer			
PVC Pressure Pipe	PVC 8"-12" DR14	Pipelife Jet Stream	DR14	501.9	AWWA C900
PVC Pressure Pipe	PVC 8"-12" DR18	Pipelife Jet Stream	DR18	501.9	AWWA C900
PVC Pressure Pipe	PVC 8"-12" DR14	Diamond Plastics Corporation	DR14	501.9	AWWA C900
Corporation Stop		Manufacturer			
Corporation Stop	Quick Joint Ball Valve	Ford	FB-1000-x-Q-NL	502.5	AWWA C800
*MUST HAVE STAINLESS STEEL COMPRESSION GRIP RING.					
* MUST HAVE FULL BODY GLANDS AND RED PRIMER.					
Fittings		Manufacturer			
Fittings, Full-Bodied	Ductile Iron	American Cast Iron Pipe		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Star Pipe		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Tyler/Union/UFC O		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	Sigma		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	US Pipe & Foundry		502.6	AWWA C110
Fittings, Full-Bodied	Ductile Iron	SIP		502.6	AWWA C110
COMPACT FITTINGS AND GLANDS ARE NOT ALLOWED.					
Manhole Appurt.		Manufacturer			
Manhole Ring & Lid	32-Inch	Bass & Hays	V1420		
Manhole Ring & Lid	40-Inch	Bass & Hays	#1225-40		
Manhole Ring & Lid	40-Inch	Neenah	R-1901		
All Manhole lids will be visible as to non-potable water					

Polywrap					
Polywrap	WRAP MUST BE LLDPE, t= 0.008", MANUFACTURED AND LABELED IN ACCORDANCE WITH AWWA C105, and shall be purple for reuse water				AWWA C105
Service Saddles		Manufacturer			
Service Saddles	Bronze Double Strap	James Jones	J-979	502.10	AWWA C 800
Service Saddles	Bronze Double Strap	Ford	202B	502.10	AWWA C 800
Service Saddles	Bronze Double Strap	McDonald	3825	502.10	AWWA C 800
Service Saddles	Bronze Double Strap	Mueller	BR2B	502.10	AWWA C 800
Service Saddles	Bronze Double Strap	Smith-Blair	323	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel	Manufacturer			
Tapping Sleeve	Stainless Steel*	Cascade	CST-EX	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Ford	FTSS	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	JCM	432	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Mueller	H304	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Romac	SST III	502.10	AWWA C 800
Tapping Sleeve	Stainless Steel*	Smith-Blair	665	502.10	AWWA C 800
*OUTLET FLANGES MUST ALSO BE STAINLESS STEEL. SHALL BE RESTRICTED TO PIPE SIZES 8" AND SMALLER AND SHALL NOT BE USED FOR TAPS GREATER THAN 75% OF THE MAIN SIZE.					
Air Valves		Manufacturer			
Valves, Air	Air Release and Vacuum Breaker	Cla-Val		502.6	
Valves, Air	Air Release and Vacuum Breaker	Val-Matic		502.6	
Valves, Air	Air Release and Vacuum Breaker	Valve & Primer (APCO)		502.6	
Valves, Air	Air Release and Vacuum Breaker	Vent-O-Mat	RBX Series	502.6	
All valves will be marked visible as to non-potable water					
Valve Stack	Covers and Lid	Manufacturer			
Valve Covers	Lid	Bass & Hays	#340-1	806.4	
Valve Covers	Lid	Sigma	VB1651M	806.4	
Valve Covers	Cover	Sigma	VB1652M	806.4	
Valve Covers	Lid	Accucast	115002	806.4	
Valve Covers	Stack	Accucast	115000	806.4	
Valve Covers	Lid	Star Pipe Products	Shorty	806.4	
Valve Covers	Stack	Star Pipe Products	Shorty	806.4	
All Valve lids will be marked visible as to non-potable water					

Valves	Vertical Resilient	Manufacturer			
Valves, Gate*	Vertical, Resilient Wedge	Mueller	2360, 2362	502.6	AWWA C509
Valves, Gate *	Vertical, Resilient Wedge	M & H	4067, 4068	502.6	AWWA C509
* MUST HAVE 316 SS BOLTS. BOTH RED EPOXY OR BITUMINOUS EXTERIOR COATINGS ARE ALSO ACCEPTABLE. AWWA C-515 RSGVs WITH REDUCED WALL THICKNESSES ARE NOT ACCEPTABLE.					
Water Meter Vaults	Large Services	Manufacturer			
Water Meter Vaults	Concrete	American Industrial Precast	Must have approved lifting devices		
Water Meter Vaults	Concrete	Brooks/Oldcastle Precast	Must have approved lifting devices		
Water Meter Vaults	Concrete	Concrete Products (CPI)	Must have approved lifting devices		
Water Meter Vaults	Concrete	Hanson	Must have approved lifting devices		
Water Meter Vaults	Concrete	New Basis (Dalworth Quickset)	Must have approved lifting devices		
Water Meter Vaults	Concrete	Turner	Must have approved lifting devices		
Engineer of record shall confirm the applicability of these approved products in AOA and CTA area. This standard product list has been developed only for projects located outside the AOA and CTA area. When technical specification for specific products are included as part of the construction contract documents, the requirements of the technical specifications will override these approved product list. It is engineer of record (EOR)'s responsibility to approve the product submittal. DFW utility group recommend EOR to coordinate with DFW Utility group prior to approval.					

5.1 Earthwork

5.1.0 Site Clearing

5.1.0.1 Clearing and Grubbing

Site work includes clearing, grubbing, grading, drainage, paving, and special site development structures. All sitework shall be designed and conducted to improve the overall aesthetics of the Airport and to promote future development.

5.1.1 Earth Moving

5.1.1.1 Grading

The site shall be prepared preserving the natural character of the terrain by minimum disturbance of existing ground forms, with the objective to develop an attractive, suitable, and economical project site. Surface and subsurface flow from stormwater shall be diverted away from buildings and pavements to prevent undue saturation of the sub-grade that could damage structures and weaken pavements.

5.1.1.2 Excavation and Fill

5.1.1.2.1 Excavation for Structures

Boulders, logs, or any other objectionable material encountered in excavation shall be removed. All rock or other hard foundation material shall be cleaned of all loose material and cut to a firm surface either level, stepped, or serrated. All seams or crevices shall be cleaned out and grouted. All loose and disintegrated rock and thin strata shall be removed. When concrete is to rest on a surface other than rock, special care shall be taken not to disturb the bottom of the excavation, and excavation to final grade shall not be made until just before the concrete or reinforcing is to be placed.

All bracing, sheathing, or shoring shall be performed as necessary to implement and protect the excavation and the structure as required for safety or conformance to governing laws.

Unless otherwise provided, bracing, sheathing, or shoring involved in the construction shall be removed after the completion of the structure. Removal shall be affected in a manner which will not disturb, or mar finished masonry.

5.1.1.2.2 Excavated Soil Materials

Excavated materials must be managed and disposed of in accordance with the DFW Contaminated Media Management Plan. Projects that involve subsurface drilling or the excavation, stock piling or movement of soils require a soil management plan be submitted and approved by EAD that details procedures to be employed to ensure proper handling and disposal. No excavated material or concrete rubble may be removed from the Airport without approval from EAD. Submit requests through the Building Official.

5.1.1.2.3 Backfill for Structures

The area around a structure shall be filled with approved material, in horizontal layers not to exceed 8 inches in loose depth. For areas under pavement and safety areas, compact to 95 to 100 percent of the Standard Proctor Density, in accordance with ASTM D 698. For all other areas, compact to 93 percent of the Standard Proctor Density. Moisture content shall be held to a range of from 1 percent below to 3 percent above optimum. Each layer shall be deposited all around the structure to approximately the same elevation.

Reuse of suitable excavated soils into structural designs and general landscaping is encouraged to reduce the amount of materials to be landfilled. Organic material is not permitted in the backfill.

The use of compost materials shall be used in the top layer of all fill materials to improve stormwater infiltration, promote ground water recharge, and reduce the need for fertilizers and other chemicals to stimulate plant growth.

In the case of concrete, backfilling shall not be placed against any structure until the concrete has attained sufficient strength to provide a factor of safety against damage or strain in withstanding any pressure created by a backfill or until laboratory tests establish sufficient strength. Sufficient strength shall be interpreted as being 75% of the design strength or the strength expected to be obtained in 7 days, whichever is greater.

5.1.2 Erosion and Sediment Controls

5.1.2.1 Storm Water Pollution Prevention Plan

For projects disturbing greater than 1 acre, a Storm Water Pollution Prevention Plan, including an erosion control plan, must be submitted to the Building Official along with a signed Notice of Intent before a construction permit will be granted. Erosion control measures shall be designed and implemented to effectively prevent discharge of sediments to the storm drain system and receiving waters. Care shall be taken in the design and implementation of such measures to ensure that a safety hazard, such as ponding water on a roadway, does not occur. Erosion control measures shall be checked each week or immediately after a storm event to be certain that they are functioning properly. Any corrective measures necessary to repair the non-functioning erosion control measures shall be performed within 24 hours of notification to the Contractor.

5.1.3 Permitting

5.1.3.1 Preliminary Actions

In order to ensure compliance with the above-described objectives, the Designer shall provide information for submittal of the following documents along with application for the construction permit to construct or renovate a facility from which there will be stormwater discharge:

1. Documents describing the type and nature of all activities to occur at the site that could potentially impact stormwater quality.
2. Documents detailing the structural controls to be constructed at the facility to impel stormwater discharges to meet EPA benchmark standards, by reducing the discharge of both point and non-point source pollutants (e.g., oil and grease, metals, sediment, and trash).

3. The use of stormwater filtration devices, oil/water separators, infiltration swales, and rain gardens shall be used to reduce stormwater runoff and improve the quality of runoff.
4. A certification, sealed by the design engineer, stating that “Based upon the above representations, the proposed structural controls will reduce the discharge pollutants to the DFW stormwater collection system, and/or receiving waters.”
5. A certification, sealed by a Licensed Professional Engineer in the State of Texas and signed by the owner/operator of the facility, stating that all structural erosion controls supporting the facility will be maintained in manner to ensure effective operation and minimize the discharge of pollutants; and operators will develop and implement a Pollution Prevention Plan upon occupancy of the facility to identify all activities with the potential to impact the quality of stormwater discharges (or general environment.- mental compliance) and the best management practices to be implemented.

5.1.3.2 Spill Prevention Control and Countermeasures Plan

Facilities may be subject to Spill Prevention Control and Countermeasures Plan (SPCC) regulations of 40 CFR 112. The design of any new or renovated facility must include a submittal by the Operator to DFW EAD, including a formal determination as to whether a SPCC is required; and if so, the design must incorporate the measures specified at 40 CFR 112 or any subsequent applicable federal or state regulation. Under the provisions of the SPCC 112 regulations Part II (73 FR 74236) effective January 14, 2010, completely buried oil storage tanks are exempt from the rule. In addition, the rule redefines “facility,” that is, the owner or operator now has discretion in identifying which building, properties, parcels, leases, structures, installations, pipe, or pipelines make up the facility.

5.1.4 Special Foundation and Loadbearing Elements

5.1.4.1 Bored Piles

Drilled piers shall be straight shaft type only. Use no belled piers.

5.1.4.2 Tunneling and Mining

5.1.4.2.1 Tunnel Construction

Tunnels shall include all below grade, enclosed structures used by pedestrians and vehicles or to hold utilities.

Due to the varied applications of tunnels, the design criteria shall be established on a project-to-project basis.

Items of particular interest which should be addressed are waterproofing, ventilation, lighting and utilities, drainage, exiting, fire protection, cathodic protection/ corrosion control, and overburden loading.

5.2 Cathodic Protection Systems

5.2.0 General Information

These criteria shall be utilized in the design of corrosion control systems for new installations and/or maintenance and repairs to existing cathodic protection installations.

5.2.1 Design Guidelines

5.2.1.1 General

The design of new or refurbishment of existing cathodic protection systems shall be performed or supervised by a Corrosion Engineer. Corrosion Engineer refers to a person who, by reason of knowledge of the physical sciences and the principles of engineering and mathematics, acquired by professional education and related practical experience, is qualified to engage in the practice of corrosion control. Such people may be a licensed Professional Engineer or may be a person certified as qualified by NACE International, hereinafter referred to as NACE, if such licensing or certification includes suitable experience in corrosion control on buried or submerged metallic piping systems and metallic tanks.

Should alternative materials or design concepts be deemed advantageous, they shall be presented to DFW Energy & Utilities Services for review and comment prior to approval.

5.2.1.2 Electrical Grounding Systems and DC Powered Equipment

Buried electrical ground rods shall be stainless steel, stainless steel-clad or galvanized steel. Buried cable shall be insulated, and cable-to-ground rod connections shall be coated.

DC powered equipment shall be insulated from other structures so that no DC current flows through the earth to other structures.

5.2.1.3 Nonmetallic Materials

In some instances, metallic and non-metallic materials are acceptable for the same service. In general, preference should be given to the non-metallic material if its durability is approximately equivalent to that of the metallic material.

Immersed or buried composite structures such as steel reinforced concrete should be made electrically continuous by appropriate bonding between sections. A minimum of 2 inches of concrete shall cover all ferrous materials used as reinforcement.

5.2.1.4 Design Life

The following criteria have been established as a guide:

A maximum internal corrosion rate of 1 mil per year based on maximum depth of penetration is acceptable to permit realization of at least 50 years of service life free from major replacements for piping systems.

Corrosion control of exterior surfaces of equipment, piping, structural and miscellaneous steel shall be accomplished by providing a high-quality paint system. Buried piping and fittings shall be protected by electrically insulating coatings supplemented with cathodic protection. Concrete encasing shall not be an alternative to coating or cathodic protection.

5.2.1.3 Painting

In general, surfaces of equipment, piping, structural and miscellaneous steel shall be painted with a protective coating. All surfaces shall be primed with a final coat applied in the field. Aluminum, stainless, or galvanized steel shall not be painted except as indicated otherwise.

5.2.2 Process Piping and Equipment Protection

5.2.2.1 Process Piping and Equipment Insulation

5.2.2.1.1 Electrical Isolation

Structures requiring protection may require isolation from other underground metals by physical separation or by suitable dielectric materials. Above ground piping, fittings or equipment requires electrical isolation from underground connected and cathodically protected piping, fittings, or other structures by the use of isolating pipe flanges.

In general, dielectric fittings shall be installed to be accessible after backfilling. The preferred location is on the vertical riser 12 to 24 inches above grade, before wall penetration into a building. If it is not practical to install the isolating fitting in an accessible location, one set of two test wires shall be thermit welded to each side of the isolating fitting and extended to an at-grade test station to facilitate testing. The cathodically protected side of the isolating fitting shall have white test lead wires, the unprotected side shall have green test lead wires.

When necessary to install dielectric fittings inside a building, the pipeline shall be isolated from reinforcing steel, masonry, concrete, etc., by passing through a non-conducting sleeve of plastic, or a metal sleeve with an insulating modular rubber seal to support, center and isolate the pipe from structural steel. Dielectric fittings should be inside the building as close as practical to the building wall. These points shall be clearly labeled and accessible for inspection and repair.

Where it is necessary to interconnect dissimilar metals underground, dielectric fittings shall be installed at the connection point. All buried insulating fittings shall be coated with approved coating material. The final decision of proper coating shall be made by a Corrosion Engineer. Painting of above-grade insulating fittings shall be done with electrically nonconductive paints. Aluminum or zinc-rich paints are not approved for this application.

All pipes entering meter vaults, penetrating building floor slabs, and building walls shall be sleeved with modular rubber links to electrically isolate the pipe from contact with building/vault steel.

5.2.2.1.2 Bonding

Insulated copper cables for RCCP shall be welded or brazed across all mechanical joints in underground ferrous piping systems and at all isolated ferrous metal fittings such as 45s, 90s, Ts, gate valves, and fire hydrants that will be cathodically protected. Bolted connections are not acceptable.

Bonds shall consist of a #8 stranded copper wire with HMWPE insulation and shall be 12 to 18 inches in length with ample slack after all welds have been made.

Bonds may be installed between adjacent or crossing piping systems to allow protection of several systems from the same current source. Drain bonds shall be installed on any structures that may be subject to cathodic interference effects.

In the area of influence of an impressed current ground bed, reinforcing steel in buried nonmetallic structures should be bonded so that the structure is electrically continuous.

These structures include reinforced concrete pipe, prestressed concrete pipe, and RCCP.

5.2.2.1.3 Casings

When a cathodically protected pipeline passes under a roadway, runway, or taxiway, it shall be placed inside a casing as required for mechanical strength. The carrier pipe (cathodically protected pipe) shall be electrically isolated from the casing. Casing insulators shall be specified for installation on the carrier pipe prior to placement in the casing.

Casings or rigid galvanized conduits shall be provided for all cathodic protection header cables under a roadway, runway, or taxiway. Casing end seals shall be specified to prevent moisture from entering either end of the casing.

Casings shall not be coated. Ferrous metal carrier pipes inside the casing shall be coated with the same quality coating as is applied to the remainder of the piping system. Ferrous metal casings shall cathodically protect. Utilization of PVC casings is encouraged.

Test stations shall be installed at one end of each ferrous metal casing containing a cathodically protected pipe. Specify the test leads for the end of the casing that will be most accessible for future testing. Each test station shall have two white insulated lead wires from the carrier pipe and two red insulated lead wires from the casing.

5.2.2.2 Underground Piping Systems

5.2.2.2.1 Underground Piping Systems Protective Coatings

All buried, pressurized, ferrous metal piping systems (except pre-insulated pipe) shall be externally protected with a bonded pipeline type system of high dielectric strength. The purpose of this coating is to isolate the pipe from the soil.

Cast iron and ductile iron pipe, fittings, valves, and fire hydrants shall be coated specifically with 20 to 30 mils (dry) of a pipeline quality asphalt-based coating. In areas where the possibility of soil contamination by jet fuel exists, a mastic coating combining coal tar with asphalt materials shall be applied to the fitting to the same thickness. Coatings with one hundred percent coal tar mastics are incompatible with the asphalt primers found on the fittings and are therefore not acceptable.

Field joint repair on jet fuel lines shall be coated with a mastic coating, two-component systems such as coal tar epoxy or the pipe coating manufacturer's recommended repair coating. All coatings submitted for approval must have documentation on coating characteristics and application thickness.

Unbonded or loose-fitting coatings such as poly bagging are not approved coatings.

Equivalent coatings shall be reviewed by DFW Energy & Utilities Services for approval.

5.2.2.3 Corrosion Control of Process System Equipment and Piping

5.2.2.3.1 Systems

All equipment and piping in the following systems shall incorporate appropriate corrosion control methods:

1. Boiler Feed Water
2. Wastewater
3. Steam and Condensate
4. Condenser Water
5. Potable Water
6. Fire Protection Water
7. Chilled Water
8. Gas and Compressed Air
9. Hot Water
10. Fuel Oil

In the above services, copper base or aluminum alloys must not be attached to ferrous materials, or to each other.

Where the interconnection of different metals is necessary, they shall be electrically isolated using approved dielectric materials, or the cathodic metal shall be required internally coated where isolation is not practical.

Material selection shall be reviewed by a Corrosion Engineer to ensure that acceptable durability shall be achieved.

5.2.2.3.2 Corrosion Inhibitors and Additives

Process systems such as boiler feed water, condensate, condenser water and chilled/hot water require the addition of chemical additives that help to reduce internal iron pipe or tank corrosion. Provisions shall be made for conveniently and safely injecting these inhibitors and additives in these systems for maintaining the required protective residuals. Provisions shall also be made for sampling these processes for testing of residual levels. All corrosion inhibitors or additives shall be specified by or approved by a Corrosion Engineer. These systems shall not discharge to the storm sewer.

5.2.2.4 Impressed Current Systems

5.2.2.4.1 General

Impressed current systems shall consist of transformer-rectifier power sources, anodes placed in suitable backfill, and appropriate wiring to connect the rectifier(s) to the structures and the anodes. A complete, coordinated system must be provided.

5.2.2.4.2 Rectifier

The rectifier(s) selection shall be based on the field survey data and design calculations performed by a Corrosion Engineer during the design phase of the project. The rectifier shall be selected to operate at efficient settings but shall provide surplus capacity for reasonable future expansion. As a minimum, provide at least 20 percent surplus capacity for protected structures.

The rectifier shall contain internal circuit breakers, an output ammeter and voltmeter, and shall be mounted in a suitable cabinet or enclosure.

All rectifiers shall be provided with a properly sized, National Electrical Manufacturers Association (NEMA) 3R rated AC service breaker, which shall be placed in proximity to the rectifier.

5.2.2.4.3 Anodes

Impressed current anodes shall consist of either high silicon, chromium-bearing cast iron or linseed oil treated graphite, complete with high molecular weight polyethylene lead wires installed with the cable-to-anode connection properly sealed or encapsulated by the anode manufacturer.

Specialty anodes shall be approved by a Corrosion Engineer if conditions require their use.

5.2.2.4.4 Ground Beds

Anodes are installed in drilled, augured, or trenched holes at depths commensurate with soil resistivity, water tables and the structure geometry. Anode holes shall be filled with the required quantity of a well compacted granular, low resistance coke breeze backfill, as specified, installed so as to uniformly surround the anode.

Where the ground bed is to be installed under concrete or asphalt, the anode holes shall be vented to permit the release of gas generated at the anode surface. Backfill above the coke breeze shall consist of pea gravel to assist in venting of generated gases and to permit percolation of water.

5.2.2.5 Galvanic Anodes

Galvanic anodes shall be installed at all isolated fittings, such as 45s, 90s, tees, gate valves and fire hydrants, clusters of up to four isolated fittings located within 10 feet of each other, short piping sections, small, well-coated structures and in areas where interference with other structures might result from the use of impressed current systems. The anodes shall consist of galvanized steel cored, magnesium or zinc rods packed in cloth bags containing a specially prepared backfill material.

All galvanic anodes shall be attached to the structure requiring protection through the connection of the anode lead wire to a calibrated 0.1-ohm shunt to one of the structure lead wires that are thermite welded to the structure. The top of all galvanic anodes are buried a minimum of 12 inches below the structure either vertically or horizontally.

5.2.2.6 Protective Potentials

5.2.2.6.1 Ferrous Metals

NACE Standard RP 01 69-92 shall be used to determine protective potentials for ferrous metals. While this standard lists three primary criteria as acceptable, DFW Airport has selected only one that will meet their requirements.

The method selected for determining a protective potential shall be as follows:

1. A negative polarized potential (immediate “off” potential) of at least 850 millivolts relative to a saturated copper-copper sulfate reference electrode.

Soil contact points (pavement inserts) over ferrous structures such as welded pipelines, valves and fittings shall be placed at intervals of 50 feet in areas where concrete or asphalt pavements prevent direct contact with the soil. If necessary, a more definitive spacing shall be determined on an individual basis by a Corrosion Engineer, where external factors that have an adverse influence on the line. Soil contact points shall be a molded polyethylene pavement insert 1 ¾ inches in diameter by 6 inches long.

Where temporary placement of a reference electrode in soil directly over the structure is not possible, a copper-copper sulfate permanent reference electrode shall be considered for installation.

5.2.2.6.2 Non-Ferrous Metals

Protective potential for nonferrous metals shall be established by a Corrosion Engineer and shall be in accordance with NACE Standard RP 01 69-92.

5.2.2.7 Cathodic Process Corrosion Protection

The airport utilizes both impressed current and galvanic anode cathodic protection systems (CPS) to supplement coatings used for corrosion control. Impressed CPS shall have a design life of greater than 20 years. Maximum design life for galvanic anode systems utilizing magnesium or zinc anodes is greater than 15 years. Cathodic protection systems for immersed service shall be designed for a minimum of 10 years. In all cases, provisions shall be made for replacement of anodes and reference electrodes.

The selection of the type of cathodic protection system to be employed shall be specified by a Corrosion Engineer following preparation of preliminary piping layout drawings. Such selection shall be reviewed by DFW Energy & Utilities Services for approval.

All buried pressurized or immersed ferrous metal piping systems (except pre-insulated pipe) shall be properly coated, electrically isolated, bonded if necessary, and cathodically protected to prevent electrolytic corrosion.

Systems to be protected include but are not limited to:

1. Jet Fuel Lines
2. Interior of Aboveground Water
3. Natural Gas or Heating Fuel Lines
4. Storage Tanks (to include underground steel tanks and piping used for fueling)
5. Potable Water Lines
6. Fire Protection Water Lines
7. Compressed Air Lines

8. Pneumatic Lines
9. Waste Oil Storage Tanks and Piping Systems
10. Underground Fuel Storage Tanks and Appurtenances
11. Hydraulic Elevators/Lifts and Piping Systems

Impressed current anodes shall be located to minimize stray current pick-up on unprotected or foreign metal structures. Care shall be taken to not place impressed current anodes near pre-stressed concrete pipe or reinforced concrete pipe, which could be damaged by excessive levels of cathodic protection current.

A sufficient number of test lead wires shall be installed on ferrous metal structures such as pipelines, so that interference situations can be analyzed and corrected. Cathodic protection interference problems and their solutions are rarely similar, precluding “rigid” or “set” specifications for mitigation. Multiple structures in an area of interference can be very complex requiring extensive coordinated testing under the direction of a Corrosion Engineer.

5.2.2.8 Water Storage Tanks

Steel water storage tanks shall be internally coated and provided with cathodic protection. The design of the cathodic protection system must not be detrimental to the applied coatings in the wetted area of the internal surfaces when the system is properly adjusted. The design life for the cathodic protection anodes shall be a minimum of 10 years.

Anodes shall be installed in the tank to provide uniform current distribution to all immersed surfaces.

DC current shall be supplied by a suitable transformer - rectifier power supply located adjacent to the tank. The rectifier shall be either a constant current, constant potential or 100% manually controlled unit, at the option of a Corrosion Engineer. Manually adjusted rectifiers are the least expensive to install, operate and maintain. Manually adjusted rectifiers shall be specified unless special conditions dictate otherwise.

A permanent copper-copper sulfate reference electrode(s) shall be installed close to the wall of the tank (between anode strings) for monitoring purposes.

The anodes shall be suspended from a roof mounted fiberglass deck mounts designed so that the anodes may be inspected or replaced without entering or draining the tank.

5.2.2.9 Above Ground Storage Tanks

5.2.2.9.1 Bottom Exterior

Tanks should be set on self-draining concrete or asphalt pads. If this is not done, external tank bottoms shall be sandblasted and coated with an approved coating system, and provisions shall be made to apply cathodic protection. All tanks not set on self-draining concrete or asphalt pads, shall have a minimum of one permanent reference electrode installed beneath the center of the tank.

When tanks are to be located on properly drained pads, one coat of an approved inorganic zinc coating shall be applied to the externally sandblasted surface prior to erection. The dry film coating thickness range shall be from 2.5 to 4.0 mils. Otherwise, cathodic protection shall be provided for the external tank bottom.

5.2.2.9.2 Tank Interior

The interior of all storage tanks shall be sandblasted and protected with an approved protective coating. If the tanks are equipped with floating roofs, the coating system shall have sufficient abrasion resistance to withstand movement of the roof.

5.2.3.9.3 Tank Exterior

The exterior surface of the tanks shall be coated with an approved coating.

5.2.2.10 Underground Storage Tanks

All metallic underground storage tanks (UST) or fittings shall be coated and cathodically protected or installed in containment piping systems in conformance with local, state, and federal regulations.

Ferrous metal USTs and appurtenances shall be coated with a protective coating.

If an approved (holiday free) factory fiberglass cladding system of at least 100 mils thick is applied, no cathodic protection is required. The appurtenances attached thereto shall be coated with a compatible coating and cathodically protected.

As an alternate, fiberglass reinforced plastic tanks may be installed. Ferrous metal appurtenances attached thereto shall be coated and cathodically protected or installed in containment piping systems. Where submerged turbine pumps are employed, a polarization cell shall be installed in the grounding circuit of the electrical power supply.

5.2.2.11 Hydraulic Elevators and Lifts

All direct buried hydraulic elevators or lifts shall be coated with a protective coating. The lift shall be isolated from all electrical equipment with insulating fittings. Cathodic protection shall be applied.

5.2.2.12 Fuel Hydrant Boxes

Flush mounted steel hydrant boxes to be located below grade shall be externally coated with a protective coating and cathodically protected. Cathodically protected piping entering and leaving the box shall be electrically isolated from the hydrant connection. All piping within the box shall be bonded together. Cathodic protection shall be provided to both the piping and fittings in contact with the soil.

5.2.2.13 Piling

Any steel in buried piles, directly exposed to the earth, shall be coated with a nominal 16 mil DFT single coat coal tar epoxy and cathodically protected if the strength of the full steel cross-section has been used in meeting structural requirements. The steel component of piles, if directly exposed to the soil, shall be bonded together by welding a reinforcing bar between all components or by provision of adequate bonding cables.

5.2.3 Process Control Inspection and Test Stations**5.2.3.1 Test Stations**

Test stations shall be installed flush to grade in a concrete slab, at predetermined locations to facilitate inspection of the system. Two test wires shall be connected to each buried structure or cluster of fittings by thermal welding and brought to the surface in an appropriate terminal box. All test stations shall be filled with clean native soil free of rocks, asphalt, or concrete.

When foreign structures are adjacent to protected structures, test wires shall be attached to both structures to facilitate interference testing and/or mitigation bonding, if necessary.

Aboveground test stations shall be specified wherever their use is permissible and will not conflict with aircraft or vehicular traffic. At-grade test stations shall be specified in locations where aboveground test stations are not appropriate.

5.2.3.1.1 Non-AOA Test Stations

All test stations installed that are not in the AOA, shall have a lockable cast iron lid with the cast-in legend "CP Test". A nonmetallic extension tube (minimum length of 18 inches) shall be attached to the head of each test station. Provisions shall be made for connecting the anode and structure lead wires inside the test station on a nonconductive terminal board.

5.2.3.1.2 AOA Test Stations

Test stations for installation in the AOA shall be an L-868 Class II ground support light base. The fixture shall include an adjustable base sufficiently long to extend through the concrete and into native soil. The fixture shall include a lockable steel lid with the welded legend "CP Test" in 1-inch-high letters.

5.2.3.2 Test Station Lead Wires

The test station structure and anode lead wires shall be black in color and a No. 12 AWG stranded copper wire with NFPA 70 type THW or equivalent insulation.

Copper sleeve adapters shall be used when thermit welding #8, or smaller, wires to structures.

Reference electrode lead wires shall be No. 14 AWG stranded copper wire with NFPA 70 type RHH-RHW insulation.

The following color code shall be used to identify test station lead wires:

Table 24 - Required Color Code for identification of Lead Wires

Structure	Wire Color
Protected Structure	White
Reference Electrode	Yellow
Casing	Red
Unprotected Side of Insulator	Green
Anode Lead Wire	Black
Foreign Structure	Blue

5.2.3.3 Cathodic Protection Interference and Cooperative Testing

Where more than one independently cathodically protected structure is in the same area, currents flowing around one structure may affect another. This is particularly true when impressed current systems are used because of their greater operating current capacity and driving voltage. To overcome this problem, two lead wires shall be attached to each structure and brought to a surface test station.

Care must be taken to insulate DC powered equipment from ground or provide low resistance metallic paths for current return. Provision shall be made in the specification for interference testing by a Corrosion Engineer.

Coordination tests shall be carried out through local committees where they exist. These groups, representing all concerned area utilities and industries, coordinate testing and arbitrate solutions as required. As a rule, the organization owning a current source is responsible for the expenditures necessary for correction of a problem that it creates.

5.2.3.4 Access for Inspection

All equipment that may be subject to corrosion or may require periodic maintenance, shall have reasonable access for inspection.

Boilers shall be equipped with firebox doors and access openings to superheat and preheat sections.

Flue gas ductwork shall be provided with reasonable access openings.

5.3 Exterior Improvements

5.3.0 Curbs, Gutters, Sidewalks and Driveways

5.3.0.1 Sidewalks

Pedestrian concrete sidewalks shall be constructed between buildings and other essential locations where such a need may occur. The minimum standard width for sidewalk pavement shall be 4 feet with proper cross slope for adequate drainage. The minimum standard walkway pavement shall be of 5 inch-thick concrete reinforced with No. 3 rebar at 12 inches each way on a minimum 2 inch sand cushion. Rolled wire fabric will not be permissible as walkway or other reinforcing. Provide contraction joints spaced every 4 feet (approximate). Pre-molded ½ inch expansion joint material spaced at 32 feet is required. Sidewalks that are directly adjacent to curbs of streets and roadways should be 5 feet wide minimum. Sidewalks adjacent to curbs or pavement should have a thickened edge and expansion joint material installed. Unless allowed otherwise in the adopted Building Code, sidewalks terminating at building doors shall be constructed as landings flush with the finish floors to the interior of the building. Slopes at landings shall not exceed 2 percent (1/4 inch/foot) slope.

5.3.0.2 Pavement Markings

5.3.0.2.3 Fire Lane Markings

Fire lanes shall be marked with a 6 inch painted red stripe. The words "FIRE LANE NO PARKING" shall be stenciled in white paint, 4 inch-high letters with a $\frac{3}{4}$ inch stroke. The interval between stenciled signs shall be adequate to inform the public of the existence of the fire lane but in no event shall the interval be greater than 20 feet. Markings shall be on each side of the designated fire lane. The shade and type of paint shall comply with State of Texas specifications for traffic paint.

5.3.1 Site Improvement

5.3.1.1 Fences and Gates

5.3.1.1.1 Leased Property Fencing

All fencing on leased property is the responsibility of the Tenant and shall be aesthetically pleasing. This can be accomplished by use of material matching or similar to adjacent structures. Chain-link fencing shall be screened with plantings where appropriate.

5.3.1.2 Retaining Walls

5.3.1.2.1 Design Criteria

Wherever slopes must be steeper than a slope of 4:1 (four horizontal units to one vertical unit), the use of retaining walls is required. Either vertical or battered wall faces are acceptable. Exposed concrete shall be buff color with surface texture matching that of adjacent buildings. Trenching or sprinkler systems shall not be allowed in the passive soil area.

Retaining wall design and materials shall comply with the applicable sections of the Building Code and the geotechnical investigation report for that project. Applicable TxDOT standards shall be adhered to unless directed otherwise by DFW.

5.3.1.2.2 Expansive Soils

The effect of wall movement caused by expansive soils shall be taken into account and, when necessary, appropriate design steps shall be taken to minimize them. Geotechnical borings shall be performed at all proposed wall locations to determine the presence of expansive soils.

5.3.1.2.3 Factor of Safety

The factor of safety for overturning shall be 1.5 minimum and 3.0 maximum. The factor of safety for sliding and circular soil arc failure shall be a minimum of 1.5. Expansion joints shall be provided every 90 feet maximum, and contraction joints every 30 feet maximum.

5.3.2 Pedestrian Bridges

5.3.2.1 General

Pedestrian bridge length, width, height, and construction materials shall comply with the Building Code.

5.3.2.2 Materials

Construction material specifications, strengths, handling, storage, and testing shall comply with the TxDOT "Standard Specifications for Construction of Highways, Streets and Bridges."

5.3.2.3 Design Loads and Loading Combinations

The minimum live load shall be 100 pounds per square foot (psf). Where equipment and small vehicles are anticipated to use this structure, live loads shall be increased accordingly.

In addition to live and dead loads, the following loadings shall be taken into account:

1. Earth Pressure
2. Buoyancy
3. Wind (Including Jet Blast and Uplift)
4. Shrinkage
5. Temperature
6. Stream Flow
7. Construction Loads
8. Any Special Loads

Loads shall be applied in such a manner as to produce the maximum stresses.

Loading combinations shall be the same as those described in the American Association of State Highway and Transportation Officials (AASHTO) guidance "Standard Specifications for Highway Bridges" and interim specifications.

5.3.3 Screening Devices

5.3.3.1 Exterior Mechanical Equipment Screening

All visible equipment, whether roof- or ground- mounted, must be painted alike and screened from view wherever possible. Color for such equipment, including roll-up doors, mechanical equipment, metal canopies, piping, electrical equipment, etc., and any other equipment of specialized function, shall match. Specially designed screens, suitable plant materials, and architectural enclosures can be used for screening purposes depending on the facility and location. Roof-mounted equipment must be concealed behind parapet walls or in a screened enclosure of approved materials. Equipment should be grouped in clusters, preferably a single cluster to minimize the number of visible screens. Screen wall supports shall be designed with an enclosed, sealed shape with minimum 16 inch vertical clearance to the first obstruction above the roof surface to allow for proper flashing installation.

5.3.4 FAA Vegetation Cover Requirements

Construction projects within the boundary of the Airport Operations Area must follow the FAA requirement of establishing vegetation cover of 97% with no individual bare area larger than 1 square foot and an aggregate for bare areas of not more than 3% of the total vegetated area. For more information, refer to the FAA Advisory Circular 150/5200-33 (current edition) for Hazardous Wildlife Attractants on or Near Airports and FAA Advisory Circular 150/5370-10 (current edition), Standard Specifications for Construction of Airports (specifically Part 10, Turfing).

5.3.5 Allowed Vegetation Seed and Sod Types

To prevent wildlife attractants as required by the FAA and promote consistent vegetation coverage on airport property, the following seed and sod types are to be incorporated into the design of specifications and drawings. This is not intended to replace landscape designer plans for developed sites where decorative landscaping involving trees, shrubs, mulch, hardscapes, planting beds and decorative grasses are used.

Deviations or changes to the listed types or the use of any native flowering plants such as naturalization with wildflower attempts must be pre-approved by the airport's Wildlife Manager before receiving a building permit. For more information, refer to FAA Advisory Circular 150/5200-33 (current edition), Hazardous Wildlife Attractants on or near Airports and FAA Advisory Circular 150/5370-10 (current edition), Standard Specifications for Construction of Airports (specifically Part 10, Turfing), and the airport's Seed and Sod Specifications (32 92 19).

For DFW Airport, the following definitions describing use and mowing frequency shall control what vegetation, seed or sod is allowed.

1. Airside: Upon completion of the project, should any construction area fall within the AOA fenced area, those areas will need to be extended by 20 feet from the AOA fence. This 20-foot clearance also applies to all Runway Protection Zones (RPZ)
2. Public areas: All non-Airside areas within 30 feet of a leased area, mowed, or landscaped.
3. Non-Public areas: All non-Airside areas greater than 30 feet from a leased area, not mowed, and not landscaped.
4. Cool Season, Temporary Type: October 1 through March 31
5. Warm Season, Permanent Type: April 1 through September 30

5.3.5.1 Airside - Warm Season Permanent

1. Sod only, Bermuda Grass Cynadon dactylon

5.3.5.2 Public - Warm Season Permanent

(Select one option):

1. Unhulled Bermuda Cynadon dactylon
2. Hulled Bermuda Cynadon dactylon
3. Zoysia Grass Zoysia japonica

5.3.5.3 Non-Public - Warm Season Permanent

(Select one option):

1. Bermuda Grass Cynadon dactylon
2. Buffalo Grass Bouteloua dactyloides (AKA Buchloe dactyloides) and Blue Grama Grass Bouteloua gracillis
3. Buffalo Grass Bouteloua dactyloides (AKA Buchloe dactyloides) and Sand drop Seed Sporobolus cryptandrus
4. Buffalo Grass Bouteloua dactyloides (AKA Buchloe dactyloides) and Hooded Windmill Grass Chloris cucullate
5. Blue Gramma Grass (native) Bouteloua gracillis
6. TXDOT Permanent Urban Seed Mix (District 18 Clay Soils) Green Sprangletop- Leptochloa dubia, Sideoats Grama (El Reno)- Boutelous curtispindula, Buffalo Grass (Texoka) – Bouteloua dactyloides, Bermuda Grass – Cynadon dactylon

5.3.5.4 Airside - Cool Season Permanent

1. Sod only, Bermuda Grass Cynadon dactylon

5.3.5.5 Public - Cool Season Permanent

1. Sod only, Bermuda Grass Cynadon dactylon

5.3.5.6 Non-Public - Cool Season Permanent

1. Sod only, Bermuda Grass Cynadon dactylon
2. Sod only, Zoysia japonica

5.3.5.7 Non-Public - Warm Season Temporary

1. Teff Grass Eragrostis tef

5.3.5.8 Non-Public - Cool Season Temporary

1. Teff Grass Eragrostis tef
2. Creeping Red Fescue Festuca rubra
3. Western Wheatgrass Passcopyrum smithii

5.3.5.9 Special Conditions: Shade or Partial Shade Areas

Sod Types:

1. Zoysia 'Palisades'

Ground Covers:

1. Asian Jasmine Trachelospermum asiaticum
2. Horseherb Calyptocarpus vialis

5.3.5.10 Special Conditions: Landscaping

For a list of approved landscape plants, refer to the Landscape Plant List, available through DFW Commercial Development.

5.4 Irrigation**5.4.0 General Information**

The Irrigation system within the DFW International Airport boundary shall conform to the requirements set forth by the following section (Commercial Development DCM should comply with this DCM/DFW). If a conflict is found between this DCM and other local, state or federal regulations, the most stringent criteria shall be used during design and installation. The designer shall notify the project manager DFW Irrigation department and DFW Code in writing of any such conflicts in these sections.

If a conflict arises between DFW and the contractor, the needs of DFW shall be deemed more important. The irrigation system shall be designed by a person licensed to do so and the designer must also have a landscape Irrigation license per 30 TAC, Chapter 30, Subchapter D.

1. All irrigation equipment and/or appurtenance, such as but not limited to pipes, valve box, controller sprinkler head, and fitting shall be purple in color.
2. All irrigation systems will be managed by DFW or the tenant. The DFW irrigation supervisor will have current copies of any irrigation schedule regardless of who the system belongs to.

3. The DFW Irrigation Supervisor will have TECH level RainBird IQ access to all systems installed either DFW or Commercial Development. This will allow real time control over leaks, irrigation running while raining and freezing weather system shut off.
4. All irrigation pressure pipes need to be installed with trace wires.
5. All drip irrigation will need prior approval from the DFW Irrigation Supervisor.
6. The irrigation system shall be designed to prevent incidental ponding or standing water and shall be managed to minimize the inadvertent contact of water with the public. It shall be designed so that the irrigation spray does not reach any privately-owned premises outside the designated irrigation area or reach public drinking fountains or storm drains.
7. The contractor is responsible for all utilities locations. Any damage from not getting locates will be the responsibility of the contractor to repair.
8. Reclaimed water users must comply with all DFW, county, state, and federal health regulations.
9. All retrofitted systems shall be dye tested before being put into service.

5.4.1 Cross-Connection Control Plan

A licensed landscape architect or other licensed professionals in a related field shall submit a cross-connection control plan to DFW DCC. This is for the contractor to follow to ensure no connection between potable and reclaimed water.

5.4.1.1 Design Plans & Submittals

Submit initial and all subsequent Irrigation Documentation Package to the DFW Project Manager, DCC Code and irrigation supervisor for review.

The package shall consist of the following:

1. Irrigation Design Plan
2. Grading Plan
3. Water Efficient Landscape Worksheet
4. Soil Management Plan

All plans are to be checked by the Engineer Landscape Architecture, and licensed irrigator of work for consistency, accuracy, clarity, and conformity with DFW, local, state and federal regulations and standards.

5.4.1.2 Approved Drawings

All irrigation plans must receive DFW Code and the DFW irrigation supervisor's approval prior to installation.

5.4.1.3 Permitting

Permitting Requirements for DFW irrigation systems prior to issuance of a permit, a letter from the Registered Landscape Architect must be on file confirming that irrigation design is in accordance with Texas Commission on Environmental Quality (TCEQ) Chapter 344 (Landscape Irrigation). In accordance with TCEQ Chapter 210 (Reclaimed Water) and Chapter 344 (Landscape Irrigation).

Prior to any new irrigation system installation, testing Static and dynamic water pressure/flow rate during multiple times of day needs to be established. Multiple irrigation systems can reduce available water supply while running. These issues need to be planned for prior to submitting any plan documentation. Any backflow devices should have any permits prior to installation. Any backflow devices need to be tested, documented and have the paperwork turned in prior to system startup.

5.4.2 Irrigation Design Plans

An irrigation system must apply a consistent, even, measurable amount of water to the landscape over a set period of time. It is necessary that the system design considers water cost and conservation, long term durability, and maintenance costs, safety issues, aesthetic issues, and site-specific requirements. Provide a complete automatic landscape sprinkler irrigation design for all landscape areas as required as a part of the project development.

The landscape irrigation system shall be designed and operated to prevent runoff and discharge of irrigation water onto roadways, storm drains, sidewalks, driveways, and adjacent properties as much as possible.

Included on the irrigation drawings shall be a complete and comprehensive irrigation legend, indicating all equipment and materials utilized in the design indicating the following:

1. Manufacturer
2. Model number
3. Size
4. Brief description.

Indicate locations of irrigation water meters, irrigation points of connection, and electrical points of connection for automatic sprinkler controllers, splice boxes, valve boxes, sleeve locations, electrical meter, master valves, irrigation valves, manual operational valves and flow measuring devices and backflow prevention devices on the irrigation drawings.

The following information shall be provided for each irrigation water meter or irrigation point of Connection:

1. Static and dynamic water pressures
2. Meter size
3. Peak irrigation demand in gpm
4. Finished grade at backflow and highest head served
 - a. Provide construction details indicating installation procedures and materials required for the installation of all major components used in the irrigation design.
 - b. Provide check valves and/or anti-drain valves to prevent drainage of irrigation water from the sprinkler system due to changes in elevation.
 - c. Provide anti-drain sprinkler heads along all sidewalks and driveways.
 - d. Zones are to provide uniform coverage and gpm from control valves in each system.
 - e. Provide a shut-off valve to isolate the irrigation system from the water main.
 - f. Provide separate control valves for sprinkler lines operating systems of slopes (slope condition).
 - g. Sprinkler lines shall run parallel (or as close as possible) to contour lines.
 - h. Design systems to the lowest static pressure available, less 20% cushion for further system fluctuations.
 - i. No 4-way crosses will be used in planning or construction.
 - j. No flex tubing will be used unless written authorization from the irrigation supervisor.
 - k. No pipe shall be under a 1" in size. If a pipe needs to be smaller, it will need approval from the DFW Irrigation Supervisor.
 - l. No drip irrigation will be used at DFW unless prior approval in writing has been received from the Irrigation Supervisor.

The maximum potential pressure should be considered in the design. If water pressure exceeds 80 psi, a pressure reducing valve(s) or heads may be used to maintain water pressure at no more than 20% higher than system design pressure. System pressure should be tested prior to any design.

5.4.3 Water Efficient Landscape Worksheet

A project applicant shall complete the Water Efficient Landscape Worksheet that contains the following information and meets the design criteria.

1. It shall contain general project information (Project Name, Project Address and location, total Project area, total Irrigated landscape area)
2. It shall contain the Water Use Efficiency Statement, which is a narrative summary of the water use efficiency practices applied in the landscape project.
3. It shall contain a water budget calculation for the project.

5.4.4 Grading

A grading plan shall indicate existing and proposed grades with contours and spot elevations. No slope in turf or planter area shall exceed 5:1 or 20%. The minimum grade within landscaped areas should be 2%.

5.4.5 Pipe

The velocity of flow in piping must not exceed 5 feet per second. Pipe alignment must take site elevation changes into consideration to minimize low head drainage.

Pipe sizing and routing must include pressure loss calculations to ensure that the required pressure will be delivered under all circumstances and that pressure variation within the lateral is at a minimum.

All irrigation pipe and will be purple schedule 40 with glue bell and spigot; fittings will be painted purple as needed.

All pipe labeling should be face up and readable while standing over the trench. No glue joints will be allowed in a sleeve that goes under a roadway. The use of Poly type pipe is preferable in these cases.

Solvent welded type pressure mainline piping for sizes four (4) inch and smaller shall be ASTM D 1785 Schedule 40 PVC. Pipe shall be made from NSF approved Type I, Grade I PVC compound conforming to ASTM Resin Specification 1785. All pipes must meet requirements as set forth in Federal Specification PS-21-70 (Solvent-weld Pipe).

PVC solvent weld fittings shall be Schedule 40, NSF approved, conforming to ASTM Test Procedure D2466. Solvent cement and primer for PVC solvent weld pipe and fittings shall be of type and installation methods prescribed by the manufacturer.

All PVC pipe must bear the manufacturer's name, nominal pipe size, schedule or class, pressure rating (psi), and NSF seal of approval.

All PVC fittings shall bear the manufacturer's name or trademark, material designation, size, applicable IPS Schedule, and NSF seal of approval.

PE pressure pipe (3/4-inch through 3- inch) and tubing (3/4-inch through 2- inch) shall meet the AWWA C901 standard. The pipe materials shall conform to standard designation code PE 4710.

Poly type pipe can be used with the proper fittings and connectors.

5.4.6 Portable Water Line Separation

Reclaimed water irrigation piping shall be separated from potable water piping by a horizontal distance of at least nine (9) feet. Where the 9 - foot separation distance cannot be achieved, the reclaimed water piping must meet the line separation requirements of TCEQ Chapter 210, Subchapter B, Rule 210.25 and TCEQ Chapter 290 (relating to Water Hygiene).

5.4.7 Sewer Line Separation

Where a reclaimed water line parallels a sewer line, the reclaimed water line shall be constructed in accordance with TCEQ Chapter 210, Subchapter B, Rule 210.25, subsection (e) or (f). Where a reclaimed water line crosses a sewer line, these requirements are in §290.44(e)(4)(B).

Avoid piping through heavy and numerous tree roots, shrubbery, and ground cover areas whenever possible.

Avoid odd angles, curves, and cross trenching; specifically, do not exceed the radius of pipeline curvature allowed by the manufacturer. Locate mainline piping in a separate trench than lateral piping.

5.4.8 Sleeve Pipe

Sleeve piping is to be schedule 40 PVC pipe or better pressure rating, two sizes larger than the sleeved pipe. The designer shall verify if the change in stress due to traffic load can be sustained by the pipe specified above. If the pipe specified above is not adequate, the designer shall specify the pipe type, pressure class, etc.

The sleeve pipe under 20 - feet shall not have joints and/or bell. Place the wire in a separate sleeve from that of the water line sleeve. Wire sleeves use 2 inch or larger schedule 40 PVC purple pipe. Glue fittings in sleeve pipe are not allowed.

The sleeve pipe shall have programmable location markers on both ends of the crossing. It will be not less than 2 inches in diameter. Omni Marker TM type location balls/disks will need to mark both sides of any sleeve. Substitutions will need approval by DFW Code and the irrigation supervisor. Any substitutions will need to be tested and verified to be compatible with the currently used system.

5.4.9 Effect of Wind on Irrigation Design

The designer shall consider the effect of wind while designing the irrigation system. Wind patterns at DFW need to be checked over the last Three (3) years to get an average wind speed and direction, this will help design a system with better wind resistance.

5.4.10 Slope Surface

Slopes and changes in elevation create a challenge for the selection, location, and operation of sprinklers. The designer shall consider land slope, soil type, plant type, and infiltration rates when specifying application rates. Sloped planting areas above a hardscaped area shall be avoided unless there is a drainage swale at the toe of slope to direct runoff away from hardscape. The swale areas may be planted with turf, ground cover or low shrubbery and shall be watered separately.

5.4.11 Obstructions

The designer shall plan for all obstructions in the design of the irrigation system. Such obstructions or differences should be brought to the attention of the DFW Project Manager, DFW Code and the DFW irrigation supervisor. In the event this notification is not performed, the Contractor shall assume full responsibility for any revisions necessary.

5.4.12 System Zoning

The irrigation system will be numbered in a following direction except for Zone 1 needs to be the closest to the water source. Zone 2 needs to be the closest to the controller.

5.4.13 Sprinkler Heads

No mixing of different kind of heads will be allowed. All heads will be RainBird.

1. The spacing of heads shall not exceed the maximum indicated on the drawings. In no case shall the spacing exceed the maximum recommended spacing by the manufacturer for wind speeds of 3-5 miles per hour.
2. Sprinklers shall not be installed any closer than four (4) inches from hard surface edges and lawn edges.
3. Install sprinkler heads in the vertical position on flat landscaped areas. In areas where installing sprinklers on slopes, keep the sprinklers positioned perpendicular to the slope surface; the sprinkler should never be more than 10° away from normal.
4. All pop-up sprinklers shall be connected to the pipe by an adjustable swing joint assembly that is sized to meet the flow requirements of the sprinkler. No flexible pipe will be used for swing joints. Rainbird SA "flexible" series swing joints are approved for use.
5. A swing joint assembly shall be either a PVC triple swing joint consisting of a Schedule 40 street elbow (MIPT x FIPT), a Schedule 80 nipple and two Schedule 40 street elbows at the head end.
6. All sprinklers with an inlet size larger than 3/4" shall use a P.V.C. triple swing joint assembly or a pre-assembled swing joint only.
7. Swing joint assemblies for valve-in- head sprinklers shall have compatible threads and an O ring sealing mechanism.
8. The bottom street elbow of the swing joint assembly shall be connected to an outlet on the side of the lateral line pipe and the nipple shall be of sufficient length to be installed at an approximate 45-degree angle when the sprinkler is mounted at finished grade.
9. All stationary shrub risers shall be installed with two PVC street elbows connected to the side of the lateral line pipe and a Schedule 80 nipple that is long enough to suit the plant material.

The designer shall group the sprinklers so the total gpm for a zone does not exceed the available safe flow. All sprinklers grouped into a zone must have the same precipitation rates. Rotating style heads should consider arcs of coverage to ensure matched precipitation. On slopes, group sprinklers from side to side across the slope and not up and down the slope. The potential for low head drainage must be minimized for each zone. The pressure variation within each zone from the first to the last head must not exceed 15%. The plan design shall show a minimum of 0.75 (75%) distribution uniformity.

The designer shall specify the spray angle. (i.e., 90°, 120°, 180°, 240°, 270°, and 360). Select nozzles with radius to the design spacing; do not allow for adjustment of more than 20 percent reduction in radius to prevent distortion of the spray. Use head-to-head spacing when designing the system.

The designer shall include the manufacturer, model number, radius, required operating pressure, flow rate (gpm), and location, nozzle size and type of each sprinkler on the design plan.

The designer shall prepare a detailed drawing showing the installation, depth of burial from lateral lines, and required fittings of all sprinklers. Where spray heads are specified, use a minimum of a six- inch spring-loaded pop-up. If smaller heads are need get permission from the irrigation supervisor. Nozzles should be of a low angle or flat trajectory. All spray heads should have filter screen devices to prevent clogging. Where spray heads are specified along sidewalks, driveways, or in areas where damage from foot traffic or vandalism can occur, use anti-drain heads.

The use of micro-irrigation or pressure compensating bubblers may be used in some cases. Small, narrow (less than 8 feet), irregularly shaped or sloping areas shall be irrigated with micro-spray. Bubblers in turf areas shall be on a separate station to provide proper deep watering. Group rotors that have matched precipitation rates where the radius of the throw is maintained at a constant, regardless of the pattern of spray.

The designer shall include the manufacturer, model number, radius, required operating pressure, flow rate (gpm), nozzle size, type and location of each sprinkler on the design plan. The designer shall prepare a detailed drawing showing the installation, depth of burial from lateral lines, and required fittings of all sprinklers. Where rotors are specified along sidewalks, driveways, or in areas where damage from foot traffic or vandalism can occur, use anti-drain heads.

Use head-to-head spacing when designing the system. Select nozzles with the closest practical radius to the design spacing; do not allow for adjustment of more than 20 percent reduction in radius to prevent distortion of the spray.

5.4.14 Irrigation Meters

Provide an approved dedicated irrigation water meter on all irrigation systems. Badger Meter with Badger "Beacon" HR-E LCD encoders and LTE cellular endpoints are required for both reclaimed and potable water. The meter register must be purple in color and properly stamped for reclaimed water. Approval can be obtained from the DFW Code. Consult DFW Water Services to department for the latest specs on the meter and register usage.

5.4.15 Temporary Irrigation

Water meters will need to be used for all temporary irrigation. If the temporary system is not removed after notification of establishment, it will be deemed abandoned. The system will become the property of DFW and disposed of as seen fit. Airfield grass areas do not require sprinkler systems. However, truck watering of areas may be necessary for the first two (2) years after initial planting. Approval by the DFW irrigation department, Operations Department and DFW Code is necessary prior to installation.

5.4.16 Irrigation Schedule

The designer shall include an irrigation schedule for twelve months including runtimes per station, cycles per day, and days per week for each station. The designer shall coordinate with the DFW irrigation supervisor and the Grounds department prior to establishing an irrigation schedule. The irrigation schedule shall comply with DFW, local and state water conservation plan. Irrigation schedules shall have each season irrigation run times listed. This schedule needs to be emailed to the irrigation supervisor.

5.4.17 Permits and Fees

The contractor shall obtain and pay for all necessary permits and fees as required.

5.4.18 Contractor Responsibility

It is the responsibility of the Contractor to familiarize themselves with all grade differences, location of walls, locates and utilities. The Contractor shall repair or replace all items damaged by its work. The Contractor shall coordinate its work with other contractors and DFW DCC for the location and installation of pipe sleeves and laterals under roadways and paving, etc. The contractor will schedule their operations as to not cause delays with other contractors when possible.

5.4.18.1 Materials List

The Contractor shall furnish the articles, equipment, materials, or processes specified by name in the drawings and specifications. No substitution will be allowed without prior written approval by the DFW irrigation supervisor and DFW Code. Equipment or materials installed, buried, or furnished without prior approval of the DFW will be rejected and the Contractor may be required to remove such materials from the site at its own expense.

Warranty: The manufacturer's warranties shall not relieve the Contractor of its liability under the guarantee. Such warranties shall only supplement the guarantee. Any warranty begins after final completion and acceptance by DFW. If the warranty runs into late fall or winter the warranty will be suspended and will resume in spring when systems get turned back on.

5.4.18.2 Design Records and GIS Files

Licensed irrigator of the work shall submit any and all of the following: Computer-Aided Design, KLM, Drafting (CADD) files and GIS Shapefiles to the DFW DCC, and DFW irrigation supervisor. Files must include irrigation pipes and their appurtenance, including but not limited to Backflow Prevention Unit, Isolation Valve, Automatic Control Valve, Pressure Regulating Valve, Controller, Sprinkler Head, Rain Sensors, Freeze Sensor, Wind Sensor, Flow Meters.

The include table in the files shall have at minimum the following field:

1. Year Designed/Installed
2. Type (Pipe or their appurtenance); if appurtenance, specify the appurtenance type
3. Material
4. Size/Diameter

5.4.18.3 Record and As-Built Drawings

The Contractor shall provide and keep up-to-date complete "As-Built" drawings indicating locations, sizes, and kinds of equipment installed. This set of drawings shall be kept on the site and shall be used only as a record set.

These drawings shall also serve as work progress sheets and the Contractor shall make neat and legible annotations daily as the work proceeds, showing the work as actually installed.

These drawings shall be always available for inspection and shall be kept in a location designated by the DFW Construction Manager.

5.4.19 Existing Water Lines Converting to Reclaimed Water

Existing buried piping which will be converted to reclaimed water use need not be marked unless the piping becomes exposed, such as during installation of new pipeline, modification of the system, or maintenance of existing pipe. All systems will be dye tested for cross-connection.

Pipe Materials and Fittings: All irrigation pipes and components, unless specified otherwise, shall be purple in color regardless of water source, either potable or reclaimed water. If available, fittings and heads should also be purple.

5.4.20 Thrust Blocks

Concrete thrust blocks for all specified piping shall be the size and type required by the manufacturer's installation guide. Form thrust blocks in such a manner to prevent any concrete from coming in contact with the pipe. Solid pipe shall be between the thrust block and the fitting to prevent direct contact of thrust block and fitting. Thrust blocks shall be installed on all pressure lines over 2" in diameter whenever pressure line changes direction.

Thrust blocks are required at the backflow prevention device.

5.4.21 Backflow Prevention Units

All potable irrigation systems shall require backflow prevention. Reclaim system do not require backflow devices.

1. At a minimum, the backflow assembly shall meet all TCEQ, DFW, State and local requirements.
2. Provide backflow prevention at all cross-connections with water supplies in accordance with 30 TAC, Part 1, chapter 344. Locate backflow prevention before any outlets from the irrigation system and in locations that allow for inspection, testing, and servicing
3. Backflow preventer type shall be per 30TAC, Part 1, Chapter 344 and shall be of a size as indicated on the drawings.
4. Where possible, a reduced pressure backflow preventer assembly should be installed down flow the meter. It shall be installed in the vandalism and freeze protected enclosure or box with heat trace wire with a crush-proof box with lids that are easily accessible by one (1) person. If this not possible because of field conditions, a backflow preventer along with a pressure-reducing head may be used.
5. If a backflow prevention device is used instead of RPZ, it must be located no more than three (3) feet from the water meter. If reclaim and potable irrigation share the same vicinity the RPZ must be used.
6. Wye strainers at backflow prevention units shall have a bronzed screwed body for sizes two (2) inches and smaller and 125 lbs cast iron flange body for sizes two (2) inches and larger. All wye strainers shall have a minimum 30 mesh screen and shall be similar to Bailey #100B or approved equal. Wye strainers shall not have a hose bib and shall be installed as per standard details.
7. Smaller mesh screens may be required as necessary.
8. Backflow prevention devices shall have a minimum size equal to the size of the water meter.

5.4.22 Isolation Valves/Ball Valves

The designer shall include the manufacturer, model number, size, and location of each valve on the design plan.

The designer shall prepare a detailed drawing showing the size and type of connecting pipe, fitting needed, and valve box.

All valves of three (3) inch or four (4) inch size shall have a standard cube head on the stem. The valve stack shall be standard cast iron or equivalent with appropriate cast iron lid. The valve must be clockwise closing. DFW standard is a (4) inch main line tap. 3-inch systems will not be designed without written permission from the irrigation supervisor and DFW Code.

Every automatic valve will have an isolation valve prior to the valve. The recommendation is a ball valve.

Looped mains are preferred and should have a minimum of three isolation valves. These should be at each tap to isolate the system and the middle of the loop. Install isolation valves to the same depths as the mainline. Where piping is 2½ inches or less, use ball type isolation valves.

Gate valves 2½ inches and larger shall be iron body, bronze stem, flanged, full port, resilient seat, or wedge shut-off which can be serviced from the top while the valve is in line.

Gate valves four (4) inches and larger shall have two (2) inch square operating nuts, with arrow cast in metal indicating direction of opening. The valves shall be right turn close.

5.4.23 Automatic Control Valves

These shall be Rainbird valves. The electric control valve shall be a normally closed, 24-volt valve. The valve shall have a slow uniform closure to eliminate water hammer or chatter. All valves shall have a manual flow adjustment. Valves shall be pressure rated to 200 psi. All control valves will have Omni Marker TM type locator balls or disks install at each valve box or similar location device. Check with DFW Irrigation for current devices and specs. Substitutions will need approval by DFW DCC and the irrigation supervisor. Any substitutions will need to be tested and verified to be compatible with the currently uses system. Locate control valves near the middle of the zone to balance pressure loss from the valve to the end of the circuit. Incorporate separate zones for slopes from top to bottom or in some cases, multiple zones: top, middle, and bottom. Group zones with consideration to plant root depths and type of vegetation. A Rainbird master valve and flow meter shall be installed and wired to the controller on the discharge side of the backflow prevention device on all installations. All irrigation will be designed to use a 2" size unless prior approval by the DFW Irrigation Supervisor. If a pipe needs to be reduced in size it should be done after the valve.

5.4.24 Pressure Regulating Valves

Where the static pressure is greater than 80 psi, Pressure reducing heads, or an adjustable brass pressure-regulating valve may be installed.

The designer shall prepare a detailed drawing showing the size and type of connection pipe, fittings needed, and valve box for pressure regulation valve.

Where systems require a pressure regulating valve, install the valve at the point of connection before the backflow preventer.

Assemble the pressure-regulator assembly in accordance with the manufacturer's instructions, DFW Code and the latest edition of the Uniform Plumbing Code.

Locate the pressure regulator where there is sufficient side clearance for testing and maintenance within the pressure-regulator assembly.

5.4.25 Control Wiring

All systems are to be standard wire design. No 2-wire path, coders or decoders will be used unless written approval from the DFW irrigation supervisor.

Underground electrical wiring used to connect an irrigation controller to any electrical component of the irrigation system must be listed by Underwriters Laboratories as acceptable for burial underground.

Underground electrical wiring that connects an irrigation controller to any electrical component of the irrigation system must be buried with a minimum of twelve (12) inches of select backfill.

All 24-volt single wires from the controller to automatic valves should be type UF, solid copper, single conductor, PVC insulated, UL approved underground feeder wire for direct burial in the ground.

Include 1 separate common wire from the controller through each zone valve to terminate at the last valve in the system. Common wires shall be white with a different color stripe for each automatic controller if more than 1 controller used in the area.

The electrical system shall be installed in accordance with all state, local, DFW, Federal and the National Electrical Code most recently adopted by the DFW. Connections between the automatic controllers and the electric control valves shall be made with direct burial copper wire AWG-UF 600 volts. Pilot wires shall be a different color wire for each automatic controller. Pilot wires for single controller installations shall be black, 12 gauge minimum.

All wires shall be installed in accordance with the valve manufacturer's specifications and wire chart. In no case shall wire size be less than #12 gauge. Wires will be correct size and type for distance and voltage drop.

Cables are desired for longer runs. These cables have individually numbered wires inside the cable and are multi strand.

Wires for the Rainbird flow meter shall be shielded from electric noise. Wires for the master valve will be different colors from any other colors used. Maxi TM type cable is acceptable.

4 12-gauge wires will be installed in each controller and terminate outside of the controller in pull box for future connections.

Wiring shall occupy the same trench and shall be installed along the same route as pressure supply or lateral lines wherever possible. See sleeving requirements.

Where more than one (1) wire is placed in a trench, the wiring shall be taped together at intervals of ten (10) feet.

An expansion curl should be provided within three (3) feet of each wire connection, valve and at each change in direction. All splices shall be made with waterproof grease filled type connectors.

Field splices between the automatic controller and electrical control valves will not be allowed without prior approval of the DFW irrigation supervisor. Any splice boxes that are approved need to use the marker ball/disk system already mentioned.

Every controller must have at least three (3) spare additional wires pulled through each valve and terminated at the farthest point in the project and coiled in a pull box for future use, these should be colors different from any other control wire color used.

5.4.26 High-Voltage Wiring for Automatic Control

The contractor shall use a 120-volt power connection to the automatic controller. All electrical work shall conform to local codes ordinances, and authorities having jurisdiction.

All wire shall be sized and specified, as shown in the design. Line voltage wire shall be CSA approved, either NMWU or TWU. Line voltage wire should be installed by a person with the proper qualifications.

White wire shall only and always be used as the common wire. All connectors shall be approved for the watertight application and assembled according to the manufacturer's recommendations. No wire smaller than 18 gauge shall be used. Install all 120-volt wiring that is above ground in gray Schedule 40 PVC electrical conduit, flexible metallic conduit, or electrical metallic conduit.

Install all 120-volt wiring below ground in conduit per electrical code.

There needs to be an electrical outlet installed in the bottom of the enclosure. The controllers will be wired with an electrical plug and not directly wired to the power source. Controllers and enclosures will be properly grounded.

5.4.27 Controllers

These shall be the current version of Rainbird LXME controllers (or compatible) with IQ flow smart modules. They will have the 12 station modules unless otherwise approved by the irrigation supervisor.

A controller housing or enclosure is required that protects the controller from hazards of the environment at the installed location and should be vandal resistant. These can be stainless steel, painted cabinet and/or pedestal combinations unless substitutions are approved by the irrigation supervisor. Show the power supply/connection point, line location, panel, and breaker number on drawings and inside the controller. Preferably these should be installed to observe the system running from the controller.

All irrigation satellite controllers must be connected to the Rainbird IQ-cloud system. The connection card should already be purchased & installed before completion. Get with DFW Irrigation department for assistance in choosing which communication module is needed. Each controller must have a compatible wired rain/freeze sensor.

The Contractor shall install the controller as per the manufacturer's instructions.

A minimum 2-year manufacturer's warranty on the irrigation system controller and all components must be provided.

Soil Moisture Sensors and Wind Sensors may be installed but are not mandatory.

5.4.28 Pre-Installation

Stake out the location of each run of pipe and all sprinkler heads and valves before trenching. Verify area dimensions while staking sprinkler head locations. Stake sprinkler head locations and measure from permanent objects, buildings, etc., or survey benchmarks. The sprinkler layout and spacing of the staked irrigation system design in the field should meet with the accuracy and approval of the Licensed irrigator/system designer.

The contractor will approve each run of the system before the start of actual installation; conflicts between irrigation system layout and proposed plant material, drainage systems, light standards, etc., will be resolved in the field by the contractor at the time of field staking. Do not deviate from the plans and specifications without written approval from the DFW Code and the irrigation supervisor.

5.4.29 Site Condition

The Contractor shall exercise extreme care in excavating and working near existing utilities. The Contractor shall be responsible for damages to utilities which are caused by its operations or neglect. The Contractor shall:

1. Check existing utility drawings for existing utility locations.
2. Call out mark-out crews for each utility and/or 811 prior to excavation. Get with DFW to locate our owned utilities that are not located by 811 or others.
3. Carefully check all grades to satisfy themselves that they may safely proceed before starting work on the sprinkler irrigation system.
4. Inspect the site for existing conditions that will affect the irrigation system design or installation and develop a plan to minimize disturbance of existing structures and landscapes.
5. Coordinate installation of sprinkler irrigation materials, including pipe, so there shall be no interference with utilities or other construction, or difficulty in planting trees, shrubs, and ground cover.

5.4.30 Pipe Deflection

Pipe joint deflection shall not exceed one (1) degree in any direction. Standard fitting shall be used for all deflection greater than one (1) degree.

5.4.31 Trenching

Dig trenches straight and support pipe continuously on the bottom of the trench. Lay pipe to an even grade. Trenching excavation shall follow the layout indicated on drawings and as noted. The trench bottom should be uniform, free of debris, and of enough width to properly place pipe and support it over its entire length. Debris, refuse, trash shall not be left in the trenches. Provide a minimum of 24" of cover for all pressure supply lines 1½" and smaller.

Provide a minimum cover of 24" (or directly below mainline where possible) for all control wiring.

Any open trench or other excavations shall be barricaded and marked with high visibility flagging tape, safety cones, and flashing light as required by DFW, local, state federal regulations. Contractor takes all liability with any of their digging or trenches.

5.4.32 Pipe Bedding

In-situ soil can be used as a bedding material if the licensed designer and pipe manufacturer approves.

Pipes shall be laid with labeling on pipe facing up. Bell holes shall be formed and, if required, a trough scooped out to grade, and the pipe laid and jointed as specified. Density shall be at least 90 percent of maximum density as determined by ASTM D 698. Moisture content shall be within -2 to +4 of optimum.

5.4.33 Backfilling

The trenches shall not be backfilled until all required tests and inspections have been performed.

Trenches shall be carefully backfilled with the excavated materials approved for backfilling, consisting of earth, loam, sandy clay, sand, or other approved materials, free from large clods of earth or stones.

The backfill shall be compacted in landscaped areas to a dry density equal to adjacent undisturbed soil in planting areas. Backfill will conform to adjacent grades without dips, sunken areas, humps, or other surface irregularities.

All backfill material will be subject to approval by the project manager and DFW Code.

If settling occurs, the Contractor shall make all required adjustments to the pipe, valves, sprinkler heads, lawn or plantings, or other construction as necessary for a period of at least 6 (six) months after completion.

5.4.34 Existing Trees

Adjacent to existing trees, the Contractor shall use all possible care to avoid injury to trees and tree roots. Excavation in areas where two (2) inch and larger roots occur shall be done by hand. When possible, trenches should not be placed where tree roots will damage irrigation infrastructure.

5.4.35 Valve Boxes

All valves shall be installed vertically and centered in the box to be easily accessible for servicing.

All valve boxes shall be installed flush with finished grade and located in shrub areas where possible.

All valve boxes shall be installed with adequate clearance above the pipe and on a firm base so as not to contact the pipe with the settlement or upon being depressed. Valve boxes shall be supported and of a type of material such as to be able to support the weight of expected traffic.

Install a manual shut-off valve at every valve box location; place shut-off valves before each automatic valve in the valve box or one shut-off valve to control all automatic valves in a valve box. Install a shut-off valve with enough clearance for proper operation.

Valve box size shall be as shown in the drawings.

Purple valve boxes or purple valve box lids shall be used and shall have an advisory label or nameplate permanently molded into or affixed onto the lid with rivets, bolts, etc., and constructed of weatherproof material with the wording IRRIGATION permanently stamped or molded into the label.

5.4.36 Testing and Inspection

All projects will be tested and inspected by the DFW Code Inspector and/or a designated authorized representative. The inspections will ensure that the installation complies with the design intent, specifications, and these guidelines.

Any deficiencies noted during the inspection and testing will be rectified promptly and signed off by the DFW DCC or an authorized representative.

No water supply system or portion thereof will be covered or concealed until it first has been tested, inspected, and approved.

The test will require that the system be operated sequentially with the controller, in the presence of the owner's representative.

There are no circumstances under which the system mainline should be subjected to the application of compressed air as a static pressure test. Hydrostatic pressure testing should be used to detect the leak and test pipes.

Bleed air out of the system before testing. Backflow assembly testing results shall be turned into DFW Code. Submit a copy of the inspection log to the DFW Code upon completion of the landscaping.

5.4.37 Site Restoration/Clean-up

Clean-up shall be made as each portion of the work progresses. Refuse shall be removed from the site, all paving shall be broomed or washed down, and any damage sustained shall be repaired to original conditions. Restore all existing landscape, pavement, and grade of areas affected by work to the original condition or to the satisfaction of the engineer or owner.

5.4.38 Final Inspection and Acceptance

All materials, equipment, and documentation shall be provided according to specifications, as shown on the plans and as specified herein. The scope of work by these guidelines:

1. Warranty requirements
2. Operations and maintenance manual
3. As-built record drawing
4. Operating instructions
5. Maintenance schedules
6. Watering schedules

A final "walk through" with the DFW Utility Group will be required to explain the operation of the system. The "walk through" must include, but not be limited to, visual inspection of all zones and emission devices operating for not less than two minutes. A review of the current program, as well as seasonal, watering schedule. Prior to final acceptance, a letter from the licensed irrigator must be on file confirming that irrigation construction and operation conform to the approved design documents, applicable state and local laws.

The letter shall say, "This irrigation system has been installed in accordance with all applicable state and local, DFW, laws, ordinances, rules, regulations or orders. I have tested the system and determine that it has been installed according to the irrigation plan and is properly adjusted for the most efficient application of the water at this time."

The letter shall include the irrigator's name, license number, company name, telephone number, and date of the warranty period.

The Licensed Irrigator and/or Engineer of record with irrigation license must certify the placement and operation of the system.

5.4.39 Extra Equipment and Material

Provide two of each item of extra equipment, tools, and special wrenches as required for the maintenance of the irrigation system. Extra material that was bought for the job will be turned in to the DFW Irrigation department.

5.5 Signage

5.5.0 Irrigation/Reclaim Water Advisory Signs

All irrigation sites must post a clearly visible sign at least 12-inch by 9-inch indicating the use of reclaimed/non-potable water. These must be installed at all entrances to the customer's facility where they can be easily seen to the extent necessary to advise passersby.

The sign may be supplied by the DFW Utilities group.

5.5.1 Above Ground Temporary Water Lines

Above ground irrigation lines, must be labeled with the words "Non-potable Water - Do Not Drink - No Tamar" and color-coded purple. If purple identification tape is used to label the pipe it must be adhesive, permanent, and resistant to the environmental conditions. Purple may also be painted around the circumference of the pipe at 5-foot intervals for color-coding.

Chapter 6 Architecture

6.0 Thermal and Moisture Protection

6.0.1 General Criteria

6.0.1.1 Energy Conservation

All DFW International Airport projects shall comply with the currently adopted edition of the International Energy Conservation Code and the DFW Green Building Standards.

6.0.2 Roofing Systems

6.0.2.1 General

6.0.2.1.1 Roof Type

Many different types of roofing systems have been utilized at the Airport. The acceptable choice and application of a particular roofing system will depend on the type, use, location, and configuration of the building. Roof systems meeting the minimum criteria outlined below may be specified on new construction or as roof replacement on existing buildings. Set up a meeting with the Roofing and Waterproofing Department for the start of project design development.

6.0.2.1.2 Acceptable Roof Systems

Acceptable roof systems for consideration include but are not limited to on low slopes (up to 1 ½ inches per foot); 2-Ply Modified Bitumen (MFR: Siplast, Soprema with maximum roof slope approved); Standing Seam Metal 22 ga minimum (MFR: McElroy 238T Panels with minimum slope 1/2 inch per foot); Single Ply PVC/KEE 50 mil minimum – fully adhered (MFR: Fibertite, Carlisle); Coatings (MFR: Hydro-Stop, Siplast, SIKA).

For steep slopes (over 1 1/2 inches per foot), properly designed and fastened: Standing Seam Metal 22 ga minimum (MFR: McElroy 238T Panels with minimum slope 1/2 inch per foot); PVC - KEE 50 mil minimum – fully adhered (MFR: Fibertite, Carlisle); Coatings (MFR: Hydro- Stop, Siplast, SIKA).

Standing seam metal roof systems may be selected with prior approval from ETAM.

6.0.2.2 Roof Load

The roofing assembly and its components must be capable of withstanding and accommodating all of the airport service conditions to which they will be exposed, including rain, snow, hail, ice, wind, sun, thermal shock, service traffic and applied loads. Roof levels designated as future floors shall be designed for a live load of 100 psf minimum and a dead load of 25 psf minimum.

The roof system components must provide optimum thermal resistance to heat gain or loss within the building, consistent with good roofing practices.

6.0.2.3 Roof Drainage

Ponding is defined as standing water on roofs for more than 24 hours. Ponding is the frequent source of leaks and the cause of premature failure of roof systems. Roof systems manufacturers recognize the seriousness of this common problem and generally exclude leaks or failure caused by ponding water from their warranties. Ponding water is not acceptable. Minimum drainage to roof drains shall be not less than $\frac{1}{4}$ inch per foot roof surface slope in accordance with the currently adopted edition of the IBC.

6.0.2.4 Minimum Standards and Recommendations Included by Reference

Roof system selection, design detailing, and specification shall, at a minimum, comply with the requirements and recommendations of the following standards:

1. National Roofing Contractors Association (NRCA), Roofing and Waterproofing Manual, latest Edition (low slope and steep manuals).
2. Architectural Sheet Metal Manual, Latest Edition, Sheet Metal and Air Conditioning Contractors National Association (SMACNA).

6.0.3 Roof System Design Criteria

6.0.3.1 Roof Replacement of Existing Building

6.0.3.1.1 Guidelines

Roofs adjacent to aircraft ramps (terminals, cargo and hanger structures, electrical vaults, etc.) shall be smooth surfaced or smooth coated. Gravel, rock, or paver ballasted roof surfaces are not permitted because high winds may displace gravel, rock, or paver material onto operational surfaces, causing damage to aircraft and aircraft engines.

Roof attachment shall equal or exceed FM Global 1-120 (based on ASCE 7 – 05) wind uplift rating for high exposed roofs or roofs subject to jet blast must be calculated from FMG Loss Prevention Data Sheets.

Roof materials and assemblies shall be listed by Underwriter Laboratories as Class A and by Factory Mutual as a Class 1 roofing material or roof assembly.

Roof coping and metal edge materials assemblies and securement shall equal or exceed the current edition of the ANSI/Single Ply Roofing Industry (SPRI) ES-1 (all roof slopes) and FMG 4435 wind design standards for all roof slopes.

Provide above roof deck rigid board insulation thickness in multiple stagger layers to achieve thermal resistance of R-25 minimum, but in no case less than that required by the currently adopted edition of the International Energy Conservation Code (IECC). Cover fastener plates/fasteners with cover boards/insulation to prevent thermal bridging of fasteners.

Where different roof types join together, provide proper seams, parapets, area dividers, or expansion joints.

Provide roof foot traffic protection at all parapets, around equipment and at all areas subject to frequent wear.

Minimum flashing heights shall be 8 inches above roof surface to the extent possible by existing design conditions, but in no case shall flashing height be less than required by the roof system manufacturer for the applicable warranty. Roof surface is defined as the roof membrane surface exposed to the exterior side elements.

6.0.3.1.2 Internal Gutters

Internal gutters are not allowed.

6.0.3.2 Warranties and Guaranties for Airport Maintained Roofs

Require Roof Manufacturer's "Total System, No Dollar Limit (NDL)" warranty for maximum time limited available (15-year for liquid applied roofing systems and 20- to 30- year for all other acceptable roof systems, depending on roof system).

5. The warranty shall cover wind speeds up to 100 mph for peak gusts measured 10 meters above ground level.
6. The warranty shall cover related sheet metal copings, edge strips and metal edges.
7. Warranty shall also include 36-man hours of labor per year throughout the term of the warranty for puncture repair work.

Roofing Contractor shall provide a 5-year guarantee/warranty against leaks and defects in workmanship in the installed roofing system.

Roof related sheet metal, copings, edge strips and metal edges: Contractor shall provide a 5-year guarantee against leaks and defects in materials and workmanship. Sheet metal exposed to public view shall have a factory-applied finish with a 20-year warranty covering fading, discoloration, peeling or other defects.

6.0.4 Energy Rating

"Energy Star" roofing materials (i.e., white in color, solar radiation reflective) are required for all new construction and major renovation projects. Reflective roof system finishes must comply with FAA guidelines.

6.0.5 New Construction

6.0.5.1 Slope

Provide a minimum $\frac{1}{4}$ inch per foot slope designed and built into the structure of the facility. Where the minimum $\frac{1}{4}$ inch per foot slope is not designed nor built into the structure of the facility, the $\frac{1}{4}$ inch slope shall be achieved using a fully tapered insulation system and/or crickets, saddles, or tapered insulation system.

6.0.5.2 Roof Drains

Size and quantity are as required by the greater requirements of the currently adopted edition of the International Plumbing Code and this Design Criteria Manual.

Locate drains a minimum of 36 inches from equipment and perimeters to allow proper slope and flashing details. In addition, any new or reconfigured roof drains from terminal, hangar and air cargo building roofs shall not be allowed to drain on the ramp.

Roof drains will be compatible with the roof system and will not be located within the structure. The Designer will use 6 inches per hour as a minimum rainfall intensity guideline for sizing roof drains.

6.0.5.3 Roof Mounted Equipment

Accurately locate, size, and measure existing flashing height above roof surface as defined above.

As designated by the designer of record, all obsolete equipment shall be removed, and penetrations through the deck shall be properly framed, infilled, and sealed.

6.0.5.3.1 Penetrations

Less than or equal to 6 inches in least dimension shall be covered with 10 gauge galvanized sheet metal extending 6 inches beyond the opening in each direction. Fasten metal sheet over penetration to deck with approved fasteners 6 inches on center, minimum two per side.

Penetrations larger than 6 inches in both directions shall be capped with a prefabricated metal curb, the opening covered by an approved deck material, the curb opening infilled the insulation in the required thickness to achieve a thermal resistance of R-25 and the curb and curb opening flashing with a minimum 60 mil High Temperature Modified Bitumen membrane, and covered with 24-gauge galvanized metal cap, sloped to drain ¼ inch per foot with 24-gauge galvanized metal cap, sloped to drain ¼ inch per foot.

6.0.5.3.2 Flashing

All equipment with flashing height of less than 8 inches above the new roof surface shall be raised to a flashing height of 8 inches minimum.

6.0.5.3.3 Shutdown Time

Limit shutdown of roof mounted equipment to hours specified by designer of record.

6.0.5.3.4 Metal Counterflashing

Metal counter flashing covering the top flashing edge by 4 inches minimum must protect all curb-mounted equipment flashing.

6.0.5.3.5 Walkway Pads

Walkway pads shall be placed around all roof access hatches, roof-mounted equipment requiring periodic service and at all areas subject to frequent wear.

6.0.5.4 Antennas

Antennas mounted on the roof shall be mounted to bases (non-penetrating where practical) designed for this purpose or to roofing system penetration support post(s) securely anchored to the building's structure below and properly flashing to the roofing system.

Antennas attached to piping or other equipment shall be removed.

6.0.5.5 Roof Penetrations**6.0.5.5.1 Piping and Conduit**

All pipe and conduit penetrations shall be securely anchored to the structure below the roofing system. All single pipe and conduit penetrations shall be flashed to the roofing system with roofing system manufacturer approved penetration materials and details. All multi-pipe and conduit penetrations shall be through covered metal pipe enclosures similar to SMACNA Figure 4-15A.

For membrane roofing systems: roofing manufacturer's prefabricated multi-pipe penetration flashings of the proper size and configuration for the specific condition may be used. The manufacturer approved fluid applied penetration flashing may be used only with approval of the Roofing and Waterproofing Department. Metal penetration dams (pitch/sealant pans) will be permitted only with the approval of the Roofing and Waterproofing Department and must have a minimum of 24-gauge galvanized metal umbrella counter flashing covers.

6.0.5.5.2 Lightning Protection

Lightning protection down leads shall be flashed with polyvinyl chloride (PVC) pipe enclosures and capped with PVC domed caps.

6.0.5.6 Solar Readiness

All roof renovation or replacement, and new construction should include readiness for future installation of solar photovoltaic systems if the finished surface is flat or south or west sloping. Where suitable orientation conditions exist, designs should follow the requirements of IECC Appendix CB: Solar-Ready Zone-Commercial.

*This requirement supports LEEDv4/5 Energy Metering and Reporting.

6.0.6 Windows

Dynamic glazing is required for new construction and major renovation projects. The installation shall target windows with south- or west-facing orientation.

*This requirement supports LEEDv4/5 Energy Metering and Reporting.

6.1 Metals

6.1.0 Structural Steel Framing

6.1.0.1 Structural Steel Detailing

Detailing for structural steel shall comply with the latest edition of American Institute of Steel Construction (AISC) "Detailing for Steel Construction." All loads shall be identified on the plans.

6.1.0.2 Connections

All steel connections shall be designed using Load and Resistance Factor Design (LRFD) in accordance with AISC 360 - Specification for Structural Steel Buildings (latest edition). Design loads shall be identified on the plans.

All connections shall be designed and sealed by a professional structural engineer registered in the State of Texas.

6.1.0.3 Structural Steel Framing

Structural steel shall conform to the latest publications for structural steel framing and designed using LRFD unless approved otherwise. Material specifications shall conform to, but not be limited to, the following:

1. W, WT Shapes: ASTM A992.
2. S, M, and C Shapes: ASTM A36
3. HP Shapes: ASTM A572 Grade 50
4. Angles and Plates: ASTM A36

5. Pipes: ASTM A53, Grade B.
6. Tubes: ASTM A500, Grade B.
7. Erection Bolts: ASTM A 307.
8. High Strength Low Alloy (HSLA) Steel: ASTM A588 steel will be allowed with prior approval for high-mast light poles and base plates. HSLA steel shall not be used in areas of high moisture or water unless a proper surface treatment is utilized. Concrete pier pedestals for high-mast light poles shall be at least 36 inches above finish grade.
9. Applied fire protection (i.e. Intumescent paint and insulation) shall be detailed on structural steel drawings. Applied structural steel fire protection applications shall submit third party inspection documentation to DFW Building code.

6.1.0.4 Floor Framing Systems

Floor systems at Terminal buildings shall be designed to eliminate excessive vibrations from pedestrian movement, Skylink, and mechanical equipment installed.

6.2 Openings

6.2.0 Entrances Storefronts and Curtains

6.2.0.1 Entrances

Automatic sliding doors at two-way traffic areas where the public is carrying baggage or boxes shall be used. Automatic swinging doors are acceptable in one-way traffic areas.

Additionally, permanent entryway systems are to be installed at exterior entries to capture dirt and particulates entering the building at regularly used exterior entrances

*This requirements supports LEEDv4.1 Enhanced Indoor Air Quality Strategies credit.

6.2.0.2 Turnstiles

Turnstiles should not be used except where required for security purposes.

6.2.0.3 Vestibules

Entries and openings into the building should be sealed by the use of vestibules or other means to be reviewed by DFW Code and DFW International Airport Stakeholders.

6.2.0.4 Curtains

Openings for baggage carriers, service openings, etc., shall be provided with curtains to prevent loss of conditioned air. A double row is suggested, with a minimum 5 foot separation.

6.3 Hardware

6.3.0. Door Hardware

Door locks for all DFW Airport-maintained buildings shall have a core approved by ETAM.

1. Manufacturer: Cormax for Lock Cores

6.3.0.1 Lock Sets

Locksets to be installed on doors in permanent buildings or semi-permanent portable buildings, to include Department of Public Safety (DPS). Temporary construction cores approved by DFW Board will be utilized until the project is complete. The installation of the final core will be made by ETAM.

1. Manufacturer: Cormax for Lock Sets

6.4 Public Restrooms in Terminals/FIDS

6.4.0 Public Restrooms in Terminals

Holistic Airside Restroom Program (HARP) Criteria shall be followed on all restroom designs. Follow the most current revisions for HARP provided by DFW Customer Experience (CX).

6.4.0.1 Related Documents

Reference Off-Site Construction, for planning, design, fabrication, inspection, regulatory compliance, and delivery requirements of infrastructure that will be manufactured off-site.

6.4.0.2 Location

Restroom facilities should be evenly and conveniently distributed throughout the concourse areas. A maximum walking distance of 300 feet to a restroom. Each location should include men, women, and unisex (family) toilet rooms. These three rooms should always be grouped together; the entries should be adjacent to each other so that they are visible to the passengers.

6.4.0.3 Accessibility Standards

Comply with accessibility standards in the adopted IBC, the Texas Accessibility Standards, and the Americans with Disabilities Act.

6.4.0.4 Entry

Entries should be designed to prevent line-of-sight from the public areas. Entries to the men's and women's rooms should not have doors, but switch-back or "T" access halls that are wide enough for two people to pass. The "T" access is preferred for rooms with larger number of fixtures because of the improved circulation. A switchback is acceptable for rooms with a smaller number of fixtures. The materials used on the switchback walls shall be durable and able to be cleaned; carpeting is not acceptable.

6.4.0.5 Circulation

The entrance of the restroom is required to be accessible from the main terminal circulation corridor.

6.4.0.6 Fixtures

The fixture counts within each room should be calculated based on the function and occupancy of the area/function the restroom supports in accordance with the IPC and the IBC, as adopted with amendments by DFW.

1. The unisex toilet room will always contain one lavatory counter, one toilet, a diaper changing station, and a bench/shelf for nursing mothers or an attendant to wait. It shall contain an adult changing station that is powered and height adjustable, with end and side clearances. Clear signage shall be placed at the entry identifying adult changing station available.

Table 25 - Minimum Number of Required Plumbing Facilities

Occupancy	Water Closets (Urinals see Section 6.4.0.6.3)		Lavatories	Drinking Fountains (50% hi-low combo per building)	Others
	Male	Female			
Public assembly areas of passenger terminals and transportation facilities	1 per 100	1 per 100	1 per 200	1 per 1,000	1 Service Sink per floor of a building

Minimum Numbers of Required Plumbing Facilities for Terminal Passenger Airside/Landside Restrooms Shall Be Coordinated with The DFW Customer Experience Department.

- All new restrooms shall have fully accessible plumbing chases when using back-to-back fixture assemblies. Wall sill/floor plates shall have minimum 36 inches, measured between the interior plates. Access door to chase shall be metal framed with opening dimensions of 2 feet width by 6 feet 8 inches height minimum.
- Water closets and urinals shall have automatic flush with manual flush button.
- Lavatory faucets shall be automatic with pre-set warm water. Hot water temperature shall not exceed 110 degrees Fahrenheit or in accordance with adopted IECC.

6.4.0.6.1 Lavatories

Sinks shall be monolithic units with fixtures spaced at 42 inches on center to allow adequate counter surface. The solid surface countertops shall be designed to minimize the ponding of water on them.

6.4.0.6.2 Water Closets

Toilets shall have elongated bowls and non-contouring seats. The supports for the toilets shall be of sufficient strength to keep the fixture from rocking.

6.4.0.6.3 Urinals

For urinals, provide extended bowl wall-mounted units.

6.4.0.6.4 Electric Water Coolers

Locate electric water coolers in conjunction with restrooms.

6.4.0.7 Countertops

Countertops shall be solid surface material to match backsplash on all walls.

6.4.0.7.1 Backsplash

The backsplash shall be a minimum of 4 inches high.

6.4.0.8 Accessories

6.4.0.8.1 Hand Dryers

When possible, air type hand dryers in lieu of paper towels should be used. Locate center pull paper towel dispensers on the mirror wall above the trash holes with clear visibility. Provide an additional sidewall unit if this location is not Americans with Disabilities Act (ADA) compliance.

6.4.0.8.2 Toilet Tissue / Feminine Product Receptacles

Provide stainless steel accessories: coreless mini jumbo roll toilet tissue and feminine product receptacles.

6.4.0.8.3 Trash Receptacle

A 6 inch diameter trash hole should be located in the countertop between the sinks, with a trash receptacle located in a lockable under counter cabinet. A semi-recessed wall trash receptacle shall be located at the exits.

6.4.0.8.4 Mirrors

Provide large wall mirrors in front of lavatories. Provide a full-height dressing mirror out of the main circulation path.

6.4.0.8.5 Hooks

Provide hooks adjacent to lavatory for hanging bags, purses, and briefcases. Provide an 8 inch-deep purse shelf at the back of the counter.

6.4.0.8.6 Access Panels

Provide easily operated access panels to conceal pipes and water shut-off valves under each sink.

6.4.0.8.7 Foam Soap Dispenser

Provide touchless foam soap dispenser. The dispenser shall be mounted such that the spout drips into the sink.

6.4.0.8.8 Ground Fault Interruption (GFI) Receptacle

Provide one ground fault interruption (GFI) duplex receptacle at the far end of each of the counters at 10 inches above the counters.

6.4.0.9 Toilet Partitions

Partitions should be a solid surface material. Partitions should be floor mounted. The partitions shall be supported to withstand the specified weight of carry-on luggage and purse/coats being hung on the doors. Accessible toilet partitions shall follow ADA and Texas Accessibility Standard (TAS) requirements.

Partition shall be 3 inches above floor and extend up to 78 inches for privacy. Latches should be easy sliding bars (ADA compliance). Stainless steel grab bars shall be installed within accessible stalls.

6.4.0.9.1 Cubicle Dimensions

The cubicle shall have minimum clear dimensions of 36 inches wide, 84 inches tall from the floor to the ceiling and 72 inches deep.

6.4.0.9.2 Purse/Coat Hooks

The purse/coat hook on partition side panel behind door shall be 60 inches above floor level. The hook shall be heavy duty to be capable of supporting carry-on luggage. The hook shall be mounted closer to the hinge side of the door to reduce the torque on the door.

6.4.0.9.3 Wall Shelf

Provide an 8 inch-deep baggage wall shelf behind toilets and urinal bank. The wall behind the shelf shall be tile (hard surface). The 8 inch shelf to provide enough room in front of the toilet to maneuver the door and carry-on.

6.4.0.10 Urinal Partition

Provide privacy partitions between each urinal shall be mounted to the walls with full height "L" brackets.

6.4.0.11 Flooring Materials

The flooring shall be porcelain tile rated for heavy duty commercial use. Reference the Floor Management Program guide for flooring material/ installation recommendations. Any semi-porous material such as terrazzo shall not be used in restrooms. Tiles should be large; 18 inches by 18 inches minimum to reduce grout lines. Accent tiles may be smaller.

Joints should be tight; (1/8 inch) maximum. Grout shall be a dark color. Square edge tile shall be used to minimize joint expression. Rustic or heavy cushioned edges shall not be used.

Tile shall be the type with multiple colors, veining, mottling, or specks. Solids should not be used. Very light or dark tones should not be used. Colors of tile and floor pattern should be carefully selected to reduce spotting and support a clean appearance.

Place an aluminum transition at the floor and wall transition.

Grout shall be an epoxy grout that shall be sealed.

The base should be a minimum height of 6 inches and shall be the same material and color as the floor.

Provide a floor drain(s) for maintenance of spills in the toilet areas. Floor drains are to be located equidistant from wall to wall and shall not be located in the main walkway. Slope interior floors to floor drain at 0.5 percent grade.

All plumbing chases shall have a waterproofing system installed.

6.4.0.12 Walls

Walls shall be commercial grade non-porous materials (ceramic tile, solid surface, hard stone, etc.) from floor to 84 inches minimum. Above 84 inches, paint is acceptable.

Color, patterns, and finish of wall tile should maximize a clean-looking, well-lighted appearance. An aluminum or tile transition shall be considered, where the tile stops, if the tile is not floor to ceiling.

All restrooms shall be finished with the same materials, colors, and patterns. If a distinction between men and women is provided, it should be in the transition area.

6.4.0.13 Entries

The entries from the terminal should reinforce “way finding” by expressing a “restroom” cue. Materials should be coordinated with the terminal finishes and shall not be the same as the restroom materials.

Inside the “T” access corridor shall be a transition area between the terminal and the toilet room. These transition walls shall be durable, cleanable materials that can tolerate rolling bags and provide acoustical treatment. In this transition area at eye level, the finishes should reinforce the overhead signage package to confirm a passenger is headed into the correct room.

6.4.0.14 Ceilings

Ceilings must be accessible board type with appropriate regular edge, 2 by 2 acoustical lay-in drop tile features.

6.4.0.14.1 Trim or Molding

A “trim” or molding feature above niches, such as crown molding or wall molding element.

6.4.0.15 Lighting

Rooms should be well lit, especially at the lavatories. Refer to Illuminating Engineering Society of North America (IESNA) Lighting Handbook (most recent edition) and ANSI Standard 90.1 (most recent edition) for recommended lighting levels and allowable lighting power density. Spacing of lighting fixtures at 6 foot intervals. Provide linear lighting above the water closets. Provide 120 Volt electrical service outlet (GFI) adjacent to light switch for servicing of equipment. The lighting should be linear with 30 foot candle lighting in stalls and 50 foot lighting in walkways. All lighting fixtures and related components shall be UL approved.

1. Use light-emitting diode (LED) lamps throughout the restrooms, in concealed lighting coves, down lights and accent fixtures. The light level should be even throughout the room to reinforce the clean/ airy appearance.
2. Emergency lighting shall be provided.

6.4.0.16 Other Amenities

Provide a built-in stainless steel diaper changing station, 42 inches long by 24 inches deep in every men’s, women’s, and family rooms. Install it in a remote, recessed but visible location. Provide a trash receptacle with lid and a paper towel dispenser at this location.

1. Provide potable water spigot, using ball valve with service hose attachment. Locate between 12 to 24 inches above finished floor (AFF). Locate cleanout for battery of fixtures at door entrance and slope drain piping to clean this out.
2. Provide music system speakers in restrooms.
3. Toilet room exhaust and ventilation rate shall meet or exceed ASHRAE Standard 62.1-2013 (70 cfm (35 L/s) per water closet/urinal.
4. Linear slot diffusers are required.

6.4.1 FIDS (Flight Information Display System)

All FIDS shall be centrally located and placed at eye-level.

6.4.2 Information Kiosks

1. Signage
 - a. All Kiosks shall have signage depicting all disability services at DFW – Hearing, Mobility, Cognitive, Visual, Self-Care.
2. Audio
 - a. All Kiosks shall incorporate audio features for visually impaired customers incorporating more languages.

6.5 Flagpoles and Trash Containers

6.5.0 Flagpoles

Flagpoles shall be cone tapered aluminum, with all standard fittings. Flagpoles shall be adequately supported and provided with lightning protection.

6.5.1 Trash Handling Devices & Containers

Space shall be provided for trash handling devices and containers depending on the size, location, and type of trash that is to be disposed of. Design plans shall indicate the proposed method(s) for trash disposal. All equipment used for handling, storage, or compaction of trash that may be in public view, shall be screened. Equipment shall be finished in a color to match other painted building equipment.

6.5.1.1 Dumpster-type Containers

1. Shall be oriented for ease of approach by truck.
2. All trash containers shall be covered or otherwise enclosed to prevent access by wildlife and high winds.
3. No open topped trash containers are permitted within the Air Operations Area, unless thoroughly protected from jet blast and high winds.

6.5.1.1.1 Volatile Organic Compounds (VOCs) limits

Specify and install paints, coatings, adhesives, sealants, flooring, walls, ceilings, insulation, and composite wood products that meet low-emitting criteria. Products must adhere to Volatile Organic Compounds (VOCs) limits for products, which must be tested to the CDPH Standard Method v1.2 (2017) and modeled using the private office scenario.

VOCs and their acceptable limits are found in CDPH Standard Method v1.2 (2017) in table 4-1. Some products are certified to the CDPH Standard, a list of which are included here: List of Certifications and Programs that Use CDPH SM v1.2 (2017).

*This requirement supports LEEDv4/v5 Low-Emitting Materials.

6.5.2 Exterior Metal Benches

Exterior metal benches shall be bolted to the ground.

6.6 Finishes

6.6.0 Interior Finishes

6.6.0.1 Guidelines

Guidelines for building finishes are also specified elsewhere. No custom finishes shall be used to ensure future availability of materials/colors unless approved by appropriate stakeholders.

6.6.0.2 Replacement Materials

Use materials that allow for future replacement.

6.6.0.3 Paint

No flat sheens of paint are allowed. Corner guards shall be installed for any painted surfaces in high traffic areas and shall be floor to ceiling preferably or at a minimum of 72 inches high.

6.6.0.3.1 Cove Base

Where cove base is used, a dark color (DFW gray or black) shall be used unless approved by appropriate stakeholders.

6.6.0.4 Terrazzo

Terrazzo flooring shall be used in all main high traffic walking areas within the terminals. The terrazzo shall be polished to a 3000 grit and shall not be wax- coated.

6.6.0.5 Carpet

Where carpet is installed, carpet tile shall be used instead of rolled carpeting so that areas can be replaced as necessary. Prefer dark/patterned/textured carpet that will more effectively hide stains.

6.6.0.6 Tile

Where tile is installed, prefer that dark colored grout be used.

6.6.0.7 Wall Covering

Where a wall covering is installed, use solid surface as opposed to laminates unless approved by DFW Code.

6.6.0.8 Door Finishes

Doors in high traffic areas shall have a stainless-steel kick plate and shall be powder coated, not painted.

6.6.0.9 Protective Covering

Halls in high traffic areas shall have protective covering installed from the floor to wainscot height.

6.6.0.10 Interior Room Signage

All rooms will be labeled with a room number sign. Please see the following guidance for the numbering scheme:

Room Numbering Scheme

First Element – Terminal

Second Element – Nearest Gate

Third Element – Area Designator

- A = Airside
- L = Landside
- AS = Airside Stairs
- LS = Landside Stairs
- AEV = Airside Elevator
- LEV = Landside Elevator
- AES = Airside Escalator
- LES = Landside Escalator
- AFD = Airside Fire Door
- LFD = Landside Fire Door
- LL = Entry Vestibule Lower Level
- UL = Entry Vestibule Upper Level

Fourth Element – Level

Fifth Element – Room Number Designator

Sixth Element (If required) – Suffix for Rooms within Other Rooms

Example:

<u>Element</u>	<u>A</u> 1 st	<u>29</u> 2 nd	<u>AS</u> 3 rd	<u>2</u> 4 th	<u>04</u> 5 th	<u>A</u> 6 th	= Room # A29AS204A
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Room numbers shall increment by 4 to allow for future additions. Gate numbers shall pass across all three levels.

All rooms on the airside of corridors shall be even numbers.

All rooms on the landside of corridors shall be odd numbers.

Elevators will include column / grid in number (example: A22LEV78A or A22AEV78C).

Where required signs shall follow ADA and TAS requirements.

Chapter 7 Fire Protection Systems

7.0 Fire Alarm

Fire Alarm System for buildings occupied or maintained by the DFW Airport Board.

7.0.1 General Requirements

Fire alarm system (FAS) shall be in accordance with adopted NFPA 72, adopted Fire Code and shall report to DFW Central Fire Alarm System (CFAS), or an approved Fire Alarm Monitoring Station, as directed by the Fire Marshal's Office. Activation of any device shall:

- Audibly and visibly annunciate the alarm condition at the local FAS.
- Report to the DFW Airport TAC Comm.
- Provide and notify the occupants in the event of an alarm.

7.1 Fire Detection and Alert Notification System

1. The FAS shall function as an integral component of the DFW CFAS. The panel shall be listed as UL 864 UOJZ and UUKL.
 - a. Systems to be approved are Honeywell XLS3000 or an AHJ-approved system.
2. The FAS shall supervise each individual device on an intelligent loop circuit such that alarm trouble and supervisory conditions are individually annunciated. All panels should be installed in a uniform workmanship manner.
3. A dedicated 120-Volt AC circuit will supply power to the fire alarm panel and its subpanels. Inside the Fire Detection and Alert Notification System, a label must be affixed to indicate the panel designation, the breaker number for the 120-Volt AC power source, and a lockable circuit breaker.
4. The panel shall contain batteries to provide standby emergency power to maintain the local fire alarm system's functionality upon losing primary power. The batteries shall be able to operate the system under standby conditions for 24 hours plus 15 minutes under alarm conditions. Transfer from normal to battery power shall be automatic. When a transfer occurs, the panel shall report a trouble alarm to the CFAS. The panel shall provide a float/equalizing charge for the batteries.
5. The panel shall provide ground fault detection for the panel and device initiating circuits and shall report ground faults to the CFAS.
6. Fire Alarm System Panel shall not be used for junction boxes or pull boxes. Splicing shall not be permitted.
7. When terminating stranded wiring in the panel for all initiating and indicating devices, "STA-KON" type lugs shall be used for the proper lug size and wire size. For devices that are not designed to be used with spade terminals, use manufacturer's suggested wiring termination technique.
8. The Fire Alarm System shall be installed in a location as approved by the AHJ.
9. The Fire Alarm System shall communicate with the DFW CFAS via fiber optic cable.

10. Space within the panel must be allocated for equipment necessary to accommodate a 25 percent expansion of zones. Fire Alarm Control Units must comply with NFPA 72 for activating Suppression Systems. The following shall be met:
 - a. Listed releasing devices, such as solenoids or relays, must be monitored for integrity (supervised).
 - b. A physical disconnect switch must be installed to interrupt the output circuit for each releasing device (e.g., solenoid, relay) and to initiate a supervisory signal.
Additionally, switches should be provided for Pre-action Sprinkler Systems.

7.2 System Devices

1. Addressable devices will receive power and communicate over the same pair of wires. SLC circuits must be Class A or X. Isolator Modules (X) must be used when the SLC serves different fire areas.
2. Additional devices shall be added to the intelligent loop circuit in such a way that the loop is continuous, with No T-Tapping (parallel tap into the loop) from any point in the circuit.
3. Each device shall contain screw terminals on rising plates for positive termination of wire up to 12 AWG.
4. Duct Smoke detectors and area smoke detectors shall be provided with remote test and reset panels when the detector is not readily accessible from floor level.
5. No intelligent devices are allowed in confined spaces. Should a device be needed in a confined space, a remote monitor module shall be provided in an approved location by AHJ.
6. All intelligent device covers shall be labeled with device address and panel loop number. The type of module shall be identified as either a control module or monitor module.
7. All intelligent sensors shall mount on a common base to facilitate a change of sensor type if building conditions change. The base shall be incompatible with conventional detectors to prevent the mounting of nonintelligent devices.
8. Each sensor shall contain an LED, which blinks each time the device is scanned by the Fire Alarm System. If the device is in alarm, the LED shall remain on to indicate the alarm condition.
9. All devices and associated circuits shall be coordinated with DFW Board.

7.3 Monitor Modules

1. The intelligent monitor module shall provide an addressable input for normally open or normally closed contact devices.
2. The intelligent monitor module shall provide a supervised initiating circuit, able to connect to either Class B supervised or Class A circuits.
3. The intelligent monitor module shall contain an LED, which shall blink upon Fire Detection and Alert Notification System scan. The LED shall latch on upon determination of an alarm or a Supervisory alarm condition
4. The intelligent monitor module shall mount in a standard 4 inch by 4 inch-deep electrical box.

7.4 Control Modules

1. The intelligent control module shall provide an addressable output for both the output circuit and the control relay, which can be connected to either Class B supervised or Class A circuits. An open circuit fault shall be announced on the fire alarm system.
2. The control relay module shall provide a controlled relay. The relay shall have a Single Pole-Double Throw (SPDT) form "C" contact, rated at 2 amps at 28-Volt DC.
3. The intelligent control module shall have an LED, which shall blink on Fire Alarm System poll. Upon activation of the module, the LED shall be latched on.
4. The intelligent control module shall be mounted in a standard 4 inch by 4 inch-deep electrical box.

7.5 Initiating Devices

1. Ionization Duct Smoke Detectors shall be intelligent and have two SPDT alarm contacts and one SPDT trouble contact rated 2 amps at 120-Volt AC and be listed for applications involving air handling systems. Detectors shall be installed in accordance with its listing. The initiating circuit shall be either Class A or B.
2. Area Smoke Detectors shall be intelligent photoelectric type and be installed in accordance with their listing and NFPA 72. The initiating circuit shall be either Class A or B.
 - a. Air Sampling-Type Smoke Detector, shall be installed in accordance with their listing and NFPA 72. The maximum transport time shall be 120 seconds.
3. Manual Pull stations shall be intelligent, of die cast metal construction and designed for semi-flush mounting. The initiating circuit shall be either Class A or B.
 - a. Each non-intelligent Manual Pull station shall be connected to an intelligent monitor module.
4. Wet sprinkler systems shall have vane type sprinkler flow switches. Flow switches shall have retards adjustable up to 2 minutes. The device shall have an SPDT, which shall close upon water flow. The initiating circuit shall be Class A and be connected to an intelligent monitor module.
5. Dry sprinkler system alarm switches shall be pressure activated and be connected to an intelligent monitor module.
6. Supervisory switches for fire protection systems shall be installed in accordance with their listing and be connected to an intelligent monitor module. Lanyard type supervisory switches are not permitted except as approved by DFW AHJ.
7. Dry system air pressure switches shall be installed to monitor both high and low air pressure conditions and be connected to an intelligent monitor module.
8. Supervisory devices may be connected to a single intelligent monitor module when located in the same room. Supervisory circuit shall be two-wire supervised circuits (Class B), with the appropriate end of line resistor installed at the last device on the loop.

7.6 Wiring

1. Wiring shall comply with the National Electric Code (NFPA 70), National Fire Alarm and Signaling Code (NFPA 72), NECA 1, project specifications, and the approved wiring diagram. If there are conflicts, the most stringent method shall be followed.
2. Fire Alarm conduit shall only include wiring for fire alarm system.

3. Prior to any connections to the FAS, wiring shall be tested for stray voltage, short circuits, and ground faults in its entirety. Stranded wires shall terminate at both the device and module with spade terminals sized to fit both the wire and screw terminal. Spade terminals are optional if the terminal is fitted with a wire pressure plate.
4. Color-coding of device indicating loops shall be as follows: Two conductors are of one color and the other two conductors are of a different color. Colors shall be continuous throughout the entire loop. Where more than one initiating loop is routed in a single conduit, the colors associated with any loop contained in the conduit shall be different from the colors of any other initiating loop contained in the conduit. Except for the SLC circuit which shall be labeled as needed.
5. All loop wiring shall be identified by their "in" and "out" circuits and loops. In is defined as coming from the panel. Labels for SLC circuits shall show panel number, loop number, and device number on the inside of the circuit. Indicating circuits labeling shall show power supply number and circuit number on the inside of the circuit. Labels shall be Self-Laminating of permanent type (wrap around) wire numbers are not acceptable.
6. Red and Black must be used for a 24-Volt panel power.
7. No voltage supply from any other source than the primary power 120-Volt AC and the panel 24-Volt DC power supply shall be utilized.
8. Intelligent loop circuits shall be labeled at all junction locations by the panel number and loop number.
9. Intelligent loop circuits shall be provided with adequate junction boxes to be expandable and provide a means for connecting to the loop in the junction box.
10. Control and other panels shall be mounted with sufficient clearance for observation and testing. Fire alarm junction boxes shall be clearly marked for distinct identification.
11. Wiring shall be in EMT conduit, minimum 3/4 inch. Flexible conduits, mounting boxes, junction boxes, and panels shall be securely fastened with appropriate fittings to ensure positive grounding throughout the entire system. Conduits shall enter the panels from the sides or bottom. Where flexible conduits are used to connect device loop wiring to alarm device, the Contractor is permitted to use 1/2 inch flexible conduit.
12. All fire alarm junction boxes shall be mounted in approved locations for ease of maintenance from floor level.
13. Backbone termination boxes should be of sufficient size to allow for termination.
14. All junction boxes shall be installed in a uniform workmanship manner.

7.7 Conduit / Raceway

1. Circuit shall meet Pathway survivability Level 1 per NFPA 72.
2. The following raceway types shall be permitted:
 - a. EMT conduit (3/4 inch minimum).
 - b. RIGID conduit (3/4 inch minimum).
 - c. Non-Metallic conduit for wet locations (3/4 inch minimum).
 - d. Metal Clad (MC) cable is permitted in concealed spaces for horizontal fire detection and alert notification branch circuits and connections to devices and fixtures.

3. All raceway types shall be new. Installing used raceway is unacceptable.
4. Using existing raceway is unacceptable without prior written permission from DFW ITS Life Safety or AHJ.
5. Boxes, supports, and other accessories for the raceway installation shall be listed for the intended application.
6. All wiring shall be installed in conduit.
7. Fire detection and alert notification system wire shall be installed in conduit or approved raceway, parallel to existing building structure when possible.
8. Riser wiring and wiring between floors shall be installed in conduit.
9. Cables and wires shall be strapped or bundled inside equipment enclosures and cabinets, parallel to the enclosure sides.
10. All EMT conduit fitting shall be steel or malleable steel compression type. All rigid conduit fittings shall be threaded with plastic insert or bushing.
11. Flexible conduit and associated junction boxes connecting sprinkler water flow and supervisory switches shall be water resistant.
12. Fire alarm conduit and junction boxes shall be RED in color. Flexible conduit between fire alarm junction boxes and device mounting boxes that are less than 6 feet in length are not required to be RED. Device mounting boxes are not required to be RED.
13. Flexible conduit to devices shall be supported between the junction box and device box.
14. Intelligent sensors shall be mounted in the ceiling of the protected area and no closer than 4 feet from any air conditioning register. Contractor shall define actual device locations in accordance with the manufacturer recommendations and NFPA approved methods. Where manual stations are installed on the building exterior, the associated monitor module shall be mounted in a transparent enclosure to maintain the environmental limitations of the module. The manual pull station shall be mounted in an approved enclosure in the immediate vicinity of the module.

7.8 Qualifications

1. Contractor Qualifications: The system contractor shall have demonstrated experience and expertise in the design, installation, and commissioning of Fire Alarm systems in similar environments and complexity to that of the DFW International Airport. The Contractor shall meet the requirements of the State of Texas Fire Alarm Rules.
2. Personnel Qualifications: All personnel involved in the design, installation, testing and commissioning of the system shall provide documentation of their qualification by one or more of the following:
 - a. Registration, licensing, or certification by a state or local authority.
 - b. Certification by an organization acceptable to the AHJ.
 - c. Manufacturer's certification for the specific type and brand of system provided.
3. Equipment Qualifications: All equipment used in the system must be listed for its intended use.

4. System Qualifications: The system must be fully compliant with the latest edition of NFPA 72 and other relevant codes and standards.

7.9 Documentation, Shop Drawings, Acceptance Documentation and Acceptance Testing for All Systems

7.9.1 Shop Drawings

Submit shop drawings and product literature for approval to DFW DPS Fire Prevention for review and approval before work can begin.

1. Shop drawings shall strictly follow the adopted NFPA 72, and all documentation shall meet these requirements.
2. Shop drawings shall be on site upon request for inspection.
3. Documentation must be completed and submitted before system acceptance, in strict accordance with the adopted NFPA 72 documentation criteria.

7.9.2 As Built

1. Upon completion, provide As-Built's drawings of the installation. One electronic copy to Records Management. One electronic copy shall be furnished to DFW Facilities Maintenance Fire Alarm Section for inclusion in Facilities Maintenance records.
2. All final as-built's shall show conduit routing, junction boxes as well as termination boxes. Wire flow and end of line (EOL) resistors shall be clearly marked at device and on the print.

7.9.3 Maintenance and Operations Manuals

1. The intent of this Section is to require complete documentation of the FAS for the purpose of system O&M during and after the warranty period. It is intended that the O&M manuals be exhaustive in the coverage of the system to the extent that they may be used as the sole guide to the trouble shooting, identification, and repair of defective parts.
2. The Contractor shall provide the DFW DPS Fire Prevention and Airport Board with three complete electronic drawings and O&M manuals on the completed system. These manuals shall include basic wiring diagrams, schematics, and functional details. Such that any component, wire, or piece of equipment in the system may be easily identified by going to the actual equipment and referring to this manual. It is required that everything in the system be neatly labeled and easily identifiable. Every terminal, wire, component, or piece of equipment, relay, and other such items shall have a number or letter designation. All these identification characteristics shall be included in the O&M manuals. The maintenance manual requirement of this Section is in addition to Shop Drawing requirements. Maintenance manuals and drawing sets shall be compiled after system fabrication and testing and shall incorporate any changes made after Shop Drawing submittal. The maintenance manuals and drawing books shall be permanently bound in hard plastic covers.
3. Provide Maintenance Manuals and manufacturer's standard literature, covering all equipment included in the system. The maintenance manuals shall contain specifications, adjustment procedures, circuit schematics, component location diagrams, and replacement parts identification. All references for equipment not supplied on this project shall be crossed out.

4. The drawing book documents shall be produced with the latest version in use at DFW. The electronic files shall be provided to the Airport at the completion of the project. Provide component identification and cross reference on the Drawings to allow the maintenance department to understand the function of each item (the block diagram), find the room where the device is mounted (Contract Document plans), find its location in a rack (elevation drawings), find how it is wired (wiring diagrams), its detailed Specifications (vendor data sheets), and how to repair it (spare part lists).

Include the following drawings as a minimum:

1. Functional Block Diagram: Provide overall block diagrams showing the major interconnections between subsystems.
2. Elevation Drawings: Provide Drawings showing the physical arrangement of all major system components.

This shall include:

1. Elevation drawings of all equipment racks showing the location of each component in the racks. Components in the racks shall be identified as in the functional block diagrams.
2. Wiring diagrams showing all field installed interconnecting wiring.
3. Wire identification on the diagrams shall agree with the wire markers installed on the equipment.

7.9.4 Acceptance Test

1. Following completion of the wiring and prior to termination of devices, an installation inspection is required.
2. Upon completion, the Contractor, in the presence of DFW Fire Alarm Contractor, DFW Fire Prevention representative, and appropriate DFW Development representatives, shall conduct such tests and inspections necessary to verify that the installation is complete and fully operational to the given intent and has been installed in accordance with the Specifications and approved Drawings.
3. All intelligent device covers shall be labeled with device address and panel loop number. The type of module shall be identified as either a control module or monitor module.
4. All equipment and materials necessary to conduct these tests shall be furnished wholly by the Contractor.

The following device keys shall be provided by the Contractor upon completion and acceptance:

1. One key for each duct smoke detector installed
2. One key or tool for each manual pull station installed
3. One key or tool for each water flow device installed
4. One key for each supervisory device installed
5. One key or tool for each panel installed.

Following completion of the installation and progressively during installation, the contractor shall remove all trash, debris, and surplus material always occasioned by this operation so that the environment presents a safe, neat, and orderly condition conducive to other activities.

7.9.5 Warranty

1. The system contractor shall provide a warranty covering the FAS System and its components, with a minimum duration of one year from the date of final acceptance of the system by the Owner Representative.

7.10 Public Address and Voice Evacuation (PAVE)

7.10.1 Purpose

The purpose of this manual is to provide the design criteria and requirements for the Public Address and Voice Evacuation (PAVE) System at DFW International Airport. The PAVE System is critical for communication during emergency situations and shall meet the highest standards of reliability, functionality, and ease of use.

7.10.2 Scope

This manual covers the design and installation of the PAVE System at DFW International Airport, including, but not limited to, all related components such as speakers, amplifiers, and control equipment.

7.11 General Requirements

7.11.1 System Performance

The PAVE system shall meet or exceed the requirements specified in NFPA 72 National Fire Alarm and Signaling Code, as well as any local codes and regulations. All PAVE System requirements shall be coordinated with the DFW Board.

7.11.2 System Architecture

The PAVE system shall be fully integrated with the Campus-Wide PAVE System to allow Emergency Notification, Mass Notification, and general paging to and from the designated PAVE Systems.

7.11.3 System Integration

The public address and voice evacuation system shall be integrated with the fire alarm system.

1. Evacuation initiated by the Fire Alarm System shall send an Evacuation signal to the PAVE System to initiate an Evacuation Message.
2. Evacuation initiated by the PAVE System shall send an Evacuation signal to the Fire Alarm System to initiate strobes.
3. Evacuation initiated by the Fire Alarm System requiring a Mute of the PAVE System shall send a Mute signal to the PAVE System to mute specified areas.
4. Mass Notification initiated by the PAVE System shall send a Mass Notification signal to the Fire Alarm System to initiate strobes.
5. Failed or abnormal performance of any active system component shall generate a supervisory signal at the Primary ACS and send a Trouble signal to the Fire Alarm System.

7.11.4 System Components

The system shall include the following minimum components:

1. Primary Announcement Control System (ACS) with redundant and backup power supply
2. Lifeline ACS with redundant and backup power supply
3. Digital Mainframe Power Amplifiers with backup power supply
4. Microphone Paging Stations with redundant data cabling
5. Loudspeakers and wiring listed and rated for intended use
6. Loudspeakers circuits with standby backup amplifier switching
7. Ambient Noise Sensors
8. Ambient Noise Collectors
9. Rackmount Microphone Paging Station in Communication Rooms
10. Enterprise Server and Enterprise System Software
11. Flight Announcement System where required
12. Text-to-Speech Courtesy Announcement System (T-CAS) where required
13. Other work and accessories required for a complete and operational system

In the event of a failed main amplifier card, a backup amplifier card shall automatically activate to maintain system integrity. Amplifier outputs shall be protected so they can survive a shorted output line while reporting an abnormal condition to the system ACS.

7.11.5 System Programming

System Programming: All hardware and software requirements for PAVE System functionality shall be coordinated with the DFW Board. This includes, but is not limited to, network connectivity, paging priorities, digital message assembly, system access, microphone paging, and paging station button functionality and screens.

7.11.5.1 Priority Levels

All announcements and messages shall be assigned a priority class to ensure that the most critical communication is delivered. Priority is assigned such that emergency paging function immediately cancels all other audio announcements or messages in the affected zones. Local paging functions have a higher priority than background music, and recorded messages in the local paging zone. Recorded messages override background music in all zones.

7.11.5.2 System Logging

The system shall provide complete logging/archiving and exporting of User Activity, Announcements, Security, and Faults.

7.11.5.3 Microphone Paging Stations

1. A passcode input shall be required to unlock a microphone station. The microphone station shall be automatically locked after a specified time, configurable by the system administrator.
2. Announcements shall be able to be distributed live if the destination zone(s) is not in use. If not available, the page shall be able to be recorded and played back in queue when the destination zone(s) is available. The microphone's display shall indicate busy when input resources are not available.

3. Microphone Paging Stations shall be capable of paging to a zone or zone groups based on the user's privileges and the microphone station configuration.
4. A passcode shall be established for Emergency Responders that allows authorized users to make announcements during emergency events from any microphone station.

7.12 Design Requirements

7.12.1 Software

System control software shall be end-user friendly and be able to be used without detailed knowledge of the native programming language. All operator input shall be checked to ensure that any input error will not enable or disable any system functions. The system software shall provide multiple levels of password protection. Permissions shall be assigned by the system administrator.

7.12.2 Coverage

The public address and voice evacuation system shall provide clear, intelligible voice announcements and alarms to all areas of the building.

7.12.3 Speaker Placement

The speakers shall be placed in a manner that ensures even coverage and minimal dead spots. The speakers shall be mounted in a manner that protects them from damage and vandalism.

7.12.4 Speaker Zoning

1. Speaker zones shall be contiguous and sectioned logically in regard to room/area function.
2. Each zone shall be comprised of a single speaker type.
3. Speaker zones shall be coordinated with the Fire Alarm System such that fire alarm zones are not split among more than one speaker zone.
4. DFW DPS Fire Prevention and ITS Life Safety Departments shall approve any changes to speaker zones.

7.12.5 Ambient Noise Sensors (ANS)

1. Ambient noise sensors are required in areas that have varying levels of ambient noise.
2. Ambient noise sensors shall be installed in a manner that ensures accurate readings of the ambient noise levels in the associated speaker zone.
3. The readings shall be used to automatically adjust the volume of the speakers as needed to ensure clear and intelligible voice announcements.

7.12.6 Monitoring

The system shall also include automated monitoring and testing capability to ensure proper operation.

1. Monitor Test System Operation. The PAVE System shall include self-diagnostics that operate in real time under software control. This self-testing shall include testing of audio operation, power supplies, power amplifiers, speaker circuits, and network communications.
2. Ground Faults: Each speaker line shall be tested for a ground fault. All ground faults shall generate a supervisory signal at the Primary ACS.

7.12.7 Maintenance

The system shall be designed for easy maintenance and repair, with readily accessible components and clear documentation.

7.13 Installation Requirements

7.13.1 Code compliance

The installation of the PAVE system shall be in full compliance with all relevant codes and standards, including NFPA 72, National Electric Code (NEC), and International Building Code (IBC). Installation shall be performed by licensed, factory trained and certified technicians skilled in the installation of the type, brand, and style of equipment to be installed.

7.13.2 Coordination

The contractor shall coordinate in advance with the DFW DPS TAC Comm, DFW Integrated Operations Center (972-973-3112), and DFW ITS Life Safety Department before performing any work affecting the existing PAVE System. An impairment request shall be submitted for any abnormal condition, during either a planned or emergency event, where a system, component, or function is inoperable. Operating, programming, modifying, or impairing the PAVE system without approval of DFW Airport is strictly prohibited.

7.13.3 Conduit / Raceway

1. The following raceway types shall be permitted:
 - a. EMT Conduit (1 inch minimum)
 - b. RIGID Conduit (1 inch minimum)
 - c. Non-Metallic conduit for wet locations (1 inch minimum)
 - d. MC cable is permitted in concealed spaces for horizontal fire detection and alert notification branch circuits and corrections to devices and fixtures.
2. All raceway types shall be new. New conduit and raceways shall be contractor provided. No existing conduit, paths, raceways, or supports shall be repurposed. Conduits shall be attached to boxes using lock washers. Cable protection shall be in the form of a plastic bushing installed on the end of the conduit.
3. Conduit shall be ran concealed in all finished spaces.
4. PAVE conduit shall be RED. Couplers, box covers, and connectors shall be painted white.

5. Conduit between boxes shall not go over (1) 90° bend and (1) changeover. (2) changeovers shall be considered the same as 90° bend. If a pathway requires that the bends exceed the above mentioned constraints, additional box(es) shall be installed.
6. Conduits shall include expansion fittings wherever conduits cross building expansion joints, for every 300' of run, and where indicated.
7. Flexible metallic conduit (FMC), maximum 6 feet in length, shall be permitted for device drops. FMC and the device shall have a separate support system. FMC shall be supported at the midpoint. The building structure shall be used for all supports. PAVE conduit shall not be used as a support system.
8. Weatherproof NEMA4 boxes with liquid-tight flexible conduit shall be used in exterior PAVE installations.
9. Pull string shall be provided in conduits.
10. Where extension of conduits occurs through fire rated partition, the rated partition shall be fire-stopped to maintain the original rating in accordance with the specifications and as required by referenced codes and standards.
11. A four-square box shall be provided for every ANS and be located in such a manner that the device will be able to be mounted directly to it.
12. Conduit pathways shall be inspected by DFW Code Enforcement after installation. Conduit shall be approved by DFW Code Enforcement before installing any cabling.

7.13.4 Cable Management

All PAVE cabling shall be installed in conduit.

1. All PAVE cabling and wiring shall be neatly organized and properly labeled, with 12 inches of excess cable provided at each termination point and at the PAVE System equipment cabinets.
2. All cabling shall be ran without breaks, splices, or terminations unless terminating into a device, End on Line terminator, or terminal block in a Zone Tie Box located in a communications room or in the field.
3. Input cabling shall be installed in a separate conduit from output cabling to ensure proper signal transmission and to minimize electromagnetic interference.
5. Install all cabling no closer than 12 inches from any horizontal/backbone cabling, power system cable/raceway, and fluorescent/ballasted light fixtures.
6. Cabling shall not be installed with a bend radius less than that specified by the cable manufacturer.
7. Speaker and ANS cabling shall not terminate directly on amplifiers or ANS collectors without first terminating on terminal strips in the zone tie box. Zone tie boxes shall be in DFW communication rooms. Cable runs shall be wired in a single run, without paralleled branches (No T-Tapping).

8. Cabling with different voltage shall not share conduit paths.
9. Cabling shall be inspected by DFW Code Enforcement after installation. Cabling shall be approved by DFW Code Enforcement before installing any devices.

7.13.5 Speakers

1. Speakers shall have a full housing that encloses the interior of the speaker and conceals all wiring.
2. Speakers shall be installed with an independent primary and secondary support system. The building structure shall be utilized for all support systems.
3. Speakers located in a hard ceiling shall have an access panel for the purpose of accessing the junction box above the drywall.
4. Speakers shall be inspected by DFW Code Enforcement after installation. Speakers shall be approved by DFW Code Enforcement before final acceptance.

7.13.6 Microphone Paging Stations

1. Microphone paging stations cabling shall be secured with a locking RJ-45 connector.
2. A redundant cable shall be installed and be left available for use by the microphone paging station.
3. Microphone paging stations shall be secured in place or to mounting equipment with at least (1) T-10 security screw.
4. Microphone paging stations shall be installed and mounted in accordance with manufacturer's best practices and recommendations.
5. Microphone paging stations shall be inspected by DFW Code Enforcement after installation. Microphone paging stations shall be approved by DFW Code Enforcement before final acceptance.

7.13.7 Labeling

1. All PAVE system devices, including speakers, amplifiers, control equipment, and conduit runs, shall be clearly labeled, and identified according to industry standards and best practices. Labels shall be easily visible and located in a convenient location for maintenance and troubleshooting purposes. The labeling scheme shall be consistent throughout the entire PAVE system and shall include the device type, location, and any relevant information such as wiring diagram references or device identification numbers. The labeling shall be completed before final acceptance of the PAVE system.
2. Conduit shall have a "PA/VE" label at every box, maximum of one foot on both sides of a penetration, and every 20 feet of a long run of conduit.
3. Conduit labeling shall be one inch tall. Label lettering shall be red with a white background.
4. Junction boxes, 4-11, and Zone Tie Box, 12x12, shall be labeled.

5. The PAVE zone name shall be included in the label on junction boxes and zone tie box.
6. Cabling terminations shall be labeled in the zone tie boxes with the type of wire, zone name, and zone description.
7. A label shall be placed on the visible face of every speaker that states “Life Safety Device – DO NOT TOUCH”
8. A label shall be placed on the visible face of every speaker with the associated zone name.

7.13.8 Capacity

At each equipment location, provide 25 percent spare system capacity including, but not limited to all amplifiers, amplifier card slots, inputs, outputs, terminal strip positions, etc.

7.13.9 Accessibility Requirements

All PAVE components shall be installed in a manner that allows for easy accessibility for maintenance and servicing. This includes, but is not limited to:

1. Ensuring that all components are mounted at a height and location that allows for easy access and visibility.
2. Providing adequate clearance and space around all components for maintenance and servicing.
3. All conduit and pathways shall have a maximum height of 5 feet above the ceiling, or 5 feet above the lowest fixture if there is no ceiling.
4. Access panels shall be installed as required to ensure easy access to all components and wiring.
5. All conduit and pathways shall be installed in a neat and organized manner as well as be labeled clearly to ensure easy identification.
6. The installation shall comply with all applicable codes and regulations, including fire safety and electrical safety codes.

7.14 Modification Requirements

1. Any modifications made to the system during construction or retrofitting shall be approved by the Life Safety System Specialist to ensure that they meet the design requirements and do not compromise the safety of the occupants.
2. If modifications are made to the building structure, such as changes to walls or ceilings, the impact on the speaker coverage and intelligibility of the system shall be assessed and addressed as needed.
3. If new equipment or devices are added to the system, they shall be compatible with the existing hardware and software and not cause interference or degradation of system performance.
4. Any modifications made to the system software or configuration shall be documented and tested to ensure that they do not introduce new vulnerabilities or affect the functionality of the system.
5. The zoning and placement of speakers shall be reviewed and modified as needed to maintain even coverage and minimize dead spots, considering any changes in the building layout or usage.

6. Any modifications made to the ambient noise sensors shall be tested to ensure that they accurately measure the ambient noise levels in the associated speaker zone and adjust the volume of the speakers as needed to ensure clear and intelligible voice announcements.
7. Any changes made to the system during modifications shall be documented and communicated to the relevant personnel, such as the system administrator or maintenance team, to ensure proper O&M of the system.
8. Existing input and output conduit and pathways feeding an area within a zone may remain in the design, provided they meet the required specifications. Existing conduit and pathways to remain shall be approved by DFW ITS Life Safety and shall not compromise the overall integrity and functionality of the PAVE system.

7.15 Testing and Commissioning

7.15.1 Testing

The PAVE system shall undergo comprehensive testing to ensure the proper functioning of all components, features, and functions. This testing shall include full-scale voice evacuation and mass notification events, as well as other functions of the system. The testing shall be performed in accordance with NFPA 72 and manufacturer instructions. Ongoing periodic testing and maintenance shall also be conducted to ensure the continued reliability and functionality of the system. All testing shall be witnessed by a DFW ITS representative and a DFW Fire Prevention representative.

7.15.1.1 Intelligibility Test

Perform intelligibility tests in compliance with NFPA 72 and this manual. Intelligibility shall meet the minimum of 0.60 STI (speech transmission index).

7.15.1.2 Acoustic Coverage Test

Feed pink noise into the system using octaves centered at 4,000 and 500 Hz. Use a sound level meter with octave band filters to measure the level at approximately 40 foot spacing intervals in each zone. For spaces with seated audiences, the maximum permissible variation in level is ± 2 dB and the levels between locations in the same zone and between locations in adjacent zones shall not vary more than ± 3 dB.

7.15.2 Commissioning

The system shall be commissioned by a qualified professional with experience in public address and voice evacuation systems. The commissioning process shall include a final inspection and acceptance test, as well as the preparation of a detailed report documenting the system's design, performance, and compliance with applicable codes and standards.

7.16 Qualifications

7.16.1 Contractor Qualifications

The system contractor shall have demonstrated experience and expertise in the design, installation, and commissioning of voice evacuation systems in similar environments and complexity to that of the DFW International Airport. The Contractor shall meet the requirements of the State of Texas Fire Alarm Rules.

7.16.2 Personnel Qualifications

All personnel involved in the design, installation, testing and commissioning of the system shall provide documentation of their qualification by one or more of the following:

1. Registration, licensing, or certification by a state or local authority
2. Certification by an organization acceptable to the authority having jurisdiction.
3. Manufacturer's certification for the specific type and brand of system provided.

7.16.3 Equipment Qualifications

All equipment used in the system shall be listed for its intended use.

7.16.4 System Qualifications

The system shall be fully compliant with the latest edition of NFPA 72 and other relevant codes and standards.

7.17 Documentation**7.17.1 Shop Drawings**

Before installation can begin, the System Designer shall provide Fire Alarm Planning Superintendent (APS), or Professional Engineer stamped shop drawings that accurately reflect the design and installation of the public address and voice evacuation system. Shop drawings shall be approved and stamped by DFW Fire Prevention.

7.17.2 Data Submittals

The system contractor shall provide complete and up-to-date data for all PAVE equipment, including manufacturer's specifications, catalog cut sheets, and product manuals.

7.17.3 Red-Line Drawings

The system contractor shall provide red line drawings that reflect all changes made to the PAVE system during installation and commissioning. All red line mark-ups shall be clear and easily legible, using color coding to distinguish between different types of changes.

7.17.4 As-Built Drawings

The system contractor shall provide a complete set of as-built drawings that accurately reflect the design and installation of the public address and voice evacuation system.

7.17.5 Acceptance Test Results

The system contractor shall provide complete and detailed acceptance test reports for the PAVE system. These reports shall demonstrate that the PAVE system meets all design criteria and requirements outlined in this manual. The documentation of tests, measurements and adjustments performed shall include a list of personnel and the list of certified test equipment used.

7.17.6 Operation and Maintenance Manuals

The system contractor shall provide detailed O&M manuals that include all relevant information on the system's design, installation, testing, and maintenance.

All documentation shall be provided and made accessible to all stakeholders electronically.

7.18 Warranty

The system contractor shall provide a warranty covering the PAVE System and its components, with a minimum duration of one year from the date of final acceptance of the system by the Owner Representative. The warranty shall include provisions for repair and replacement of any defective components and a telephone response to Owner's questions within 4 hours and on-site assistance within 24 hours.

This manual provides the design criteria and requirements for the Public Address and Voice Evacuation system at DFW International Airport. All stakeholders involved in the design and construction of the system shall comply with the requirements specified in this manual to design requirements per NFPA 13, 26.25 Airport Terminals, Fueling, Ramp Drainage and Loading Walkways shall be implemented.

7.19 Fire Suppression

7.19.1 General Requirements

7.19.1.1 Fire Sprinkler System Design Requirements

The fire sprinkler system in airport passenger terminals and Board occupied, operated, and maintained buildings shall be designed for a minimum ordinary hazard type occupancy. All equipment and materials shall be Underwriters' Laboratories (UL) or Factory Mutual (FM) approved and listed and shall bear the appropriate stamp or label.

7.19.1.2 Sprinkler Systems

Sprinkler systems shall be designed and built to facilitate the capture of fire sprinkler water and drained into the facility's sanitary sewer system as permitted by the DFW Environmental Affairs Department (EAD). All fire sprinkler water not plumbed to a sanitary drain shall be tested, collected, and disposed of properly per DFW Environmental requirements.

1. All sprinkler systems shall be constructed to allow for water quality sampling. Except for an activated discharge, the system should have a means to drain directly to sanitary sewers. The rate of discharge must be compatible with capacities at the point of entry into the sewer system. Sprinkler systems must be constructed to facilitate the collection of the entire volume of water generated during testing, expansion, modification, or repair. Discharge of anything other than sanitary sewer is not allowed.
2. Implement MIC monitoring program according to current National Fire Protection
3. Association (NFPA) 13, Installation of Sprinkler Systems
4. All sprinkler pipes that penetrate masonry or concrete walls or floors shall be sleeved with steel pipe.
5. The inspector test valves shall terminate to the exterior of the building. Discharge shall not be near any pits.
6. Provide a bypass around the check valve in the fire department connection line with a control valve in the normally closed position. The bypass is required for the performance of a full flow test of the system demand through the back flow preventer.

Exception: If the main drain can achieve the flow demand of the system, no bypass is required.

7.19.1.3 Sprinkler Riser Access

Additional exterior doors shall be provided when immediate exterior access is not provided to fire sprinkler riser rooms. Coordinate with DFW the location of additional exterior doors.

1. All sprinkler pipes that penetrate masonry or concrete walls or floors shall be sleeved with steel pipe.
2. The inspector test valves shall terminate to the exterior of the building. Discharge shall not be near any pits.
3. Provide a bypass around the check valve in the fire department connection line with a control valve in the normally closed position. The bypass is required for the performance of a full flow test of the system demand through the back flow preventer.

Exception: If the main drain can achieve the flow demand of the system, no bypass is required.

4. All riser rooms must be readily accessible without any obstructions or blockages. The janitor rooms should be planned and designed separately from these fire suppression purpose rooms to ensure compliance and safety.

7.19.2 Water-Based Fire Suppression System

7.19.2.1 Facility Fire Department Connection

All Fire Department Connections (FDC) shall be equipped with a 5-inch quick connect fitting with a 30 degree down angle.

All CRs must include a double-interlocked pre-action fire suppression system. The system must be located outside of the CR and as close as practical to the CR.

1. Fire Alarm
 - a. UPS units supporting life safety systems must be monitored by the Fire Alarm system.
2. Fire Alarm panels must not be located within CRs, to reduce access and traffic into the CRs.
3. Visual fire alarm strobe notification must be provided within the CRs.

7.19.2.2 Fire Suppression Sprinkler Freeze Protection

UL Listed Sprinkler Head Trace is permitted.

7.19.2.3 Dry-Pipe Sprinkler System

Dry pipe systems shall meet one of the following options:

1. Dry sprinkler systems shall consist of Schedule 40 galvanized for 3 inch pipe and larger, Schedule 40 for 2.5 inch pipe and smaller, and the sprinkler system attached to a nitrogen generator.
2. Dry sprinkler systems not connected to a nitrogen generator shall consist of Schedule 40 piping. Each dry-pipe system shall have its own air pressure supervisory switch to monitor and report both high and low air pressure conditions. The switch shall be between the air supply check valve and sprinkler alarm valve.
3. Dry pipe sprinkler systems with air compressors shall be connected to the existing piping system via stainless steel mesh connectors and installed with no bends. All air compressors shall be installed on spring vibration isolation pads.

4. A pressure gauge shall be installed and located at the furthest branch line.
5. Remote Inspectors' Test Valves shall be installed per NFPA 13 for Water Delivery verification.

A manual shut-off valve shall be provided between the hi/low switch and the main air supply line leading to the compressor. The air compressor shall be hardwired directly to a lockable disconnect box or to a dedicated branch circuit.

7.19.2.4 Sprinkler System Flushing

Sprinkler pipes shall be thoroughly flushed each time the system is expanded or modified.

7.19.2.5 Concealed Low Point Drain

All concealed low point drains shall be visually identified and provided with a sign to identify the system.

7.19.2.6 “Looped” Standpipe Systems

Terminal Buildings shall have a looped standpipe system that allows for fire department support from any FDC.

7.19.2.7 Pre-action Sprinkler System

System piping may be supervised by air or nitrogen. Piping shall be galvanized. Pre-action Valve assemblies shall not be installed in public areas or ceiling plenums. All pre-action System drains shall terminate to a suitable drain that can accommodate the removal of system water.

A physical disconnect switch shall be provided that interrupts the output circuit to each releasing device (e.g., solenoid, relay) and initiates a Supervisory Signal. Switches shall be provided for Pre-action Sprinkler Systems.

7.19.2.8 Sprinkler Isolation Valves

Each Tenant space shall have an isolation valve that will allow for maintenance and remodeling to happen without compromising the entire system, which is accessible and approved by the Building Official. Each gate will have an isolation valve that will isolate the sprinkler protection in the gate area including the hold room, ticketing area, and boarding walkway, which are accessible at 3'-5' off floor level with proper signage and approval from the Building Official.

7.19.2.9 Wet System Sprinklers

A sight glass is required to verify water in the system; These shall be located at the end of the farthest branch line.

7.20 Special Considerations

7.20.1 Floor Penetrations for Conveyors

Where conveyors penetrate rated assemblies or floors, provide closely spaced sprinklers in combination with draft stops as follows:

7.20.1.1 Draft Stop

The draft stops shall be immediately adjacent to the opening, at least 18 inches deep, and of noncombustible material that will stay in place before and during sprinkler operation. Sprinklers shall be spaced approximately 6 feet apart and placed 6 to 12 inches from the draft stop on the side away from the opening. An area smoke detector shall be placed in the ceiling above the floor opening and wired to the fire alarm system.

7.20.1.2 Shutters

Conveyor openings may be provided with fire/smoke shutters that can be manually closed or automatically closed by smoke detectors installed in accordance with NFPA 72 in lieu of method described above. The preferred method is to use a fire rated “dog house” consisting of sheetrock walls and roof with combination fire/security door that drops through the conveyor. This provides security between ramp and concourse levels and reduces outside air infiltration into the concourse.

7.20.1.3 Baggage Conveyor Systems in Terminal Buildings

Baggage conveyor belts shall be protected with sprinklers spaced no closer than 6 feet and no farther than 8 feet on centers in above ceiling areas. Sprinkler heads shall clear baggage and other items. Sprinkler head guards shall be installed.

7.20.2 Safe-Type Box and Lock System

A “safe” type box shall be provided for all facilities and buildings and located as directed by the Fire Marshal.

1. The Fire Marshal recommends using Knox Box 3200 or 4400 for building keys and access badges. When a padlock is required, use Knox Packlock Model 3771. Terminal Concession spaces will require a Master Lock Key Vault Model #5441D.

7.20.3 Fire Lanes

1. All fire lanes shall be at least 24 feet wide and capable of supporting at least 85,000 pounds. They will follow the currently adopted IFC Index D. All fire lanes will be required to have an inside turning radius of 35' and an outside turning radius of 57'.
2. Construction Fire Lanes- Anything less than 24' wide will require Fire Marshal approval.
3. Traffic calming devices. Refer to the Standard Markings document for further information.
4. Terminal Roadways- Anything less than 20' for roadways during construction in the terminal area is subject to Fire Marshal approval.
5. Gates across Fire Lanes- Any gate that crosses a fire lane must be equipped with a Fire Marshal-approved Opticom, model 711 or 721, and manual release with a Knox Lock.

7.20.4 Polyfluoroalkyl Substances (PFAS)

PFAS Foam is prohibited from being used in Fire Suppression Systems.

7.20.5 Concession Sprinkler Isolation Valve

An isolation valve is required within all Concession Spaces.

7.20.6 Gas Detection

Gas Detection Systems are required in all Concession kitchens which have Natural Gas (NG) cooking equipment and must be integrated with the FA System.

7.20.7 Commercial Kitchens

All gas and electric ovens including but not limited to standard, bake, roasting, revolving, retherm, convention, combination convention/steamer, countertop conveyorized baking/finishing, deck and pastry shall be considered Medium-Duty Cooking Appliance.

7.21 Fire Marshal Standard Markings Document

This document has additional requirements and specific procedures that the Fire Marshal's Office requires for the DFW Board and the non-board buildings.

7.22 EV Chargers

All EV charging locations will be equipped with an easily identifiable quick disconnect. The Standard Markings document will discuss specifics.

Chapter 8 Plumbing

8.0 General

8.0.1 General Requirement

Reference Chapter 3 Off-Site Construction for planning design, fabrication, inspection, regulatory compliance, and delivery requirements of infrastructure that will be manufactured offsite.

8.0.2 Plumbing Design

The Plumbing System Designer shall analyze and design the domestic hot water recirculating system such that the circulating temperature is maintained above 122 degrees F for Legionella control. Ensure that lengths of pipe runs, circulator(s) control, insulation effectiveness, master mixing valves, and system balancing, at a minimum, have been analyzed.

8.0.3 Common Work Results for Plumbing

8.0.3.1 Hangers and Supports for Plumbing Piping and Equipment

All pipes must be adequately supported throughout and shall withstand the effect of gravity loads and stresses. Generally, hangers shall be split ring or clevis type. However, trapeze hangers constructed of steel channels with welded spacers and steel rods may be used. An engineered system may use pre-engineered hangers and supports. All hangers and supports shall comply with the Manufacturer's Standardization Society (MSS) standards. Vertical pipes must be supported on each floor with pipe clamps.

Provide pipe saddles fabricated from galvanized metal (for insulated pipe) extending at least 12 inches in length and covering a minimum of half- pipe circumference.

Generally, the gauge shall be as follows:

Table 26 - Pipe Saddle Gauge Specification for Plumbing

Avenir LT Std 35 Light	Avenir LT Std 35 Light
Up to 3 inches	No. 22
3"-6"	No. 16
>6"	No. 12

8.0.3.2 Vibration and Seismic Controls for Plumbing Piping and Equipment

8.0.3.2.1 Vibration Isolation

To prevent excessive noise or transmission of vibration to the building structure due to the operation of machinery or equipment, or due to interconnected piping, ductwork, or conduit, proper vibration isolation shall be provided.

Static deflection of vibration isolators generally shall conform to minimum design criteria recommended in the latest American Society of Heating, Refrigeration, and Air-Conditioning Engineers (ASHRAE) Guide and Data Book for the actual floor spans involved.

A single vibration isolation manufacturer shall supply vibration isolation equipment for any one project.

The vibration isolation manufacturer and his representative shall have been engaged in the business of vibration isolation for no less than 5 years.

The design shall include lateral bracing with pipe hangers and supports to prevent swaying.

8.0.3.3 Hammer Arrestor

The flow velocity of the water distribution system shall be controlled to reduce the possibility of water hammer.

A water-hammer arrestor shall be installed where quick-closing valves are utilized.

Water-hammer arrestors shall be installed in accordance with the manufacturer's instructions. Water-hammer arrestors shall conform to ASSE 1010. Hydraulic shock absorbers may be used in accordance with Water Hammer Arresters.

8.0.3.4 Identification for Plumbing Piping and Equipment

All piping in buildings shall be identified by the use of pipe marker bands with direction of flow arrows at 10 foot intervals in concealed spaces; 20 foot intervals in exposed areas and on each side of any penetrated wall, ceiling, or floor. Pipe marker color coding shall follow industry practice ANSI/ American Society of Mechanical Engineers (ASME) A13.1 "Scheme for Identification of Piping Systems."

8.1 Plumbing Insulation

8.1.0 Minimum Thickness

Insulation thickness shall comply with DFW Airport's currently adopted version of the International Energy Conservation Code.

8.1.1 Insulating Unions and Adapters

Provide dielectric insulating unions or adapters as required between copper and steel pipe, and equipment. Dielectric insulators/adapters shall contain nylon insulation.

Plumbing Piping Insulation shall not contain asbestos, lead mercury or mercury compounds and must conform to ASTM standards.

8.1.2 Pipe Sleeves

Provide pipe sleeves for all pipes passing through masonry and concrete construction. The annular space between pipes and sleeves must be permanently sealed and sleeves below grade must be watertight. The sleeve shall be two pipe sizes greater than the pipe passing through the wall.

Install link seals at outer sleeve and the sleeve where piping rises upward through a foundation.

Pipe joints should not be made closer than 12 inches to a wall, ceiling, or floor penetration, unless such pipe is welded.

PVC larger than 2 inches which penetrates through a 2-hour fire rated upper floor shall be equipped with 2-hour intumescent fire protection coupling.

8.1.3 Heat Tracing

Heat tracing for freeze protection shall use electric heating cables that are self-regulating and parallel resistance.

Piping exposed to the elements are to be both heat traced and insulated. Such piping shall also have a protective jacket of aluminum.

Piping in unconditioned spaces and inside exterior walls shall be both heat traced and insulated. Dry vent piping does not require insulation, heat tracing, or jacketing of any type.

8.2 Plumbing Piping

8.2.0 Prohibited Installation

Plumbing systems of any type shall be prohibited from being installed in pier caps. Piping cannot be installed transversely within the beam, or any other structure component of a foundation.

8.2.1 Domestic Water

Domestic cold water (inside) – Type K or L copper with solder (95-5); no lead. Copper systems using mechanical joining, such as “ProPress” by Rigid/Viega may be used in lieu of “solder” joining method. Chlorinated polyvinyl chloride (CPVC) may also be used to meet return air plenum restrictions.

Type K or L Copper, (rigid if fittings are brazed, or soft), Cross-linked polyethylene/aluminum/cross-linked polyethylene (PEX-AL-PEX) pipe, and Schedule 40 PVC are permitted within 5 FT of the building before transitioning to copper.

All Terminals require copper piping for domestic water. Underground copper piping shall be installed with brazed joints only.

8.2.1.0 Plumbing Valve Installation

1. All valves shall be installed in an upright position.
2. A valve schedule shall be provided prior to a project receiving a Plumbing final.
3. Additionally, all valves shall be within 10 feet of the finished floor.

8.2.1.1 Roof Hydrant Valve Install

Frost proof roof hydrants shall have a full port valve within 10 foot above the finished floor.

8.2.2 Gas

Gas pipe fittings located in ceiling spaces or walls shall be welded or an approved pressed fitting.

Gas pipe systems shall be tested at twice the working pressure.

All gas piping on Terminal roofs shall have lockable stops.

Gas piping to kitchen appliances exposed to direct heat (i.e. above cook lines) shall be hard piped only. Flexible connectors are not allowed for this application. All kitchen equipment final connections shall be attached with quick-disconnect connectors or approved equal.

8.3 Plumbing Specialties

8.3.0 Backflow Preventers

Provide double check backflow preventer when domestic service is connected to service water main. An RPZ shall be installed on any domestic water line serving other closed or chemically treated systems that could foreseeably contaminate the potable water line. All backflow preventers shall be installed in a readily accessible location, 1 foot to 5 foot above the finished floor.

Unless an ice maker is equipped with an internal air gap, an RPZ must be installed.

Unless a dish machine is equipped with an internal air gap, and RPZ must be installed.

Water filter systems require an RPZ before the filter.

8.3.1 Domestic Water Pumps

Pumps shall be close-coupled, horizontally mounted, in-line centrifugal domestic water pumps.

Pumps shall be all bronze or stainless steel for domestic water service. Provide a line size ball valve on suction and discharge side of pump. Provide unions or bolted flange connection on each side of pump. Pressure taps and thermometer wells are not required on in-line circulators. Sleeve type bearings are acceptable for in-line pumps.

The plumbing system designer shall study water usage periods and shall operate pumps just prior to usage periods and limit operation of pumps as much as possible. A 7-day, 12- hour timer should be installed to control such pump operation, especially during peak demand periods as an energy reduction measure. Pumps shall be monitored and /or controlled via the Building Automation System. Applicable also to pumps described in the International Plumbing Code under Facility Drainage.

8.4 Sanitary Sewer, Storm Sewer, and Grease Waste

8.4.0 Sanitary Drains

Above Grade Shall be cast iron with cathodic protection, copper, or PVC.

Sanitary Sewer, grease waste, and storm sewer piping Below Grade shall be PVC or stainless steel.

Sanitary and grease waste piping which is subject discharge temperatures above 140 degrees shall be stainless steel or cast iron with cathodic protection above grade; stainless steel below grade.

All existing piping which will be connected to new sanitary sewer, storm, and grease waste piping shall be video inspected and reviewed by DFW Code prior to covering the piping.

New sanitary sewer and grease waste piping shall be video inspected and reviewed by DFW Code prior to covering the piping.

All drainage piping below the floor within a crawl space or ramp area shall have floor or wall cleanouts.

Piping exposed to the elements shall be both heat traced and insulated. Such piping shall also have a protective aluminum jacket.

PVC larger than 1 1/2 inches which penetrates through a 2-hour fire rated assembly shall be equipped with 2-hour Intumescent fire protection coupling.

PVC routed within a return air plenum shall be wrapped with 3M Number 5A wrap or approved equal.

There shall be clear water only in combination waste and vent.

Dish machines, which drain into PVC shall either be low temperature types or encompass a mixing valve, which cools the discharge waste temperature to 120 degrees prior to entering the drainage system.

8.4.1 Grease and Oil Removal Devices

Wastewater from sinks, dishwashers, floor drains, mop sinks, and floor sinks in food service facilities shall drain to a grease collection system or through a grease trap or grease interceptor serving one or more facilities. Installation shall comply with the applicable plumbing code. All grease waste piping shall be insulated; exposed piping shall have an outer metal jacket.

Grease interceptors shall be sized in accordance with the TCEQ Grease Interceptor Sizing Worksheet and shall utilize the following factors:

- 10. Meal Factor - 1.33
- 11. Waste Flow Rate - 6 Gallons
- 12. Retention Time - 2.5 Hours
- 13. Storage Factor - 2.00

Sanitary sewer systems conveying, or potentially conveying grease shall have cleanouts at 50 foot intervals.

For grease waste systems, the designer shall consider a Grease Eliminator System. This system shall employ non-toxic, non-pathogenic bacteria to metabolize grease and oil into a clean waste product. This system shall be installed at the furthest point of use from the grease interceptor.

8.4.2 Submersible Sewage Pumps

Sewer ejector pump design and selection design criteria are the same as those listed for "Sump Pumps" except sewer ejector pumps shall be of the standard 3 inch, non-clog type specifically designed and installed for their purpose intended.

All pumps shall be inspected prior to installation by the DFW Code Electrical Inspection department.

8.4.3 Roof Drains

Roof drains shall be compatible with roof system. The Designer shall use 6 inches per hour as a minimum rainfall intensity guideline for sizing roof drains. Roof drain and overflow drain systems shall have cleanouts installed at 12 inches above finished floor if tied directly into the storm discharge system.

8.4.4 Floor Drains

All floor or trough, drains must be accessible. All restrooms shall have at least 1 floor drain for each 400 square feet of floor area or percentage thereof. Such drains shall have an outlet of not less than 3 inches in diameter.

Trap seals/guards and/or trap primers shall be installed. Trap primer(s) are the preferred method

All trap primers shall be accessible and shall include a ball valve and a union above the trap primer for maintenance purposes.

8.4.5 Wet Pit Mounted, Vertical Sump Pumps

Generally, duplex sump pumps are required when located in a mechanical/electrical equipment room containing high voltage switchgear or motor control panels. A simplex pump may be used if an area does not contain critical equipment.

Provide a mechanical alternator on duplex pumps and provide a separate circuit and circuit breaker for each pump. Provide check valves, and bypass pipe work and valves as required (in-line check valves are not recommended). Pumps shall be complete with automatic float switch with rod, rod guide, and copper float.

A check valve and a full open valve located on the discharge side of the check valve shall be installed in the pump or ejector discharge piping between the pump or ejector and the gravity drainage system. Access shall be provided to such valves. Such valves shall be located above the sump cover required by IPC or, where the discharge pipe from the ejector is below grade, the valves shall be accessibly located outside the sump below grade in an access pit with a removable access cover.

Pumps shall be of the wet-pit type complete with gas tight sump cover, curb ring, and grease lubricated, including alemite fittings extended to the pump base plate.

Pumps shall be heavy-duty, explosion-proof if underground wet pit type, vertical centrifugal, open non-corrosive impeller type with vertical drip-proof type motor with anti-friction grease lubricated bearings.

Where sump pumps are installed to provide protection for mechanical/electrical equipment, a high-water alarm bell shall be provided in the area and alarm contacts shall be provided for a central monitoring system.

8.4.6 Submersible Sump Pumps

Generally, submersible pumps are avoided where possible except electric power manholes where high voltage switches or tap boxes are installed. Diaphragm actuated pumps are preferred rather than float actuated pumps.

8.4.7 Discharge of Sump Pumps

Discharge of sump pumps to the storm sewer system is prohibited.

8.4.8 Sump Pit

The sump pit shall be not less than 18 inches (457 mm) in diameter and not less than 24 inches (610 mm) in depth, unless otherwise approved. The pit shall be provided with access and shall be located such that all drainage flows into the pit by gravity. The sump pit shall be constructed of concrete, steel, plastic, or other approved materials.

The pit bottom shall be solid and provide permanent support for the pump. The sump pit shall be fitted with a gastight removable cover that is installed not more than 2 inches (51 mm) below grade or floor level. The cover shall be adequate to support anticipated loads in the area of use. The sump pit shall be vented in accordance with Chapter 9 of the IPC.

8.5 General Service Packaged Air Systems

8.5.0 General Service Packaged Air Compressors and Receivers

House service should not be taken from the instrument or control air distribution line or instrument or control air receiver unless specifically approved, except for doors and dry system fire sprinklers in Terminal buildings.

Provide an ASME pressure rated receiver and pressure relief assembly for the working pressure involved. Provide a non-cycling refrigerated air dryer when required. Provide an automatic drain on receiver with drain line piped to floor drain or hub drain. Air compressor assembly shall be provided with adequate vibration isolation. Adequate sound isolation shall also be provided. Generally, a horizontal tank-mounted type unit, with motor and compressor arranged for V-belt drive and mounted on a common steel base supported for the receiver is preferred.

Compressors that are equipped with cylinder unloading devices that will unload compressor on stopping and prevent cylinders from loading or starting until the rated motor speed is attained are preferred.

8.6 Plumbing Equipment

8.6.0 Electric Domestic Tank-Type Water Heaters

1. Water heaters shall be glass lined, storage type.
2. Electric water heaters shall be UL listed, and FM approved when possible.
3. All standard water heaters shall have a 10-year limited warranty. The designer shall coordinate all warranty requirements with ETAM. All energy saver water heaters shall meet ASHRAE Standards for Energy Efficiencies, latest edition.
4. Water heater drains shall be plumbed indirectly to a floor drain or mop sink with copper piping.
5. All water heaters shall be accessible.
6. Electric water heaters located in ceiling/attic spaces shall be accessible through an access door or by removing ceiling tile. No ACM shall be used.
7. Expansion tank required.
8. Any water heater which is installed above a ceiling shall be accessible and limited to a maximum of 10 feet above the finished floor.

8.6.1 Electric Tankless Water Heaters

1. Electric water heaters shall be UL listed, and FM approved when possible.
2. All standard water heaters shall have a 10-year limited warranty. The designer shall coordinate all warranty requirements with ETAM. All energy saver water heaters shall meet ASHRAE Standards for Energy Efficiencies, latest edition.
3. Water heater drains shall be plumbed indirectly to a floor drain or mop sink with copper piping.
4. All water heaters shall be accessible.
5. Electric water heaters located in ceiling/attic spaces shall be accessible through an access door or by removing ceiling tile. No ACM shall be used.
6. Any water heater which is installed above a ceiling shall be accessible and limited to a maximum of 10 feet above the finished floor.

7. Domestic hot water systems exceeding 34,000 BTU/hr (10kW) must have a weighted average efficiency of 2 UEF (COP of 2) achieved by utilizing air source heat pump water heater technology; or a domestic hot water solar fraction of at least 0.4 for solar thermal hot water systems.
 - a. If electric, heat pump, or solar hot water systems, or smaller electric resistance point-of-use equipment is not practical, natural gas fired hot water systems shall meet the requirements of 8.6.3.

8.6.2 Thermal Expansion Tanks

A thermal expansion tank shall be supported in accordance with the manufacturer's instructions. Thermal expansion tanks shall not be supported by the piping that connects to such tanks.

8.6.3 Commercial Gas Domestic Tank-Type/Tankless Water Heaters

Water heaters shall adhere to the following requirements:

1. Water heaters shall be glass-lined, storage type.
2. Gas water heaters shall have automatic gas shut-off device and be equipped with an American Gas Association certified draft hood. Water heaters shall utilize electric ignition devices.
3. All standard water heaters shall have a 10- year limited warranty.
4. All energy saver water heaters shall meet ASHRAE Standards for Energy Efficiencies, latest edition.
5. Water heater drains shall be plumbed indirectly to a floor drain or mop sink with copper piping.
6. All water heaters shall be readily accessible by access panel or removable ceiling tile.
7. Expansion tank required.

8.6.4 Domestic Water Heat Exchangers

Domestic water heat exchangers shall adhere to the following requirements:

1. Water heaters shall be completely copper lined.
2. Heads shall be cast bronze, coiled tubes shall be copper, and all other internal parts shall be copper and bronze. Heat exchanger tubes shall be double-wall type.
3. The tank shall be ASME stamped at or above the scheduled working pressure. Factory-furnished and approved accessories, including ASME temperature and pressure relief valve, shall be used.

8.6.5 Thermal Expansion Tanks

A thermal expansion tank shall be supported in accordance with the manufacturer's instructions. Thermal expansion tanks shall not be supported by the piping that connects to such tanks.

8.7 Plumbing Fixtures

8.7.0 General Information

Plumbing designers are required to use high-efficiency fixtures and valves, such as automatic sensors, aerators on lavatories, and WaterSense™ or equivalent-certified fixtures. Fixture fittings must meet the maximum install flush or flow rates provided below:

1. Toilet (water closet*): 1.28 gpf
2. Urinal*: 0.5 gpf
3. Public lavatory (restroom) faucet: 0.5 gpm
4. Private lavatory faucets*: 1.5 gpm

5. Kitchen faucet: 1.8 gpm
6. Showerhead*: 2.00 gpm

*The WaterSense label is available for this fixture type. Private lavatory faucets do not have to be WaterSense labeled but must comply with the maximum 1.5 gpm / 5.7 lpm WaterSense limit.

** For dual-flush toilets, the full-flush volume shall be equal to or less than the maximum installed flush rate of 1.28 gpf/4.8 lpf.

All exposed metal work at fixtures shall be brass with chromium plate. All faucets, fittings, supply stops for fixtures, and similar devices shall be from one manufacturer unless otherwise required. Each fixture shall contain standardized interchangeable operating units made up of separate renewable stem, seat, washer retainer, and nut. All faucets and fittings must close with the water pressure. All fixtures shall be installed with supply stops/valves accessible at the fixtures.

All fixtures and accessories listed apply to Board owned, operated, or maintained buildings. Some fixture and accessory preferences may change over time depending upon current maintenance warehouse stocking. Tenants may have different preferences and shall be consulted.

On renovation projects, an effort shall be made to match existing fixtures and trim. On renovation projects where fixtures and trim cannot be matched and on new projects, fixtures shall be water conserving American Standard or an approved equal.

8.7.1 Commercial Water Closets

Wall-hung water closets are preferred. Water closets shall be white, vitreous china, siphon jet, elongated bowl, with white open-front seat without cover.

Flush valves for water closets in Terminal buildings shall be as follows:

1. Toilet shall have concealed rough brass hydraulically operated flush valve, 1 inch wheel handle back-check stops, adjustable tailpiece, solenoid motor operator, sensor, vacuum breaker, elbow flush connection and spud coupling for 1.5 inch concealed back spud.
2. Automatic sensor for operation of each water closet, with required transformers, controls, and complete wiring diagrams for separate operation in each toilet, all as recommended and approved by flush valve manufacturers, are required.
3. Flush valves as described above are approved, or an approved equal may be used.
4. For recommended flush valves (Manufacturer: Sloan).

8.7.2 Commercial Urinals

Wall-hung urinals are preferred. Urinals shall be white, vitreous china, wash-out type.

Flush valves for urinals in Terminal buildings shall be as follows:

1. For recommended flush valve system (Manufacturer: Sloan)
2. Urinal shall have concealed rough brass hydraulically operated flush valve, $\frac{3}{4}$ inch wheel handle back-check stops, adjustable tailpiece, solenoid motor operator, sensor, vacuum breaker, elbow flush connection and coupling for $\frac{3}{4}$ inch concealed back spud, wall, and spud flanges for each urinal.
3. Automatic sensor for operation of each urinal with required transformers, controls, and complete wiring diagrams for separate operation in each toilet, all as recommended and approved by flush valve manufacturers, are required.

4. Flush valves as described above are approved, or an approved equal may be used.
5. Meet accessibility requirements. Make provisions for dispersion of chemical treatment for public banks of urinals.

8.7.3 Commercial Lavatories

Commercial Lavatories in Terminal buildings shall be as follows:

8.7.3.1 Requirements

Faucets and fixture fittings shall conform to ASME A112.18.1/CSA B125.1. Faucets and fixture fittings that supply drinking water for human ingestion shall conform to the requirements of NSF 61, Section 9. Flexible water connectors exposed to continuous pressure shall conform to the requirements of IPC.

8.7.3.2 Faucets and Supply Fittings

1. Faucets and supply fittings shall conform to the water consumption requirements of IPC.
2. Lavatories shall comply with IPC and the DFW Airport Customer Experience Standards.

8.7.4 Mop Basins

Mop basins shall be one-piece mop service basin, size 24 square inches by 12 inches high outside, with Type 304 stainless steel, 20-gauge cap, continuous on all sides.

Basins shall have wall flashing on back and sides as required. Provide silicone base for full seal at floor. Grout entire installation level.

Service faucet shall be chrome plated with vacuum breaker, integral stops, adjustable wall brace, pail hook, and 0.75 inch hose thread on spout, 8 inch spread. Hose and hose bracket shall be 30 inches long flexible, heavy-duty 0.75 inch rubber hose, cloth reinforced, with 0.75 inch chrome coupling at one end. Five-inch-long bracket by 3 inch wide, with rubber grip.

Mop hanger shall be 24 inches long by 3 inches wide, 18-gauge No. 302 stainless steel attached with flat head, slotted machine screws.

8.7.5 Service Sinks

Service sinks shall be white enameled cast iron, 20 inches by 22 inches, blank back with wall hanger supports.

Trap shall be adjustable standard for 3 inch pipe connection with cleanout plug and strainer, enameled inside. Consideration shall be made for the use of trap primers and/or trap seals. The "Green Drain" is an effective solution in most cases. Rim guard shall be 9 inches and 12 inches stainless steel rim guard, front and sides.

8.7.6 Electric Water Coolers (EWC)

Wall hung water coolers are preferred. Electric water coolers shall meet accessibility requirements. Some Terminal buildings have a central water-cooling system. The Designer shall investigate the possibility of connecting to this system where it is available. Where filters are installed at the water coolers, RPZs shall be installed per IPC. The RPZ shall discharge to a floor drain, floor sink, or mop basin.

8.7.7 Water Refill Stations

Bathroom renovations, new terminal construction, and /or major plumbing upgrades, install water bottle refill stations in all terminals, ensuring strategic placement. Specific placements include outside every restroom, past security checkpoints, near food courts, and near baggage claim area.

8.7.8 Appliances and Process Water Efficiency

Newly installed appliances, equipment, and processes must meet the following requirements:

1. Commercial clothes washers are to be ENERGY STAR-certified
2. Commercial dishwashers, ice machines, and combination ovens must be ENERGY STAR-certified
3. Pre-rise spray valves must be less than or equal to 1.3 gpm (4.9 lpm)

Chapter 9 Heating, Ventilating, Air-Conditioning (HVAC)

9.0 General Information

9.0.1 General Requirements

Reference Chapter 3 Off-Site Construction for planning design, fabrication, inspection, regulatory compliance, and delivery requirements of infrastructure that will be manufactured offsite.

9.0.2 Scope

This Chapter defines general design criteria that apply to the design of heating, ventilation, air conditioning (HVAC) at DFW, as well as those HVAC systems to be served by the Central Utility Plant (CUP), eCUP, or East Utility Plant. Chapter 1 should be consulted for specific Airport regulations and standards that also apply.

It shall be the Designer's responsibility to verify locations or the adequacy of record information prior to design and construction of HVAC systems. The Designer shall coordinate the development of the design at all stages with DFW Energy, Transportation and Asset Management (ETAM).

All areas used primarily to accommodate people-oriented activities such as offices, concessions, concourses, cafeterias, etc., shall be air conditioned and heated. Areas classified as storage or manufacturing shall be mechanically ventilated and conditioned to minimum requirements of the Building Code and MEP/Machinery/Communication equipment operation temperature to guarantee the proper operation and maximized lifecycle. The criteria for a particular HVAC system will vary somewhat from building type to building type, or project to project, which may change certain parameters of initial design considerations.

9.0.3 Restrictions

Design of refrigerant-using or refrigerant-containing equipment shall be self-contained with no field installed piping for refrigerant systems with GWP above 700. For systems with GWP below 700 with field-assembled joints, perform leak check, vacuum check, and pressure check prior to charging with refrigerant.

Designs are encouraged to specify an "automatic leak detection" system in fully enclosed spaces with equipment that has an overall refrigerant charge exceeding 100 tCO₂e. Leak check. Prior to substantial completion, check both new and existing refrigerant-containing equipment for refrigerant leaks and repair all identified leaks.

A table of substitute refrigerants is available from the Environmental Protection Agency (EPA) at <http://www.epa.gov/ozone/snap> or may contact the Environmental Affairs Department for more information.

Renovation of HVAC systems still using a Class I or Class II refrigerant shall incorporate into design and specifications the replacement of the refrigerant with a suitable substitute.

Accessibility for All Units: All units must be accessible for maintenance with a ladder of 12 feet or less or a compact indoor scissor lift. For equipment in high-ceiling areas, a mezzanine should be considered for adequate access. Architectural designs must ensure door sizes and ceiling heights are suitable for the equipment dimensions.

Separation for Restroom-Related Units: Supply air systems, such as those from VAV boxes, fan coil units, or FPTUs, and exhaust systems, must be separated from the standard supply and return/exhaust air ducts used for other purpose rooms.

Redundant capacity for Sensitive Operation Areas: Critical areas, including communication rooms, electrical rooms, transformer rooms, PDS rooms, vaults, and similar spaces, shall be equipped with N+1 redundant HVAC units to ensure uninterrupted operation.

Designer shall provide designs that ensure continuous operation of the plumbing chase by incorporating design elements that maintain appropriate temperature conditions to prevent freezing during winter months.

All outdoor units shall include proper hail guards to prevent potential damage to the components.

9.0.4 System Types

9.0.4.1 Terminal Buildings

System design for new Terminal facilities should conform to the basic design parameters and equipment and material criteria described herein. Heat pumps, roof top units, and plenum-mounted condensing units are not desired and require approval from DFW's Energy Transportation & Asset Management (ETAM) Department.

9.0.4.2 Buildings Other Than Terminal Buildings or Other ETAM Operated Facilities

New construction, major renovation, and replacement of existing heating system should use heat pumps as the primary source of heating, targeting a weighted average coefficient of performance (COP) of at least 2.0. They should be capable of providing heat without combustion for outside air temperatures above 20°F (-6.5 °C). If heat pumps are not available, natural gas may be used as an alternative.

For air-source equipment, use outdoor air temperature (OAT) AHRI rating conditions of 17°F db/15°F wb (-8°C db/-9°C wb), or the next highest OAT rating conditions if not applicable. For liquid-source equipment, use design conditions.

All curb-mounted units shall be furnished with appropriate enclosed engineered curbs to provide for level unit mounting with minimum 16 inch curb height above top of the roof deck surface.

All condensate drainage shall discharge indirectly into the sanitary sewer.

The following criteria shall apply where applicable:

1. Chilled water systems are required for 50 tons and above up to a combined plant size of 300 tons.
2. Air cooled chilled water systems can be used to 100 ton unit capacities.
3. Water cooled chilled water systems are required above 300 tons combined plant size.

If available, natural gas is preferred as a source of heating. Otherwise, heat pumps shall be used.

9.0.5 Documentation

It is essential to ensure that all activities performed during the design and construction phases are thoroughly documented and submitted as formal submittals or transmittals. Proper documentation is a key factor in minimizing project risks, ensuring compliance, and maintaining clarity throughout construction and post-construction stages.

During each design review phase, the Basis of Design (BOD) should be included in the submission package. The BOD must provide comprehensive information, including load calculation results, energy consumption data, referenced codes or standards, and any assumptions made during the design process.

To minimize risks and ensure accountability, the following final deliverables should be considered and included in the project documentation upon completion:

9.0.5.1 Shop Drawing

Submit shop drawings and product literature for approval to EOR and DFW ETAM Engineering for review and approval before work can begin.

1. Shop drawings shall be stamped by an Engineer or endorsed by Approved Professional Specialist.
2. Shop drawings shall contain title blocks identifying the project name and permit number with revision log.
3. Plan drawings showing the locations of the system components, including any adjustments in the quantities.
4. Riser diagram showing system components, interconnecting wiring, air flow and connections to other building systems and equipment.
5. Shop drawing Submittals shall be marked to indicate the specific models, sizes, types, and options being provided.
6. Submittals not so marked will be considered incomplete submittals and will be rejected.

9.0.5.2 Closeout Documentation Set Including As Built

Upon completion, provide closeout document sets of the installation. One electronic copy to Records Documentation Management. One electronic copy shall be furnished to DFW ETAM Engineering for inclusion in Facilities Maintenance records. One electronic copy shall be delivered to DCC Digital Facilities & Infrastructure team.

5. Basis of Design (BOD) with Calculations: Ensure the final BOD includes all load calculations, assumptions, and energy consumption data, clearly documented and reflective of the design intent.
6. As-built: Verify that as-built drawings accurately capture the actual geometry, equipment locations, and all relevant details, including controls documentation. Contractor's shop drawings must accurately reflect the site conditions as built. All final as-built's shall show the Duct routing, Equipment location as well as controls. Air and CHW/HW flow shall be clearly marked at device and on the print.
3. Asset Equipment Sheet for Computerized Maintenance Management System (CMMS): Ensure that all equipment information is recorded in the Asset Equipment Sheet after project completion. This sheet should include LOD500 details, preventive maintenance schedules, decommissioning information, and warranty documentation, providing a comprehensive record for future maintenance and historical reference.

4. BIM Model: Where applicable, include a Building Information Model (BIM) that provides accurate geometry and the General Arrangement (GA) of each piece of equipment. All these shall be based on the Digital Facilities & Infrastructures Guideline (DFI), through the up to date version.
5. Testing, Adjusting, and Balancing (TAB) Report: Ensure that a detailed TAB report is provided, covering all testing, adjusting, and balancing activities performed on mechanical systems to verify that they meet the design specifications.
6. Commissioning Report: Include a complete commissioning report to confirm that all systems are installed and functioning according to the design intent.
7. Punch List and Closeout: Work with the contractor to prepare a punch list detailing any outstanding issues and ensure that a punch-clear report is provided once all items are resolved.
8. Quality Reports: Make sure that any Non-Conformance Reports (NCR) or Quality Observation Reports (QOR) are documented, along with corrective actions taken to resolve deviations from quality standards.
9. Operation and Maintenance (O&M) Manuals: Confirm that comprehensive O&M manuals are provided for each piece of equipment, including detailed operational procedures, maintenance guidelines, and troubleshooting information.

9.1 Common Work Results for HVAC

9.1.0 Expansion Fittings and Loops for HVAC Piping

A pipe stress analysis shall be performed and documented to ensure the system has adequate flexibility and to determine support and equipment reaction forces and support spacing. Expansion loops and/or expansion joints shall be provided where necessary to limit pipe stress or reaction forces in accordance with the applicable ASME B31 piping code. Where possible, the use of engineered expansion loops is preferred to expansion joints. Joints shall be piston-ring, internally guided, double-expansion joint or a pack less expansion joint.

Expansion joints shall provide 200 percent absorption capacity of piping expansion between anchors. All chilled and hot water piping joints shall be welded; therefore, special consideration must be given to pipe layout for expansion and contraction.

9.1.0.1 Drain and Vents

Low-point drains, and high-point air vents shall be indicated in the hydronic system.

9.1.1 Anchor Locations

A pipe stress analysis shall be performed to determine proper pipe anchor locations.

The pipe stress analysis shall be documented and submitted for review and approval to DFW Code and DFW ETAM.

9.1.2 Expansion Tank

An expansion tank shall be installed in every hot water system. For multiple boiler installations, not less than one expansion tank is required. Expansion tanks shall be of the closed or open type. Tanks shall be rated for the pressure of the hot water system.

Exception: Expansion tanks shall not be required in the collector loop of drain-back systems.

9.1.2.1 Closed-type Expansion Tank

Closed-type expansion tanks shall be installed in accordance with the manufacturer's instructions. Expansion tanks for systems designed to have an operating pressure in excess of 30 psi (207 kPa) shall be constructed and certified in accordance with the ASME Boiler and Pressure Vessel Code. The size of the tank shall be based on the capacity of the hot-water-heating system.

9.1.2.2 Open-type Expansion Tank

Open-type expansion tanks shall be located not less than 4 feet (1219 mm) above the highest heating element. The tank shall be adequately sized for the hot water system. An overflow with a minimum diameter of 1 inch (25 mm) shall be installed at the top of the tank. The overflow shall discharge to the drainage system indirectly and shall be readily observable.

9.1.3 Meters and Gages for HVAC Piping

British thermal unit (BTU) metering is required for all chilled and hot water applications served by the EP or the East Side Plant (ESP). Metering must provide flow, temperature, and BTU per hour values to the Energy Plaza (EP) control system.

9.1.3.1 Revenue-Grade Metering Devices

All facilities must be metered using revenue-grade metering devices.

9.1.4 Hangers and Supports for HVAC Piping and Equipment

A pipe stress analysis shall be performed to determine pipe hanger support system and spacing requirements. All hangers and supports shall comply with MSS standards. Vertical pipes must be supported at each floor with pipe clamps.

9.1.4.1 Pipe Saddles

Pipe saddles shall be galvanized metal (for insulated pipe) extending at least 12 inches in length and covering a minimum of half-pipe circumference. Protection shields shall be provided for all insulated pipes.

Generally, the gauge shall be as follows:

Table 27 - Gauge Specification for HVAC Pipe Saddle

Pipe Diameter (inches)	USS Gauge
Up to 3	No. 22
3 to 6	No. 16
Above 6	No.12

9.1.5 Vibration Controls for HVAC

To prevent excessive noise or transmission of vibration to the building structure due to the operation of machinery or equipment, or due to interconnected piping, ductwork, or conduit, engineered vibration isolation shall be provided.

Static deflection of vibration isolators shall conform to minimum guidelines recommended in the latest ASHRAE handbooks for the actual floor spans involved, and NFPA 90A.

Consideration shall be given to sound transmission by following ASHRAE Guidelines in Design.

The vibration isolation system shall consist of foundation, base, spring isolators, and rubber and shear pads, as necessary to provide maximum isolation conforming to ASHRAE guidelines and standards.

9.1.5.1 Noise Control

Mechanical noise levels shall be controlled by proper design of the noise producing mechanical and electrical equipment such as fans, mixing boxes, diffusers, pumps, transformers, emergency generators, etc., so as to not exceed acceptable levels as set forth by industry standard criteria. The acceptable noise level shall be described in terms of Noise Criteria as defined by the ASHRAE Handbook, Systems Volume, (Sound and Vibration Control Chapter) latest edition).

9.1.6 Identification for HVAC Piping and Equipment

All piping in buildings shall be identified by the use of pipe marker bands or pipe insulation sleeving, with direction of flow arrows. Pipe marker and sleeving color coding shall follow standard industry practice ANSI/ASME A13.1 "Scheme for Identification of Piping Systems."

9.1.7 Testing, Adjusting and Balancing for HVAC

The balancing, testing, and adjusting of the heating, ventilating and air conditioning systems shall be performed by an independent technical firm or balancing agency not involved in the design. Balancing firm shall be Associated Air Balance Council- or National Environmental Balancing Bureau-certified. The balancing agency must stamp the report.

The balancing, testing, and adjusting of the heating shall be approved by the engineer of record with either a stamped letter or by a stamp on the report, prior to the report being submitted to DFW Code for acceptance. All tests shall comply with certification agencies standards and practices.

9.1.8 Commissioning for HVAC

All HVAC commissioning processes must comply with the Board policy issued by ETAM. Both the designer and contractor are responsible for being familiar with this policy and are required to collaborate with Commissioning Agents to ensure the delivery of high-quality work to the Owner.

9.2 Insulation

9.2.0 Ductwork Insulation

Ductwork shall be labeled in a manner which will identify what type of system the duct is serving while providing the direction of the flow with 4 inch labeling every 10 feet in ceilings and 20 feet when exposed.

Labels shall be provided which signify whether duct mains are low, medium, or high pressure.

9.2.0.1 Internal Insulation

Generally, all ductwork except exhaust ductwork shall be externally insulated in accordance with temperatures involved and current IECC and International Fire Code. Ductwork insulation materials shall be selected for the function involved, considering sound absorption coefficients, velocities, etc. (Manufacturer: Thermaflex).

9.2.0.2 External Insulation

Where external rectangular ductwork insulation is required, particular attention shall be given to joints, terminating edges, operation of air control devices, etc.

External duct wrap may be used where insulation is not exposed to abuse. Where insulation may be subject to abuse, insulation shall be 2 inches thick, 3-pound density glass fiber rigid board duct insulation complete with reinforced foil-kraft integral heavy vapor proof covering on the outside surface. Insulation shall have a minimum compressive strength of 140 psf at 10 percent deformation.

Securely fasten all edges, joints, etc., to provide a vapor proof duct. Rigid insulation shall be mechanically fastened to the duct.

9.2.0.3 Insulation for High Velocity Ductwork

Generally, high velocity ductwork requiring external insulation shall be insulated with blanket wrap fiberglass insulation, 2 inches thick, 1-pound density, complete with scrim kraft jacket. Facing overlapping joints shall be at least 2 inches and held in place with outward clinching staples on approximately 4 inch centers.

9.2.0.4 Insulation for Ducts Exceeding 24 Inches in Diameter

Underside of ducts exceeding 24 inches in diameter shall be mechanically fastened.

This ductwork shall be UL 181, Class I, with rating to meet or exceed NFPA 90A-90B and reinforced with a perforated sheet metal inner jacket. Maximum length of flexible duct shall be 5 feet. Flexible duct shall be composed of an inner duct of woven and coated fiberglass permanently bonded to a spring steel helix with 6.0 R-value fiberglass insulation and fiberglass reinforced metalized film outer jacket.

9.2.1 HVAC Piping Insulation

All insulations, jackets, adhesives, coatings, vapor barrier mastics, etc., shall meet the requirements of NFPA Bulletin 90-A, ASTM E 84, and UL 723, and shall have a flame spread of 25 or less and smoke developed rating of 50 or less. For all piping systems except steam, fiberglass insulation and cellular glass insulation are required. Use of flexible cellular insulation must be approved by DFW Code.

All piping and vessels with a surface temperature less than ambient temperature shall have a vapor barrier covering. The vapor seal shall be continuous, unbroken, and adhere to surface so that the insulation is airtight in order to minimize the possibility of vapor draining into the insulation material.

Form fitted polyurethane insulation shall be used on all coil header piping to the extent necessary to include all valves, including flow control valves, and other appurtenances utilized to evaluate the performance of the coil.

9.2.1.1 Heat Tracing

Heat tracing for freeze protection shall use electric heating cables that are self-regulating and parallel resistance. The system shall be controlled by a heat trace system that provides operational data and system health back to the respective energy management control system. Heat Tracing shall connect to a GFI breaker at the panel.

9.2.1.1.1 Piping Exposed to the Elements

Piping exposed to the elements are to be both heat traced and insulated. Such piping shall also have a protective aluminum jacket.

9.2.1.1.2 Piping in Unconditioned Spaces & Inside Exterior Walls

Piping in unconditioned spaces and inside exterior walls shall be both heat traced and insulated.

9.2.1.2 Chilled Water Piping Insulation

Chilled water pipes shall be covered with minimum 2 inch thick fiberglass or cellular glass insulation. ASHRAE Standard 189.1 shall be followed if greater than 2 inch thickness occurs. Fittings shall be insulated with 2 inch thick pre-molded or shop-fabricated fiberglass, or cellular glass insulation with PVC fitting covers. Insulation on lines in concealed areas shall be jacketed with white reinforced foil-kraft vapor barrier.

Insulation on lines exposed in equipment rooms or in occupied areas shall be jacketed with pre-sized glass cloth vapor barrier jacket. Jacket laps and butt strips should be adhered with vapor barrier adhesive or position sealing system. Fittings shall be vapor sealed with vapor barrier mastic.

9.2.1.2.1 Insulation at Pipe Supports

Higher density insulating materials should be used at pipe supports, if required to prevent crushing/cutting of insulation.

9.2.4.2.2 Exterior Exposed Pipe Insulation

All exterior exposed pipes shall have aluminum metal jacket as specified below. Direct buried chilled water piping shall be pre-insulated with urethane foam, 2 inches thick.

9.2.1.3 Hot Water Piping Insulation

Follow description above for chilled water piping except that vapor barrier is required on direct buried pipe only.

9.2.1.4 Drain, Refrigeration Suction, and Chilled Water Piping 2 Inches and Smaller

Follow description above for chilled water piping except thickness shall be 1 inch.

9.2.1.5 Tunnel Piping

All chilled water, and hot water piping, in tunnels or exposed, shall have a smooth finish aluminum metal jacket on polyurethane insulation or small rib texture aluminum metal jacket on calcium silicate. Minimum jacket thickness shall be .016 inch.

9.2.1.6 Chilled and Hot Water Pump Insulation

Chilled and hot water pumps shall be insulated. Provide removable insulated enclosure over pump casing to facilitate pump maintenance.

9.3 Instrumentation and Control for HVAC**9.3.0 DDC Control System for HVAC BACnet Compatible Direct Digital Control (DDC)**

Controls shall be used to control all HVAC equipment in facilities with a total cooling capacity greater than 20 tons. The controls shall be interfaced with the Building Automation System (BAS) front-end system that is already in use at the Airport and must interface with the Enterprise Integrated Operation System (EIOS). Design of the DDC system must be coordinated with the Airport's BAS Administrator. Cooling systems that are dedicated to data or communications rooms will be connected to the HVAC control system for monitoring, alarms, and control. Section 2.2 Building Automation System should be consulted for additional information, including examples of other systems to be monitored.

9.3.0.1 Sequence of Operations for HVAC Controls

A clear Sequence of Operations (SOO) shall be provided by the design team in the design submittal. The record drawings shall reflect any changes made during construction and startup. A dedicated controls engineer must develop the SOO for any chilled water system over 200 tons.

In addition to industry best practices and code requirements, the following key points shall be included in the Sequence of Operations for HVAC controls:

1. Supply air temperature and pressure reset schedule according to outside air temperature system response to humidity conditions exceeding 60 percent relative humidity.
2. Nighttime and weekend setup/setback schedule shall be patterned to match the building usage for both terminal boxes and air- handler units.
3. Provision should be included to allow for a single point control of each AHU (for non-terminal box units, FCU, etc.) and Terminal Box systems (i.e. all the boxes served by an AHU) room temperature setpoints as well as another for all the systems in a building for load shedding in the event of usage reduction requests.
4. Coordination of the use of return fans. Turning the fan off when supply air pressure can be effectively maintained by the supply air fan only.
5. Areas that require dedicated cooling support, such as data or communications rooms, must provide constant (24/7/365) cooling.

9.3.0.2 Individual Space Control System for HVAC

Individual space control is desired for each individual totally enclosed office space or room. Individual Variable Air Volume (VAV) units or a two-position damper may be utilized which shall have independent controls.

9.3.0.3 Indoor Air Quality Monitoring

For new construction and major renovation projects, provide indoor air monitors for all of the following parameters in densely occupied spaces (greater than 250 people per 1,000 square feet):

8. Carbon Dioxide (CO₂)
9. Particulate Matter (PM_{2.5})
10. Total Volatile Organic Compounds (TVOC)
11. Temperature
12. Relative Humidity

Monitors must be building grade or better (for example Reset Air Accredited Monitors Grade B or UL 2905 grade B) and located between 3 and 6 feet above the floor.

9.4 HVAC Piping and Pumps

9.4.0 Hydronic Hot Water Piping

Hot water piping shall be black steel pipe ASTM A53 Grade B, seamless or welded, Schedule 40 with welded joints and must be in accordance with ASTM B 36.10M or ASTM B88 Type L. For pipe diameters greater than 12 inches, standard wall piping is acceptable.

9.4.0.1 Hot Water Piping - Inside

Hot water (inside) piping shall be steel or K copper.

9.4.0.2 Hot Water Piping - Underground

Underground hot water piping is not allowed without the approval of the Energy Transportation & Asset Management Department.

9.4.1 Chilled Water Piping

Chilled piping shall be black steel pipe ASTM A53 Grade B, seamless or welded, Schedule 40 with welded joints and must be in accordance with ASTM B 36.10M or ASTM B88 Type L. For pipe diameters greater than 12 inches, standard wall piping is acceptable.

9.4.1.1 Chilled Water Piping - Inside

Chilled water (inside) piping shall be steel or K or L copper.

9.4.1.2 Chilled Water Piping - Underground

Underground chilled water piping is not allowed without the approval of the Energy Transportation & Asset Management Department.

9.4.1.3 Chilled Water Piping from Central Plant

All control valves shall be two-way valves. For design purposes of new connections to the secondary loop, supply and return pressures must be obtained from the EP Manager.

Facilities requiring greater than this value will need to include booster pumps for the chilled water.

For new installations, heat exchangers are required on the chilled water service from the Energy Plaza secondary.

All components of the system cooling load shall be designed to return 62 degrees F water when the supply water temperature is 38 degrees F, and the load is operating at design flow rates. All loads shall be provided with design flow rate control. Manufacturers of flow rate control device: Griswold/Flow Design/Approved Equal. 40 degree supply with a 62 degree return is acceptable in applications where a heat exchanger is installed to hydronically separate the facility from the EP.

9.4.1.4 Chilled Water from ESP

Design for additional facilities attached to this system must include consideration of existing plant pressures and capacities. Modifications to the plant pumping system must be included in the design to maintain necessary flow rates to existing facilities. Manufacturer of maximum flow control device: Griswold/Approved Equal.

The chilled water supply temperature is reset on ambient air temperature ranging from 40°F to 53°F. Design loads should be based on this range of water temperature. Spaces with high loading, such as computer rooms, will require special design considerations for adequate cooling with 53°F chilled water supply temperatures. In certain instances, supplemental cooling for interior high load density areas may be necessary. All control valves shall be two-way valves, and all cooling coils shall be provided with a design flow rate control device.

Any new facilities planned for construction adjacent to East Airfield Dr. should be considered for chilled water service from the ESP. Conceptual design development must be coordinated with DFW stakeholders.

9.4.1.5 Hydronic Piping Fittings

All hydronic piping must be equipped with vent valves and drains in a proper position to prevent air pockets that could impact system performance. Pressure condition and sizing shall be noted in the design drawing.

Each branch of the hydronic piping should have appropriate isolation valves to minimize drainage during maintenance activities.

9.4.2 Hydronic Pumps

“Stand-by” pumps are required on most facilities, especially for facilities containing computer rooms.

Pumps that are connected to the EP or ESP chilled water loops must be outfitted with suction and discharge pressure transmitters that will tie into the respective control system. The control system must be interfaced to the pump controls to provide full control and visibility of the pump operation.

Booster pumps for chilled and hot water systems shall be controlled via tie-in to the EP Distributed Control System or the Airport enterprise control system. All requirements must be verified and approved by the BAS Administrator.

Impellers shall be one piece, hydraulically and statically balanced, and keyed to the shaft. Impeller size shall not be more than 90 percent of the maximum impeller size allowed for the pump casing.

Pumps shall be of a premium efficiency design. The pump motor assembly shall be mounted on a common steel or cast iron base.

Pump and motor bearings shall be grease-lubricated, complete with alemite fittings.

Variable volume pumping systems with variable frequency electric drives are required.

Provide pressure gauge taps with stop cocks and gauges on suction and discharge sides of pump. Provide thermometer wells on suction and discharge sides of pumps.

9.4.3 Aboveground Steam and Condensate Piping General Requirements

The existing system capacity must be checked before each addition. The location of heating water pump rooms must be considered when adding new heating units at the terminal buildings.

Hot water coil sizing shall be based on a minimum 35°F delta temperature at 160°F supply.

New heating water systems shall be variable pumping. Work in adjacent areas shall consider increasing existing coil capacity to minimize pumping requirements.

9.4.3.1 Condensate Flanges

Condensate flanges shall be ASTM A 105/A 105 M, forged, weld neck 150 pounds flat or raised faced, bored to Schedule 80, in accordance with ANSI B16.5.

9.4.3.2 Condensate Pumps and Receivers

Condensate units shall be of the duplex type with two bronzes, fitted, close coupled centrifugal pumps with pressure gauge taps with stop cocks on both suction and discharge sides. Maximum pump speed shall be 1,750 rotations per minute (rpm).

The receiver tank will be fabricated of cast iron or steel, as applicable, and equipped with required float switches and alternator to provide automatic alteration of pumps. The receiver shall be provided with condensate return, vent, overflow, and valved drain connections.

9.4.4 Refrigerant Piping

Refrigerant piping shall be Type ACR copper, capped and cleaned. Joints shall be made with silfoss and purged with nitrogen.

9.5 Hydronic Piping Testing and Cleaning Procedures

9.5.0 Hydronic Pipe Testing

All hydronic piping shall be hydrostatically tested prior to placing in service. The Contractor shall fill the system with water and hydrostatically test to a pressure 150 percent of the operating working pressure. During this process, all major equipment such as air handling units (AHU)s and pumps shall be isolated. Once the test pressure is achieved, it must maintain the test pressure for a duration of 4 hours after the hydrostatic pump is removed. The test is successful with a zero pressure drop for the duration of the test.

9.5.1 Hydronic Pipe Cleaning and Flushing

At the successful completion of the hydrostatic testing process, the system shall be flushed for approximately 2 hours at a rate of 7 feet per second to remove any construction debris. During the flushing process, make-up water shall be added at a rate equal to the flush water. A temporary trash pump shall be used and all system pumps and major equipment such as AHUs shall be isolated and bypassed.

Once the system is flushed, the Contractor shall refill, circulate, and add an approved chemical cleaner to the system at a concentration recommended by the chemical manufacturer. The system shall be allowed to circulate at a minimum rate of 7 feet per second for a duration of 24 hours. At the completion of the 24 hours, the Contractor shall flush the system with potable water.

The Energy Transportation & Asset Management department's Water Treatment Specialist or his/her designee shall deem the test satisfactory when the flush water has reached the water quality satisfactory when the flush water has reached the water quality standards to that of the potable water. If the test fails, the Contractor will be required to repeat the chemical cleaning process until required results are achieved.

The potable water parameters will be measured at the beginning of the test. The Operator shall submit documents detailing the design criteria utilized to ensure effluent from fire suppression systems can either be released into sanitary sewer systems or easily collected and not discharged into the storm water collection system during required maintenance and flushing activities. The Operator shall submit documents detailing the design criteria utilized to ensure effluent from fire suppression systems can either be released into sanitary sewer systems or easily collected and not discharged into the storm water collection system during required maintenance and flushing activities.

9.6 HVAC Air Distribution

9.6.0 Terminal Building HVAC System

The Terminal buildings shall be served by VAV AHUs with fan powered VAV terminals and hot water heating. Consider the use of parallel fan-powered terminal units at exterior zones and regular terminal units for interior zones.

For kitchens in terminals, ensure adequate air distribution is achieved. Provide supply and return air in volumes necessary to maintain and established setpoint, independent of make-up air for hoods and other points of exhaust.

9.6.0.1 Existing HVAC Systems

All existing VAV AHUs and equipment shall be verified to be in compliance with a project's design prior to construction.

9.6.1 Metal Ducts

All ductwork systems shall be constructed and installed in accordance with SMACNA and ASHRAE Standards and Guidelines.

Duct material shall be zinc-coated sheet steel or aluminum of the thickness of the metal and stiffeners as indicated in the SMACNA Manual. All ductwork installations shall meet SMACNA Class A duct sealing requirements. All medium pressure ductwork (equal to or greater than 2 inch w.g. design static pressure) shall be pressure tested in accordance with SMACNA standards, with measured leakage not to exceed 1 percent of design flow rate.

9.6.1.1 Flexible Connections

Wherever ductwork is connected to fans, AHUs, or other equipment that may cause vibration in the duct, the connection to the equipment shall be by means of a flexible connection constructed of fire-resistant flexible canvas or other approved material. The connection shall be suitable for the pressures at the point of installation.

9.6.1.2 Duct Insulation

All ducts, except general exhaust ducts, shall be externally insulated. Internal duct liner of any type shall not be used for any duct system, except for applications where the acoustic value of internal liners should be considered.

9.6.1.3 Underfloor and Crawl Space Duct

All ductwork installed below the floor in crawl spaces or below grade shall be constructed with watertight joints and shall be tested and proved tight before floors are poured. The under-floor duct system shall be constructed of fiberglass, PVC, or other approved non-metallic material.

Flexible ductwork shall comply with UL 181 Class 1 and shall meet or exceed NFPA 90A- 90B rating. Flexible duct shall be composed of an inner duct of woven and coated fiberglass permanently bonded to a spring steel helix with 6.0 R-value fiberglass insulation and fiberglass reinforced metalized film outer jacket. Maximum length of flexible duct shall be 5 feet.

9.6.1.4 Flexible Duct Connections to Grilles and Registers

All flexible duct connections to grilles and registers shall have a radius forming brace installed at the connection.

9.6.2 Variable Air Volume Units

The preferred system in the Terminal buildings is the use of fan-powered Variable Air Volume (VAV) box with a hot water coil. VAV boxes shall be double wall construction, exposed internal insulation shall not be allowed. The primary cold air damper will be closed to 0°F or to a minimum position of 30% as a default analysis value to maintain ventilation rates when cooling is no longer required. However, total air supply shall not be less than that required by currently adopted International Mechanical Code. The wall-mounted thermostat/sensor (all thermostats/sensors should be mounted on the walls) will control the volume damper and a floating or modulating hot water valve in sequence. For heating, the fan motor shall start and operate to satisfy the thermostat/sensor setting; upon a further need for heating, the floating or modulating hot water valve will open to satisfy the thermostat/sensor setting. When the thermostat/sensor is satisfied, the floating or modulating valve will close, and the fan motor will cycle off in sequence. There is a 2°F dead band between closing of cold air damper and the starting of the fan motor and another 2°F dead band before opening of control valve. A filter will be included on the inlet to the fan powered box to protect the heating coil and fan from dirt.

Consideration should be given to static pressure drop. Static pressure controls should be sufficient to allow boxes to operate with stability. The effect of the box addition on the remainder of the ducted system must be evaluated to ensure the required static pressure from the AHU is not impacted. All variable volume terminal units shall be equipped with at least a 5-diameter length of straight rigid duct immediately upstream of the volume control devices.

Terminal boxes located above finished ceilings will have adequate ceiling access panels or other means of access to box for maintenance and removal. Except for lift out ceiling installation, all access panels will be hinged.

9.6.3 Equipment Connected to Energy Plaza

The EP operating conditions will have a significant influence on the design of new or modified HVAC systems for buildings served by the EP. Variable seasonal Chilled Water System operating supply temperatures will have a significant influence on mechanical design of new HVAC systems.

9.6.4 Equipment Connected to East Side Plant

All buildings served with chilled water from the ESP should use a locally based boiler system. The boiler should be natural gas fired and meet the requirements of the currently adopted energy code. Boiler configuration and sizing should be such that the failure of one boiler will provide no less than 85% of the design peak load. The smaller of the boilers should be capable of a high turn-down ratio to permit efficient operation during short periods of light loading.

9.6.5 Heat Exchangers for HVAC

Stainless Steel Plate type heat exchangers shall be used.

9.7 Central HVAC Equipment

9.7.0 Indoor Central-Station Air Handling Units

AHUs shall adhere to the following requirements:

9.7.0.1 Variable Frequency Drives

Provide Variable Frequency Drive (VFD) with a hand operated Hand-Off-Auto (HOA) switch and remote-control terminations at each air handler location. VFD shall be provided with integral bypass. Provide a fused disconnect switch at each location.

Fan array type configuration shall be used for all VAV-type AHU in excess of 5,000 cubic feet per minute (CFM).

9.7.0.2 Drain Pan

Provide a double wall insulated stainless steel drain pan sloped in two or more planes to eliminate stagnant water with a drain line not less than $\frac{3}{4}$ inch in diameter or size of tap on drain pan. Use a plugged tee for all changes in direction rather than a 90-degree ell. Condensate shall be drained to the sanitary sewer.

9.7.0.3 AHU Casings

AHU casings shall be double-wall construction, galvanized steel with insulation between the walls. Under no circumstances will it be acceptable to have insulating material exposed to the air stream. The casing leakage shall not exceed 1 percent of total airflow at design static pressure.

9.7.0.4 Filters

Air handling equipment should be equipped with filters. High efficiency filters shall be provided on equipment over 152,000 CFM. High efficiency filters shall be MERV 13 rated cartridge type with MERV 7 rated, 2 inch thick prefilters. Use of UV-C lights is required where applicable.

9.7.0.5 Bearings

Field lubricated ball bearing equipment is preferred over sleeve bearings. All air handling equipment shall be selected and installed such that bearings can be replaced with minimum demolition of equipment or surrounding structures. Bearing lubrication points shall be extended to a central external accessible point and fitted with alemite fittings.

9.7.0.6 Electric Motor Speeds

Electric motor speeds in excess of 1,800 rpm are discouraged. Fans shall be selected for the highest efficiency; airfoil type is preferred. Fan wall technology should be considered and evaluated.

9.7.0.7 Chilled and Hot Water Coils

Chilled water and hot water coils shall be of the continuous copper tube type with copper or aluminum fins. Tubes shall be a minimum 0.5 inch outside diameter (O.D.) and a minimum 0.025 inch wall thickness. All coils shall be of the cleanable and drainable type. Each tube shall be accessible without piping disconnect. Headers shall be removable at the opposite end. All water coils shall have maximum flow rate control devices in the return line. Coil design shall be based on the following criteria:

1. 24°F minimum L.W.T. Cooling, 38°F supply
2. 35°F minimum delta T heating, 160°F supply
3. Air friction loss across coil shall not exceed 0.7 inch W.C. (Faces velocities below 450 feet per minute are desired).

9.7.0.8 Access Covers to Water Coils

Access covers to water coils on the AHU housing shall be readily removed for access without piping disconnect or demolition of surrounding structures.

9.7.0.9 Hinged Access Doors

Provide hinged access doors with gaskets to AHU sections with view windows. Provide marine lights in each AHU section with timer-driven light switches on outside of unit.

9.7.0.10 AHU Motor

Air handling units with a 7.5- horsepower motor or larger shall be selected with a keyway shaft and keyed fan hub. Standard size key and keyway are preferred.

9.7.0.11 Chilled Water & Hot Water Control Valves

Chilled water and hot water control valves shall be pressure independent type with flanged or set in unions for easy removal.

9.7.0.12 Coil Connections

All Hydronic piping supply piping shall connect to the unit at the leaving air side of the coil.

9.7.0.13 Ventilation Fans

Restrooms in terminals shall be designed to operate with a ventilation rate of 20 air changes per hour, which should be accounted for during the design phase.

9.8 Decentralized HVAC Equipment

9.8.0 Packages Air Conditioners

Where building use, type, and DFW stakeholders review justifies the use of package equipment, the following shall apply:

9.8.1 Roof Top System

Rooftop systems shall be completely self-contained, with factory wired controls and factory assembled components and piping. Equipment shall have 2 inch-thick pleated replaceable media filters. Compressors shall have a 5-year warranty, including parts and labor (5 tons and under). All curb mounted units shall be furnished with appropriate enclosed engineered acoustic curbs to provide for level unit mounting with minimum 16 inch curb height above top of the roof deck surface. An alternative is a structural steel support frame with a minimum of 40 inch clearance above the finished roof. All rooftop systems require approval from DFW Code.

9.8.2 Split System Air Conditioner

Split systems shall consist of furnace, coiling section plenum with direct expansion cooling coil, air-cooled condensing unit or heat pump, piping, controls, etc. All components shall be factory wired and assembled. Furnaces shall have filter rack complete with 1 inch-thick throw-away filters. Compressors shall have a 5-year warranty, including parts and labor (5 tons and under).

Split system air conditioners shall have a Seasonal Energy Efficiency Ratio (SEER) rating of 13 or meet the latest version of the IECC requirement, whichever is greater.

9.8.3 Fan Coil Units (Floor & Wall Mounted Equipment)

FCUs shall have a high-medium-low-off switch where adjustment can be made without removal of the access door or unit housing. This switch shall be easily accessible for room or area occupant's personal adjustment. FCUs shall be double-wall construction as described for Modular Indoor Central-Station AHUs. FCUs shall be equipped with replaceable pleated filters, minimum MERV 7.

The use of FCUs shall be limited to sizes below 2,000 CFM and have prior approval from DFW Code.

9.8.4 Chillers

9.8.4.1 Site Selection and Installation Location - Noise and Vibration Control:

The chiller must be installed in a location that minimizes the impact of noise and vibration. Design should incorporate sound attenuation measures to comply with DFW noise restrictions. This may include installing acoustic barriers or placing the chiller in locations that reduce sound transmission.

Ensure that anti-vibration mounts or isolators are installed to minimize mechanical vibrations that could affect the building or surrounding structures.

9.8.4.2. Maintenance Accessibility

Adequate Space for Maintenance: Ensure adequate space for maintenance and future repairs, including clearances around the chiller for servicing and airflow requirements. The location should provide easy access for maintenance personnel.

Spare Parts Management: Recommend the inventory list of critical spare parts to have quick access to replacements to minimize downtime during repairs or part replacements.

Flange and Tee consideration: Consider the installation of a Tee with a flange into the hydronic line near the chiller. This is intended to facilitate easier connection of a temporary unit for future maintenance or replacement work, and accessibility must also be taken into account.

9.8.4.3. Space and Sizing Requirements

Adequate Space: Large air-cooled chillers require significant space, including both the physical footprint of the equipment and the space for proper airflow. Ensure the area is sufficient to avoid heat buildup, which can reduce chiller efficiency.

Maintenance Access: Allocate space for regular maintenance activities such as coil cleaning, compressor checks, and part replacements. Design should allow for ease of access to key components.

9.8.4.4. Power and Electrical Supply and code compliance

Power Capacity: Verify that the building's electrical system can support the high power demands of a large-capacity chiller. Electrical load calculations must be performed, and upgrades to transformers, switchgear, or panels may be necessary.

Electrical Codes: Ensure compliance with the National Electrical Code (NEC) for chiller installation, including proper grounding, wiring, and circuit protection.

9.8.4.5. Condensate and Drainage Environmental Considerations:

Check for compliance with water management regulations to prevent runoff from affecting nearby properties or water sources.

9.8.4.6. Energy Efficiency and Environmental Compliance

Energy Efficiency Standards: Large chillers must comply with energy efficiency standards, such as those outlined by ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers). Consider selecting Energy Star-certified chillers to ensure maximum energy savings and potential tax credits.

Refrigerant Compliance and Environmental Regulations: If the chiller uses refrigerants, ensure compliance with the Environmental Protection Agency (EPA) guidelines on refrigerant management, specifically regarding the type of refrigerant used (e.g., CFC, HFC) and leak detection requirements.

9.8.4.7. Low-Temperature Operation

Freeze Protection: Implement freeze protection strategies such as the use of glycol mixtures or electric heat strip for pipes, and critical components prone to freezing.

Cold Start Mechanisms: Install delayed start mechanisms or pre-heating systems for compressors and pumps to prevent damage or malfunction during low-temperature start-ups.

Insulation: Ensure proper thermal insulation of piping and other exposed components to minimize heat loss and protect against freezing.

9.8.4.8. Humidity and Corrosion Prevention

Corrosion-Resistant Materials: Utilize corrosion-resistant materials, such as stainless steel or aluminum, for external components, and consider applying anti-corrosive coatings to further enhance durability.

9.8.4.9. Hail Protection

Protective Louvers and Screens: Incorporate protective louvers, grills, or hail guards over the condenser coils and other vulnerable areas to prevent direct impact from hail.

Durable Housing: Design the exterior housing and fan guards with robust, impact-resistant materials that can withstand hail strikes without compromising system performance.

9.8.5 Cooling Towers**9.8.5.1 Site Selection and Installation Location**

Adequate Ventilation: Cooling towers must be installed in a location with sufficient airflow to ensure optimal performance. Ensure adequate spacing around the tower to prevent restricted airflow and performance degradation.

Noise and Vibration: Cooling towers can generate noise and vibrations, which should be mitigated with sound barriers and anti-vibration mounts to minimize the impact on surrounding structures.

Structural Stability: Cooling towers must be securely installed, taking into account the structural load of the building or installation platform. The design should also consider external forces such as earthquakes or strong winds.

9.8.5.2 Power and Electrical Systems

Power Capacity: Ensure the electrical system has enough capacity to support the power demands of the cooling tower and accessories. If necessary, upgrade the electrical infrastructure.

Electrical Code Compliance: The installation must comply with the National Electrical Code (NEC) for proper grounding, wiring, and circuit protection.

9.8.5.3 Water Management and Drainage

Water Quality Management: Implement proper water treatment systems to prevent scaling, corrosion, and biological growth within the cooling tower. Consider the room for the regular essential chemical treatment to maintain water quality.

Drainage System: Design the drainage system to handle condensate and runoff safely, ensuring compliance with environmental regulations and minimizing the impact on surrounding areas.

9.8.5.4 Energy Efficiency

Energy Efficiency Standards: Ensure the cooling tower and its accessories meet ASHRAE and other relevant energy efficiency standards. Selecting high-efficiency equipment can reduce operating costs.

Operational Optimization: Install automatic control systems to monitor and adjust the cooling tower's operation, ensuring optimal performance and energy use.

9.8.5.5 Low-Temperature and Weather Conditions

Freeze Protection: Installation of freeze protection devices such as heat tracing or glycol mixtures in pipes and system components to prevent freezing during winter months shall be considered.

Hail and Wind Protection: Design consideration to equip cooling towers with durable exterior materials and protective guards to withstand hail and high winds, minimizing weather-related damage is necessary.

9.8.5.6 Maintenance Accessibility

Adequate Space for Maintenance: Ensure sufficient space around the cooling tower and accessories for routine inspections and maintenance. Key components such as fans, pumps, and heat exchangers should be easily accessible.

Spare Parts Management: Recommend the inventory list of critical spare parts to have quick access to replacements to minimize downtime during repairs or part replacements.

9.8.5.7 Corrosion Prevention and Durability

Corrosion Prevention: Apply anti-corrosion coatings and use corrosion-resistant materials (e.g., stainless steel, aluminum) to protect the cooling tower from direct exposure to harsh environment.

Durability Enhancement: Use durable, weather-resistant materials for the cooling tower exterior and install protective grilles or covers to guard against environmental damage.

9.8.5.8 Safety Compliance

Safety Standards: Comply with OSHA and other safety regulations during the installation of cooling towers, ensuring that proper safety measures are in place for current and future workers.

9.8.6 Pumps**9.8.6.1. Site Selection**

Optimal Location: The pump shall be installed in a location that minimizes noise and vibration, thereby reducing the impact on nearby facilities or residential areas.

Optimized Piping Layout: The installation location should allow for short and direct piping routes to maximize efficiency and prevent performance degradation.

Maintenance Space: Adequate space must be provided around the pump and its accessories to facilitate regular maintenance and repairs. The area should remain free of obstructions for easy access.

9.8.6.2. Noise and Vibration Control

Noise Reduction: Noise mitigation measures, such as soundproofing panels or insulation, should be considered to minimize operational noise. Installing noise-dampening equipment around the pump is also recommended.

Vibration Control: Anti-vibration mounts or vibration isolation devices must be installed to reduce the impact of pump vibrations on nearby structures and equipment.

9.8.6.3. Fluid Flow and Piping Design

Flow Optimization: The piping design must account for the pump's performance characteristics and ensure proper flow conditions. Pipe sizing should be appropriate to maintain the desired flow rate and operational efficiency.

Material Selection: Piping materials should be selected based on the fluid and operating conditions, with consideration for resistance to corrosion, scaling, and other potential issues.

9.8.6.4. System Balancing and Pressure Control

System Balancing: The system design must include control valves and balancing devices to maintain consistent flow rates and ensure proper system balance.

Pressure Control Devices: Pressure control devices and monitoring equipment should be considered to maintain stable system pressure and prevent issues such as over-pressurization or improper flow.

9.9 Commercial Kitchens**9.9.0 Commercial Kitchen Appliances**

All gas and electric ovens including but not limited to standard, bake, roasting, revolving, retherm, convention, combination convention/steamer, countertop conveyORIZED baking/finishing, deck and pastry shall be considered Medium-Duty Cooking Appliance.

9.9.1 Type I Hoods

Type I hoods shall be installed where cooking appliances produce grease or smoke as a result of the cooking process and shall be installed over cooking appliances in a commercial kitchen.

Type I Hoods shall be installed over electric cooking appliances, light-duty cooking appliances, medium-duty cooking appliances, heavy-duty cooking appliances and extra-heavy-duty cooking appliances.

Commercial kitchen ventless hood system(s) shall not be allowed.

Chapter 10 Electrical

10.0 General Information

10.0.1 General Requirements

Reference Chapter 3 Off-Site Construction for planning design, fabrication, inspection, regulatory compliance, and delivery requirements of infrastructure that will be manufactured offsite.

10.0.2 Scope

This Chapter defines general design criteria that apply to the design of electrical systems at DFW. Chapter 1 should be consulted for specific Airport regulations and standards that also apply. The electrical design criteria in this section are organized into three distinct chapters: interior electrical, exterior electrical, and airfield electrical. Electrical design should comply with the National Electrical Code (NEC) as amended by the Airport Board.

10.0.3 Short Circuit Studies

The maximum available short-circuit current should be calculated for the different system configurations (including open and closed transition, from normal and alternative power sources) at each bus, and the results used to verify the interrupting ratings of the overcurrent protective devices, complying with Institute of Electrical and Electronics Engineers (IEEE) guidance IEEE3004. Summarize the results from the calculations in the Equipment Evaluation Report. The results of the study are to be implemented within the design. All calculations shall be provided to the Owner in a file format compatible with Electrical Transient Analyzer Program (ETAP) software. Electrical One-Line diagram is required to be identified open or closed transition operation.

10.0.4 Predictive Device Coordination

Study Perform coordination study for all the protective devices in the scope of the project, using the results of the fault study, and complying with IEEE3002. Prepare a written report with all findings and recommendations. Complete a table with the recommended settings for all protective devices within the scope of work.

10.0.5 ARC Flash Study

Perform an arc flash analysis per the requirements of the IEEE 1584 equations that are presented in NFPA70E. The Arc-Flash Hazard Analysis shall include all significant locations in 240-volt and 208-volt systems fed from transformers equal to or greater than 125 kilovolt amps (kVA) where work could be performed on energized parts. The short-circuit calculations and the corresponding incident energy calculations for multiple system scenarios must be compared and the greatest incident energy must be uniquely reported for each equipment location. Calculations must be performed to represent the maximum and minimum contributions of fault current magnitude for all normal and emergency operating conditions. Use the worst-case scenario for the equipment labeling. The labels shall include the following information at a minimum: Location Designation, Nominal Voltage, Flash Protection Boundary, Hazard Risk Category, Incident Energy, Working Distance, Required Personal Protective Equipment (PPE), and Issued Date. All calculations shall be provided to the Owner in a file format compatible with ETAP software.

10.0.6 Preservation of Power Quality

All low voltage power and distribution system designs of facilities with computers and other sensitive electronic loads shall comply with IEEE3002. Supporting documentation to be provided by contractor.

10.0.7 Harmonic Current Limits

The harmonic current distortion of any individual device or piece of equipment specified in the design documents shall not exceed values given in IEEE3002.8. All calculations shall be provided to the Owner in a file format compatible with ETAP software.

10.0.8 Utilities

The Airport's 25-kilovolt (kV), three-phase, medium voltage distribution system is owned and maintained by Oncor Electric. The Designer shall confer with the DFW Code and Oncor Electric to determine service arrangement for each project.

10.0.8.1 Redundant Services

Two independent electrical power sources shall be used for all critical use facilities. Redundant services are available from the Airport electrical distribution system and are normally provided for terminal buildings, FAA buildings, emergency facilities, crash/fire/rescue stations, APM, the Central Plant, and other critical use facilities. Other alternate emergency sources may be considered for true uninterruptible power supply. Terminal building service entrance switchgear is double-ended drawout-air-circuit-breaker-type. Transformers, main breakers, and bus on each side of the tie breaker are to be sized to serve the total service load. The Airport utility service provider provides the main services switches and transfer switch (circuit breakers). The Airport provides all distribution switchgear from this point to the building electrical systems.

Buildings not consisting of critical loads shall be served with single radial feed. The Designer should verify service type with Airport Maintenance Engineering.

10.0.8.1.1 Uninterruptible Power System

A single or three-phase UPS shall be provided where it is required, to provide continuous, regulated alternating current (AC) power to the critical loads under normal and abnormal conditions, including loss of the utility AC power. The UPS shall contain a full rated input rectifier, output inverter, battery charger, bypass static switch, maintenance free replaceable battery plant, liquid crystal display (LCD) interface and controls, options for remote monitoring, and maintenance bypass circuit to isolate UPS for safety maintenance, without disruptions to the critical loads. Comply with the current editions of UL1778 (UPSStandard) and NFPA70 (NEC).

10.0.8.1.2 Generators

Where standby generators are provided, they shall comply with Construction Specifications Institute (CSI) standards. Provide automatic transfer switch as part of design. Manual transfer switches shall be approved by DFW Code.

10.0.8.2 Sub-metering

New construction and major renovation should install electrical submetering to record and monitor electrical load (kW) and usage (kWh) for each of the following:

1. Electrical distribution panelboards
 - a. HVAC systems, including chillers, fans, heaters, cooling towers, package units and circulation pumps
 - b. Lighting - Interior and Exterior
 - c. Non-HVAC loads rated at 25 kVA or greater
 - d. Motor control centers
2. Domestic and service water pumps
3. Receptacle circuits
4. Refrigeration systems
5. Electrical vehicle charging stations

6. Renewable energy systems
7. Baggage handling systems and other airport process equipment rated above 25 kVA
8. Tenant spaces

Location of submeters shall be reviewed and approved by ETAM.

*This requirement supports LEEDv4/5 Energy Metering and Reporting.

10.0.9 Electrical Distribution

All electrical distribution shall be underground unless otherwise authorized by DFW.

10.0.10 Enclosures

Provide all ductwork, equipment pads, and metering enclosures in accordance with Oncor Electric specifications and standards. Provide issued address label on all meter enclosures.

10.0.11 Protection of New and Existing Electrical Equipment

All new and existing transformers, Oncor switch gear and switches, and DFW Switch gear and electrical service equipment within the work zone as shown on the construction documents shall be protected from physical damage during construction by installing orange safety fence around the equipment. Orange safety fence shall be installed no closer than 10'-0" from the equipment and shall encircle the entire equipment.

A safety fence shall be installed before any work is performed in the work zone.

Engineer of record to include this information on all drawings for DFW Airport Electrical Infrastructure.

10.0.12 Unused Electrical

Unused electrical or equipment designed for "Future" is not permitted. All electrical equipment must have an associated load for the equipment being installed. Electrical Service design can allow for future expansion.

10.0.13 Electric Fences

Electric fences are not permitted on DFW Airport Property.

10.0.14 Energized Electrical Work

DFW International Airport has a "No Energized Electrical Hot Work Policy".

All electrical work performed at DFW International Airport shall be placed in an electrically safe/ secure state prior to any electrical work being performed.

10.1 Identification for Electrical Systems

10.1.1 Nameplates and Labels

Basis of Design requirements for the terminal in which the work is occurring for labeling of the switchboards, panelboards, transformers, transfer switches, and safety switches. The nameplate to include voltage, equipment identification, electrical room feed from, switchboard/ panelboard feed from, circuit numbers feeding equipment.

10.1.2 Wire Markers

All ungrounded conductors shall be identified for each electrical circuit using conductor (wire) markers.

10.1.3 Conduit Markers

All Conduits 2" and larger shall be labeled every 50 ft. and after entering another room.
Color code for labels is as follows:

Table 28 - Color Code Requirement for Wire Labeling

Type	Label Color	Legend	Spacing
480V System-Conduit	Yellow	BLACK LETTERS "277/480V"	50' OC
480V System	Yellow w/black letters	Tag conduit with origination/destination location & contract #	Origination/Termination Points & Junction Boxes
208V System	Blue	WHITE LETTERS "120/208V"	50' OC
208V System	Blue w/white letters	Tag conduit with origination /destination location & contract #	Origination/Termination Points & Junction Boxes
FA System – Conduit	White w/red letters	Red Conduit/Boxes	25' OC
LV Conduit – Pathways for Communication Systems	Orange w/black letters	Tag "COMMUNICATIONS" & origination/destination location & Contract Number	50' OC
LV System	Orange w/black letters	Tag "COMMUNICATIONS" & origination/destination location & Contract Number	Termination Points & (Junction Boxes)
DFW Cable Tray Pathway/Cable Tray Communication Systems	Orange w/black letters	"DFW-COMM-ITS" & Contract Number	50' OC
All Other Cable Tray Pathway/Cable Tray Communication Systems	Orange w/black letters	"COMMUNICATIONS" & Contract Number	50' OC
MATV – Conduit	Orange w/black letters	Follow LV Conduit Guidelines	50' OC
MATV System	Orange w/black letters	Follow LV System Guidelines	50' OC
Safety Radio DAS System	Orange w/black letters	"PS-DAS" & Contract Number	50' OC

10.2 ONCOR/DFW Duct Bank and Service Entrance Requirement

10.2.1 Service Entrance

Unprotected secondary system from the ONCOR transformer to the building or structure will require concrete encasement if the location of the transformer from the building or structure exceeds 50 ft. or is subject to possible damage during future excavations.

10.2.2 Primary (25kv) Duct Bank

All 25-kV concrete encased duct bank is owned and maintained by DFW; 25-kV cabling system and controls are owned and maintained by ONCOR Electric.

See DFW Airport Concrete Encased Electrical Duct Bank Detail (Exhibit 1) This detail shall be included in IFC DFW Code approved plans. This duct bank detail applies to all concrete encased electrical duct banks installed at DFW Airport. This duct bank detail shall be used for all conduit sizes.

10.2.3 Utility Boring Requirements

All Bores for DFW Electrical Utility Infrastructure shall be a minimum of 5'-0" deep and shall be installed in steel casing. Grouting of the bore is at the discretion of Oncor.

10.3 Medium Voltage Electrical Distribution (600 Volts To 5000 Volts)

10.3.1 Manholes and Handholes

Manholes and handholes shall be provided with heavy duty traffic covers. All circuits shall be labeled with stamped brass tags. Manufacturer of heat-sealing tags: Scotch.

10.3.2 Medium Voltage Open Conductors

Conductors shall be copper with insulation suitable for wet locations, sized based on NEC recommended maximum allowable voltage drops.

10.3.3 Medium Voltage, Single and Multi-Conductor Cables

Direct burial cable is not permitted. All cables shall be placed in conduit (PVC Schedule 40 as minimum). Bends in underground conduit system are not permitted without a junction box. All cables shall be tagged in manholes with heat sealing tags, Scotch HB-21 or approved equal, imprinted with the circuit number on each side of L- 823 connectors. 5-KV series circuit cables shall be color coded as to circuit type (i.e., Runway CKT, Taxiway CKT, Sign CKT, etc.). Verify color coding requirements with the Airport ETAM Department.

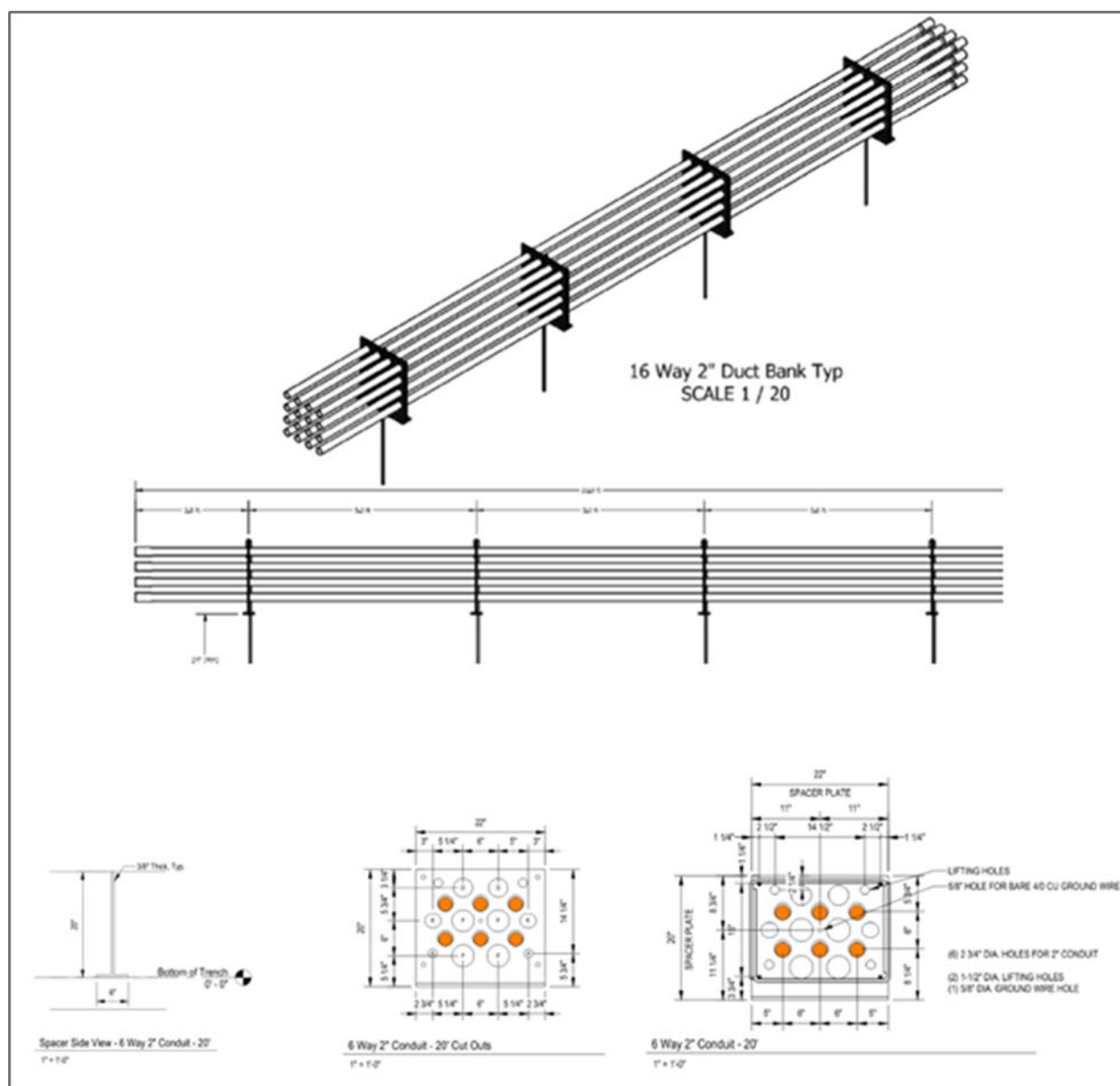


Figure 13 - DFW Airport Concrete Encased Electrical Duct Bank Pre-Fabrication

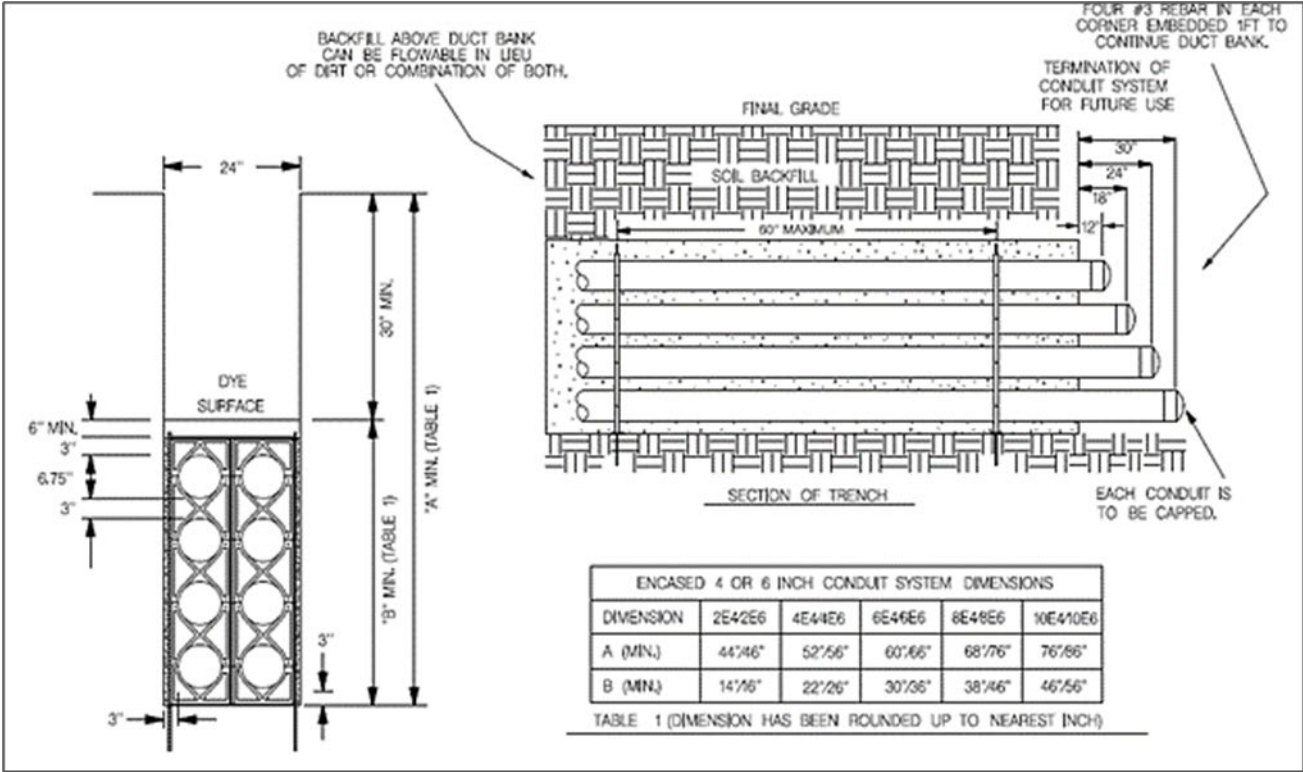


Figure 14 - Primary Concrete Encased Electrical Duct Bank Detail

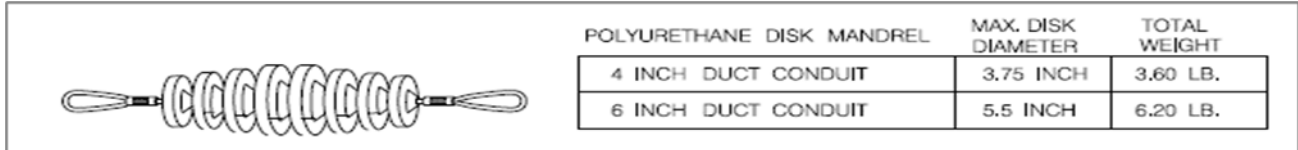


Figure 15 - Trench View of 2WX4H Conduit System

Notes:

- 1. All primary electrical duct banks must be field inspected by DFW Electrical Code and Oncor prior to concrete placement.
- 2. Conduit for concrete encasements is to be solid wall PVC SCH-40 and shall not be cellular core PVC product.
- 3. All concrete of the encasement is to have a compression test strength of 3000 PSI at twenty eight days. Use of a concrete vibrator must be used during concrete placement.
- 4. Install duct spacers "Carlson (Snap-Loc) Type" or equal, every 5 feet along the entire length of the duct. DFW Code preapproved sheet type spacers may substitute the Carlson spacers. Sheet spacer must have additional holes to allow for concrete flow providing for monolithic concrete placement. All dimensions shown on this detail must be compiled with prefabrication in lengths not to exceed 20'(ft)-0" (in) is acceptable.
- 5. The duct spacers are to provide 3 inches of vertical and horizontal separation between conduits.
- 6. The duct line shall be secured to earth at each spacer location prior to pouring concrete to prevent the conduit from floating or buckling.

7. All conduit/duct spans should slope toward adjacent manholes to provide drainage to a pumpable location.
8. All backfill of a conduit/duct trench shall be replaced to a min compaction of 95%. Flowable backfill is acceptable in lieu of dirt.
9. Red and/or blue powder concrete dye or concrete paint is to be placed on top of the “pattern finished” encasement. Place red dye on “all” 25KV primary duct banks. Place blue dye on secondary duct banks where applicable.
10. Conduit ends for incomplete duct lines (stubs) are to remain exposed from the encasement for future retrieval. The ends shall be capped watertight.
11. Each conduit of concrete encased duct shall be inspected by pulling a disk mandrel of proper size thru its entire length as soon as possible after encasement to verify integrity.
12. Each 4 inch and 6 inch conduit shall have mule tape installed (see chart); except where energized equipment is connected.
13. Concrete encase with a minimum of 3 inches of concrete on all sides and a minimum of 6 inches on top.
14. SCH-40 PVC bends are allowed with minimum of 36 inch radius.
15. When entering flat surfaces of pull boxes and manholes. Dowel rebar ends at minimum 3 inch, if possible, into sides to prevent shear points.
16. For continuation of duct banks, provide 45 degrees slope for the cold joint to prevent shear points.
17. Add four #3 rebar, one in each corner, embedded minimum 1 foot on all cured duct banks if altering or continuation from existing stub.
18. Conduit joints shall be staggered at 6 inch and constructed so that a mandrel can be pulled through its entire length.
19. Allow a minimum of 12 hours before backfilling concrete encased duct banks (CED).
20. Oncor comms – some concrete encased duct banks (CED) may require 2-3 inch conduits.
21. All 25KV civil infrastructure is owned and maintained by DFW Airport.

APPROVED BY ONCOR

10.4 Low-Voltage Electrical Distribution (0 Volts To 600 Volts)

10.4.1 Switchboards and Panel Boards

Panelboard/ Switchboard main bus may be copper or aluminum.

Panelboards/Switchboards shall have conductor color code identification. Panelboard circuit breakers shall be bolt on type. Stab-in type panelboards are prohibited.

Fused switches shall utilize current limiting fuses with “rejection” type pins where applicable. A spare fuse panel and a set of three spare fuses of each size used on a project will be furnished.

Freestanding switchboard construction shall be specified for bus sizes larger than 1,200 amp. Freestanding equipment shall be installed on a concrete pad.

All new electrical panels shall have Surge Protector Devices (SPD), which are designed to protect electrical devices from voltage spikes, shall be installed in accordance with the project specifications.

Power Meters, which measure the energy consumption of the electrical panels, shall be installed on all electrical panels and connected to the Building Automation System (BAS) to provide real-time data for the digital twin initiatives.

Mains Lug-only panels shall be permitted if both subpanel and main panel are in the same room.

Switchgears:

The manufacturer of this equipment shall have produced similar electrical equipment for a minimum of twenty (20) years. When requested by the Engineer, an acceptable list of installations with similar equipment shall be provided to demonstrate compliance with this requirement.

Manufacturers: EATON, Siemens, ABB, Schneider.

The continuous thermal monitoring solution, when integrated with the Building Automation System (BAS), in alignment with digital twin initiatives, ensures real-time detection of potential overheating, thereby enhancing system reliability and providing reassurance to the stakeholders.

The switchgear's arc rating shall be Type 2B as defined by IEEE/ANSI C37.20.7, such that the operator is protected around the entire perimeter of the equipment with the low-voltage control, instrumentation, and breaker secondary customer termination compartment doors open.

Breakers shall be provided in drawout configuration, and all breaker cell sizes shall have a common height and depth, ensuring the flexibility and ease of maintenance. Breaker frames of the same size shall be fully interchangeable, providing confidence in the system's adaptability.

The breaker shall be ready to accept the connection of a remote racking device without modification of the breaker, cell, or door.

Switchgear shall have a safety shutter in the cell when the circuit breaker is withdrawn, which automatically covers the line, and load stabs and protects against incidental contact. Provide a key-lockable breaker door to prevent access to the shutter when the breaker is removed from the cell.

The circuit breaker door design shall allow the following functions to be performed without opening the door: levering the circuit breaker between positions, operating the manual charging system, closing and opening the circuit breaker, examining and adjusting the trip unit, and reading the circuit breaker rating nameplate.

The electronic trip unit shall be capable of monitoring the following data:

1. Instantaneous, minimum, and maximum phase, neutral, and ground current values.
2. Instantaneous, minimum, and maximum of line-to-line voltage and line-to-neutral voltage.
3. Power metering, including demand power and power factor.
4. Energy metering.

An infrared window shall be installed in the switchgear, allowing noninvasive, real-time inspection of the switchgear's internal components. The IR window needs to be compliant with UL 1558 and must be positioned to allow full visibility of all critical components, including but not limited to Circuit Breakers, Busbars, and connections.

Required Documents for Electrical Plan Review:

DFW International Airport shall receive a single line (or one line diagram) with all the information's below:

1. A detailed legend
2. Identification of equipment
3. Incoming service voltage
4. Transformer ratings
 - a. Voltage ratio
 - b. Impedance
 - c. Winding connections
5. Feeder cable phase, neutral, and ground sizes
6. Conductor material and conduit size and type
7. Switchgear, switchboards, MCCs, fuses, panelboards, circuit breakers, automatic transfer switches, and continuous current ratings
8. Protective relays with device numbers, and CTs with associated ratios
9. Equipment rated current.
10. Bus voltage, frequency, phases, and short circuit current withstand ratings.
11. Cable number, lengths, and sizes
12. Transformer connection type, voltages, kVA, and impedance
13. Generator voltage and kW
14. Motor HP
15. Current and voltage ratios of instrument transformers
16. Relay device numbers.

Single line diagram needs to be updated when:

1. A new installation in the facility
2. A modification to the existing system
3. Change in utility or source.
4. Change in system configuration, and loading.
5. Change in protection devices or settings.

Every circuit needs to be identified on the lighting and power plans.

Details needs to be included on how the panels or electrical items will be mounted.

Existing equipment details need to be included.

10.5 Airfield Lighting, Signs, and Equipment

10.5.1 Saw Kerfs

Sawkerfs – (Shallow Groove Pavement Cutting) are permitted from terminal building or terminal building drip edge not to exceed the end of the jet bridge. Only one, 1-inch PVC coated RMC or IMC conduit per saw kerf is allowed. Saw kerfs shall not exceed 5-inch depth. Minimum coverage of 3 inches above conduit is required. Sawkerfs in locations other than described above are prohibited.

10.5.2 Airfield Lighting

Runway and taxiway lighting and visual aids shall be designed based on the design standards and requirements of the most current FAA Advisory Circulars, as appropriately supplemented by the National Electrical Code as it pertains to DFW International Airport vault work and the power load side of the vault equipment.

System layout configuration and fixture utilization and design shall be specified by all current applicable FAA Advisory Circulars.

Airport operated lighting systems shall be designed for the most critical operational criteria (CAT II/III).

10.5.3 Standard Lock Out Procedures

Designer shall instruct Contractor that no construction or maintenance shall be allowed on Airfield lighting circuits while the circuits are energized. The published circuit “Lock-out and Lock-in” procedures will be required to be followed for all work performed on the airfield. Designer shall instruct contractor to confirm with DFW Operations Department the need for temporary circuits for extended outages (more than one work period, 8 to 12 hours). Standard electrical circuit lockout procedures are available from Airport Energy, Transportation and Asset Management (ETAM) Electrical Shop.

10.5.4 Fixtures

In all specifications for lighting and electrical equipment, the item and manufacturer must be listed as approved for use by the FAA and be UL rated. In some instances, requirements above those required for FAA approval will be stipulated. These must be specified precisely; for example, “Signs (L- 858) – To withstand high wind velocity up to 200 mph regardless of location.”

Fixtures and lamps shall be specified to match existing equipment whenever possible. Fixtures shall be installed on deep bases (cans) housing the isolation transformers. Inset (shallow) bases shall not be. Runway Distance Remaining Signs and Taxiway Guidance Signs shall be internally lighted to match existing equipment.

Minimum coverage for wet and dry bores for duct bank installation is 10 feet under taxiways and runways.

10.5.5 Counterpoise

All direct buried counterpoise group wire used at the Airport shall be #6 solid, tinned, soft drawn, copper wire conforming to ASTM #B- 33.

10.5.6 Ground Rods

All ground rods on DFW Airfield Operations Area shall be ¾ inch by 10-foot stainless steel.

10.5.7 Junction Can Plazas, Junction Boxes, and Pull Boxes

These structures shall be located outside of runway and taxiway safety areas (as defined in AC 150/5300-13) where DFW Maintenance personnel can service them without closing runways or taxiways. Structures shall sufficiently be raised above the surrounding grade to prevent ponding water on the structure and the top cover sloped to drain. A concrete apron shall be constructed around all electrical manholes located in turfed areas. No manholes are allowed.

Particular attention shall be given to stormwater drainage plans to prevent placement of electrical structures in areas channeled for drainage. These structures shall have all joints and openings completely sealed and vermin-proof with secure covers with bolts.

Structures, covers, and frames in the runway and taxiway safety areas shall be heavy duty designed for aircraft at 250 psi tire pressures and wheel loads of at least 40,000 pounds. Homerun manholes and pull boxes shall be located at a maximum spacing of 500 feet.

10.5.8 Abandoned Conduit

Fill abandoned conduit with Polywater FST-250 seal Kit. Contractor shall identify all abandoned conduit on as built (redlines) drawings.

10.6 Conduits

Conduit shall be galvanized rigid metal conduit (RMC) or intermediate metal conduit (IMC) steel, except as outlined below.

All RMC and IMC conduit couplings shall be threaded type only.

Minimum conduit size shall be $\frac{3}{4}$ inch for home runs and feeders and $\frac{1}{2}$ inch for branch runs.

10.6.1 Electrical Metallic Tubing

EMT may be used where permitted by NEC. Use RMC or IMC where subject to severe mechanical or physical damage 6 feet or less above floor, if exposed. EMT fittings shall be steel compression or steel set screw type.

10.6.2 Direct Buried Metallic Conduit

Direct buried metallic conduits require PVC coating or approved equal coating to protect against corrosion.

10.6.3 PVC Conduit

PVC conduit shall be minimum Schedule 40 or thicker, and may be used underground, direct buried, or concrete encased. Conduit bends shall be IMC, PVC coated.

PVC 90 Degree bends are permitted for conduits 2 inches or smaller, with runs less than 150 feet long, installed with no more than two 90° bends in the conduit run.

10.6.4 Flex Conduit

Steel flex or liquid-tight flex (for damp or wet locations) shall be used for connection to all movable, rotating, or vibrating equipment, including dry type transformers. Flex connection shall not be used as an equipment grounding conductor.

Flex metal or MC type cable, where concealed by walls, floors, or ceilings is acceptable wiring method installed per NEC 330 requirements. MC Cable may be installed exposed were used for fixture whips to lighting fixtures only, not to exceed 10'-0".

MC Cable shall be installed in a neat and workmanship like manner per NEC 110.12.

MC cable is not permitted as an exposed wiring method other than noted in this Section.

When crossing building expansion joints, seal-tite (3-foot minimum), may be used to span the expansion joint. Pre-approved seal-tite design is used in all parking garages to allow for vertical and horizontal movement. Flexible Metal Conduit will not be allowed.

10.6.5 Underground Ducts and Raceways for Electrical Systems

Underground electrical circuits shall be installed in Schedule 40 PVC. DFW Concrete Encased Electrical Duct Bank Detail shall be used for all electrical duct banks installed at DFW Airport. Dyed concrete only applies to primary and secondary electrical systems. All other systems shall not have dye placed on top of the concrete.

10.7 Low Voltage Cables (0 Volts to 600 Volts)

10.7.1 Low Voltage Open Conductors

Copper shall be used for all wiring, aluminum wire 1/0 and larger may be used for temporary service conductors.

Exception: Aluminum is permitted for exterior pole mounted fixtures.

Minimum size #12 American wire gauge (AWG) conductors. WAGO type wire connectors are not permitted.

Total voltage drop shall be less than 5%. Limit feeder voltage drop to less than 2% and branch circuit voltage drop to less than 3%.

Control wiring shall be stranded. WAGO type wire connectors are not permitted.

10.7.2 Exterior Lighting

All exterior lighting circuits serving high intensity discharge (HID) lighting fixtures shall be limited to a loading of 60 percent or less of the "Full Load" capacity of the circuit (including circuit breakers, switches, relays, etc.). HID fixtures include high pressure sodium, metal halide, and mercury vapor lamps.

10.7.3 Counterpoise

All direct buried counterpoise group wire used at the Airport shall be #6 solid, tinned, soft drawn, copper wire conforming to ASTM #B- 33.

10.7.4 Interior Power Wiring

Interior power wiring for power and lighting shall be color coded as follows:

Table 29 - Color Code Requirement for Power and Lighting Wire

480Y/277V, 3□, 4W	208Y/120V, 3□, 4W	240Y/120V, 1□, 3W
A□ - Brown	A□ - Black	A□ - Black
B□ - Orange	B□ - Red	C□ - Red
C□ - Yellow	C□ - Blue	N - White
N - Gray	N - White	Grnd - Green
Grnd - Green	Grnd - Green	Iso. Grnd - Green/Yellow
Iso. Grnd - Green/Yellow	Iso Grnd - Green/Yellow	

10.8 Grounding and Bonding

10.8.1 Grounding and Bonding for Electrical Systems

Consideration shall be given to local conditions affecting grounding methods. Ground resistance shall not exceed limits as established by IEEE Standard 142 for type of facility. Ground rods shall be ¾ inch by 10 feet copper clad minimum.

Exception: All ground rods on DFW Airfield Operations Area shall be ¾ inch by 10-foot stainless steel.

10.8.2 Ground Connections

All ground connections shall be bolted (where accessible) or by the exothermic process. All Exothermic welds shall be coated against corrosion.

An equipment grounding conductor shall be routed with all conduit installations and shall be sized per NEC 250.122. Conduit shall not be used as a ground path.

10.8.3 Counterpoise

All direct buried counterpoise group wire used at the Airport shall be #6 solid, tinned, soft drawn, copper wire conforming to ASTM #B- 33.

10.9 Lightning Protection

10.9.1 Certification

Specify that upon completion of construction, the Prime Contractor is required to transmit on his letterhead an affidavit bearing the notarized signature of the Lightning Protection Institute (LPI) Certified Master Installer that the lightning protection system complies with NFPA 780 and, in building additions, that it has been bonded to the existing system. Installation shall be made under the direct supervision of a Certified Master Installer, whose certification has been granted by the LPI.

10.9.2 Lightning Protection for Buildings

A Lightning Protection System shall be provided for all DFW Airport building structures. The lightning protection system shall comply with the latest edition of NFPA 780, for the Installation of Lightning Protection Systems of the National Fire Protection Association. Except for cable fasteners, all components of the lightning protection system shall be listed and labeled by an Approved Testing Laboratory recognized by Nationally Recognized Testing Laboratories (NRTL).

10.9.3 Lightning Protection for Building Additions

For additions to buildings already having a lightning protection system, provide lightning protection in the new construction only, and properly bond to the existing system.

10.10 Lighting

10.10.1 Interior Lighting

The most currently released edition of the IES NA Lighting Handbook and IECC shall be used to determine design lighting levels. The following light conditions should be used as design levels during the area's predominate period of occupancy. Describe the intended ballast factor of the ballast to be used. LED light fixtures shall be used when possible.

Table 30 - Design Light Level by Space

Terminal waiting areas	20 Hfc
Terminal Concourse	10 Hfc
Private office area (maximum level)	50 Hfc at the work surface (Task lighting is encouraged)
Common office area (cubicle)	40 Hfc at the work surface
Corridors in office area	25 fc
Ticket counters and TSA Documentation review	30 Hfc
Baggage claim	15 Hfc
Jet Bridge	10 Hfc
Rest Rooms	10 Hfc, 5 Vfc
Equipment maintenance	50-75 Hfc on task plane
Storage	30 Vfc on shelf
Parking garage	5 Hfc, 3 Vfc, 10:1 Max/min uniformity
Surface parking	0.5 Hfc, 0.25 Vfc, 15:1 max/min uniformity

10.10.2 Fixture and System Voltage

Fixture and system voltage shall match existing for maintenance purposes.

10.10.3 Ballasts/ Drivers

Specific ballasts that are manufactured exclusively for a particular fixture, or fixtures that require a particular ballast are not permitted. Avoid use of unusual fixtures, tubes, voltages, etc.

10.10.4 LED Lighting

LED sourced lighting is acceptable and preferred. LED recessed down lighting with less than 8-inch apertures installed in gypsum-board or inaccessible ceilings, shall have all components and wire connections accessible per NEC110.26 and 300.14 and shall extend a minimum of 3 inches through the aperture.

10.10.5 Exterior Lighting Systems

All airport light sources shall be aimed in a manner that will not interfere with FAA Air Traffic Control Tower operations or piloting activities. Where lighting sources are directed toward runway ends or FAA towers, they shall be shielded as necessary to deflect the light source down and away from those areas. Lighting systems shall be designed in accordance with IES recommendations to provide an energy efficient system with minimum maintenance. Refer to the latest edition of the IESNA Lighting Handbook.

10.10.6 Lighting Poles

Poles shall be located to provide adequate working clearance for maintenance without disrupting roadway traffic. Lighting Poles installed in non-accessible areas, shall be equipped with mechanical fixture lifting device.

All high poles modified with fixture lowering devices shall be inspected by the manufacturer of the device and the manufacturer shall issue a certificate as to the proper installation to factory specification with a 5-year warranty period to cover installation and factory defects.

10.10.7 Mounted Fixtures

The Designer shall obtain a list of approved lamps from DFW Airport Energy, Transportation, and Asset Management (ETAM) Department.

10.10.8 New Underground Lighting Systems

All street and high mast lighting shall be designed using Tx Dot Standards. New underground lighting systems shall be installed in direct buried Schedule 40 PVC. Alternate conduits require DFW Airport approval. Splices shall be made in pole base or concrete splice box with heavy duty traffic cover.

10.10.9 Roadway Lighting

All street and high mast lighting shall be designed using Tx Dot Standards. The roadway lighting system is 480-volt, single- phase, with power extending to pole bases underground in Schedule 40 PVC duct. Sleeves shall be installed for future Airport growth. Roadway lighting shall be per IES Type II or Type III with distribution obtained by reflectors or in some cases by refractors if approved by DFW Code. Roadway lighting systems shall be controlled.

10.10.10 Lighting and Sign Base Grounding

All street and high mast lighting shall be designed using Tx Dot Standards.

10.10.11 Anchor Bolts and Poles – Design Standards

1. ACI 318, most current edition.
2. AISC, most current edition.
3. AASHTO, Specification for Design and Construction of Structural Supports for Highway Luminaries

10.10.12 Corrosion Protection

The top 10 inches of anchor bolts and their associated nuts shall be hot dipped galvanized. All lighting pole base plates shall be exposed to air, no grout allowed.

10.10.13 Anchorage

Anchor bolts shall have full embedment as required by ACI. Anchor bolt nuts shall be torqued per manufacturers specifications.

10.10.14 Apron Lighting (Ramp Lighting)

Apron lighting shall be in accordance with IES Recommended Practice RP-14 “Airport Service Area Lighting.”

Exceptions:

1. General lighting near the Terminal buildings is specified at approximately two maintained foot- candles.
2. In the service areas, approximately 50 feet to 300 feet from Terminal buildings, average illumination levels, at 5 feet above the apron are specified at 3- to 4-foot candles horizontal and 7- to 9-foot candles vertical. The uniformity ratio is in the range of 3:1 (average/minimum) and 6:1 (maximum/minimum). The fixtures have a specified cutoff at 400 feet from the Terminal building to prevent direct spill onto the taxiways. In addition, internal louvers shall be provided for maximum bare lamp shielding.

3. The Designer shall call for submittals of the manufacturer's lighting analysis to include point by point foot-candle calculations to demonstrate the proposal meets the design criteria.
4. Prior to the final acceptance, the Contractor must submit a report, of testing accomplished by an IES-certified agency, showing that the installation complies with the specified criteria and the proposal calculated values. These tests shall indicate areas of unacceptable glare as indicated by FAA air traffic controllers or pilots. All testing shall comply with IES recommendations. All adjustments in fixture aiming shall be accomplished by the Contractor at no additional cost to the Airport.

10.11 Lighting Controls

10.11.1 Lighting Controls

Lighting controls shall be installed to meet the adopted IECC.

For new construction, major renovation, and interior finish-out projects with applicable scope, installation dimmable or multilevel lighting shall be provided in regularly occupied spaces.

*This requirement supports LEEDv4.1 Interior Lighting credit.

10.11.2 Energy Code

Lighting systems shall be installed to meet the adopted IECC.

Chapter 11 Information Technology (ITS)

11.0 Administration

11.0.1 Document Scope

This chapter defines general design criteria that apply to DFW's Information & Communications Technology (ICT) systems. ICT design must comply with the National Electrical Code (NEC) as adopted and amended by the Airport Board, as well as BICSI's Telecommunications Distribution Methods Manual and referenced ANSI/TIA standards.

11.0.2 Related Documents

This Design Criteria Manual section is not a stand-alone document. Refer to the References section at the beginning of this manual for additional DFW manuals and design guides which may impact ICT system designs.

Refer to relevant design standards issued by United States Customs and Border Protection, Transportation Security Administration, American Airlines, and other tenants that may be impacted by construction projects.

11.0.3 Abbreviations

ABF – Air Blown Fiber
AFF – Above Finished Floor
ANSI – American National Standards Institute
BICSI – Building Industry Consulting Services International
CFCI – Contractor Furnished, Contractor Installed
CFOI – Contractor Furnished, Owner Installed
CMP – Communications Plenum Cable
CR – Communications Room
CRAC – Computer Room Air Conditioning
DAS – Distributed Antenna System
ITS – Information Technology Services
MATV – Master Antenna Television
MCR – Main Communications Room
Min. – Minimum
OFCI – Owner Furnished, Contractor Installed
OFOI – Owner Furnished, Owner Installed
OSP – Outside Plant
PDS – Premise Distribution System
POE – Power Over Ethernet
SADS – Satellite Antenna Distribution System
TIA – Telecommunications Industry Association
TDMM – Telecommunications Distribution Methods Manual
WAP – Wireless Access Point

Additionally, refer to the Abbreviations section at the beginning of this manual.

11.0.4 Requirements

11.0.4.1 Design process

This manual provides typical minimum requirements for the design of DFW IT infrastructure. The design team must engage DFW-ITS and other stakeholders to confirm any additional requirements or deviations which may be required by a specific project.

The design process must include DFW-ITS engagement, to include project introduction and review as well as design team “page turns” at the 35%, 70% and 100% Design phases – or as further defined by the project scope of services documents.

11.0.4.2 Manufacturer/distributor qualifications & certifications

Designers shall require that the firm must be regularly engaged in the manufacture of products of the types, rating, and capacities required by the project.

Designers shall require that the firm’s products must have been satisfactorily used in similar service for not less than 5 years.

Designers shall require that the firm must have production capabilities per applicable NEMA standards.

11.0.4.3 Contractor qualifications & certifications

Designers shall require that the Contractor must be a firm regularly engaged in the installation of ICT systems and that has at least five years of installation experience with systems similar to that in use at DFW Airport.

All individuals, who will be in an on-site supervisory capacity, shall be required to attend the pre installation conference. This includes project managers, site supervisor and lead installers. Individuals who do not attend the conference will not be permitted to supervise the personnel that install, terminate, or test communications/low voltage cables on the project. The Contractor’s RCDD that will oversee the installation is required to attend the pre-installation conference Per DFW Requirements

Designers shall require that the Contractor must have at least three sites with actively similar systems that have been in operation for at least the proceeding twelve months.

Designers shall require that the Contractor must provide a BICSI Registered Communications Distribution Designer (RCDD) for project oversight, and the RCDD must be a full-time employee of the contractor.

Designers shall require that the Contractor must be certified by the manufacturer to sell, install, and maintain the products, as well as facilitate an extended warranty between the Owner and manufacturer.

11.0.4.4 Technician qualifications & certifications

Designers shall require that the contractors must submit their installation and service personnel’s qualifications and certifications including, but not limited to:

1. Installation and service personnel must hold a current and valid DFW Airport badge.
2. Manufacturer certified to install the proposed product.

11.0.4.5 Product data submittals

Designers shall require that the contractors must submit manufacturer's product information for all components including, but not limited to:

1. Product data sheets
2. Installation instructions
3. O&M information suitable for inclusion in maintenance manuals

Designers shall require that the submittals must be marked to indicate the proposed product, where multiple products may be listed.

11.0.4.6 Shop drawings

Designers shall require that the contractors must generate and submit for approval their original, detailed installation shop drawings including, but not limited to:

1. Equipment, device, and infrastructure locations and arrangements
2. All pathway cable fill ratios to ensure compliance with the latest release of ANSI/TIA-569
3. Assembly drawings of racks, cabinets, and enclosures
4. Wiring diagrams which show all field connected wiring, termination points, and labeling.

11.0.4.7 Test Results

Designers shall require that 100% of all cables, conductors, and strands must be tested and certified according to the manufacturer's recommendation. These test reports shall be provided to the Owner.

11.0.4.8 As-Built Drawings

Designers shall require that the Contractor provide updated shop drawings which reflect accurate as-built information including, but not limited to, cable routing and cabinet elevations.

11.0.4.9 Warranty Certificates

Designers shall require that the Contractor provide the manufacturer's extended warranty certificate.

11.0.4.10 O&M Manuals

Designers shall require that the Contractor provide operational, maintenance, and warranty data.

11.1 Labeling

1. Comply with latest release of ANSI/TIA-606 and DFW labeling standards.
2. Conduit Color Code
 - a. Reference Table 30 below, which depicts the approved DFW conduit labeling requirements. Refer also to Chapter 10 herein.
 - b. Where conduits are painted to match an architectural finish, the labels must be installed with masking prior to painting and then the label exposed after painting. Conduits with labels which are not properly adhered to the conduit are not acceptable.
3. Communications Data Outlet Identification
 - a. Reference Figure 16 below, which depicts the approved DFW communications data outlet identification requirements.

Table 31 - Typical DFW Conduit Labeling

Type	Label Color	Legend	Spacing
FA System – Conduit	White w/red letters	Red Conduit/Boxes	25' OC
LV Conduit – Pathways for Communication Systems	Orange w/black letters	Tag “COMMUNICATIONS” & origination/destination location & Contract Number	50' OC
LV System	Orange w/black letters	Tag “COMMUNICATIONS” & origination/destination location & Contract Number	Termination Points & (Junction Boxes)
DFW Cable Tray Pathway/Cable Tray Communication Systems	Orange w/black letters	“DFW-COMM-ITS” & Contract Number	50' OC
All Other Cable Tray Pathway/Cable Tray Communication Systems	Orange w/black letters	“COMMUNICATIONS” & Contract Number	50' OC
MATV Conduit	Orange w/black letters	Follow LV Conduit Guidelines	50' OC
MATV System	Orange w/black letters	Follow LV System Guidelines	50' OC
Safety Radio DAS System	Orange w/black letters	“PS-DAS” & Contract Number	50' OC
PA/VE Conduit Boxes & Couplers	White w/red letters	Red Conduit Tag “PA/VE” and Contract Number	Termination Points & Junction Boxes – White Couplings
PA/VE Conduit – PA/VE (Speaker Zones)	White w/red letters	Red Conduit Tag “PA/VE S Zone” (Speaker Zone Number) & Contract Number	25' OC
PA/VE Conduit – PA/VE (Ambient Microphones)	White w/red letters	Red Conduit Tag “PA/VE A Zone” (Ambient Microphone Zone Number) & Contract Number	25' OC
AACS Conduit – DFW Security System (AACS)	White w/red letters	Tag “AACS” on conduit only	25' OC
AACS System – DFW Security System (AACS)	White w/red letters	Tag “AACS” & Origination location only (see details below) & Contract Number – all enclosures	Termination Points & (Junction Boxes)
CCTV Conduit – DFW Security Systems (CCTV)	White w/red letters	Tag “CCTV” on conduit only	25' OC

CCTV System – DFW Security Systems (CCTV)	White w/red letters	Tag “CCTV” & Origination location only (see details below) & Contract Number – all enclosures	Termination Points & (Junction Boxes)
Label Sizes for Voltage Identification - All conduits 4-inch trade size and larger – 3-inch-high labels with 2-inch-high lettering. - All conduits 2-inch trade size and larger, but smaller than 4-inch – 1-inch-high labels with $\frac{3}{4}$ inch high lettering. - Low-voltage conduits smaller than 2-inch trade size (LV, MATV, PA/VE, AACS, CCTV) – 1-inch-high labels with $\frac{3}{4}$ inch high lettering.			
Supplementary Information (DEST/ORIG, Contract #) - Primary reasoning for labeling the voltage is for differentiation of different conduit systems (i.e., safety). Therefore, the label sizes for voltage must meet the above criteria. All other information on label is to aid maintenance personnel in identifying the systems in the field. - Sizes of labels and letting for the supplementary information can be smaller than the voltage ID sizes. - Origination should be labeled with a distribution equipment label ID (i.e., Panel #, DP #, Switch Gear #) and Room Name. - Destination should be labeled with the equipment label ID (Switch Gear #, Panel Board #, AHU #, etc.) and Room Name (if applicable). - Contractors to submit label to design teams for approval prior to installation.			

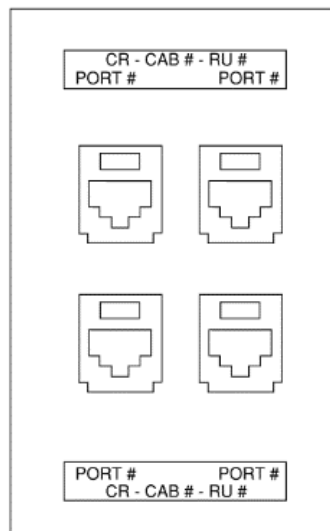


Figure 16 - Typical DFW Communications Data Outlet Identification

11.2 Telecom Grounding & Bonding

10. Comply with latest release of ANSI/TIA-607
11. Reference Figure 17 below, which depicts a typical telecom grounding and bonding system in a multi-story building.
12. Additional Requirements:
 - a. Metallic telecommunications pathways must be bonded to the Primary Bonding Busbar or Secondary Bonding Busbar, per ANSI/TIA-607.
 - b. Structural metal must be used in place of the Telecommunications Bonding Backbone and/or Backbone Bonding Conductor where the structural metal is electrically continuous and provides acceptable resistance. Refer to ANSI/TIA-607 for further requirements and information.

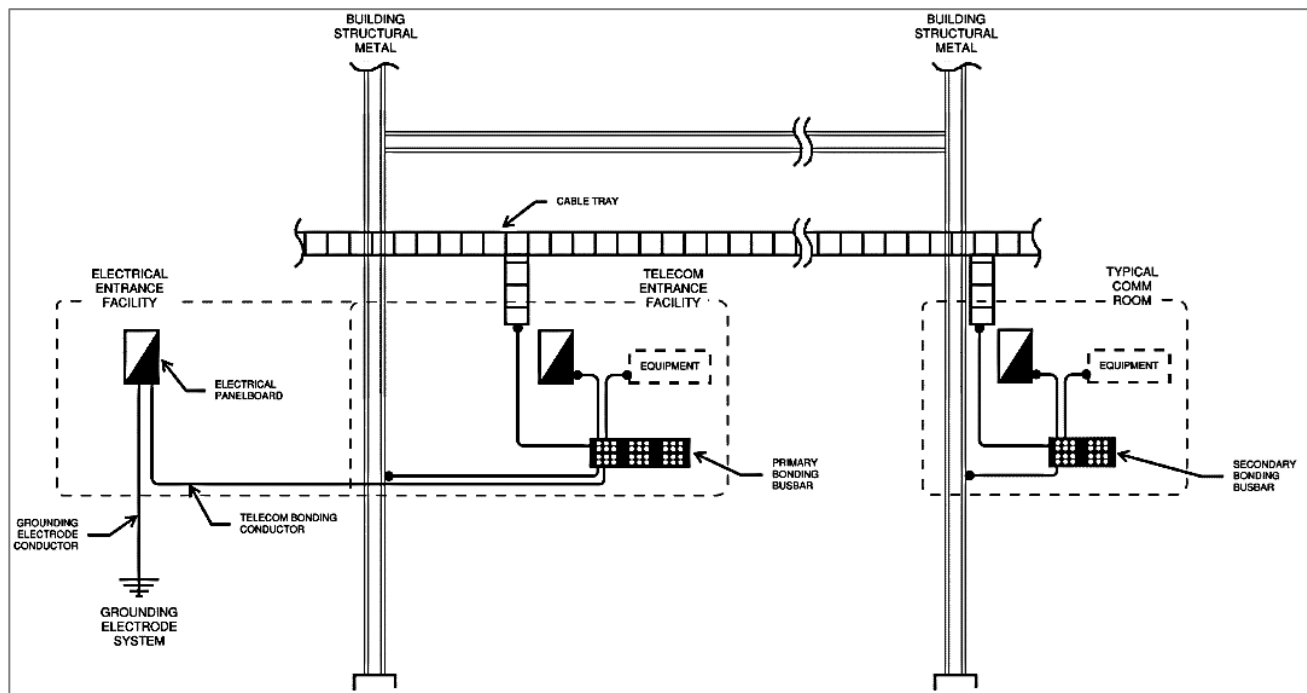


Figure 17 - Typical Telecom Grounding & Bonding System

11.3 Telecom Spaces & Pathways

1. Comply with latest release of ANSI/TIA-569
2. Communication Rooms
 - a. Space planning
 - i. Types
 - (1) DFW Terminal MCR/CR
 - (2) DFW not-Terminal MCR/CR
 - (3) DFW quad closets
 - (4) DFW quad enclosures
 - (5) Shared tenant Comm Rooms
 - (6) Tenant Distribution Panels (TDPs)
 - (7) Jet Bridge Interface Boxes (JIBs)

- b. Sizes

The size of an MCR/CR must be based upon the required equipment as well as projected future equipment growth.

- i. A typical Terminal MCR is a primary point of entrance for outside plant telecommunications infrastructure, as well as main distribution point for data infrastructure to other CRs within the building. A typical terminal MCR houses minimum (14) network cabinets for systems cabling infrastructure and equipment. Cabinet space allocations must include space requirements to serve initial system's needs, as well as space for projected future growth. Reference Figure 18 below, which depicts a typical MCR configuration in a terminal building. Design teams shall coordinate exact layout, systems space assignments, and other requirements with DFW ITS early in the design of a project planning a new MCR.

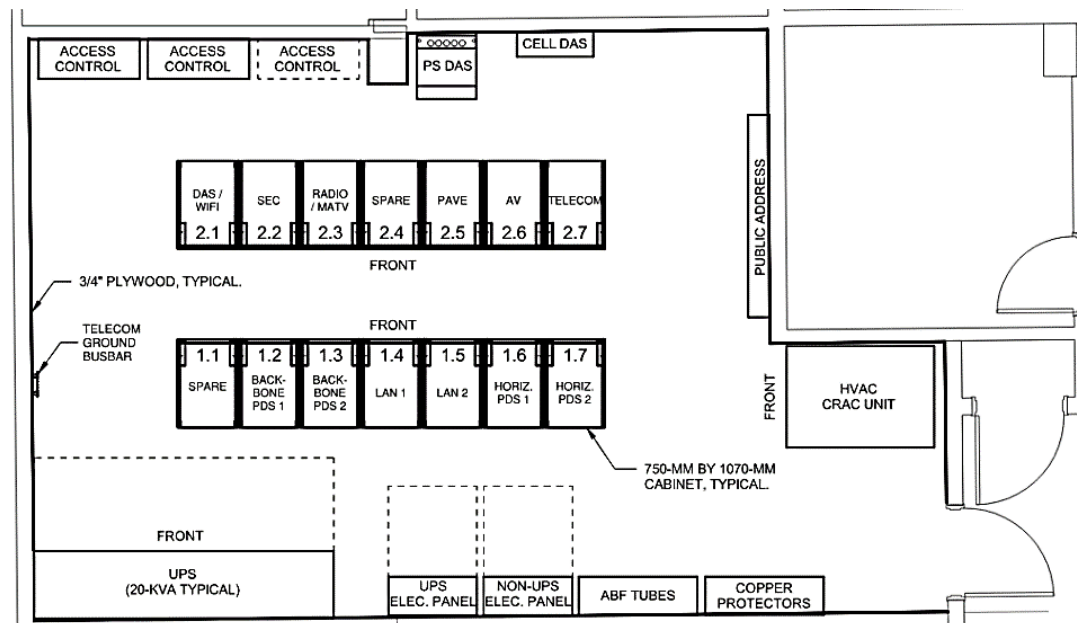


Figure 18 - Typical Terminal Main Communications Room (MCR) Layout

- ii. A typical Terminal CR serves as data distribution point to DFW data network and IT end devices within the Ethernet service boundary of the room. CRs should be placed centrally within the footprint of the building, where feasible, to optimize horizontal cabling distribution to devices served from the room. A typical DFW CR includes a single door with electronic access control, as well as camera coverage inside and outside of the room. In a terminal, a typical DFW CR contains minimum six network cabinets and equipment mounted to wall fields such as Cellular DAS, Public Safety Radio DAS, PAVE, ABF distribution, UPS, and electrical panelboards, and CRAC unit. Cabinet space allocations must include space requirements to serve initial system's needs, as well as space for projected future growth. Reference Figure 19 below, which depicts a typical CR configuration in a terminal building. Design teams shall coordinate exact layout, systems space assignments, and other requirements with DFW ITS early during the design phase of a project planning a new CR.

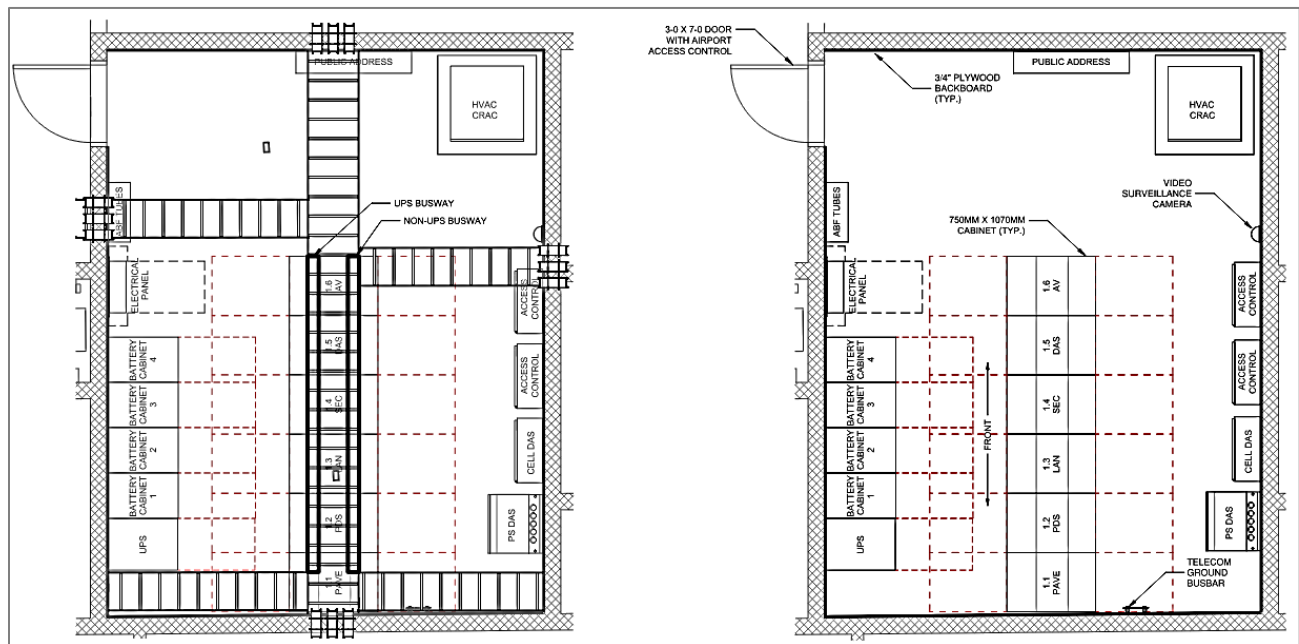


Figure 19 - Typical Terminal Communications Room (CR) Layout

- iii. A typical Quad Closet includes data infrastructure intended to support limited security services for DFW systems. A typical Quad Closet includes one wall-mount open-frame network rack mounted on plywood backboard. This space requires redundant ABF single-mode fiber cables from each MCR, UPS and non-UPS power extended from the nearest CR, double door with electronic access control, and camera coverage outside of the room. A dedicated HVAC unit is not required, but cooling shall be provided by an independent HVAC zone. A telecom grounding busbar and general cable management shall be included on the plywood backboard. Reference Figure 20 below, which depicts a typical Quad Closet configuration in a terminal building. Design teams shall coordinate exact layout, systems space assignments, and other requirements with DFW ITS early during the design phase of a project planning a new Quad Closet.

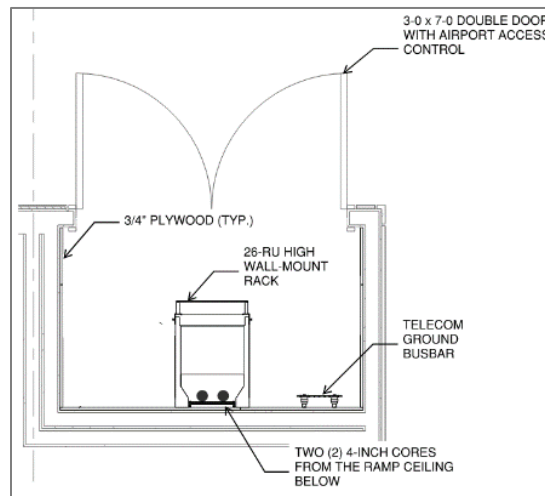


Figure 20 - Typical Terminal Quad Closet Layout

- iv. A typical Shared Use Tenant CR configuration includes data infrastructure intended to support IT services for both DFW and Tenant systems. The standard DFW CR construction supporting DFW systems should be provided for DFW exclusive use. In addition, the designer should provide additional tenant cabinets in a space directly adjacent to the DFW rooms & cabinets. Tenant colocation cabinets must be physically separated from the DFW CR via partition to maintain separate badged access levels between the two spaces. Partitions between the DFW and Tenant communications rooms may be a hard wall or may be constructed with chain link fence. The exact shared use communications room configuration must be coordinated with DFW ITS during early design phase to gain consensus on the room configuration based on the project requirements and long-term use of the facility.

c. Dimensional Clearances

- i. The room size must accommodate all required equipment, planned future equipment expansion, and minimum working clearances around all equipment.
- ii. All wall fields must accommodate equipment which is no less than 6 inches in depth, except where wall-mounted equipment is known to exceed that depth such as DAS equipment.
- iii. All cabinets must include no less than 42 inch clearance in the front and back, to allow equipment installation and service. Cabinets configured in a bayed configuration, where multiple cabinets are installed in a physical row, shall include no less than 42 inch clearance around one end of the row.
- iv. Mechanical, electrical, and fire protection equipment must include the required clearances, and must be coordinated during design.

d. Water Mitigation

- i. CRs must not be located adjacent to or below wet facilities, including concessions, restrooms, and mop sinks. Where wet utilities must be located adjacent to a CR, there must be floor drains in the adjacent spaces and concrete curbs must surround the CR to provide further protection.
- ii. Wet plumbing must not route across or through a CR. Where wet plumbing must route into a CR to serve equipment within the space (e.g. CRAC units) there must be drip pans and water sensors installed to alert in the event of a leak.

e. Location

- i. Terminals must receive redundant MCRs, at opposite ends of the facility to provide diversity and resiliency.
- ii. Non-Terminal buildings may require redundant MCRs. Coordinate these requirements with DFW on an individual basis.
- iii. Terminal MCRs and CRs must be on the ramp level.
- iv. Where CRs are provided on multiple levels of a building (such as a non-Terminal office building), the CRs must be “stacked” and allow straight vertical cable risers.
- v. CR entrances must be accessible from a corridor or ramp area and must not be located within tenant or mechanical/electrical spaces.
- vi. CRs must be secure, and must limit access and traffic through a CR. This includes limiting equipment which will be located inside the room to only equipment required to serve the room (CRAC unit, UPS unit, etc). Fire alarm and electrochromic glazing panels must not be located within CRs.

f. Proximity

- i. CRs must be placed such that the entire facility can be served within the 90-meter Ethernet cable length limitation from patch panel to work area outlet (including service slack), with cable routes parallel and perpendicular to building lines.

11.3.1 MEP

1. Electrical Distribution

- a. Panelboards supporting CR equipment receptacles shall be located within the CR. These panelboards shall not be used to support lighting, electric motors, or any devices outside of the CR.
- b. 120-volt 20-amp quadplex convenience receptacles shall be provided along every wall within the CR, spaced no more than 6-feet apart.
- c. Receptacles for wall field equipment, such as security or DAS equipment, shall be dedicated UPS circuits. These are in addition to the convenience receptacles noted above.

- d. Receptacles for network cabinets shall be located above the cabinets, oriented to the rear of the cabinets.
 - e. Each cabinet shall typically receive one L5-30R from UPS power and one L5-30R from house (non-UPS) power. Coordinate these exact requirements with the Owner.
 - f. Terminal CRs shall include two overhead electrical busway systems, one from UPS power and one from house (non-UPS) power.
2. UPS
- a. Terminal CRs shall include a modular in-room UPS, sized to accommodate the telecommunications, life-safety, and security equipment electrical loads.
 - b. UPS shall provide 24-hour runtime at full load. This may be reduced to 4-hour runtime at full load, when all the following conditions are met:
 - i. The electrical service is dual fed, from two sources.
 - ii. A connection is provided for mobile generator use.
 - iii. AHJ has approved the 4-hour runtime.
 - c. UPS equipment must include Fire Alarm monitoring when the UPS supports the DFW Production network, Security, PAVE, and/or DAS systems.
 - d. All UPS systems within DFW Comm Rooms must include a 2-port data outlet, to support remote monitoring via APC's ISX platform. Commissioning of each UPS will require connectivity to the ISX system, which in turn must communicate to DFW's EIOS system.
3. CRAC
- a. Environmental cooling and dehumidification shall be provided by a dedicated CRAC unit. This is traditionally a floor mounted CRAC unit located within the CR. Where an overhead unit is required due to project-specific constraints, that overhead unit shall be located outside of the CR and ducted into the CR.
 - b. CRAC units shall be equipped with leak detection.
 - c. A thermostat shall be located within the CR, and the system shall include remote monitoring by DFW's EIOS system.
4. Lighting (including emergency)
- a. Lighting shall be provided in the aisles in front and behind all network cabinets, providing no less than 50 foot-c measured at 3 feet AFF.
 - b. Lighting placement shall be coordinated with conduit, cable tray, and ladder rack systems.
 - c. Glass lamps shall be protected from accidental damage. Fixtures must be installed as high as practical, and no less than 12 inches higher than the highest horizontal ladder rack.
 - d. Emergency lighting shall be included within the CR.
5. Grounding
- a. Electrical grounding shall be provided for all electrical infrastructure as required by NEC.
 - b. A grounding conductor shall be provided from the electrical panelboard to the telecom grounding busbar.
6. Transformers
- a. Electrical transformers shall not be located within CRs. It is typical for these transformers to be located in an adjacent transformer room associated with each CR. All ICT infrastructure shall maintain separation from transformer per the latest release of ANSI/TIA-569.

11.3.2 Architectural finishes

1. Flooring
 - a. Floors must be anti-static vinyl composite tile, with bonding conductor for connection to ground.
 - b. Sealed concrete is not allowed.
 - c. A rubber wall base shall be included throughout.
2. Walls
 - a. Wall shall be minimum 2-hour fire-rated partitions extended to the deck above.
 - b. All walls within a CR shall include $\frac{3}{4}$ inch AC-grade fire-treated plywood, painted on all 6 sides with two coats of fire-retardant paint. Take care to mask the fire-retardant stamp on the plywood prior to painting and ensure that stamp is visible after painting.
 - c. Plywood shall be installed in 4 foot by 8 foot sections, in portrait orientation, starting at 6 inch AFF. The plywood shall be secured with toggle bolts and washers in each corner and roughly every 24 inches across the top, bottom, and sides. The plywood shall not be secured by screws.
 - d. Electrical and data outlets shall be recessed, and the plywood shall be cut to allow the receptacle faceplates to be installed on the plywood face.
 - e. Plywood and finished wall paint colors must match.
3. Ceilings
 - a. Where walls are extended to deck above, the CRs shall be open to deck with no ceiling. Any deviation must be coordinated with DFW early in the design process.
4. Doors
 - a. Doors shall be minimum hollow metal, measuring 36 inches by 84 inches, with a hollow metal frame, continuous hinges, electrified mortise handset, discrete electrical power transfer (EPT), recessed door contacts, and welded backboxes for EPT and door contacts.
 - b. Door shall open outward where code allows.
5. Columns
 - a. Columns must not be placed in CRs, in order to most efficiently use the space that is available. Where columns or similar obstructions must be placed with a CR, the room size must be increased accordingly.
6. Penetrations
 - a. The rating of the partitions shall be restored after any penetration is made.
 - b. Mechanical devices for cable pathways through rated partitions shall be used. Devices shall restore the fire (F) rating and smoke (L) rating of the partition. The installed device must accommodate future cable changes without maintenance or replacement of protective materials.

11.3.3 Security

1. Access control
 - a. Electronic access control shall be provided on all CR entrances.
2. CCTV
 - a. Video surveillance shall be provided inside and outside all CRs.
3. Wall partitions
 - a. Wall partitions must extend to the deck above. Any deviation must be coordinated with DFW early in the design process.
 - b. Windows are not allowed, including within doors.

11.3.4 Fire Protection & Fire Alarm

1. Fire Protection
 - a. All CRs must include a double-interlocked pre-action fire suppression system. The system must be located outside of the CR and as close as practical to the CR.
2. Fire Alarm
 - a. UPS units supporting life safety systems must be monitored by the Fire Alarm system. Refer to 11.3.1.2 for additional detail.
 - b. Fire Alarm panels must not be located within CRs without prior written approval by DFW-ITS.
 - c. Visual fire alarm strobe notification must be provided within the CRs.

11.3.5 Cabinets and Cable Runways

11.3.5.1 Locking cabinets

1. Typical equipment cabinets within CRs shall be 750mm wide by 1070mm deep by 48U high, rated to support no less than 3000-pounds static load.
2. Cabinets shall include ventilated front door and ventilated split rear doors, with keyed lock.
3. Cabinets shall include removable side panels, with option for brushed cable pass-through. Top panels shall also be removable, with option for brushed cable pass-through that support high density patch panels. Coordinate exact cable pass-through configuration with the Owner based upon a project's specific needs.
4. All cabinets shall include integrated vertical cable management with hinged covers on the front rails of the cabinets, to facilitate network cable management.
5. All cabinets shall include ring-style vertical cable managers on the rear rails of the cabinet, to facilitate power cable management.
6. All cabinets shall include horizontal cable management, as required based upon a project's specific needs.
7. Cabinet shall include cable management troughs which attach to the top panel of the cabinet. A trough shall be provided along the front of the cabinet for optical fiber management, and a trough shall be provided at the rear of the cabinet for power cable management.
8. Each cabinet shall include two vertical metered PDUs, with no less than (24) NEMA 5-20R outlets.
9. Cabinets shall include vertical grounding busbar.

The following products must be used:

1. APC AR3157 NetShelter SX, Server Rack Enclosure, 48U, Black, 750mm W x 1070mm D
2. APC AR7375 Split Feed-Through Side Panel (Qty. 2)
3. APC AR7252 NetShelter Networking Roof Panel
4. APC AR8442 Vertical Cable Manager, Cable Rings, for Rear Rails
5. APC AR7588 Vertical Cable Manager, for NetShelter SX 750mm Wide 48U, for Front Rails
6. APC AR7589 Vertical Cable Manager, Hinged Covers, for NetShelter SX 750mm W 48U
7. APC AR8600A Horizontal Cable Manager, 2U, Single Side with Cover
8. APC AR8602A Horizontal Cable Manager, 1U, Single Side with Cover
9. APC AP8832 Vertical PDU, Metered, L5-30P, (24) NEMA 5-20R
10. APC AR8173BLK Cable Management, Cabinet Roof Data Cable Trough

11.3.5.2 Ladder Rack

1. Minimum 18 inches wide ladder rack is required around the room, to support cabling to wall mounted equipment.
2. Minimum 24 inches wide ladder rack is required above the network cabinets.
3. Ladder rack system shall include all manufacturer-recommended accessories.

11.3.5.3 Horizontal Pathways & Backbone Pathways

1. Pathways must comply with the latest release of ANSI/TIA-569, except where this manual provides requirements specific to DFW.
2. Conduit
 - a. Refer to DCM Chapter 10 for Conduit Type (EMT, RMC) & Fittings
 - b. Flexible metallic conduit is limited to short device connections, no longer than 6-feet without approval by DFW-ITS. No flex allowed on security system except for ceiling mounted camera drops.
3. Conduit Sizes
 - a. Horizontal pathway = Minimum 1 inch unless otherwise approved in writing by DFW-ITS
 - b. Quad pathway = Minimum 2 inch (supporting ABF)
 - c. Backbone pathway = Minimum 4 inch
 - d. Conduit fill ratio must include de-rating for bends, per the latest release of BICSI TDMM, NEC.
4. Pull boxes and Junction Boxes
 - a. Refer to latest release of ANSI/TIA-569 for conduit pull box frequency.
 - b. Pull box size for a single conduit up to 1.25 inch diameter must be no smaller than 4-11/16 inch square by 2-1/8 inch deep.
 - c. Pull box size for a conduit up to 2 inches must be no smaller than 4 inches wide by 4 inches high by 24 inches long. For each additional conduit, the width must be increased by double the conduit diameter (adding a 2 inch conduit = 4 inch additional width).
 - d. Pull box size for a conduit up to 4 inches must be no smaller than 8 inches wide by 8 inches high by 36 inches long. For each additional conduit, the width must be increased by double the conduit diameter (adding a 4 inch conduit = 8 inch additional width).
 - e. Pull boxes and Junction Boxes which are on the ramp level or outdoor must meet or exceed NEMA-3R rating.
 - f. Include all removable covers shall have Safety chains installed at a minimum of 12 inches. Provided by installer.

Table 32 - Pull Boxes and Junction Box Sizing

Maximum Trade Size of Conduit (inches)	Minimum Box Size (inches)			For Each Additional Conduit Increase Width (inches)
	Width	Length	Depth	
1	4	16	3	2
1.25	6	20	3	3
1.5	8	27	4	4
2	8	36	4	5
2.5	10	42	5	6
3	12	48	5	6
3.5	12	54	6	6
4	15	60	8	8

5. Bend Radius and Installation Requirements

- a. Refer to latest release of ANSI/TIA-569 for conduit bend radius requirements.

Exception: conduit bend radius for ABF must be no less than a 40 inch radius regardless of the conduit size.

- b. New CR backbone conduits must include one empty spare conductor, for future use.
- c. All unused conduits must be capped immediately after installation to prevent debris from entering the conduit system. Conduits must be kept free from dirt, debris, and moisture prior to use.
- d. Conduits must terminate at work area outlets into a backbox which is not smaller than 4-11/16 inch square by 2-1/8 inch deep.

Exception: when approved by DFW-ITS, a retrofit project with existing structure and painted walls may utilize low voltage retrofit mounting bracket in lieu of conduit to backbox (Caddy MPLS, Caddy MP1P, Panduit LV-W-1G, or equivalent).

- e. Metal conduit outside of CRs shall be bonded/grounded at all termination points and shall be electrically bonded to the cable tray.
 - f. Metal conduits within CRs or entering CRs shall be grounded directly to the TGB and not to the ladder rack or other surfaces.
 - g. Conduits must be installed in an organized, professional, and workmanlike manner. This includes routing parallel and perpendicular to building lines and grouping together where possible.
 - h. Conduits must not cause potential safety hazards or interfere with operating and maintenance procedures.
6. Outdoor underground handholes and maintenance holes
- a. Reserved for future use.

7. Innerduct

a. Air Blown Tubes

- i. DFW utilizes air blown fiber for campus fiber distribution throughout various outside plant (OSP) fiber segments between buildings. The majority of this OSP air blown fiber is Emtelle FiberFlow™ solution. Outside Plant air blown fiber tube cables installed at DFW are intended to provide major backbone trunk cabling from existing Data Centers to various Smart Demarc (SD) and building locations across the campus. Where DFW projects require significant fiber quantities to be installed to support distribution to a large facility, or to multiple facilities in the area, air blown fiber tube cell infrastructure must be provided. Design teams shall coordinate ABF tube cell segments and quantities with DFW-ITS in early design phase of the project to discuss proposed ABF considerations and sizing needs. Other OSP air blown fiber manufactured solutions may be considered, however any solution variances must be coordinated with DFW ITS during the design phase and may be allowed on a case-by-case basis.
- ii. DFW utilizes air blown fiber for inside plant (ISP) fiber distribution throughout the terminal buildings. Most of this ISP air blown fiber is Sumitomo FutureFLEX® air blown fiber. Inside plant air blown fiber tubes are intended to provide backbone fiber connections between MCRs, as well as between MCRs and CRs. Where terminal projects are constructing new DFW Communications Rooms, redundant air blown fiber tube cell shall be provided to facilitate redundant fiber service to each CR. Design teams shall coordinate ABF tube cell segments and quantities with DFW during early design phases to properly size and quantify fiber tube cell needs. Other ISP air blown fiber solutions may be considered; however any solution variances must be coordinated with DFW ITS during the design phase and may be allowed on a case-by-case basis.
- iii. Ensure ABF conduits include min. 40 inch radius bends, regardless of the conduit size.
- iv. Follow manufacturer requirements for tube cell installation, including tube cell distribution cabinet within CR.
- v. The following products must be used:
 - (1) ABF Outdoor Solution: Emtelle FiberFlow™
 - (2) ABF Indoor Solution: Sumitomo FutureFLEX®

8. Fabric

- a. Fabric innerducts are typically used within backbone conduit systems.
- b. Designers shall require installation utilize manufacturer-approved swivel to prevent twisting of the fabric innerduct and follow all other manufacturer installation recommendations.
- c. The typical configuration for a 4in conduit consists of (9) 3 inch fabric innerducts, however the fabric innerduct configuration shall meet the project requirements.

9. Corrugated Plastic

- a. Corrugated innerduct is not a standard product for use within DFW conduits. Any projects planning corrugated plastic innerduct shall be coordinated with DFW-ITS in early design phase of the project to discuss proposed innerduct requirement.

10. Cable Tray

a. Aluminum Ladder

- i. Outdoor and ramp level cable trays must be aluminum, with two longitudinal members (side rails) and transverse members (rungs).
- ii. Tray must be constructed of extruded aluminum alloy 6063-T6 and welded together.
- iii. Tray must be supplied in standard 20 feet to 24 feet lengths, except where shorter lengths are required.
- iv. Rungs must be spaced maximum of 12 inches.
- v. Side rails must be available with 4 inch and 6 inch heights, allowing a loading depth of 3 inch and 5 inch, respectively.
- vi. Tray must be classified as an equipment grounding conductor per NEC 392.7 with a maximum 1200 ampere rating.
- vii. Include all necessary accessories such as splice plates, expansion plates, blind-end plates, ladder dropouts, conduit adapters, and support hardware.
- viii. Route above common corridors and circulation areas, where available.

b. Wire Mesh

- i. Concourse and support space cable trays must be manufactured from round steel wire that is minimum .196 inches (5mm) in diameter, welded together at intersections to form a 2 inch by 4 inch grid pattern.
- ii. The tray must be U-shaped with equal height sidewalls.
- iii. Tray must be 4 inches high, allowing a loading depth of 3 inches.
- iv. Wire mesh tray must be zinc electroplated after fabrication, and UL Classified for grounding purposes.
- v. Where cable tray is in an exposed ceiling area, the tray color must be coordinated with the environment and a tray liner must be provided.
- vi. Include all necessary accessories such as splice hardware, expansion plates, conduit adapters, and support hardware.
- vii. Route above common corridors and circulation areas, where available.

c. Non-continuous pathway (J-hook)

- i. J-hook cable pathways are not to be used for supporting primary cable pathways but are allowable for secondary cable pathways (e.g. cable tray to work area outlet) when all of the following conditions are met:
 - (1) Above acoustic ceiling with lift-out tiles
 - (2) Supporting no more than (25) cables
 - (3) J-hook must be able to support a minimum of 30lb with a safety factor of 3

- ii. J-hook cable pathways must be non-metallic, plenum rated, installed between 4 to 5 feet apart, provide 1 inch horizontal and vertical bend radius control to prevent pinch points, available in various color options, and comply with current ANSI/TIA-568, 569.
 - iii. The following solutions must be used:
 - (1) Panduit J-Pro
- d. Penetrations
 - i. Conduit
 - (1) Conduits penetrating a rated wall must be sealed following the appropriate UL Listed system.
 - (2) All metallic conduits must include an insulated ground bushing, for bonding to the nearest cable tray when located outside of the CR or to the telecommunications busbar when located inside of the CR.
 - ii. Conduit Sleeves and Mechanical Sleeve Devices
 - (1) Metallic pathways under 3 feet in length (such as floor sleeves, wall sleeves, and j-hook supports) are not required to be bonded, per ANSI/TIA-607. Metallic pathways 3 feet or longer in length must be grounded to the telecommunications busbar.
 - (2) All exposed conduit sleeves shall extend a minimum of 3-inches above finished floor, or 4-inches below finished ceiling. The Designer must locate these adjacent to a wall with sufficient space between the conduit and wall for installation of conduit hardware.
 - (3) Conduit penetrations shall be spaced with sufficient clearance from adjacent penetrations to maintain the structural integrity of the floor or wall.
 - iii. Cable Tray
 - (1) Cable tray sections must be bonded together and bonded to telecommunications busbar with a #6-AWG conductor.
- e. Security Identification Display Area (SIDA) Walls
 - i. Cable tray or sleeves shall not penetrate a SIDA wall, instead they shall transition to a minimum of 72 inches by 3 inches EMT conduit sleeve which has one 90-degree bend.
- f. Roof
 - i. Roof penetrations are highly recommended to be coordinated with CRs, such that roof penetrations are not placed directly above a CR.
 - ii. Roof penetrations must be provided by the warranting roofing contractor, to maintain roof's warranty.
- g. Floor
 - i. Conduits protruding through the floor must not interfere with circulation or cause a potential safety hazard.
 - ii. Conduit stub-ups must terminate at 3 inch AFF.
 - iii. Conduits stub-ups must be rigid galvanized steel.
 - iv. Clear space between conduits must be no less than the conduit diameter. This is required for structural stability and provides working clearances.
 - v. Conduits must be sealed immediately upon installation to prevent debris from entering the conduit system.

11.4 Backbone Cabling

11.4.1 Terminal Facilities

11.4.1.1 Cable Infrastructure

9. Multi-Pair Copper
 - a. Each CR must receive no less than 50-pair Category 3 cable from the nearest MCR.
10. Optical Fiber
 - a. Each CR must receive no less than 72-strands of single-mode ABF cabling from each redundant MCR, for a total of 144 single-mode strands. These strands must be installed continuous, with no splice or patch points. One path from the nearest MCR must be in a dedicated conduit system and one path from the farther MCR must be in the DFW cable tray system.
 - b. Redundant MCRs must be interconnected with no less than 144 strands of single-mode ABF cabling. These strands must be installed continuous, with no splice or patch points.
 - c. No less than 25 feet of service slack must be provided at each end of a traditional optical fiber cable.
 - d. The following products must be used:
 - i. ABF Outdoor Solution: Emtelle
 - ii. ABF Indoor Solution: Sumitomo FutureFLEX®
 - iii. Traditional Outdoor Solution: Corning ALTOS® or FREEDM®, Panduit
 - iv. Traditional Indoor Solution: Corning MIC®, Panduit

11.4.1.2 Rack Terminations

1. Multi-Pair Copper
 - a. Wall-mounted multi-pair surge protection blocks must be included, where required.
 - b. Rackmount 110-style termination kits with 5-pair connecting blocks and jumper troughs must be provided.
2. Optical Fiber
 - a. Each CR must receive no less than (12) LC/APC single-mode connectors for each 72-strand. The remaining quantity of strands from each MCR must terminate with LC/UPC connectors. The exact quantity of LC/APC connectors must accommodate DAS, MATV, SADS, and Concessions.
 - b. Fusion splice cassettes must be used in rack-mount installations, and fusion splice-on connectors must be used in all other installations. Field polished connectors and mechanical splice connectors are not allowed.
 - c. Backbone cable terminations must be housed in high-density fiber enclosures.
 - d. Note that horizontal fiber terminations must be housed in separate enclosures.
 - e. The following products must be used:
 - i. Panduit Fusion Splice-On Connectors
 - ii. Panduit HD Flex™

11.4.1.3 Patch Cords

6. Patching must use products to maintain the manufacturer's channel system warranty.
7. Patching must use products designed for use within high-density optical fiber enclosures.
8. Coordinate quantity, connector types, and lengths with Owner based upon a project's specific needs.
9. The following products must be used:
 - a. Panduit Opti-Core® Push-Pull duplex patch cords (UPC and APC as required).

11.4.2 Non-Terminal Facilities

Reserved for future use.

11.5 Horizontal Cabling**11.5.1 Terminal Facilities****11.5.1.1 Riser & Plenum Category 6A Cable infrastructure**

1. Horizontal cable infrastructure must comply with the existing campus-wide manufacturers extended warranty.
2. Cable must be installed per manufacturer instruction, including not exceeding the recommended pulling tension or bend radius.
3. A service loop of no less than 10 feet must be included at the CR and no less than 1 foot must be included at the work area outlet.
4. The cable must be Category 6A, with blue jacket.
5. Cable jacket must be plenum or riser rated as required.
6. The cable must be UTP, with Vari-MaTriX tape cut into segments of varying length.
7. The nominal outside diameter must be no more than .23 inches.
8. The cable must have a Liquid Petroleum (LP) Rating of no less than .7A
9. The cable channel shall be 100 meters or less.
10. The following products must be used:
 - a. Panduit PUP6AHD04BU-G

11.5.1.2 Outdoor-Rated Category 6A Cable infrastructure

1. Horizontal cable infrastructure must comply with the existing campus-wide manufacturers extended warranty.
2. Cable must be installed per manufacturer recommendations, including not exceeding the 25-lbf recommended pulling tension.
3. A service loop of no less than 10 feet must be included at the CR and no less than 1 foot must be included at the work area outlet.
4. The cable must be Category 6A, with black UV-resistant polyethylene jacket and water blocking agent to prevent water ingress.
5. The cable must be UTP, with an alien crosstalk barrier.
6. The nominal outside diameter must be no more than .33 inches.
7. Cable installation must comply with BICSI TDMM recommendations for installation within a building (e.g., must not extend more than 50 feet into the building).
8. The cable channel shall be 100 meters or less.
9. The following products must be used:
 - a. Panduit TX6A™ U/UTP OSP Cable

11.5.1.3 Extended Distance UTP Cable Infrastructure

1. Horizontal cable infrastructure must comply with the existing campus-wide manufacturers extended warranty.
2. Cable must be installed per manufacturer instruction, including not exceeding the recommended pulling tension or bend radius.
3. A service loop of no less than 10 feet must be included at the CR and no less than 1 foot must be included at the work area outlet.
4. The cable must support 1-Gbps at 200 meters.
5. Cable shall be available with blue CMP or black OSP jacket.
6. Cable jacket must be plenum, riser, or OSP rated as required.
7. The cable must be UTP, with nominal outside diameter no more than .25 inches (CMP) or .31 inches (OSP).
8. The cable channel shall be 100 to 200 meters, with DFW-ITS approval.
9. The following products may be used with DFW-ITS approval:

- a. General Cable 8131800 or 8146100

11.5.1.4 Optical Fiber Cable Infrastructure

1. Use Case for Horizontal Optical Fiber Cabling
 - a. Horizontal fiber cabling consists of 6-strand fiber cables supporting equipment which include, but are not limited to:
 - i. Fire Alarm Control Panels
 - ii. Tenant Distribution Panels
 - iii. Jet bridge Interface Boxes

11.5.1.5 Rack Terminations**11.5.1.5.1 Copper**

1. Horizontal copper cables must terminate on 24- or 48-port unloaded angled modular patch panels. Panels must include integrated label pockets for port and panel identification.
2. Horizontal cables must terminate on blue RJ45 modular jacks housed within the modular patch panel.
3. Appropriate cable management must be provided, including vertical management within cabinets.
4. Include rack-mount surge protection where required, which complies with bandwidth and POE requirements.
5. The following products must be used:
 - a. Panduit CPPLA24WBLY
 - b. Panduit CPPLA48WBLY

11.5.1.5.2 Optical Fiber

1. Horizontal 6-strand fiber cables to concession spaces must terminate with SC/APC connectors. Horizontal 6-strand fiber cables to fire alarm or jet bridge interface boxes shall terminate with SC-UPC connectors.
2. Refer to 11.4.1.2.2 for acceptable fiber terminations.
3. Horizontal cable terminations must be housed in standard-density enclosures, separate from the high-density backbone cable terminations.
4. The following products must be used:
 - a. Panduit FRME Opticom® enclosures

11.5.1.5.3 Work Area Outlets

1. Outlets within work areas must provide for the immediate equipment needs as well as provide 50% spare capacity for future use.
2. A work area outlet must provide no less than two ports.
3. Horizontal cables must terminate on blue RJ45 modular jacks housed within an Executive-series faceplate.
4. The following products must be used:
 - a. Panduit MiniCom® Executive Wallplate

11.5.1.5.4 Patch Cords**11.5.1.5.4.1 Copper**

1. Work area outlets must receive Category 6A 23-AWG or 24-AWG patch cables. Patch cables which are installed within a plenum environment must be rated accordingly.
2. Patch panels must receive Category 6A 28-AWG small-diameter patch cables.
3. Jacket color must be organized by application as follows:
4. Typical data cord = blue jacket
5. Coordinate quantity and lengths with Owner based upon a project's specific needs.
6. The following products must be used:
 - a. Panduit PanZone® CMP Patch Cords
 - b. Panduit TX6A™ UTP Patch Cords
 - c. Panduit TX6A-28™ UTP Patch Cords

11.5.1.5.4.2 Optical Fiber

1. Patching must use products to maintain the manufacturer's channel system warranty.
2. Coordinate quantity, connector types, and lengths with Owner based upon a project's specific needs.
3. The following products must be used:
 - a. Panduit Opti-Core® duplex patch cords (UPC and APC as required).

11.5.1.5.5 Tie Cables**11.5.1.5.5.1 Rack to Rack**

1. Category 6A cabling is required between cabinets within a CR. These connections, known as tie cables, provide a neat and managed structured cabling solution for patching of equipment between cabinets.
2. Coordinate rack-to-rack tie cable requirements with DFW-ITS during the design phase. Typically, no less than (24) tie cables are required from PDS Cabinet to the following cabinets:
 - a. PAVE Cabinet
 - b. Security Cabinet
 - c. MATV Cabinet

11.5.1.5.5.2 DFW CR to Shared Tenant CR

1. Category 6A cabling is required between DFW CR and adjacent Shared Tenant CR. These connections, also known as tie cables, provide a neat and managed structured cabling solution for connection of shared tenant equipment using DFW horizontal infrastructure.
2. One tie cable is required from DFW CR to Shared Tenant CR for every DFW Horizontal Category 6A cable installed from the DFW CR.

11.5.2 Non-Terminal Facilities

Reserved for future use.

11.6 Network Equipment & Electronics**11.6.1 Network Switches**

The DFW network switching hardware, software and associated licensing are furnished and installed by DFW ITS, with the project funding the equipment procured for the project. DFW typically utilizes Cisco enterprise border and edge node network products.

The designer must engage DFW ITS when planning, coordinating, and designing DFW Communication Rooms, and the network switching contained in the equipment cabinets/racks. This will allow ITS to develop the equipment list and begin the procurement process. This will allow the lead time requirements of the project to be more easily met.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.6.2 Communications Rooms

The designer is expected to use the network equipment list, provided by DFW ITS, to coordinate the design, equipment cabinet/rack elevations, wire management, spacing, growth, power, and cooling requirements within DFW communications rooms with the other design team disciplines.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.6.3 Jet Bridge Interface Boxes (JIBs)

Each jet bridge is equipped with a network interface presence to overcome the cabling distances in reaching the cab of the jet bridge.

The designer must engage DFW ITS when planning, coordinating, and designing JIBs and the network switching contained in the JIBs. DFW ITS will provide and install the network switching and perform the patching within the JIBs.

The contractor will be responsible for submitting shop drawings during construction for approval by DFW ITS to ensure coordinated and finalized cabinet/rack elevations and JIBs details.

Any network switching, patching, or supporting cabling infrastructure deviations or substitutions must receive DFW ITS approval in writing.

11.7 Master Antenna Television

11.7.1 System Overview

The existing DFW Master Antenna Television (MATV) System receives, combines, and modulates video feeds from off-air signals, DFW Board feed, and news/weather channels from Digital Broadcast Satellite (DBS). The resultant MATV signal is distributed throughout the DFW campus over a hybrid fiber and coaxial cable infrastructure. DFW's service agreement with the DBS provider limits the distribution of DBS content to DFW employee areas, therefore the signal is not suitable for public viewing or sharing among other users.

The MATV signals are extended from the head end location within Energy Plaza to each terminal's main communication rooms over redundant connectivity, with automatic optical switching if the primary connection is disrupted. The MATV distribution within the terminals must be a star topology from each main communication room to the remaining terminal communication rooms. Each communication room will become a secondary distribution point that feeds identified MATV end devices. For ease of maintenance, consistency, and flexibility MATV zones must correlate with Premise Distribution System (PDS) zone wiring.

The MATV headend includes chassis-based optical fiber transmitters. The transmitters must be hot swappable, and should be consistent in the manufacturer, model number, and transmitter output power, allowing DFW to keep a consistent attic stock and to reduce maintenance downtimes.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.7.2 Design Criteria

The backbone distribution must consist of single-mode optical fiber infrastructure with SC-type angle polished connectors. Within terminal buildings, the horizontal distribution from the communications rooms to the MATV devices must occur over 75-ohm coaxial cable infrastructure with minimum RG-6 performance characteristics.

The MATV system must mute upon activation of the fire alarm system. This is typically accomplished utilizing a coaxial switcher, connected to a fire alarm relay.

Any MATV system deviations or substitutions must receive DFW-ITS approval in writing.

Table 33 - DFW MATV

DFW MATV				
Basis of Design Manufacturer	Basis of Design Model Number	Basis of Design Description	Attic Stock	Responsibilities
N/A	N/A	Commercial Television Display	10% or quantity 1, whichever is greater	Contractor furnished; contractor installed
Olson	LP-OT-15-A-SA	MATV Laser Transmitter, 15-dB	5% or quantity 1, whichever is greater	Contractor furnished; contractor installed
Olson	OTPN-2200C	MATV Optical Node	5% or quantity 1, whichever is greater	Contractor furnished; contractor installed
Emcore	OT-DTCR-2	A/B Optical Switch	1% or quantity 1, whichever is greater	Contractor furnished; contractor installed

11.7.3 Gate Area Entertainment Television

With the discontinuation of CNN Airport television services, the entertainment television displays within the gate areas will no longer be served from the DFW MATV system. It is expected that these entertainment television displays will now be served by an IP-based television solution. Designers must coordinate exact requirements with DFW-ITS early during the design process.

11.8 Satellite Antenna Distribution Systems (SADS)

11.8.1 System Overview

Concessions and other tenants utilize Digital Broadcast Satellite (DBS) subscription services for their television needs. DFW does not allow these tenants to place DBS dishes on the rooftop, instead these services are distributed via optical fiber backbone for tenant use. This distribution system is known as the Satellite Antenna Distribution System (SADS).

The SADS consists of a headend with a DirecTV dish antenna with feeds to four L-Band (950-2150 MHz) coax to fiber converters. The outputs from the fiber converters are fed into an 8-input wave division multiplexer (WDM) that stacks the signals onto a single-mode fiber optic strand. The unused inputs on the WDM are reserved for a future DISH Network connection. The output of the WDM is fed into an EDFA before being split for distribution to the campus fiber backbone.

The fiber backbone distributes the L-Band signals from the headend to each terminal's main communication room, where they are fed into a wave division demultiplexer, which feeds into L-Band fiber to coax converters that feed into a single-wire multi-switch.

11.8.2 Design Criteria

The SADS distribution within the terminals must be a star topology from each main communication room to the remaining terminal communication rooms. Each communication room is a secondary distribution point, converting the media type from optical fiber to coaxial cabling for location distribution to the tenant spaces. Tenants are responsible for coaxial cable distribution within the leased space, as well as the necessary subscription service and satellite receivers.

11.8.3 System Muting

The SADS system must mute upon activation of the fire alarm system. This is typically accomplished utilizing a switching power module receptacle, connected to a fire alarm relay.

Any SADS system deviations or substitutions must receive DFW-ITS approval in writing.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

Table 34 - DFW SADS

DFW SADS				
Basis of Design Manufacturer	Basis of Design Model Number	Basis of Design Description	Attic Stock	Responsibilities
N/A	N/A	Commercial Television Display	N/A	Tenant furnished; tenant installed
Olson	OTCP Series	Optical Splitter	N/A	Contractor furnished; contractor installed
Olson	OT-DWDM Series	L-Band Receiver	5% or quantity 1, whichever is greater	Contractor furnished; contractor installed
Olson	LP-OLAR-X4000-75-SA	L-Band Optical Receiver	5% or quantity 1, whichever is greater	Contractor furnished; contractor installed
DirecTV	SWM16	Multi Switch	5% or quantity 1, whichever is greater	Contractor furnished; contractor installed
Emcore	OT-DTCR-2	A/B Optical Switch	1% or quantity 1, whichever is greater	Contractor furnished; contractor installed

11.9 Audio/Visual

11.9.1 Digital Signage/Content Management System

Digital wayfinding and digital signage media content will be distributed over the DFW network system utilizing the existing Synect Media solution. The Airport Enterprise Communications System (AECS) also utilizes the Synect Media solution. A media player connected to the DFW network is required at each of these displays.

Any content management system deviations or substitutions must receive DFW-ITS approval in writing.

11.9.2 Public Address/Voice Evacuation System (PAVE)

Refer to separate document – DFW ITS DCM Phase 2 Section 7.

11.9.3 Visual Paging

Visual paging integration is an existing integration between the PAVE system and the Electronic Visual Information Display System (EVIDS). This media content is distributed to the displays utilizing the Synect Media solution.

Any visual paging system deviations or substitutions must receive DFW-ITS approval in writing.

11.10 Information Technology

11.10.1 Demolition & New Construction Scope Responsibilities

11.10.1.1 General

Minimum requirements for installers must be as follows:

1. Strictly adhering to all BICSI and TIA recommended installation practices when installing communications pathways.
2. Firms that are regularly engaged in the installation of Communications infrastructure and have five (5) years of installation experience.
3. Provide verification that installation personnel responsible have been professionally trained to install the products described in this Section.

4. Provide a BICSI RCDD certified professional for oversight on this project.
5. Provide full-time project manager with a minimum of ten (10) years' field experience in installation of communications infrastructures.

Demolition drawings must be prepared which clearly indicate altered, discontinued, and removed work. Demolition drawings must indicate, but not be limited to, the following pertinent information:

1. Communications Rooms
2. Backbone Cabling and Pathway
3. Horizontal Cabling and Pathway
4. Technology devices
5. Security devices
6. Other information to clearly indicate the extent of known spaces, systems, and conditions to be demolished.
7. Symbols that are used for the demolition work must be the same as those used on the drawings for new construction.
8. Design of make-ready solutions for essential systems that are to be demolished allowing for systems interruption due to planned construction activities.

11.10.1.2 Definitions

Remove - Detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged or removed and reinstalled.

Remove and Salvage - Carefully detach from existing construction, in a manner to prevent damage, and deliver to Owner.

Remove and Reinstall - Detach items from existing construction, prepare for reuse, and reinstall where indicated.

Existing to Remain - Existing items of construction that are not to be permanently removed, and that are not otherwise indicated to be removed, removed, and salvaged, or removed and reinstalled.

11.10.1.3 Stated Tasks to be Defined

Activities must be completed prior to the commencement of primary demolition activities.

Examine record documents of existing construction provided by Owner.

Survey existing conditions and correlate with requirements indicated to determine extent of selective cable demolition required.

Identification and labeling of selected existing communications backbone and horizontal cabling and its associated pathway.

Completing a cabling inventory spreadsheet.

Completing a colored-coded diagrammatic cabling record (both floor plans and risers).

Resulting in a stated Remediation Plan of every element.

Salvaging existing items to be reused or recycled.

11.10.2 Outside Plant System (OPS)

11.10.2.1 Underground Ducts and Raceways for Communication Systems

11.10.2.1.1 General

Minimum requirements for installers must be as follows:

1. Strictly adhering to all BICSI and TIA recommended installation practices when installing communications pathways.
2. Firms that are regularly engaged in the installation of Communications infrastructure and have five (5) years of installation experience.
3. Provide verification that installation personnel responsible have been professionally trained to install the products described in this Section.
4. Provide a BICSI RCDD certified professional for oversight on this project.
5. Provide full-time project manager with a minimum of ten (10) years' field experience in installation of communications infrastructures.
6. Terminals and critical facilities, as defined by DFW, must have redundant and diversified pathways into the building.
7. Buildings greater than 30,000 square feet in size, or as defined by DFW, must have a separate and dedicated Entrance facility.

11.10.2.1.2 Duct Banks

Duct-banks must consist of no less than no less than four four-inch conduits.

Communication duct-banks must be concrete encased.

All communications concrete encased conduit systems installed on DFW Airport property, including Airfield, must comply with Figure # below.

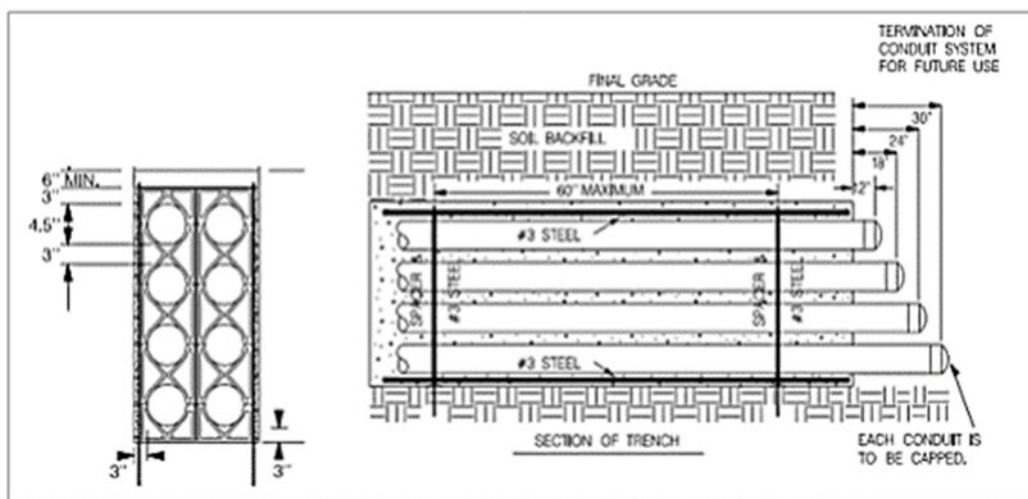


Figure 21 - Communications Conduit Encased Detail

11.10.2.1.3 Installation Notes

1. Conduit for concrete encasement is to be four-inch solid wall PVC SCH-40 and must not be cellular core PVC product.
2. Conduits bored under roadways, taxiways, and runways must be four-inch PVC SCH-80 or HDPE.
3. All concrete encasements must have a minimum compression test strength of 3000 psi at 28 days.

4. All concrete encasements installed under roadways or in the Airfield (AOA) must have a compression test strength of 4000 psi at 28 days.
5. Install spacers every five- feet along the entire length of the encasement. Sheet spacers are not acceptable.
6. Install a “bottom” spacer inverted on top of spacer assembly.
7. Install #3 reinforcing rods through the internal vertical channels at both outer edges of spacer and drive rods into earth to anchor assembly.
8. Install #3 steel reinforcing rods horizontally on the top of the inverted spacer and secure to vertical reinforcing rods.
9. Tie all reinforcing rods using #16 minimum tie-wire.
10. Reinforcing rods are NOT permitted to touch conduit at any point including the tie-wire.
11. All backfill of a conduit duct trench must be replaced to a minimum compaction of 95%.
12. Each four-inch conduit must have a 6000 lb. pull tape installed.
13. Encase with a minimum of three inches of concrete on all sides and a minimum of three-inch on top.
14. SCH-40 PVC bends are allowed with a minimum of 42 inch radius.
15. Burial depths must reflect the top of duct envelope at least 24 inches below finished grade in areas not subject to deliberate traffic, and at least 30 inches below finished grade in deliberate traffic paths for vehicles unless otherwise indicated.
16. Conduits must be arranged to drain back to maintenance-hole or pull box.
17. For continuations of concrete encased installation, provide a 45-degree slope for cold joint concrete overlap to prevent sheer points in the concrete encasement.
18. Allow a minimum of 12 hours concrete cure time prior to backfill and compaction.
19. Alternate designs must be submitted for review and must be approved by DFW.
20. Each Conduit route must be installed with the least number of bends possible. No section of conduit must contain more than two 90-degree bends (offset is a 90-degree bend) between pull points.
21. Ensure underground Warning-line Tape, bury conducting underground line no less than 12 inches above all concrete-encased duct and duct banks and approximately 12 inches below grade.

11.10.2.1.4 Kerf-cuts

Kerf-cuts are PROHIBITED, with one allowable exception:

1. From terminal building or terminal building drip edge not to exceed the end of the jet bridge, only one one-inch PVC coated RMC or IMC conduit per kerf-cut is allowed. Kerf-cuts must not exceed five-inch depth. A minimum cover of three inches above conduit is required.

11.10.2.1.5 Maintenance-holes and Handholes

These structures must be located outside of runway and taxiway safety areas (as defined in AC 150/5300-13).

Units in roadways and other deliberate traffic paths must require precast concrete with a H20 structural load rating (AASHTO HB 17).

Units in driveways, parking lots, and off-roadway locations, which are subject to occasional, nondeliberate loading by heavy vehicles must require precast concrete with a H20 structural load rating (AASHTO HB 17).

Units in landscaping, sidewalk, and similar applications with a safety factor for nondeliberate loading by vehicles must require precast concrete with a H10 structural load rating (AASHTO HB 17).

Units must be designed for flush burial and have closed bottom unless otherwise indicated.

Cover design load must not exceed the design load of the handhole or box, yet heavy-duty. Each cover must have a label molded into the cover, labeled as COMMUNICATIONS with a unique IDENTIFIER (alpha/numeric).

Duct entrances in maintenance-holes or handhole walls must be cast end-bell or duct-terminating fitting in the wall for each entering duct.

Fittings must align with elevations of approaching duct and be located near interior corners of handholes to facilitate racking of cable.

Maintenance-holds or handhole must have inserts for cable racks and pull-in irons installed before concrete is poured.

Structures must sufficiently be raised above the surrounding grade to prevent ponding water on the structure and the top cover sloped to drain. A concrete apron must be constructed around all communications maintenance holes located in turfed areas.

Attention must be given to stormwater drainage plans to prevent placement of communications structures in areas channeled for drainage. These structures must have all joints and openings completely sealed and vermin-proof with secure covers with bolts.

Maintenance holes must be a minimum of six-foot wide by seven-foot high by twelve-foot long.

Hand-holes must be a minimum of three-foot wide by four-foot high by five-foot long.

Maintenance holes must be located at a maximum spacing of 500 feet.

Handholes must be located at a maximum spacing of 300 feet.

Where conduits enter a building, a handhole must be provided:

1. Within 50 feet of the building exterior
2. At a lower elevation than the building foundation

11.10.2.1.6 Above-ground Enclosures

Outdoor network and/or security enclosures shall be Hoffman T5777ASY pad-mounted enclosure, or equivalent product with prior written approval by DFW-ITS. The enclosure must feature:

1. Aluminum construction with stainless steel fasteners
2. Pitched roof to eliminate water runoff
3. 5 inch plinth base
4. Front and rear door access
5. 3-point latching mechanism and padlocking handles

6. Foam-in place gasketing to protect against dust and moisture
7. Wind-latch mechanisms to protect door when open for maintenance
8. Removable lifting lugs
9. Adjustable rack angles for 19 inch or 23 inch equipment
10. 4000 BTU/Hr. air conditioner with heat package
11. Integrated 12 inch wide compartment for electrical service
12. Approximate dimensions of 49 inch high by 42 inch wide
13. Minimum 24 inch depth

11.10.3 Wi-Fi – Infrastructure & DFW Vendor Engagement During Design

Wi-Fi is deployed in Public and key back of house spaces throughout all levels of a terminal.

The designer must engage DFW ITS for the current third-party Wi-Fi vendor as early as possible.

The designer will coordinate with DFW-ITS and the Wi-Fi vendor to ensure that there is 100% Wi-Fi coverage in public areas and strategic key areas. Designers will provide DWG files at each stage of design to DFW ITS for the Wi-Fi vendor use to create a heat map. Some key areas include:

1. Electrical rooms
2. DFW Comm Rooms
3. Mechanical rooms
4. Apron
5. Inside of jet bridges

The designer shall include the required pathway and cabling infrastructure in the design documents to support the coordinated Wi-Fi layout.

Some Airlines deploy a separate Wi-Fi network, and it is the responsibility of the designer to ensure their specific requirements are met as well.

Refer to the responsibility matrix below to better understand the division of scope for the installing contractor, DFW ITS and the third-party vendor.

DFW non-terminal (Board) facilities will operate focused on DFW Wi-Fi only. Coordination with DFW ITS Construction is required for additional information.

Table 35 - WI-FI RESPONSIBILITY MATRIX

WI-FI Responsibility Matrix				
	Responsible Party			Remarks
	DFW-ITS	Contractor	Other	
Public (AT&T) Wi-Fi Devices			X	AT&T to install Wi-Fi antenna devices and network switches as required
Public (AT&T) Wi-Fi Pathway		X		Aggregate pathways by owner by system is allowable
Public (AT&T) Wi-Fi Cabling		X		Contractor shall furnish and install horizontal cabling infrastructure per coordinated AT&T direction
Patching Wi-Fi Within the Communications Room	X			Contractor to provide all patch cords. Third party vendor to patch to the antenna
Patching At the Device/Wi-Fi Antenna Location			X	Contractor to provide all patch cords. Third party vendor to patch to the antenna

11.10.4 DFW Timeclocks – DFW & Vendor Timesheet Management

11.10.4.1 General

Although DFW timeclocks are POE, always include an electrical outlet with each unit.

Each receives a single CAT6A cable from the nearest DFW Communications Room.

Electrical service to be placed next to the work area outlet. Follow the manufacturers' recommended placement for the DFW unit type KRONOS 9100.

Both the work-area outlet and electrical receptacle are to be 48-inch above finished floor.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.11 Voice/Telephony

11.11.1 Telephone System

DFW Airport uses a distributed Voice over IP telephone system for most voice services. The airport provides shared tenant services from the system for tenants and concessionaires who wish to use it.

The system is used in the following applications:

1. DFW Airport Board offices (VoIP)
2. Tenant and concessionaire spaces (analog)
3. Elevators (analog)
4. Terminal courtesy phones (analog)
5. Jet bridge cabs (analog)
6. Common/shared use gate counters – one per pair of positions (analog)
7. Common/shared use ticket counters – one per pair of positions (analog)
8. Emergency blue phones (analog)

All telephones, analog gateways and licenses are provided by DFW ITS but funded by the project.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.11.2 Analog Telephony Services

Analog port counts should be provided to DFW ITS during design, and rack space must be allocated for analog voice gateways, when required.

Analog gateways are equipped with RJ-11 jacks and patched much like a network switch, requiring wire management above and below the gateway. RJ-11 to RJ-45 patch cords are required for patching from the gateway to the work area outlets.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.11.3 Emergency Blue Phones

Emergency Blue Phones are required in parking structures and some surface lots and are provided and installed by the contractor.

The units are free standing towers, painted hospital blue, with the word "EMERGENCY" painted on the side in large white letters. The towers have a 120V powered blue light on top and an analog speakerphone installed in them. The phone is equipped with information and emergency call buttons and connects to the campus-wide telephone system.

The towers typically require airport CCTV camera coverage.

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.11.4 TBU Intercom System

The DFW Airport Transportation Business Unit (TBU) owns a Zenitel Alphacom intercom exchange that provides two-way communications between motorists and the TBU operations center. Intercom substations are typically deployed at parking control plazas and revenue/non-revenue area cross-over gates.

The standard intercom station is the Zenitel TCIS-2 Turbine unit.

11.12 Radio

11.12.1 Cellular Distributed Antenna Systems (DAS) – Infrastructure & DFW Vendor Engagement During Design

The Designer must coordinate with the DFW ITS during detailed design to confirm requirements of the Cellular DAS system to support 5G. Terminal Gate Hold Lounges also require 5G/MM capabilities.

The Designer must engage the DFW approved DAS System Maintenance Vendor to perform a spectrum analysis of the proposed site for Cellular signal levels. The report must consider the building materials and interior building environment and make a recommendation based on the findings. The vendor is to use the report to create a Cellular DAS design and integrations of those systems for coverage throughout the project area. The Vendor will be responsible for installation, cabling, and pathway to support the system as it is integrated into the DFW campus-wide solution.

The designer is responsible for planning, coordinating, and designing the DFW Communication Rooms to contain adequate wall space, equipment cabinet space, power, cooling, grounding, cabling infrastructure and service clearances to support Cellular DAS system equipment requirements. The designer is also responsible for coordinating any pathway accommodations through challenging areas, such as vertical chases.

If the project involves a DFW Terminal facility that includes gates operated by American Airlines, the design team and DAS vendor must consult with the American Airlines Mobility team during the design process to ensure forward planning with American Airlines enterprise requirements.

DFW Terminal Gate Hold Lounges require 5G/Millimeter wave capabilities for two carriers, AT&T and Verizon. Each carrier will need to be coordinated with and provide design, integration, and installation. Both carriers have composite single mode fiber-based solutions.

DFW non-terminal (Board) facilities will operate in a different fashion. Coordination with DFW ITS Construction is required for additional information.

11.12.2 Public Safety Distributed Antenna Systems (PDAS)

The Public Safety Distributed Antenna System (PDAS) is also known as the Emergency Responder Radio Coverage System (ERRCS).

The Designer must engage the DFW IT Radio System team to provide Public Safety Radio DAS design and integrations of those systems for coverage throughout the project area and coordinate with the Owner during detailed design to confirm requirements of the system.

The PDAS System Maintenance Vendor is responsible for performing a spectrum analysis of the proposed site for IBRS signal levels. The analysis report must consider the building materials and interior building environment and make a recommendation based on the findings. The vendor is to use the report to create the design and integrate the systems for coverage throughout the project area. The Vendor will be responsible for installation, cabling, and pathway to support the system as it is integrated into the DFW campus-wide solution.

The designer is responsible for planning, coordination, and designing the DFW Communication Rooms to contain adequate wall space, equipment cabinet space, power, grounding, cooling, cabling infrastructure and service clearances to support Public Safety Radio DAS system equipment requirements. The designer is also responsible for coordinating any pathway accommodations through challenging areas, such as vertical chases.

If the project involves a DFW Terminal facility that includes gates operated by American Airlines, the design team and DAS vendor must consult with the American Airlines Mobility team during the design process to ensure forward planning with American Airlines enterprise requirements.

DFW non-terminal (Board) facilities will operate in a different fashion. Coordination with DFW ITS Construction is required for additional information.

11.13 Building Management Systems or Building Automation Systems

11.13.1 EIOS Building Management Platform – Systems Integration for Monitoring & Control

EIOS (enterprise integrated operating systems) is the platform utilized at DFW airport to monitor equipment and devices across all systems. EIOS is spoken to throughout other sections of the DCM and how specific equipment shall interface to the DFW network. EIOS is often referred to as BMS (building management system) or BAS (building automation system).

The designer is responsible for providing pathway and cabling infrastructure in the design documents to support the equipment that is to be integrated into EIOS.

Prior to commissioning the systems, DFW BAS will provide a spreadsheet for the installation contractor to fill out to obtain network connectivity. This typically includes the following information:

Table 36 - BAS TABLE

Column	Contents	Example	Provider
Object	Name of device.	PMX-4000 Power Meter Main	DFW BAS
MAC Address	Ethernet MAC	10:1C:A6:06:99:EF	Contractor
Location	Physical location of device	Electrical Room 202	Contractor
Patch Cable Identifier	Label on the device's patch cable	TC-RS2-073 3-36-23	Contractor
Protocol	Communications protocol	BACnet IP	Contractor
Is the unit powered up and cable plugged in?	Acknowledgement that the device is ready to be connected to the network.	Y, Y	Contractor
IP	IP Address	172.16.34.248	DFW BAS
BACnet Network	Only applicable to BACnet devices.	38545	DFW BAS
BACnet Instance ID	BACnet Network ID followed by the MSTP address (a.k.a. BACnet MAC) of the device.	3854500	DFW BAS
DNS	DNS Host Name	BAS-Admin-CR1-UPS	DFW BAS
Assigned Switch Port	Switch blade and port	CR1 Blade 3 port 14	DFW IT
Network Comms Successful?		Yes	DFW BAS

Column	Contents	Example	Provider
Comments/Status	Any important status information.	Port configured. Need to restart the JACE.	

DFW non-terminal (Board) facilities will operate similarly. Coordination with DFW ITS Construction is required for additional information.

11.13.2 View Glass / View Sense Systems

VIEW Glass is deployed throughout all levels of terminal curtain walls where glazing is present and exposed to the public.

The designer must engage DFW DCC for the decision on whether VIEW Glass is required. If it is a scope element, then the designer will coordinate with DFW DCC, VIEW and the project architect to ensure that there is a design prepared by VIEW.

The designer shall prepare design work for pathway of the VIEW cabling infrastructure between the VIEW glazing elements and the electrical room housing the VIEW Controller panels. This pathway should consist of the following elements:

1. Proportionally sized Cable Tray
2. Large Pull-boxes (~30-inch by ~30-inch by ~12-inch) as the VIEW design requires, to house the Network Window Controllers
3. Sleeves into each Electrical Room identified in the VIEW design

VIEW Sense is deployed throughout all levels of terminal exposed to the public.

The designer must engage DFW DCC for the decision on whether VIEW Sense is required.

If it is a scope element, then the designer will coordinate with DFW DCC, VIEW and the project architect to ensure that there is a design prepared by VIEW.

The designer shall prepare design work for pathway and cabling between the VIEW glazing Network Window Controller and each VIEW Sense device. This pathway should consist of the following elements:

1. 0.75-inch conduit between Network Window Controller Pull-box and the backbox associated with each VIEW Sense device.
2. 4-11/16-inch by 2-1/8-inch double-gang backbox associated with each VIEW Sense device.

11.13.3 Vertical Conveyance & Moving Walkway

There is a monitoring system that is utilized for vertical conveyance and moving walkways separate from BMS/BAS monitoring. The additional monitoring system is a delegated design, and the designer needs to contact Blake Moore and Jonah Johnson with DFW ITS during the design process to confirm exact requirements.

The designer is responsible for providing pathway and cabling infrastructure in the design documents to support the monitoring system for each vertical conveyance and moving walkway.

Required cabling infrastructure:

1. 6 strands single mode fiber from the nearest DFW Comm Room typically to elevator machine room where to interface with the elevator traveler cable.

2. Elevator traveler cable is by the elevator contractor
3. (2) port WAO for required elevator phone
4. (2) port WAO for BMS/BAS

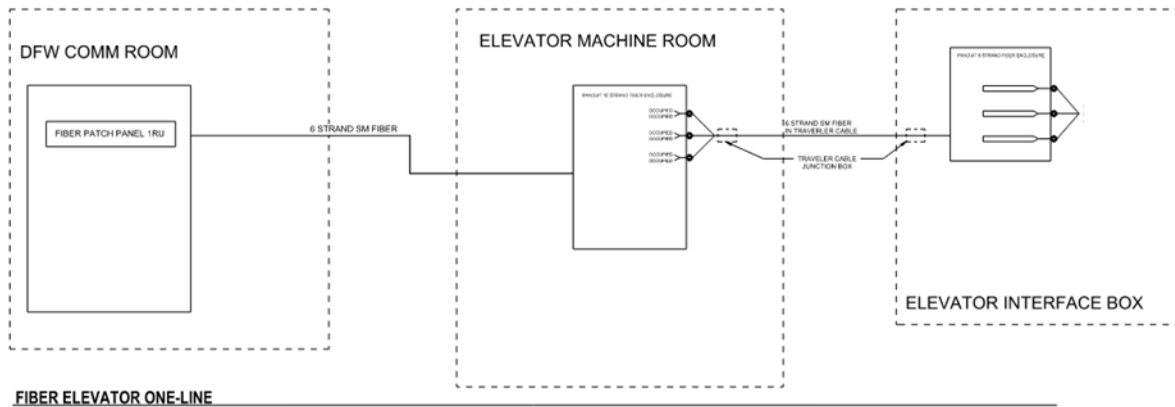
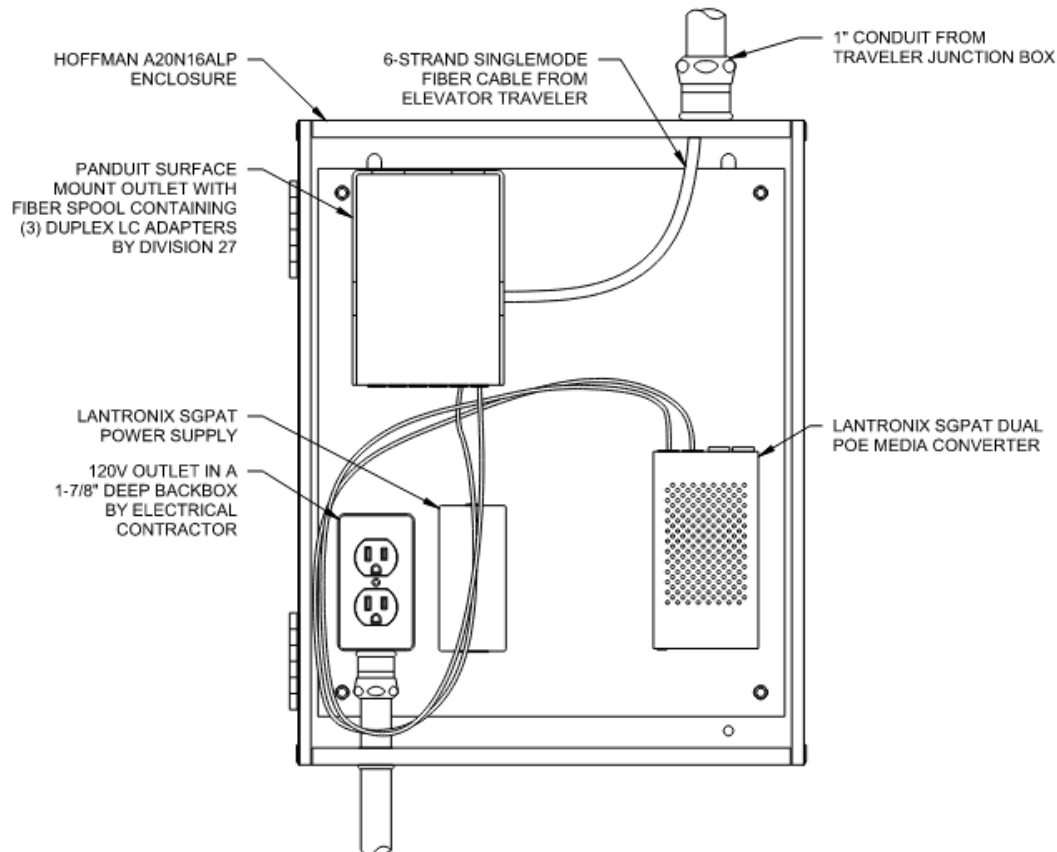


Figure 22 - Fiber Elevator One-Line



ELEVATOR INTERFACE BOX (EIB)
N.T.S.

Figure 23 - Elevator Interface Box (EIB)

11.13.4 Concessions Operation Status Sensor

11.14 Airport Specific Systems

11.14.1 Advanced Visual Docking Guidance System (A-VDGS)

The A-VDGS, sometimes referred to as the autodocking system, primarily allows pilots to park their aircraft at their assigned gate without the assistance of the ground crew. The system utilizes the ADB SAFEGATE Safedock product line and new deployments are to be added to the existing apron management system.

This system is to be provided by the contractor and requires the contractor to engage ADB SAFEGATE for licensing, system configuration and commissioning. The designer is to develop the design at each gate, based on the desired striping plan, in partnership with ADB SAFEGATE and DFW ITS.

11.14.2 Last/Final Bag System

The last/final bag system is a DFW Airport subsystem that is managed by the apron management system. The principal user of the system is American Airlines. The system is used to notify gate agents when the last bag has been loaded onto the aircraft and when the Lightning Warning and Notification system thresholds have caused the ramp to be closed to ground personnel, due to lightning being detected in the area.

The Last/final bag system is deployed in the terminals and consists of PLC output modules installed in DFW communications rooms that are wired to a multicolor indicator lamp installed in gate counters. The module is installed on a 2RU rack shelf and wired to a 24-port patch panel. Lights are wired to a single-ended patch cord and plugged in to a DFW work area outlet and patched to the I/O module's patch panel in the communications room. Figure 24 below shows the typical communications room rack elevation.

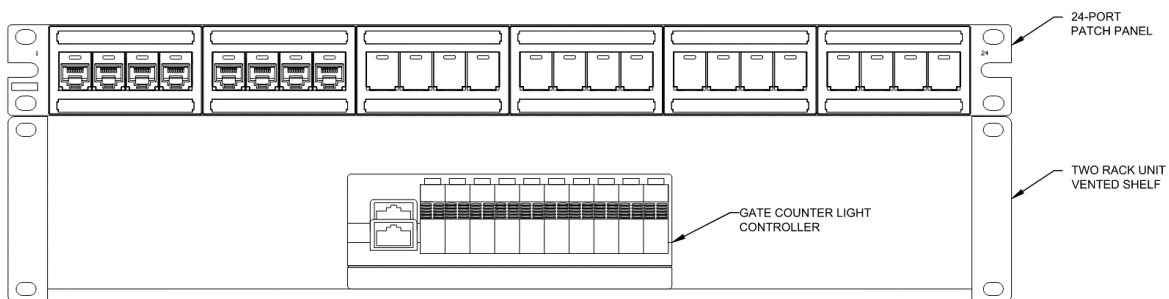


Figure 24 - Last/Final Bag System Controller

Figure 25 below shows the typical wiring between the PLC module and the patch panel.

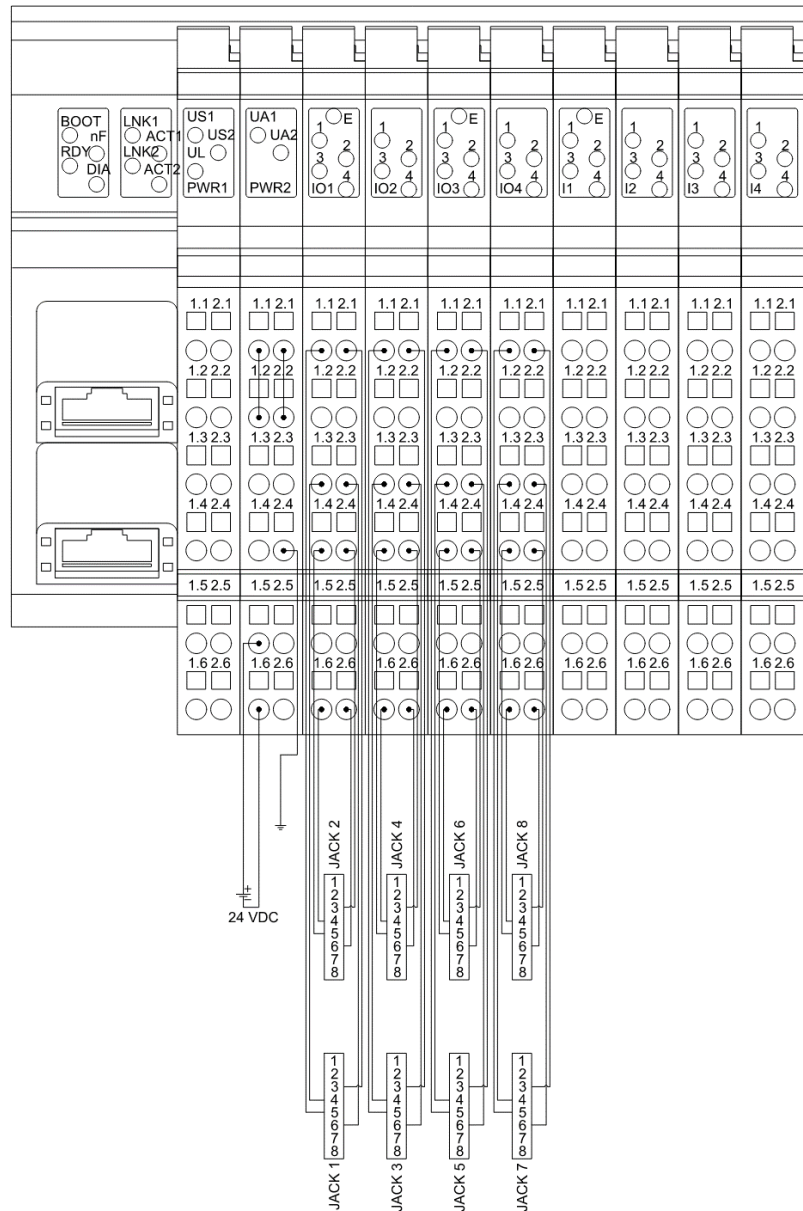


Figure 25 - Last/Final Bag System Controller

The indicator lamp at the gate counter is typically a blue and yellow LED multicolor lamp, with the blue LED wired to the last/final bag circuit and the yellow LED wired to the ramp closed circuit. The installed location of the lamp is to be coordinated with the gate counter millwork design.

11.14.3 Common Use Passenger Processing Systems (CUPPS)

The common use system shall be compliant with all IATA/ACI standards and recommended practices, including, but not limited to RP1797, RP1800, and RP753.

Common use is deployed throughout OAL terminals at the following locations:

1. Self-service kiosk
2. Self-bag drop
3. Ticket counters
4. Gate counters
5. Gate information display (GIDS)
6. Counter information display (CIDS)

All peripheral devices for common use workstations are provided and installed by DFW ITS. The designer is responsible for including:

1. Required pathway and cabling infrastructure for common use systems.
2. Mounts for GIDS and CIDS
3. Mounts for Biometric Reader at gate counters

Required cabling infrastructure:

1. (4) port work-area-outlet (WAO) for self-service kiosk and bad drop
2. (6) port WAO for per gate and ticket counter
3. (2) port WAO for paging mic at gate counter
4. (2) port WAO for each GIDS and CIDS

Refer to the PDS section in phase 1 of the DCM for additional information and infrastructure requirements.

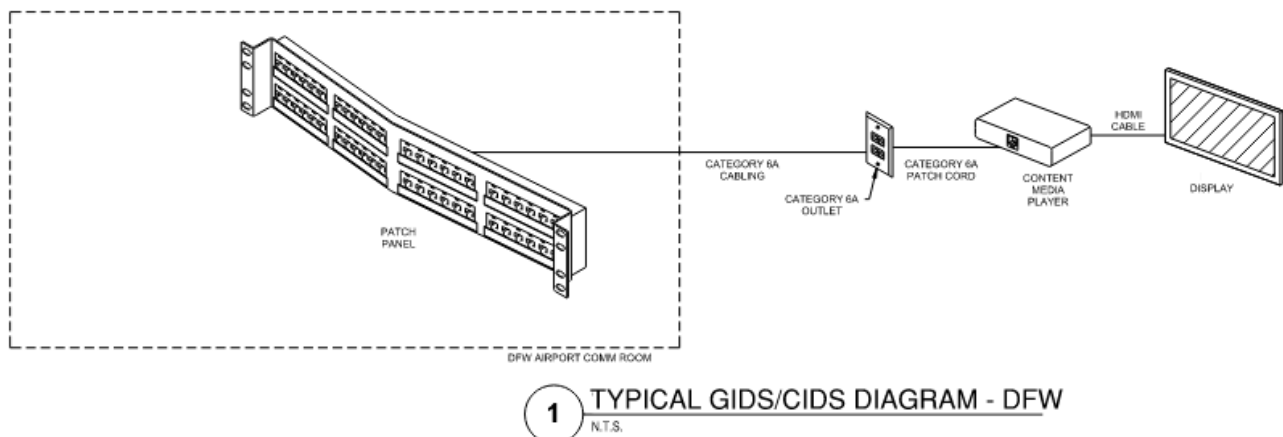


Figure 26 - Typical GIDS/CIDS Diagram – DFW

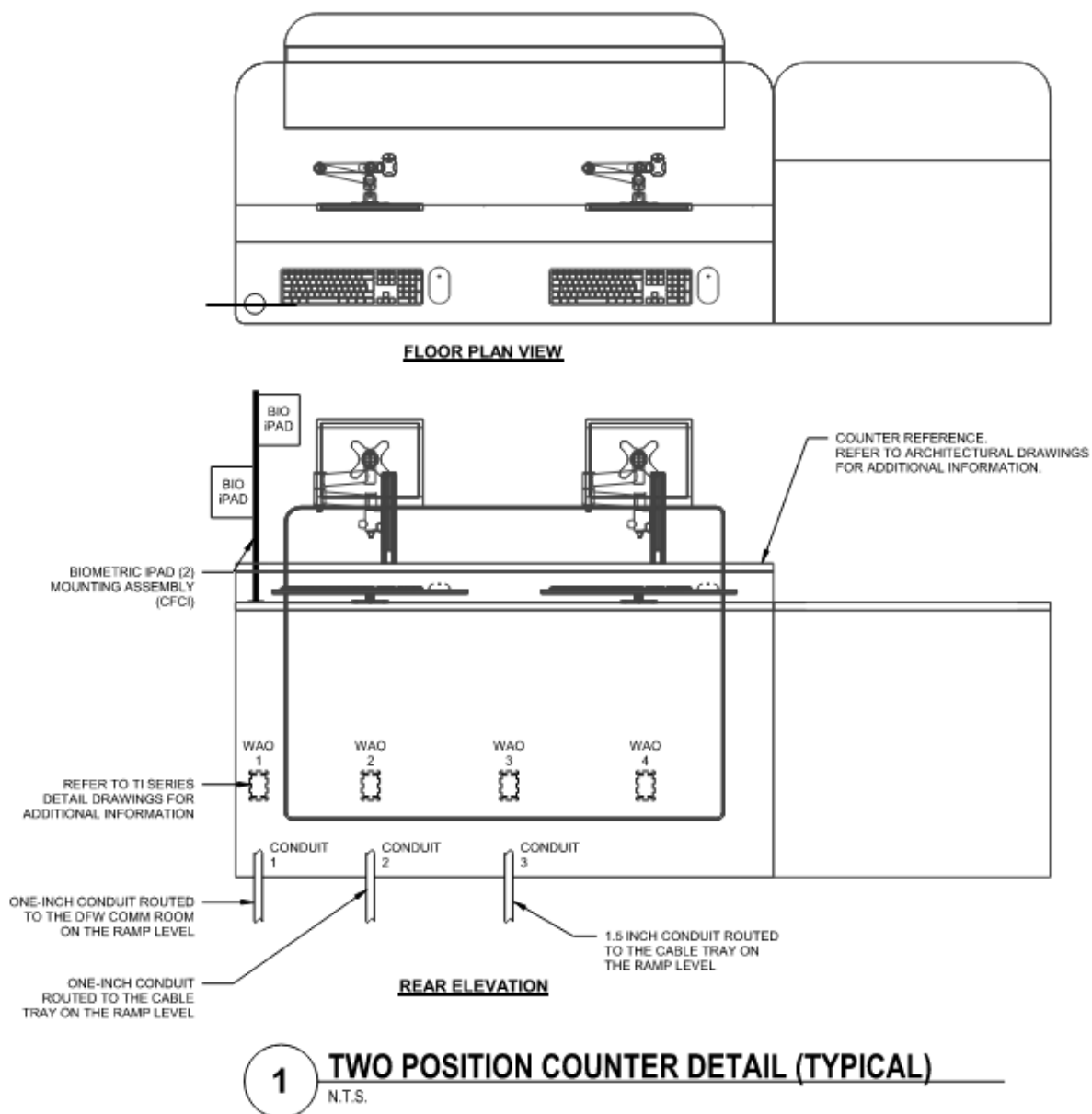


Figure 27 - Two Position Counter Detail (Typical)

11.14.4 Biometric Exit

Biometric exit is a CBP initiative that verifies a passenger, typically one who arrived in the country on a visa and left the country. This system is deployed at the passenger boarding bridge doors and is supplied and installed by DFW Airport. It typically requires a power and data outlet per unit.

11.14.5 Electronic Visual Information Display Systems (EVIDS)

DFW Airport's media content distribution to EVIDS displays occur over the DFW network system utilizing the Synect Media solution.

EVIDS are project funded and Owner-Furnished, Contractor-Installed (OFCI). Funding must include attic stock requirements as noted below.

A typical Baggage Information Display System (BIDS) display is a commercial grade 43" size diagonal monitor. Monitor size to be confirmed with DFW-ITS during design.

A typical Customer Information Display System (CIDS) display is a commercial grade 43" diagonal monitor. Monitor size to be confirmed with DFW-ITS during design.

A typical Flight Information Display System (FIDS) display is a commercial grade 75" diagonal monitor. Monitor size to be confirmed with DFW-ITS during design.

A typical Gate Information Display System (GIDS) display is a commercial grade 43" diagonal monitor. Monitor size to be confirmed with DFW-ITS during design. These may be replaced by dvLED technology.

A typical Ramp Information Display System (RIDS) display is an exterior signage display tied to the AVDGS system. Monitor size to be confirmed with DFW-ITS during design.

Direct-view LED (DvLED) displays, such as Security Screening Checkpoints and Skylink areas, receive media content from the Synect Media solution. The typical DvLED display manufacturer is LG, and DFW ITS approval is required for equivalent solutions.

A minimum of 10% attic stock is required for each DvLED display. A minimum of 10% attic stock is also required for LCD monitors.

DFW corporate content monitors are furnished and installed by the Contractor. Synect Media solution content players are funded by the project and Owner-Furnished, Contractor-Installed (OFCI).

Any EVIDS deviations or substitutions must receive DFW-ITS approval in writing.

11.14.6 Lightning Warning and Notification System

The Lightning Warning and Notification System is integrated with the DFW Airport apron management system. It triggers the RIDS to display a "RAMP CLOSED" message, and it activates indicator lamps throughout the American Airlines terminals to notify personnel that lightning has been detected in the area and when the ramp is closed. See the Last/Final Bag System section for design criteria associated with this system.

11.14.7 Passenger Flow and Queue Monitoring

Passenger flow is deployed throughout each terminal. Current areas of consideration are:

1. TSA SSCP and associated circulation paths and Ticketing Halls in proximity of the SSCP
2. Concessions
3. Independent Ticketing Halls

The designer must engage DFW ITS for concise direction of each space considered. Customer Experience shall also be engaged for the elements important to them.

There is a Service Provider that will ultimately provide a design, the sensors and associated hardware and electronics to DFW.

If it is a scope element, then the designer will coordinate with DFW ITS, DFW Customer Experience, the Service Provider, and the project architect to ensure that there is a design prepared by the Service Provider (as of 01-01-2024 is XOVIS).

The designer shall prepare design work for 1) pathway and cabling infrastructure between each sensor and the nearest DFW Communications Room housing the service providers appliances and electronics, and 2) Visual displays.

This pathway should consist of the following:

1. Either a 0.75-inch conduit or non-contiguous supports between sensor and the nearest Cable Tray

This cable will consist of the following:

1. A CAT6A cable, DFW is the owner

DFW ITS and DFW Customer Experience shall direct the designer on the visual display type; either as a traditional LCD/LED type display bank or dvLED panels.

11.14.8 Smart Restroom Systems

Smart Restroom Systems are deployed at all Airside public restroom locations.

The designer should engage DFW ITS early in the design process so that they may engage the third-party vendor responsible for deployment and final layout of smart restroom devices.

Each smart restroom includes a network hub that provides network connectivity via a typical copper DFW work-area-outlet. The network hub is where all cabling for devices associated with the smart restroom system connects.

Smart Restroom devices include monitors, throughput counting sensors, occupancy lights, and proximity beacons.

Monitors are located at the entrance of each restroom and need to be recessed so they are flush with the wall. The designer must coordinate with the architect and other disciplines for monitor mounting, electrical requirements, and any ventilation required.

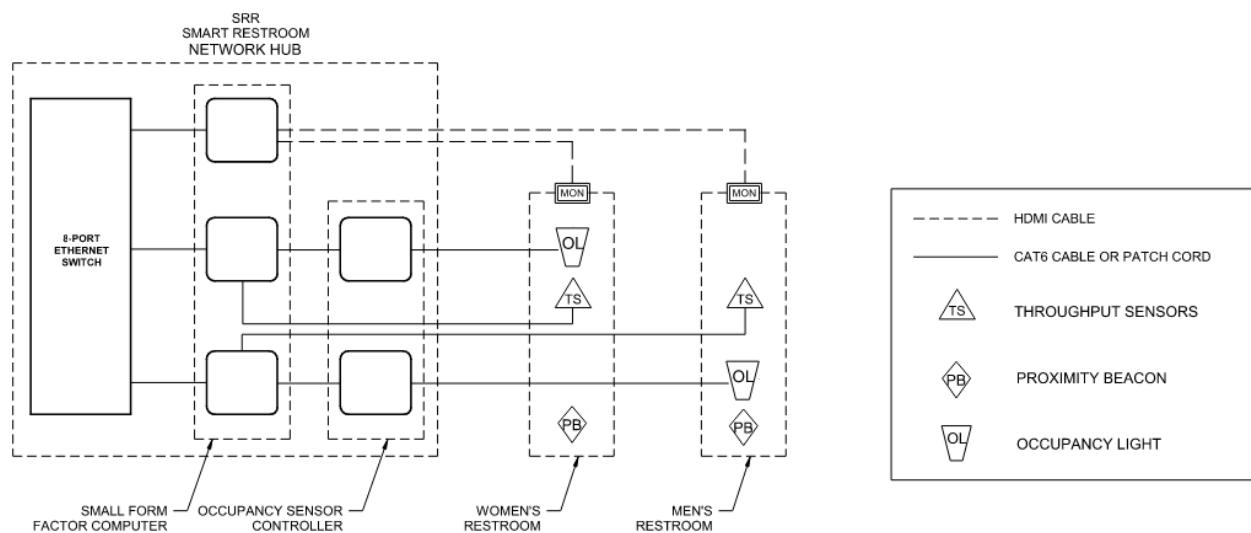


Figure 28 - Smart RR System

11.15 Digital Twin Requirements

11.15.1 Overview

The DFW Digital Twin is being developed to create a virtual representation of the airport's physical assets. Several initiatives are underway to develop the data infrastructure and standards for information ingestion. To ensure data accuracy and integrity, it is vital that these standards are communicated early to consultants and adhered to during model and data delivery to DFW. Required adherence to the DFW Airport Digital Facilities and Infrastructure Standards Manual (DFI).

11.15.2 BIM Standards Outline

On most large projects DFW Airport requires that the Architecture, Engineering and Consulting teams develop each project and each sub-discipline in Autodesk Revit and adhere to the DFW Airport Digital Facilities and Infrastructure Standards Manual (DFI). The prime architect for each team should be able to provide and coordinate this information through the DFW Design, Code & Construction department.

Each design team subconsultant is required to run the DFW DFI Model Checker XML tool to validate that their Revit project model complies with DFW's DFI standards and successfully achieves a score of 100% before submitting their model. Coordination meetings and project model checks are required at multiple stages during the design process.

Refer to these for model setups and QAQC requirements for Digital Twin conforming data.

11.15.3 Sensors Standards Outline

DFW has created sensor standards and guidelines that can be found in the "Sensor Device & Data Acquisition Standards, Principles, Guidelines" refer to ITS for current requirements:

Refer to DFW ITS for standard to determine useability and adherence to DFW data needs.

11.16 Security Systems

11.16.1 Overview

The DFW Airport security systems include the Automated Access Control System (AACS) and the Security Surveillance System (SSS). These systems are integrated with a common PSIM that provides the security staff comprehensive operations and alarm management for the entire airport.

11.16.2 Abbreviations

Trade association names and security terminology are frequently abbreviated. The following acronyms or abbreviations may be referenced within this document.

AACS	Automated Access Control System
ANSI	American National Standards Institute
AVI	Automatic Vehicle Identification
AWG	American Wire Gauge
AOA	Air Operations Area
BHMA	Builders Hardware Manufacturers Association
BHS	Baggage Handling System
CBP	Customs and Border Protection
CIB	Camera Interface Box
CSI	Construction Specification Institute
CTB	Concrete Traffic Barrier

DIB	Door Interface Box
DCC	Design Code and Construction
DPS	Department of Public Safety
DPDT	Double Pole Double Throw
EPT	Electrical Power Transfer
ISC	Intelligent System Controller
IDS	Intrusion Detection System
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Standards Organization
ITS	Information Technology Services
MOV	Metal Oxide Varistor
MSG	Manufacturers Standard Gauge
NEMA	National Electric Manufacturers Association
OSDP	Open Supervised Device Protocol
PE	Polyethylene
PIN	Personal Identification Number
PLC	Programmable Logic Controller
SAD	Silicon Avalanche Diode
SIDA	Security Identification Display Area
SDI	Steel Door Institute
SPDT	Single Pole Double Throw
SRD	Signal Receiving Device
SSI	Sensitive Security Information
SSS	Security Surveillance System
TIA	Telecommunications Industry Association
TSA	Transportation Security Administration
UPS	Uninterruptable Power System

11.16.3 Standards and Codes

In addition to other the references and standards outlined in this chapter, special attention must be made to the following specific codes, standards, and publications. References to codes and standards called for in this document refer to the latest edition, amendments, and revisions to the codes and standards currently in effect.

1. NECA 1 – Standard Practice of Good Workmanship in Electrical Construction
2. NFPA 101 – Life Safety Code
3. UL 10B – Standard for Fire Tests of Door Assemblies
4. UL 10C – Standard for Positive Pressure Fire Tests of Door Assemblies
5. UL 294 – Standard for Access Control System Units
6. UL 305 – Standard for Panic Hardware
7. UL 325 – ANSI/CAN/UL Standard for Door, Drapery, Gate, Louver, and Window Operators and Systems
8. UL 497B – Standard for Protectors for Data Communications and Fire-Alarm Circuits
9. UL 508A – Standard for Industrial Control Panels
10. UL 634 – Standard for Connectors and Switches for Use with Burglar-Alarm Systems
11. UL 1034 – Standard for Burglary-Resistant Electric Locking Mechanisms
12. UL 1638 – Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories

11.16.4 Sensitive Security Information

Portions of the information related to the DFW Airport security systems contain Sensitive Security Information (SSI), as defined in Title 49 C.F.R. Part 1520.7, and must be protected as such. This SSI must not be released to anyone without the prior written approval of the DFW Airport DPS, or their designated representative.

11.16.5 Design and Engineering

DFW Airport requires that all additions and modifications to the security systems be contained in a comprehensive set of drawings and specifications that are sealed by a Professional Engineer, licensed by the Texas Board of Professional Engineers, holding a current DFW Airport SIDA badge. These designs must be approved by the Airport Security Coordinator, then submitted to the DFW Airport Board for code review and permitting before any work begins.

Certain requirements must also be met when performing work on the AACS. These requirements are meant to document both the engineering and design aspects of the system, as well as demonstrate the quality of workmanship and operational requirements needed to meet the design and operational standards of the Airport.

Security design drawings must be organized as follows:

1. Drawing sheets should start with the discipline code of "TY" which identifies them as DFW security drawings.
2. DFW security drawings must contain only DFW security systems. Any tenant or other third-party security systems must reside on another series of drawings.
3. Every sheet in the security drawing package must be marked as SSI per Title 49 CFR Part 1520.13 and include the distribution limitation statement in the prescribed locations. The drawing series must have DHS Form 11054 affixed as a cover page to the set. The drawing series must be packaged separately from all other design documents and protected. Distribution of these documents is limited to the DFW DCC project manager, or their designated representative, DFW DPS and DFW ITS.

Security design drawings must include the following elements:

1. Floor plans, with devices, proposed pathways, door numbers, camera coverage, security boundaries and identification of security zones.
2. Enlarged plans, with proposed equipment layouts.
3. Telecommunications spaces depicting wall elevations, with proposed access control ISC panel locations, equipment layout, and other details.
4. Rack elevations depicting horizontal cabling, network switches, wire management and fiber terminations.
5. AACS door elevations, with details, devices, conduit sizes, wire and cable types
6. Sequence of operation for every door type, including entry procedures, exit procedures, monitoring, and life safety considerations.
7. Configuration of each ISC panel, including the proposed board layout and a board schedule.
8. Door schedule, with door numbers, door locations, door types, sheet references, detail references, ISC panel assignments, I/O point counts, the names of the spaces on either side of each portal, and camera IDs of the cameras on either side of the portal.
9. Security camera, intercom and duress alarm installation and pathway details.
10. The designer must ensure every AACS portal includes surveillance coverage for remote alarm resolution and forensics purposes. Video surveillance coverage must be provided on both sides of an AACS portal, except for BHS openings and vehicle gates, which may only require coverage on one side.

11. The designer must ensure the AACS receives an output from a fire alarm system that signals when the fire alarm system has been activated. This signal must be used to disable certain door power supplies powering locking hardware that has been identified to fail safe.

11.16.6 Specifications

The following CSI specification sections are owned by DFW Airport and maintained by DFW ITS. These specifications must be used on any project that includes DFW Airport security in the construction documents.

1. 28 13 00 – Automated Access Control System (SSI)
2. 28 23 00 – Security Surveillance System (SSI)

Copies of the latest editions of these specifications must be obtained through the DCC project manager at the beginning of every project. These specifications must not be altered without prior approval from DFW ITS and DPS. Copies of these specifications must not be carried over from project to project.

11.16.7 Automated Access Control System (AACS)

11.16.7.1 Overview

The DFW Airport Automated Access Control System is currently deployed to secure federally regulated doors and vehicle gates, as defined in Title 49 C.F.R. Part 1542, certain critical infrastructure areas, the perimeter of DFW Board buildings and other high value spaces. As a policy, the AACS is not used for securing tenant, concessions, or other third-party spaces.

The AACS is owned and administered by the DFW Airport Department of Public Safety (DPS) and maintained by the DFW Airport Information Technology Services (ITS) department. All work on the access control system is a modification or expansion of the existing system. All products supplied and standards of construction must be equal to that of other parts of the existing system. The system must be expected to continue to function without interruption outside of the areas of work.

The AACS is integrated with the Security Surveillance System to provide event-driven camera call-ups on the operator workstations. Most AACS deployments will require comprehensive camera coverage for alarm resolution.

11.16.7.2 Definitions

The following definitions are provided to establish a common language for terms used throughout this manual.

1. SC Panel - An access control field panel consisting of a distributed intelligence processor that can control multiple portals. The ISC panel provides the electrical interface for the credential reader, door monitor points, door devices and door control components. It also performs credential authentication and makes decisions regarding the authorized use of the portal, and it reports the various events back to the core AACS servers.
2. Portal - A secured opening monitored or controlled by an ISC panel.
3. Breach - The act of opening a monitored portal, without receiving specific authorization from the AACS.
4. Credential Reader - A device that reads authorized smart card credentials for the purpose of establishing a person's identity and access rights. Credential readers may include an alphanumeric keypad for the purpose of providing a second factor of authentication or enabling certain predefined system functions.

5. Biometric Reader - A device that establishes a person's identity based upon a physical trait. This device is typically used in conjunction with a credential reader.
6. Electrified Lock - A device to secure a portal and allow passage through the portal by means of remote control.
7. Door Contact - A device used to monitor the open/closed status of a portal. This designator refers to the purpose of the device and not its technology.
8. Duress Alarm – A device used by an individual to generate an alarm in the system when they need emergency assistance.
9. Request to Exit - A means used to allow egress through a secured portal without the need to know the identity of the person leaving, and without generating a breach alarm.
10. Latch Bolt Monitor - A device installed in a door strike that detects the latch and bolt positions when the door is closed.
11. Duress Button - A device used to generate an immediate alarm condition at designated alarm monitoring locations.
12. Audiovisual Device - A device installed at a portal that indicates an alarm condition at a portal.
13. Passive Infrared Detector (PIR) - A device used to detect a person approaching the secure side of a portal, usually for the purpose of generating a Request-to-Exit condition.
14. Key Switch - A key controlled device used to arm, disarm, or reset certain functions of the AACS.
15. Door Release - A device or software function used to remotely unlock a portal to allow a visitor through.

11.16.7.3 False Alarm Mitigation

The DFW Airport DPS mandates that all alarms generated in the access control system be investigated and responded to. Therefore, the parties involved in designing, constructing, and maintaining the access control system have a duty to ensure that all work is carefully thought out and constructed, so as not to introduce unwanted false alarms.

11.16.7.4 Specifications

The CSI Section 281300 – Automated Access Control System specification, describing the AACS, is owned by DFW Airport and maintained by DFW ITS. Copies of the specification may be obtained through the DCC project manager. The specification must not be altered without prior approval from DFW ITS and DPS.

11.16.7.5 Standard Door Types

DFW uses the door type design philosophy, where doors are tagged with an identifier that references a detail that provides a complete design of the portal, rather than enumerating individual devices on the plans. Standard door types are maintained by DFW ITS and may be obtained through the DCC project manager.

11.16.7.6 Pathways

Use of cable tray and non-continuous pathways for AACS cabling is not permitted. All cabling must be run in Contractor furnished conduit, based on criteria found in Chapter 10 of this manual. Conduit must be installed according to the latest edition of ANSI/TIA-569. All conduits must include a pull string.

All pull boxes and junction boxes must be secured with security screws. Boxes must not be left with any open knockouts.

11.16.7.6.1 Door Interface Box (DIB)

Located at each AACS door or other AACS ISC presence, as designated in the design drawings; the DIB is an enclosure that serves as a central wiring makeup point for the local devices.

A DIB associated with a door must be installed on the more secure side of a door, and within 10 feet of the door. A DIB associated with other devices, such as duress buttons, must be installed in a secured space. The DIB must be placed such that a technician on a ladder servicing the DIB will have line of sight of the door or device(s) it serves.

The DIB must include a panel for mounting terminal strips and other equipment. The enclosure must include a tamper switch and the cover must be equipped with a safety chain or hinge and be secured with security screws.

A typical DIB is shown below in Figure 29 below.



Figure 29 - Typical DIB

The DIB must have the following characteristics:

1. Fabricated from minimum 16-gauge steel, without knockouts, cutouts, or holes.
2. Continuous welded seams.
3. Powder coated finish.
4. Cover is equipped with a permanently secured oil-resistant gasket, attached with captive security screws.
5. Conforms to NEMA standard for Type 12 and Type 13.
6. Internal threaded weld studs or weld nuts are provided for mounting a panel.
7. UL 508A listed, Type 12 and Type 13.
8. Minimum 16 inches high by 14 inches wide by 6 inches deep.
9. The cover must be tethered to the enclosure with a chain.

Acceptable products:

1. B-Line 16146-12SC, with AW1614P panel.
2. Hammond 1414SCO6, with 14R1513 panel.

Other equivalent products may be submitted for approval by the DFW Airport Board.

11.16.7.6.2 In-building Home Run Conduits

Homerun conduits from the DIB to the ISC panel must comply with Chapter 10 of this manual. The minimum trade size is 1 inch and must be sized to have a fill ratio of no more than 40% and be no longer than 275 feet. A homerun conduit longer than 275 feet requires prior written approval from DFW ITS before installation.

Homerun conduits must be labeled every 25 feet with a 1-inch-high white adhesive label, with red lettering. The lettering must include the acronym "AACS," in 3/4-inch-high lettering, plus the originating communications room and the contract number for the project in smaller lettering.

Doors requiring a connection to a third-party intrusion detection system must have a second 3/4-inch homerun conduit from the DIB to the nearest IDS panel. This conduit may be unlabeled, and its length is limited only by the specifics of the IDS.

Home run conduits must be individually run and never aggregated with any other conduits without prior approval from DFW ITS. Pull box covers are to be secured with security screws.

11.16.7.6.3 Outside Plant Home Run Conduits

Under certain conditions, homerun AACS outside plant cabling may share a conduit with other low voltage systems, if the cabling is run in an innerduct, separate from the other cabling. This requires prior written approval from DFW ITS.

11.16.7.6.4 Device Conduits

Device conduits that run from the DIB to the field devices at the door must comply with Chapter 10 of this manual. The minimum trade size for device conduits is 1/2 inch and must be sized to have a fill ratio of no more than 40%. Device conduits must not have intermediate pull boxes but must still comply with the ANSI/TIA-569 standard.

11.16.7.6.5 Backboxes

Backboxes for devices installed outside of the door frame, which do not require a specific manufacturer-supplied backbox, must be 4-11/16 inches square and a minimum of 2-1/8 inches deep. Reducer rings must be provided for devices that are designed to install on a single or double gang box.

Devices recessed inside a hinged door frame, such as door contacts and electrical power transfers must be installed in custom backboxes welded to the door frame. These boxes must be fabricated with a knockout for a conduit fitting and the seams must be fully welded or caulked to prevent grout and other liquids from seeping into the box.

11.16.7.7 Electronic Access Controlled Openings

Access controlled and other monitored openings include hinged security doors, overhead doors and coiling gates, conveyor chop doors and vehicle gates.

11.16.7.7.1 Hinged Security Doors

Security doors must be constructed to limit the potential for the assembly falling out of tolerance, as the door and hardware wears. This progressive wear will eventually lead to false alarms.

Federally regulated doors facing public areas must be equipped with sweeps to prevent objects from being passed under the door, and double doors must have center mullions or astragals to prevent objects from being passed between them. These doors should also be equipped with closers.

11.16.7.7.2 Hinged Door Types

Of the hinged door types, steel doors are approved for most AACS applications and should be deployed whenever possible. Other door types may be accommodated, based on the architectural design of the space, but these are exceptions.

Regardless of type, hinged doors must be constructed with a maximum 1/8-inch gap between the top of the door and the door frame and at the strike edge of the door to the door frame, to meet the door contact tolerances.

11.16.7.7.3 Steel Doors

Except where certain ballistic requirements must be met, new steel doors must have the following characteristics:

1. Standard 1-3/4 inch.
2. Factory constructed of MSG number 14 steel, full flush, continuously welded, and seamless.
3. The edges must be square.
4. The top and the bottom of the door must each have a flush closure.
5. Rated for maximum duty.
6. Minimum factory applied primer coating.
7. Carry an SDI Level 4, Model 2 designation.
8. Meet ANSI/SDI A250.8-2017 specifications.
9. Meet ANSI/SDI A250.6-2015 reinforcement standards.
10. Meet SDI 117-19 tolerances.

Door frames are to be fabricated with welded backboxes for all devices installed in the frame, such as door contacts and power transfer devices, as described in the Pathways section above.

Doors should be equipped with a stainless-steel kick plate on the push side.

Doors identified as being extremely high use should be equipped with an approved stainless steel armored plate on the push side and door handle guards.

11.16.7.7.4 Aluminum Storefront Doors

New aluminum doors must have the following characteristics:

1. Standard 1-3/4 inch constructed from minimum .125-inch-thick material.
2. Minimum 5-inch-wide stiles.
3. Minimum 6-inch-wide top and bottom rails.
4. Minimum 6-inch-wide center rail is required.
5. The edges must be square.
6. The top and the bottom of the door must each have a flush closure.
7. Rated for heavy duty use.

Door frames are to be fabricated with MIG or TIG welded backboxes for all devices installed in the frame, such as door contacts and power transfer devices, as described in the Pathways section above.

High use AACS controlled aluminum storefront doors, such as passenger boarding bridge doors, are not permitted to be constructed as part of a window or storefront system. These doors must be installed in a separate panel constructed within the window system and supported at the floor and the deck above. The panel must be wide enough to accommodate auxiliary devices, such as credential readers and audiovisual devices.

11.16.7.7.5 Wood Doors

Wood doors are not permitted for AACS installations, without written prior approval from the DFW Airport Board.

11.16.7.8 Door Frames

For stud wall construction, door frames must be affixed to metal studs that are secured at the floor and the deck above.

Doors that are identified for extremely high-trafficked use, such as employee portals and passenger boarding bridges may require the door frame to be anchored to a specially engineered steel structure.

Door frames anchored to wood stud construction are not permitted without prior written approval from the DFW Airport Board.

11.16.7.8.1 Steel Door Frames

Except where certain ballistic requirements must be met, new steel frames must have the following characteristics:

1. Factory constructed of MSG number 14 steel; full profile welded.
2. Minimum factory applied primer coating.
3. Carry an SDI Level 4 designation.
4. Factory prepped for a continuous hinge.
5. Factory prepped for a discrete door contact.
6. Factory prepped for a discrete electrical power transfer device.
7. Factory prepped for a discrete latch/bolt monitor device in the strike.
8. Meet ANSI/SDI A250.8-2017 specifications.
9. Meet ANSI/SDI A250.6-2015 reinforcement standards.
10. Installed per ANSI/SDI A250.11-2012 recommendations.
11. Meet SDI 117-19 tolerances.

11.16.7.8.2 Aluminum Door Frames

Contact ITS Access Control Department for aluminum door frame requirements.

11.16.7.9 Door Frame Preparations

Door frames, regardless of type, must be prepared for access control devices before installation. These preparations include cut-outs, device backboxes and conduit pathway. Backboxes must be fabricated and welded to the frame. Many of the examples shown in this section are based on details provided in door frame shops and technical manuals by Curries, Mesker and others. These manufacturer manuals are to be consulted for further information.

11.16.7.9.1 Door Contact Rough-In

Door contacts are to be installed approximately 8 inches from the strike edge of the door frame.

Example details of the backbox construction are shown in the exploded view in Figure 30 below. All openings around the assembled box must be sealed to prevent liquids and debris from entering the box. All conduit fittings must be liquid tight.

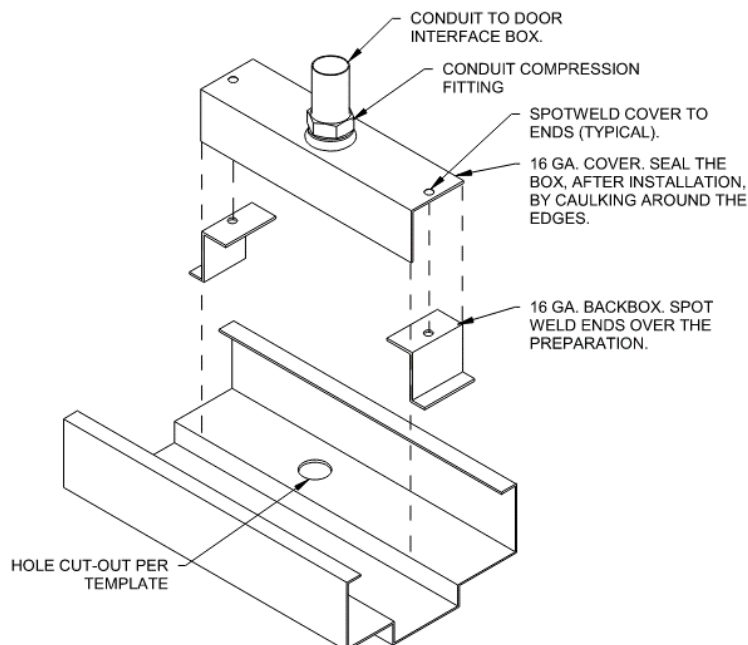


Figure 30 - Door Contact Backbox Exploded View

11.16.7.9.2 Electrical Power Transfer Rough-In

Electrical power transfers must be dedicated EPT devices concealed in the door frame.

EPTs come with a manufacturer's backbox that must be installed in a fabricated backbox in the hinge-edge of the door and frame, between the second and third hinges. An example rough-in exploded-view is shown in Figure 31 below.

Frames prepped for EPTs must include a blank plate to cover the opening when the door is not initially equipped with an EPT.

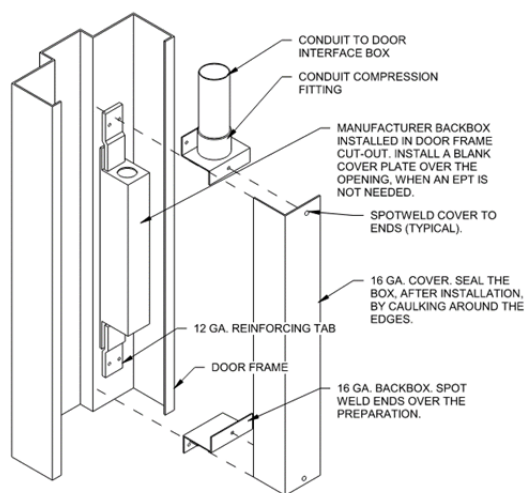


Figure 31 - EPT Backbox Exploded View

11.16.7.10 Architectural Door Hardware

All architectural door hardware must be certified ANSI/BHMA A156.13 operational and security grade 1.

11.16.7.10.1 Hinges

Access controlled doors must be equipped with continuous geared hinges, to provide better wear performance over time and better security.

Continuous hinges must have the following characteristics:

1. Certified to ANSI 156.26 Grade 1.
2. Extruded 6063-T6 aluminum alloy, with pin-less geared assembly that distributes the load uniformly along the full length of the door and frame.
3. Serves as reinforcement for the door and frame. No special hinge reinforcements are required.
4. Guaranteed for the life of the opening.
5. UL 10B and UL 10C listed.
6. Prepped for a discrete concealed EPT device.
7. Finish coordinated with the remaining door hardware.
8. Guaranteed for the life of the opening.
9. Finish must be coordinated with the other door hardware.

Approved Products:

1. Ives EPT HD Series
2. McKinney MCK-12HD, shown in Figure 8, with PT or EPT power transfer modifications.

Other equivalent products may be submitted for approval.

11.16.7.10.2 Electrified Handsets

Electrified handsets must be of the mortise variety. Cylindrical handsets are not permitted without prior written approval from the DFW Airport Board.

Handsets must be equipped with a key cylinder, with a DPS-provided core.

Handsets must have the following characteristics:

1. Certified ANSI/BHMA A156.13 operational and security grade 1.
2. Meets A117.1 Accessibility Code.
3. Meets minimum requirements of UL 1034.
4. Designed for standard 1-1/4-inch-thick doors.
5. Must have an armored front.
6. Must be manufactured with an anti-microbial finish.
7. Assists with load reduction in optimizing energy performance credit in LEED.
8. Quick reversible handing, without disassembling the lock case.
9. 2-piece mechanical latch bolt.
10. Auxiliary latch bolt.
11. 12/24VDC fail safe/fail secure operation.
12. Integrated latch bolt monitor.
13. Integrated request to exit monitor.
14. Integrated security monitor.
15. Wiring terminated with Molex connectors.
16. SFIC 7-pin cylinder with disposable temporary core.
17. Minimum 3-year mechanical warranty.
18. Minimum 1-year electromechanical warranty.
19. Trim and finish must be coordinated with the other door hardware.

Approved handsets include the following:

1. Schlage L909xEL/EU Series

Other equivalent products may be submitted for approval.

11.16.7.10.3 Electrified Exit Devices

Electrified exit devices must be equipped with pull trim, whenever possible. Electric trim may be used for some applications with prior written approval from the DFW Airport Board.

Electrified exit devices must have the following characteristics:

1. Certified ANSI/BHMA A156.13 Series 1000, operational and security grade 1.
2. UL 10C (Fire) and UL 305 (Panic) listed.
3. Non-handed.
4. Designed for standard 1-1/4-inch doors.
5. Comes in various widths.
6. Must be manufactured with an anti-microbial finish.
7. 24VDC low current draw remote latch retraction, eliminating the local power supply.
8. Integrated request to exit monitor.
9. Integrated latch bolt monitor.
10. Integrated trim monitor.
11. Wiring terminated with Molex connectors.
12. Minimum 5-year mechanical warranty.
13. Minimum 2-year electromechanical warranty.
14. Length must be coordinated with the size of the door.
15. Trim and finish must be coordinated with the other door hardware in the facility.

Approved exit devices include the following:

1. Von Duprin 98/99 Series with Options LX, SS and QEL

11.16.7.10.4 Electromagnetic Locks

Electromagnetic locks are primarily deployed for specific applications where required exit doors are to be temporarily held shut. These applications include international gate interlocks, public exits, and employee portals.

Electromagnetic locks are not permitted for use as a primary locking device without prior written approval from the DFW Airport Board.

Electromagnetic locks must have the following characteristics:

1. Minimum 1200 lbs. holding force.
2. 24VDC power.
3. Integrated bond sensor with single pole, dual throw contacts.
4. UL 10C and UL 294 listed.
5. Built-in arc suppression.
6. Instant release.
7. Lifetime warranty
8. Finish must be coordinated with the other door hardware.

Approved products include the following:

1. Alarm Controls 1200LB
2. Schlage M492P

Other equivalent products may be submitted for approval by the DFW Airport Board.

11.16.7.10.5 Electric Strikes

Electric strikes are not allowed without prior written approval from the DFW Airport Board.

11.16.7.10.6 Electrical Power Transfers

EPTs must be concealed in the door and frame. The use of surface mounted armored loops or wireless EPTs is prohibited without prior written authorization from the DFW Airport Board.

Doors equipped with EPTs must be equipped with an appropriate combination of EPT, hinges, closers and stops that prevent the door swing from over-extending the travel of the EPT.

Electrical power transfer devices must have the following characteristics:

1. Mortises into the edges of the door and frame
2. Tamper resistant.
3. All metal construction.
4. 10-wire configuration terminated with Molex connectors.
5. Designed to install above the second hinge.
6. Conduit knockouts on backboxes.
7. Lifetime warranty.
8. Finish must be coordinated with the other door hardware.

Approved EPT devices include the following:

1. Securitron CEPT-10
2. Pemko EL-CEPT
3. Von Duprin EPT10

Other products may be submitted for approval, to accommodate a specific door configuration.

An extended length EPT may be required for doors with a 180-degree swing. This must be coordinated with the architect and their door consultant.

11.16.7.10.7 Transfer Hinges

Transfer hinges typically have a short service life. As a result, the use of transfer hinges is not permitted without prior written approval from the DFW Airport Board.

11.16.7.10.8 Door Contacts

Door contacts must be of the high security variety and concealed in the door and door frame. The use of surface mounted switches on a hinged door will not be permitted without prior written approval from the DFW Airport Board.

Door contacts must have the following characteristics:

1. Meets UL 634 Standard Level 2 High Security
2. Suitable for outdoor use.
3. Two closed loop alarm contacts.
4. One closed loop tamper contact.
5. Integrated removal tamper circuit and hardware.
6. ANSI form factor.
7. Minimum 3-year warranty.

Approved door contacts include the following:

1. Magnasphere L2C-111 Series HSS

11.16.7.10.9 Electromagnetic Door Holders

Electromagnetic door holders are used for specific applications, where an AACS door must be held open for an extended period. The primary application includes aircraft boarding doors.

Door holders must be floor mounted. Wall mounted holders are not permitted without additional structural reinforcement in the wall.

AACS door holders are unrelated to smoke compartment door holders and are controlled and powered by the AACS ISC panel.

Door holders must have release buttons associated with them. Coordinate the location of the release button with DFW Airport and airline stakeholders.

Electromagnetic door holders must have the following characteristics:

1. Must be floor mounted.
2. 24VDC powered.
3. 25 to 40 lbs. holding power.
4. ANSI/BHMA A156.15 complaint.
5. Minimum 2-year warranty.

The basis of design includes the following:

1. Rixson 980M

Other equivalent products may be submitted for approval by the DFW Airport Board.

11.16.7.10.10 Door Closers with Integrated Hold Open

Door closures with an integrated electronic door hold open have a short service life. As such, these are not permitted without prior written approval by the DFW Airport Board.

11.16.7.10.11 Automated ADA Operators

Automated ADA operators must include a controller that allows the operator to be interlocked with the AACS.

ADA pushbuttons must be hard-wired. The pushbuttons and cabling must be installed in backboxes and conduit and connected to the DIB. The operator must have a backbox roughed into the door frame, with a 3/4-inch conduit to the DIB.

The system must be configured such that pressing the pushbutton will not activate the operator until the AACS has unlatched door.

11.16.7.10.12 Electrified Door Wiring Harnesses

The electrified door hardware must utilize modular wiring harnesses, terminated with Molex connectors.

The wiring harnesses must have the following characteristics:

1. Cable must be made up of twelve 20 AWG stranded conductors.
2. Cables are terminated with 8 and 4 pin Molex connectors that mate with the connectors on the EPT and door locking hardware.
3. Two cables are required for each door: one between the locking hardware and the EPT, and one between the EPT and the DIB.

Approved products include the following:

1. Allegion Connect Series
2. McKinney ElectroLynx Series

Other equivalent products may be submitted for approval by the DFW Airport Board.

11.16.7.10.13 Handing

Figure 32 below illustrates the door handing configurations, for single doors. Figure 8 illustrates handing for double doors.

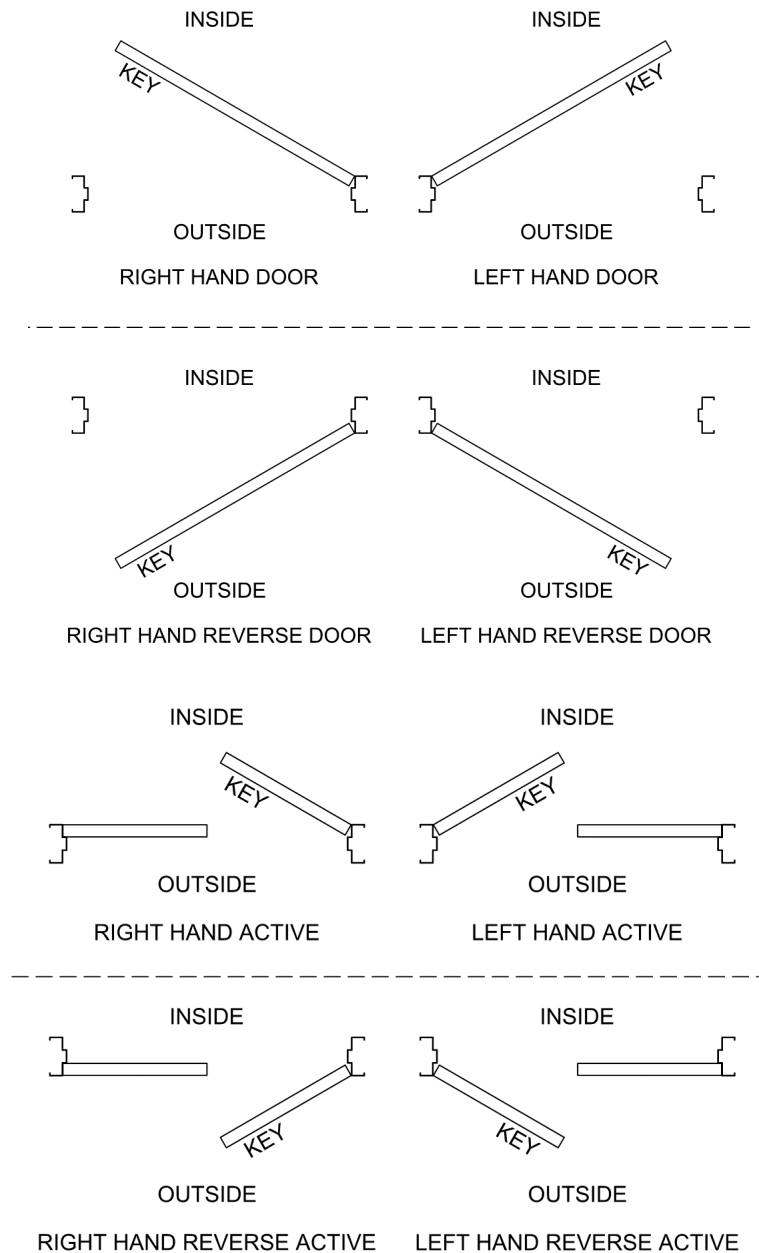


Figure 32 - Handing for Double Doors

Single doors with biometrics must be installed with left handing, either left hand or left-hand reverse, with readers on the right side of the door.

Double doors with biometrics must be installed with right handing on the active leaf, either right hand active or right hand reverse active, with the readers on the right side of the door. Readers must be positioned to not interfere with the door swing.

11.16.7.10.14 Overhead Doors

Overhead doors must be of the coiling variety. Sectional doors will not be permitted without prior written approval from the DFW Airport Board.

Overhead doors and coiling gates must be equipped with low profile floor mounted door contacts on each end. Automated doors with readers must be equipped with operators that can be controlled by the AACS.

11.16.7.10.15 Conveyor Security Doors

Conveyor coiling security doors must be controlled and monitored by the AACS through a purpose-built BHS interface. This interface must support the complete set of control and monitoring requirements outlined below.

The BHS interface has the following requirements:

1. The interface must contain a set of 6 relays that are provided by the BHS contractor. All relays must be supplied and wired to the BHS, regardless of whether they are needed by the intended control scenario.
2. The relays must have 24VDC coils, with DPDT form C contacts.
3. The relays must be installed in an appropriately rated NEMA enclosure, by the BHS contractor, near the chop door being monitored or controlled.
4. All voltages within the enclosure must be 24VDC.
5. The AACS Contractor must install a DIB at the chop door and provide a 3/4-inch conduit, containing a 12-conductor, 18 AWG cable from the DIB to the BHS interface enclosure.

The signals supplied to the 6 relays include the following:

1. Chop Door Photobeam State. Power is supplied to the relay coil when the chop door photobeam detects the chop door is in the closed position. The signal must be derived from the output of the photobeam device and not a PLC output.
2. Run Signal. Power is supplied to the relay coil when the conveyor belt motors at the chop door are running. This signal must be derived from the motor controller output and not a PLC output.
3. Run Enabled. Power is supplied to the relay coil when the BHS controls are commanding the conveyor to be in a running state. This signal is derived from a PLC output.
4. AACS Card Read. Power is supplied to the relay coil for one second when the AACS receives a valid credential read for that belt. This signal is derived from an output on the ISC panel.
5. System Start. Power is supplied to the relay coil when the BHS System Start control button, for that conveyor, has been pressed. This signal is derived from the button itself and must be the button's only electrical connection.
6. Maintenance Override. Power is supplied to the relay coil when a maintenance override condition has been indicated by the BHS controls. This signal is derived from a PLC output.

11.16.7.10.16 Automatic Vehicle Gates

Automatic vehicle gates include security gates installed as part of a fence line, such as sliding, swing and folding systems, and traffic control arms. Gates must include the following functionality:

11.16.7.10.16.1 Entrapment Protection

Automatic vehicle gates must be equipped with minimum Class III operators, and appropriate UL 325 primary and secondary entrapment protection.

11.16.7.10.16.2 Fail Safe

All automatic vehicle gates and devices must incorporate a fail-safe manual backup or release. All gates and devices must always permit the safe exit of emergency vehicles.

11.16.7.10.16.3 Physical Protection

Automated gate equipment must have appropriate physical protection, in the form of bollards or CTBs, to prevent damage from vehicles and other equipment.

11.16.7.10.16.4 Credential Readers

Vehicle presence loops are required for all credential readers. Gates equipped with driver credential readers require bollard protection around pedestals. Gates equipped with SRD readers must have bollard protection around the mounting structure.

11.16.7.10.16.5 Video Surveillance

At a minimum, pole or gantry mounted fixed video surveillance coverage of the driver and front license plate is required at each reader location. Additional surveillance coverage may be required, depending on the installation.

11.16.7.10.16.6 Remote Monitoring

The open/closed position and other functions of vehicle gates are not remotely monitored.

11.16.7.10.16.7 Fire Apparatus Facilities Requirements

All non-AOA automatic gates and devices across required fire apparatus access roads must open upon activation of the building fire alarm system and remain open until such time that the fire alarm is reset. These gates must also open with a signal from a bi-directional GTT Opticom system.

The GTT Opticom system is not approved for use on AOA vehicle gates.

11.16.7.10.16.8 Fire Apparatus Roadway Requirements

All automatic gates and devices across required fire apparatus access roads must open through one of the following methods.

1. The gate must open from a signal received from an Airport Board SRD
2. Airport Board Selected Vehicle Device.

11.16.7.11 Access Control Devices

DFW Airport currently maintains the construction specifications for the Automated Access Control System and most of its components. This specification contains SSI, and the design team must obtain a copy of the specification from DFW DCC. Any modifications to the contents of this specification must be redlined and submitted to DFW ITS for approval.

11.16.7.11.1 Credential Readers

Credential readers include badge readers, with and without keypads, and biometric terminals, which have integrated readers. The location and use of the portal will dictate the type of reader deployed, as determined by DFW DPS.

11.16.7.11.2 AVI Readers

AVI readers must have the following characteristics:

1. 865-870 MHz and 902-928 MHz capable.
2. Integrated antenna.
3. RS-232 with Wiegand compatible interface.
4. Adjustable RF output power.
5. Designed for outdoor use.
6. Tamper resistant enclosure.

Approved products include the following:

1. TransCore Encompass Series

11.16.7.12 Cables

Homerun cables must be continuously run from the ISC panel to the DIB, whenever possible. Some devices may require their cables to pass through the DIB and be continuously run to the end device.

11.16.7.12.1 Cable Splices

All cable splices are permitted only when continuous runs are not practical, or when transitioning from outside plant to inside plant cabling.

Splices must be housed in a minimum 10-inch, by 10-inch, by 4-inch-deep enclosure, rated for the environment in which it is installed. The enclosure must be equipped with a tamper switch and secured with security screws.

Splices must be made up on a terminal strip, with a minimum of 5 feet of service loop from each cable contained entirely within the enclosure.

Building entrance, or surge protection, must be treated the same as a splice.

11.16.7.13 Intelligent System Controller Panels

Intelligent System Controller panels are based on the Mercury Security hardware platform and assembled in a centralized enclosure.

11.16.7.13.1 Requirements

A typical fully assembled ISC panel must contain the following items:

1. An enclosure rated for the environment where the ISC panel will be installed.
2. AC surge protection and distribution.
3. Access control power supply and distribution.
4. Door power supply, door power controllers and distribution.
5. One (1) Intelligent controller board.
6. Up to seven (7) Reader interface boards.
7. UP to two (2) general purpose input interface boards.
8. UP to two (2) general purpose output interface boards.
9. Network data outlet.
10. Reader and I/O surge protection, where required.
11. Ventilation or cooling, depending on the environment.
12. Enclosure tamper switch.
13. Enclosure padlock.

11.16.7.13.2 Indoor Installations

ISC panels installed indoors must be housed inside of a Type 12 or 13 enclosure and located in DFW Airport Board communications rooms, when available. When a Board communications room is not available, ISC panels must be installed in non-hazardous Board-owned spaces, such as mechanical or electrical rooms.

Homerun conduits from the DIBS are routed into a 6-inch high and deep gutter. This gutter is connected to the ISC enclosure with two 4-inch conduits. This ensures the access control cabling is never exposed to tampering.

11.16.7.13.3 Outdoor Installations

ISC panels installed outdoors housed inside of a NEMA 4 enclosure, equipped with an air conditioner. The enclosure must be specified in Division 26 and installed by the electrical contractor, to ensure its NEMA rating is maintained after assembly. The enclosure is to have a plinth base and be bolted to a reinforced concrete pad. All conduit pathways must be fed into the bottom of the enclosure.

11.16.7.13.4 Batteries

Provide batteries to support the access control and door power supplies. Batteries must have sufficient capacity to support a fully loaded ISC panel for a minimum of 30 minutes.

11.16.7.13.5 Surge Protection

Each ISC panel requires full surge protection on the main AC power feed and for any device installed outdoors, away from a building.

11.16.7.13.5.1 AC Power Protection

Each ISC panel requires surge protection on the main AC service feed. The protector must have the following characteristics:

1. Designed to protect UL listed control panels.
2. Protects a 120VAC circuit, with a minimum 20A continuous current.
3. Utilizes MOV or SAD technologies for protection.
4. Protects all modes – L-G, L-N, N-G.
5. Surge current rated for 50,000A.
6. Hardwired installation.
7. Form C relay output to indicate the protection status.

Approved products include the following:

1. DITEK DTK-120SRD

Other products may be submitted for approval by the DFW Airport Board.

11.16.7.13.5.2 Device Surge Protection

Devices installed in separate buildings or outdoor locations away from where the ISC panel is installed require surge protection on the cabling.

11.16.7.13.5.3 Card Reader Protection

Provide surge protection for card readers installed at vehicle gates, guardhouses, or outdoor pedestals. Surge protector must have the following characteristics:

1. Protects a 12VDC circuit, with up to 3A continuous current.
2. Must utilize SAD technology, to limit its effect on the transmitted data.
3. Supports data rates of over 200Kbps.
4. Screw terminal installation.
5. UL 497B Approved.

Approved products include the following:

1. PolyPhaser IX-4L for OSDP readers
2. Other products may be submitted for approval by the DFW Airport Board.

11.16.7.13.5.4 Digital I/O Protection

Provide surge protection on I/O circuits that monitor or control remotely installed devices. Surge protectors must have the following characteristics:

1. Protects a 12VDC loop, with up to 5A continuous current.
2. MOV or SAD technology.
3. Line-ground protection mode.
4. Surge current rated for 20,000A.
5. Screw terminal installation.

Approved products include the following:

1. DITEK DTK-2MHLP12B

Other products may be submitted for approval by the DFW Airport Board.

11.16.7.13.5.5 Low Voltage Power Protection

Provide surge protection for remotely installed locking devices, such as electromagnetic locks, electric strikes, and powered door hardware. Surge protectors must have the following characteristics:

1. Line-ground protection mode.
2. Parallel installation.

Approved products include the following:

1. DITEK DTK-ESS

Other products may be submitted for approval by the DFW Airport Board.

11.16.7.13.6 Primary and Backup Power

Power to an ISC panel must be supplied from a UPS. Each UPS must be equipped with a network card and actively monitored.

11.16.7.13.7 Indoor Installations

A UPS supporting an ISC panel in an indoor installation must provide power for a minimum of eight (8) hours during a power failure when permanent standby power is not available.

Where permanent standby power is provided, in the form of a generator or a main-tie-main utility service, the UPS must provide power for a minimum of one (1) hour during a power failure.

The UPS output must be wired to an electrical distribution panel, to allow for support of multiple ISC panels.

11.16.7.13.8 Outdoor Installations

A UPS supporting an ISC panel in an indoor installation must provide power for a minimum of one (1) hour during a power failure.

11.16.7.13.9 Security Surveillance Coverage

The designer must ensure every AACS portal includes surveillance coverage for remote alarm resolution and forensics purposes.

Video surveillance coverage must be provided on both sides of an AACS portal, except for BHS openings and vehicle gates, which only require coverage on one side.

11.16.7.13.10 Fire Alarm System

The designer must ensure the AACS receives an output from a fire alarm system that signals when the fire alarm system has been activated. This signal must be used to disable certain door power supplies powering locking hardware that has been identified to fail secure.

11.16.8 Security Surveillance System

11.16.8.1 Overview

The DFW Airport Security Surveillance System is currently deployed to assist in AACS alarm resolution, provide situational awareness and to assist in forensic analysis for law enforcement and other airport personnel.

The SSS consists of the following subsystems:

1. Closed Circuit Television (CCTV) System
2. Security Intercom System
3. Audio Recording System

The SSS is a mostly IP-based solution with several types of IP cameras and intercom stations that are typically powered from the network.

Field devices are served from DFW communications rooms or dedicated IT presences in the form of legacy ceiling zone enclosures, known as "quads", or small closets, known as "quad closets." New DFW communications rooms and quad closets are to be designed by an RCDD and detailed in the technology infrastructure drawings, usually identified with the "TN" discipline designator. The ceiling quad model is depreciated and will require approval from DFW ITS to deploy new ones.

11.16.8.2 Pathways

Use of cable tray and non-continuous pathways for SSS cabling is not permitted. All cabling must be run in Contractor furnished conduit, based on criteria found in Chapter 10 of this manual. Conduit must be installed according to the latest edition of ANSI/TIA-569. All conduits must include a pull string.

All pull boxes and junction boxes must be secured with security screws. Boxes must not be left with any open knockouts.

11.16.8.2.1 Camera Interface Box

SSS end devices must connect to a CIB, which is a 10-inch high, 10-inch wide, and 4-inch-deep junction box, rated for the environment in which it is installed. For indoor environments, this is typically a NEMA Type 1 box. The purpose of the CIB is to hold the work area outlet and service loop.

11.16.8.2.2 In-Building Home Run Conduits

Homerun conduits from the CIB to the nearest communications presence must comply with Chapter 10 of this manual. The trade size is 1 inch and must have a fill ratio of no more than 40% and be no longer than 275 feet. A homerun conduit longer than 275 feet requires prior written approval from DFW ITS before installation. All pull box covers must be secured with security screws.

Homerun conduits must be labeled every 25 feet with a 1-inch-high white adhesive label, with red lettering. The lettering must include the acronym "CCTV," in 3/4-inch-high lettering, plus the originating communications room and the contract number for the project in smaller lettering.

11.16.8.2.3 Device Conduits

Device conduits that run from the CIB to the field devices must comply with Chapter 10 of this manual. The minimum trade size for device conduits is 3/4 inch. Device conduits must not have intermediate pull boxes but must still comply with the ANSI/TIA-569 standard. Device conduits typically only carry the patch cord from the work area outlet in the CIB to the end device.

For cameras installed in ceilings with removable panels, the device conduit may be liquid tight flexible conduit. An extra 6 feet of conduit is provided over what is required to reach the camera, to allow the camera to be moved to an adjacent tile, should it become obstructed by signage or amenities. The additional slack must be coiled up and supported from the structure.

11.16.8.2.4 Backboxes

Backboxes which do not require a specific manufacturer-supplied backbox must be 4-11/16 inches square and a minimum of 2-1/8 inches deep. Reducer rings must be provided for devices that are designed to install on a single or double gang box.

For cameras installed in ceilings with removable panels, a tile bridge is required to support the backbox and camera assembly. Heavier cameras, such as certain PTZ models, may need to be pendant mounted from the structure.

11.16.8.3 IT Presences**11.16.8.3.1 DFW Communications Rooms**

DFW Communications Rooms typically include a dedicated security cabinet, which holds the patch panels for the horizontal cabling, network switches, any power supplies and media converters, and fiber backbone connections. This must be coordinated with the overall technology infrastructure design.

Homerun conduits from the field devices are routed into the room, terminated with bushings. The cable is routed down into the ladder rack within the room, to the security cabinet.

11.16.8.3.2 Quad Closets

Quad closets are typically reserved for use in the terminals. These are located on the concourse level to reduce the vertical cabling. The closets have the following criteria:

1. Must be in the sterile area, preferably out of view of the public.
2. 6-foot wide, 3-foot-deep interior dimensions.
3. No ceiling in the room.
4. 6-0 x 7-0 double door with the following criteria:
 - a. Electrified storeroom function handset with request to exit switch on the active leaf.
 - b. Automatic flush bolts on the inactive leaf.
 - c. Must have door contacts on both leaves.
 - d. No closers.
 - e. Must have stops.
 - f. 180-degree swing for both leaves. The EPT length will need to be considered.
 - g. No sweeps or seals; only silencers
 - h. Brush style astragal.
 - i. Card reader and Local alarm device.
 - j. Dedicated CCTV coverage outside the room, on axis with the door.
5. Ventilation – preferably positive from a nearby supply duct to keep dirt from entering the room from under the doors.
6. LED light fixture with an occupancy sensor.
7. Two L5-20R and two 5-20R quad receptacles, each on dedicated circuits from the UPS in the nearest DFW communications room.
8. Open wall mount rack.
9. Two horizontal PDUs with L5-20P cords.
10. Two-inch backbone conduit from the nearest DFW Communications Room containing a 4-cell innerduct.
 - a. One tube is to be plumbed to the nearest DFW Main Communications Room.
11. Two Air Blown 24-strand single mode fiber bundles.
 - a. One bundle blown from the nearest DFW Main Communications Room.
 - b. One bundle blown from the nearest DFW Communications Room.
 - c. Fiber in the closet terminates in a 2RU fiber enclosure.
 - d. Fiber terminates in the security cabinets of the serving communications rooms.

Refer to the Telecommunications Spaces & Pathways section of this chapter for additional criteria.

11.16.8.4 CCTV System

11.16.8.4.1 Overview

The CCTV system is an IP-based solution with network video recorders and several types of IP cameras that are powered from the network, in most cases.

Cameras are served from DFW communications rooms or dedicated IT presences in the form of legacy ceiling zone enclosures, known as "quads", or small closets, known as "quad closets." New DFW communications rooms and quad closets are to be designed by an RCDD and detailed in the technology infrastructure drawings, usually identified with the "TN" discipline designator. The ceiling quad model is depreciated and will require approval from DFW ITS to deploy new ones.

Network video recorders, video storage, channel licenses and network switches are provided by DFW ITS. Therefore, the CCTV design needs only to include cameras, camera installation details, any prescribed pathways, horizontal cabling, and a camera schedule.

11.16.8.5 Security Intercom System

11.16.8.5.1 Overview

The security intercom system consists of a central exchange with master and remote substations. It is deployed in locations where two-way communication with security personnel is critical. These locations include:

1. Passenger security screening checkpoints.
2. Employee screening checkpoints
3. Certain access-controlled portals

The intercom system may be deployed in DFW Board buildings as a local telephone entry system. However, the system is separate from the intercom system deployed by the Transportation Business Unit system.

The system is integrated with the CCTV system to provide camera call-ups when a call is placed from a substation. Therefore, most substation installations will also be accompanied by camera coverage from the CCTV system.

Chapter 12 Transportation

12.0 Airfield

12.0.1 General Information

This section covers all applicable facilities within the AOA that shall be planned, designed, and constructed in accordance with current FAA standards and criteria. These consist of Federal Aviation Regulations (FARs and current editions of Advisory Circulars. Copies may be obtained from the Federal Aviation Administration (FAA) Southwest Regional Office and U.S. Department of Transportation.

An AC will establish minimum standards to be used at the airport. If there are design criteria decisions to be made which are not covered in the respective AC or this Manual, the project Designer will make recommendations to DFW Airport on a case-by-case basis.

The designer shall provide a matrix including the list of material that require testing and the frequency of the test for each project.

12.0.1.1 Construction Specifications

All airfield construction contract documents shall be prepared in accordance with AC 150/5370-10 current edition. The Division One (01) portion (including Notice to Bidders, Instructions to Bidders, Proposal Forms, Bid Schedule Forms, Bond Forms, General and Special Provisions, etc.) of the contract documents shall be prepared based on guidance and direction from DFW Airport. Designer shall submit the following at the 100% submittal: The plan set, the technical specifications, and the Division one specification for approval prior to solicitation.

12.0.1.2 Site Preparation for NAVAIDS

Design criteria for NAVAID critical areas shall conform to AC 150/5300-13 (current edition), Airport Design. FAA NAVAIDS access roads shall be a minimum of 10 feet wide. Airport facilities will be checked for compliance with FAA electromagnetic standards.

12.0.1.3 Turfing

Turf on the AOA and in areas that are planned to be AOA for which the project is intended shall consist of sod only. No seed is to be used in any designated airside surfaces including runway protection zones.

The placement of all turf adjacent to paved surfaces shall be 1 1/2" below the pavement surface. Consult with the airport's Seed and Sod Specification for soil preparation, fertilizing and other requirements. The project area's turf is to be maintained so that weed species are not permitted to persist, and grass is maintained by the contractor at 6" tall or less until the project is closed.

12.0.2 Design Aircraft

The Critical Design Aircraft (CDA) shall be identified for each project however the standard is Airplane Design Group (ADG) VI and Taxiway Design Group (TDG) 6, per, AC 150/5300-13, Airport Design (current edition).

The CDA, associated fleet mix, and anticipated operations, will be furnished by DFW Airport for airside pavements. And review by the Vice President of Operations or designee, all AOA ADG standards shall be accomplished prior to final design.

12.0.3 Geometrics

All airfield geometry shall conform to the current ALP. Detailed geometry not included or referenced on the ALP shall conform to the requirements in AC 150/5300-13 and other relevant ACs. All fillets for “Cockpit-over-Centerline” steering will be designed in accordance with AC 150/5300- 13.

Changes from AC 150/5300-13, Airport Design (current edition) standard may be made pertaining to any of the following elements:

12.0.3.1 Runway Length

The CDA will be furnished by DFW Airport.

12.0.3.2 Width, Clearances and Separations of Runways, Taxiways and Parking Aprons

The CDA, or its associated Airplane Design Group per AC 150/5300-13, Airport Design (current edition), will be recommended by the Designer based on traffic forecasts furnished by DFW Airport or Tenant Airline.

12.0.3.3 Line of Sight

All runways and runway safety areas shall conform to the line-of-sight criteria of, AC 150/5300-13, Airport Design. Taxiways under the control of the Air Traffic Control (ATC) Towers shall be in full view of the tower cab full length and width. An ATC Tower Line-of-Sight (Shadow) Study shall be prepared to determine the line-of-sight acceptability. Ramp control towers may require line-of-sight studies for aircraft parking areas and taxi-lane intersections.

Line-of-sight considerations may also be required when facilities are planned and designed near, or in the vicinity of, FAA Navigational Aids (NAVAIDS). Prior to commencement of airfield construction, a DFW Airport Board Airspace Review Application” shall be completed with appropriate information exhibits required by the FAA on which FAA can conduct an Aeronautical Study of the proposal – (reference AC 70/7460-2 and FAR Part 77).

Non-AOA projects will require a Form for staging areas, batch plants, construction cranes and other related items. Construction activities (temporary stationary objects) shall be reviewed by the Federal Aviation Administration (FAA) through the DFW Airport Code Department.

No construction activity shall commence until the Airspace Study is completed and comments have been incorporated into the project plans and specifications.

12.0.3.4 Gradients and Slopes

All paved and turfing areas on the airfield AOA shall meet the requirements of AC 150/5300-13, and as supplemented by the following criteria:

1. Side slopes on excavation (cut) and embankment (fill) areas outside of runway and taxiway safety areas shall have a slope no steeper than 4 to 1 vertical.
2. All topography and above ground objects, except those required by function for navigation, shall be clear of the imaginary surfaces of Federal Aviation Regulation (FAR) Part 77 (Safe, Efficient Use, And Preservation of the Navigable Airspace) and shaped or designed to avoid line-of-sight problems and interferences with Airport navigational instruments and facilities.

Objects that are within safety areas, and in object free areas shall comply with FAR Part 139, Certification of Airports.

3. The standard crowns (transverse slope) on runways and taxiways shall be one percent, except where flatter grades are necessary due to intersection transitions, in which case they shall be a minimum of 0.5 percent.
4. All (transverse slope) of runway shoulders and taxiway shall be paved with a minimum of one percent to a maximum of 5 percent surface gradient (cross Slope). The desirable slope is 2 percent. The maximum slope shall not be used without approval from Vice President of Operation or designee of DFW Airport. The edge of pavement to edge of shoulder shall be at the same elevation (no pavement lip).
5. Pavement gradients on aircraft parking aprons shall be 0.5 percent min., except where conforming or transitioning to existing facilities, and except for 50 feet from Terminal buildings at the gate and parking positions which shall be 1 percent to conform to National Fire Protection Association (NFPA) Standard 415 on "Aircraft Fueling Ramp Drainage."
6. Gradients, slopes, and object clearing criteria for "Obstacle Free Zones", "Runway and Taxiway Safety Areas," and "Runway Protection Zones" shall conform to the standards of AC 150/5300-13, Airport Design (current edition), for the respective critical aircraft or mix of aircraft.

12.0.3.5 High Speed Exit Taxiway (Acute Angled Taxiways)

Locations shall be as shown on the Airport Layout Plan. The geometric layout shall either match existing high speed exit taxiways on the Airport or conform to AC 150/5300-13, Airport Design (current edition), Larger-than-standard fillet radii shall be investigated where traffic "backturns" are anticipated.

12.0.3.6 Right Angle Connector Taxiway

Right angle intersections shall meet the requirements of cockpit-over-centerline steering and shall conform to the requirements of AC 150/5300-13 (current edition).

12.0.3.7 Runway and High-Speed Exit Taxiway Grooving

All runway and taxiway grooving should conform to, AC 150/5370-10 (current edition) Standard Specification for Construction of the Airport. Slurry from sawing must be vacuumed as part of the sawing operation and disposed of off the Airport property. Final cleanup shall include flushing with water.

12.0.3.8 Parking Aprons

Where parking Aprons are included in the project scope, the overall location and geometric layout will be furnished by DFW Airport. Widths, clearances, fillet radii and other details not furnished shall conform to AC 150/5300-13, Airport Design (current edition) or as recommended by the Designer and approved by DFW Airport.

12.0.3.9 Aircraft Parking Aprons

Aircraft parking aprons shall be based on an "Apron Utilization Plan". Apron utilization criteria, including wingtip clearance, shall be approved by DFW and must be within the maneuvering limits of the Aircraft Characteristics Manual of the Critical Design Aircraft. Aircraft service pits shall be located to minimize impact on PCC pavement joint performance.

12.0.4 Pavement

12.0.4.1 General

All aircraft worthy pavements (including Apron areas) shall be rigid pavement. Shoulders and blast protective pavement can be flexible or rigid pavement depending on pavement design, location and loading requirements. The Designer shall meet the Critical Design Aircraft (CDA) and operations mix requirements, regarding the type of pavement recommended. Follow AC 150/5320, Airport Pavement Design (current edition).

Each project Designer shall prepare and recommend a pavement structure for DFW Airport to review and approve. A final geotechnical report shall be submitted with the final construction documents.

Follow AC 5370-10, Standard Specification for Construction of Airport (current edition).

Blast protective pavement type shall be recommended by the Designer based on an occasional pass by the critical maintenance or ARFF equipment and approved by Vice President of Operations or designee.

For new or rehabilitation civil projects with regards to transportation, install temperature sensors where specified by DFW on the airfield or landslide.

A non-invasive pavement weather sensor monitoring system shall be implemented to monitor elevated bridge surfaces. This system must continuously assess surface parameters, including temperature, and determine whether the surfaces are dry or wet. The active components of the system shall be provided by OTT-Hydromet.

The monitoring system must utilize a cellular modem for communication, connecting to ViewMondo, the platform employed by DFW, to relay status updates to the owner

12.0.4.1.1 Aircraft Worthy Pavement Section

Surface Layer: Portland Cement Concrete (PCC) shall be designed based on 700 psi flexural strength at 28 days. All pavements shall be reinforced with steel. Keyways will not be allowed. Surface texture for all runways and high-speed exits shall be grooved by sawing.

Dowels are recommended when positive load transfer is required. Dowel Size and Spacing. Dowel bars should have a diameter consistent with the slab thickness and be placed at the mid-depth of the existing slab.

Shoulders: Shall be Flexible pavement (asphalt) designed for associated fleet mix.

For mix design proportions and criteria follow AC 150/5370-10-P-401, Standard Specifications for Construction of Airports, AC 150/5370, (current edition). All joints between concrete and asphalt shall be sawed and sealed to retard moisture intrusion and vegetation growth.

The Engineer should specify the widest paving lane to hold the number of longitudinal joints to a minimum. Additional job specific construction limitations may be added as necessary covering such items as echelon paving, hot joint construction, etc.

Base Layer: Designer can design for Cement Treated Base (Stabilized) or asphalt stabilized base.

Subbase Layer: Minimum of 8-inch crushed aggregate is required as under-drain system (under drain and edge drain), unless recommended otherwise by the Designer. System layout, elements, and design shall be designed based on soils investigation results, pavement function, and other relevant factors and parameters.

If under-drains are not recommended, the Designer shall present the basis on which they are not recommended and submit it to DFW Code Department for approval.

Subgrade: Establish minimum density requirement for cohesive soil type for subgrade.

12.0.4.1.2 Aircraft Rescue and Fire Fighting Roads

12.1.4.1.2.1 Geometrics

All ARFF roads should be minimum twelve feet (12') wide and shoulders should have a minimum width of six feet (6') continuously Reinforced Concrete Pavement (CRCP).

Load Limits: The minimum vehicle load limit for Fire Apparatus Access Roads is 53,000 pounds. The minimum vehicle load limit for AOA fire apparatus access roads is 113,000 pounds. All bridges and elevated roads shall conform to this requirement.

Turning Radius: The external turning radius (wall to wall) shall not be less than 57 feet. The internal radius shall be no less than 35 feet. The turn at no time will be less than 22 feet wide.

Grade: The maximum grade change of any portion of a fire apparatus access road shall not exceed 6%.

12.0.4.1.3 ARFF Roads Pavement Section

All ARFF road pavements shall be rigid pavement. (4,400 psi).

Surface Layer: Minimum 12 inch of Continuous Reinforced Concrete Pavement (CRCP).

Base Layer:

Minimum of 8 inch of flex base or a minimum of 9 inch Cement Treated Base (CTB) (Stabilized Base).

Geosynthetic separation material for integrity between the surface layer and the base layers is required.

Subbase Layer: Minimum of 8 inch crushed aggregate is required as under-drain system (under drain and edge drain), unless recommended otherwise by the Designer. System layout, elements, and design shall be designed based on soils investigation results, pavement function, and other relevant factors and parameters.

If under-drains are not recommended, the Designer shall present the basis on which they are not recommended and submit it to DFW Airport Code Department for approval.

Subgrade Layer: Minimum 9 inch compacted soil. Establish minimum density requirements for cohesive soil type.

12.0.4.1.4 Other Airside Pavement Section

All Other AOA roadway pavements shall be rigid pavement except the FAA access roads.

Surface Layer: Minimum 8 inch of Continuous Reinforced Concrete Pavement (CRCP).

Base Layer: Minimum of 8 inch of Flex Base. Geosynthetic separation material for integrity between the surface layer and the base layers is required.

Subgrade Layer: Minimum 6 inch compacted soil. Establish minimum density requirements for cohesive soil type.

Material: All materials shall be in accordance with the FAA requirements for airfield pavements including runways, taxiways, and access roads. Follow AC 5370-10, Standard Specification for Construction of Airport (current edition).

12.0.5 Aircraft Bridges

12.0.5.1 General

This section shall apply to all bridges, tunnels, culverts, vaults, and all other structures supporting aircraft or under runways, taxiways, or aprons. Such structures shall conform FAA AC 150/5300-13, Airport Design (current edition). Unless otherwise approved by DFW Airport VP of Operation.

Airplane Design Group

Structures shall be designed and proportioned to accommodate Aircraft Design Group (ADG) VI and Taxiway Design Group (TDG) 6 as defined in FAA AC 150/5300-13, Airport Design (current edition). Should EOR propose another section the design shall be approved by DFW Airfield Operations Department.

Each element of the structure shall be designed to accommodate the most demanding airplane under this design group. This may result in more than one airplane being used in designing a particular structure (i.e., bridge width may be controlled by the airplane with the longest wingspan, whereas another airplane may have higher wheel loads, thus controlling beam design).

Live Loads

Structures shall be designed for the following airplane loads:

- 1. Spans under 2 feet in the shortest direction, including manholes lids and grates - uniform live load of 250 psi.
- 2. Span lengths 2 to 10 feet in the shortest direction, with the greater of a uniform live load varying between 250 and 50 psi in inverse proportion to the span length or the maximum number of wheel loads for the airplane which can be applied to the structure.
- 3. Span lengths greater than 10 feet in the shortest direction, wheel loads for the design airplane.

Impact

For those elements listed in Group A (defined below), the live load shall be increased by the following percentages. This increase will account for impact loads and vibration:

- 1. 30 percent: Parking aprons and low speed taxiways
- 2. 40 percent: High speed taxiways and runways
- 3. 100 percent: Touchdown areas of runways.

Live loads shall not be increased by impact for those items in Group B (defined below).

Impact for structures covered with fill shall vary from the percentage shown at ground level to 0 percent at a depth of 10 feet.

Table 37 - Structural Elements for Group A and Group B

Group A	Superstructure, columns, and pedestals which support the superstructure with rigid, fixed or expansion bearings, or which are rigidly attached to the superstructure, and legs of rigid frames. The portions above the ground line of piers that are rigidly connected to the superstructure as in rigid frame or continuous structures.
Group B	Abutments, retaining walls, piers, pile caps and pilings which are not rigidly connected to the superstructure. Buried foundations, footings and supporting soil, and structures with 3 feet or more of earth cover.

Braking Force

Longitudinal forces due to braking shall be included in the design of all structures subject to direct wheel loads. This braking force shall be the following percentages of live load without impact:

1. 30 percent: Parking aprons and low speed taxiways.
2. 70 percent: High speed taxiways and runways.

Design Load Combinations

In addition to live and dead loads, the following loadings shall be taken into account: earth pressure, buoyancy, wind (including jet blast and uplift), shrinkage, temperature, longitudinal force, stream flow, construction loads and any special loads. Loads shall be applied in such a manner as to produce the maximum stresses.

Loading combinations shall be the same as those described in latest version of the AASHTO's "Standard Specifications for Highway Bridge's" and interim specification.

12.0.6 Drainage

Refer to Chapter 5, Section 5.0.4 title "Stormwater Drainage Utilities" for drainage design requirements.

12.0.7 Signs, Markings, and Lighted Signs**Pavement Marking**

Pavement marking of runways, taxiways, taxi lanes and other paved areas within aircraft operations areas shall conform to AC 150/5340-1 (current edition), Standards for Airport Markings, and be approved by DFW Code and DFW Operations.

Thermoplastic, in accordance with AC 5370-10, Airport Construction Standards, shall only be installed in nonmovement area boundary, intermediate holding position, and Apron Entry Points (AEPs).

Type III glass beads, in accordance with AC 5370-10, shall be applied in all white markings, and Type I glass beads shall be applied in all red markings and all yellow markings, except taxiway edge markings.

For all surface painted markings, spacing between letters and characters shall be 15 inches unless otherwise specified in the AC 150/5340-1 (current edition), Standards for Airport Markings.

Never paint black markings on top of yellow markings.

Per DFW Airport Standards, Surface Painted Holding Position Signs shall follow:

1. Dimension A: 37 inches
2. Dimension B: 4 feet and 7 inches

*Note: Dimension B can vary if necessary to meet the 2-foot minimum requirement of the distance between edge of the sign and taxiway edge marking.

Taxiway Edge Lights shall be spaced 9 feet and 6 inches from taxiway edge marking.

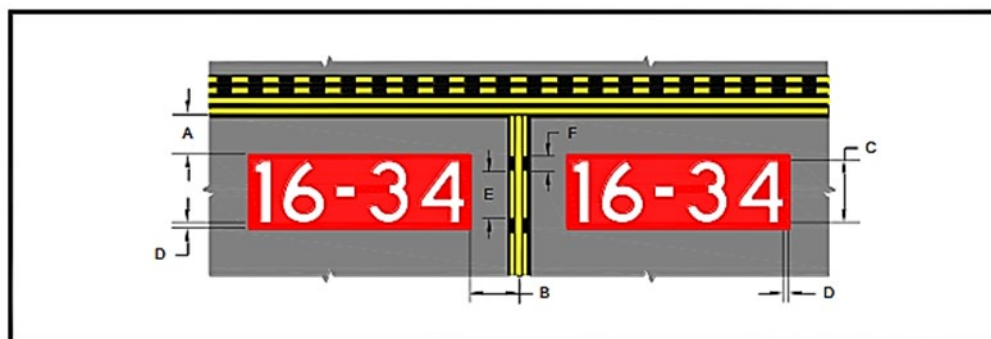


Figure 33 - Surface Painted Holding Position Sign

DFW Airport Standards for Apron Markings

Apron Vehicle Service Roads (VSR)s shall have a surface painted stop sign with stop bar at intersections.

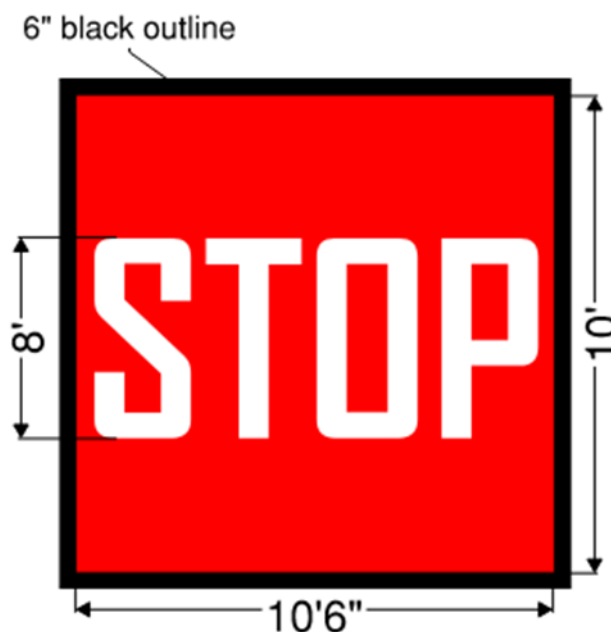
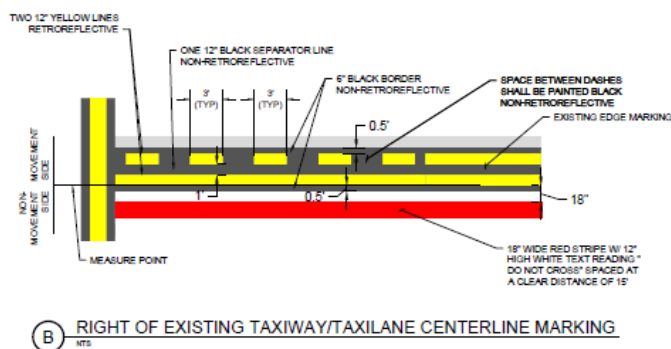
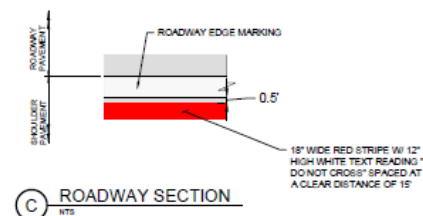
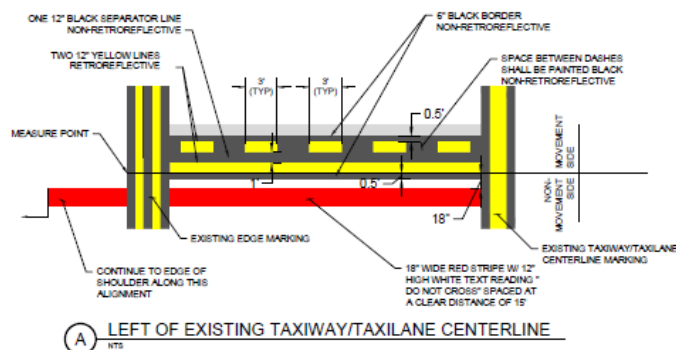


Figure 34 - Surface Painted Stop Sign for Vehicle Surface Roads

The following are not contained in FAA AC 150/5340, Airport Markings:

1. Lead-In Lines: 12 inch wide yellow lead-in line with 6" wide black outlines on both sides.
2. Taxi lane (TL) Edge Markings: two 6 inch wide yellow stripes separated and outlined on each side by 6 inch black outline stripes. (Follow FAA standard taxiway edge marking for Taxi Lane).
3. Surface Painted Gate Identification Signs: Characters will be black and 8 feet tall with spacing of 6 inches between characters, with a yellow box of 10 feet height for the background. The font will be FAA AC 150/5340-1(current edition), Standards for Airport Marking.

4. Marshaller Stop Line: 12 inch wide yellow stripe with a 6 inch black outline and a 6 foot width.
5. Stop Bar Aircraft Identification: Yellow characters of 8 inch height on a black background of 12 inch height. The font will be FAA AC 150/5340-1 (current edition), Standards for Airport Marking.
6. Aircraft Safety Envelope: White/Red/White stripes that are 4 inches in width.
7. Bridge Swing Limit (No Parking Area): Shall have red cross hatches at 45 degree angle, and 6 feet apart, with 4 inch wide red stripes for boundary.



“DO NOT CROSS” red line as defined by this detail

8. Object Free Area Limit Line: Shall have 3 feet length and 6 inch width alternating black and white stripes without outline.
9. Roadway Signs: Yield to Aircraft and 20 MPH signs shall be surface painted on the apron roadways. The letters will be white with glass beads (no outline is needed), 4 foot high and 9 inches wide. Lines of text to be separated by 2 ½ feet. Font to be used is either TXDOT Grid or "E" modified font (Follow TxDOT standards). The Yield and MPH signs to be separated by 10 feet (groupings shall be spaced 200 feet typical).

12.0.8 Construction Safety and Phasing Plan (CSPP)

The Designer shall coordinate with DFW Operations and DPS through the Project Manager, regarding all safety and security provisions of the project. Other considerations, depending on the project scope, include temporary modifications to pavement markings, airfield signage, and airfield lights, and required Special Provisions to fit the project. Provisions must be made for and included in all contract's documents pertaining to safety during construction, construction sequencing, access to the site, Contractor's staging area, haul routes, barricades, fencing, traffic control, etc. Follow AC 150/5370 (current edition) Operational Safety on Airports During Construction, and DFW Airport Rules and Regulations for Maintenance and Construction on the AOA contains the means by which construction may be accomplished within the AOA.

12.0.9 Variance from Design Criteria Manual

Any proposed variation from DCM must be submitted for review and approval by the DFW Airport prior to design submittal.

12.1 Landside**12.1.1 General**

The Roadway Design Guidelines reference the latest and most current editions of the publications listed below, but not limited to, and should be referenced to provide guidance in design methodologies and establishing project criteria:

1. Roadway Design Manual, TxDOT.
2. Bridge Design Manual, LRFD TxDOT
3. Geotechnical Manual TxDOT
4. Highway Illumination Manual.
5. Bridge Railing Manual TxDOT
6. Hydrology and Hydraulics TxDOT
7. Texas Manual on Uniform Traffic Control Devices (TMUTCD), TxDOT
8. Standard Specifications for Construction and Maintenance of Highways, Streets and Bridges, TxDOT Statewide Planning Map
9. TxDOT Access Management Manual
10. Procedures for Establishing Speed Zones, TxDOT
11. Traffic Standards, TxDOT Statewide Planning Map
12. A Policy of Geometric Design of Highway and Streets (Green Book), AASHTO
13. Roadside Design Guide, AASHTO (latest edition)
14. Transportation Impact Analysis for Site Development, Institute of Transportation Engineers (ITE)
15. Manual of Transportation Engineering Studies (ITE)
16. American Concrete Institute (ACI)
17. Concrete Reinforcing Steel Institute (CRSI)

The designer shall follow roadway classification defined by TxDOT, Roadway Design Manual (current edition) and the AASHTO Green Book for guidelines and design criteria.

A non-invasive pavement surface weather sensor monitoring system shall be implemented to monitor elevated bridge surfaces. This system must continuously assess surface parameters, including temperature, and determine whether the surfaces are dry or wet. The active components of the system shall be provided by OTT-Hydromet.

The monitoring system must utilize a cellular modem for communication, connecting to ViewMondo, the platform employed by DFW, to relay status updates to the owner

Table 38 - Roadway Classification

Classification	Pavement Width (between curbs)	***Driving Lanes	ROW	***Speed Limit
Regional Arterial	2 – 36 feet	6 – 12 feet	150 feet	45 miles per hour
Major Arterial	2 – 36 feet	6 – 12 feet	120 feet	45 miles per hour
Minor Arterial	2 – 24 feet	4 – 12 feet	100 feet	35 miles per hour
Collector	2 – 36 feet	6 – 12 feet	70 feet	30 miles per hour
Curbside Collector	48 feet	4 – 12 feet	60 feet	20 miles per hour
Terminal Collector	24 feet	2 – 12 feet	25 feet	15 miles per hour
Freeway	Multi-Lanes	12- foot lanes		55 miles per hour
Ramp		14-foot lane		20 – 25 mph

*** Freeway - International Parkway is designated as a Freeway within the Airport. International Parkway is a divided highway for moving through traffic with full control of access. The posted speed limit on International Parkway is 55 mph.

*** Speed Limit - The speed limit or posted speed limit is the maximum (legal) speed limit set by local authorities for a certain roadway classification based on appropriate traffic studies. The design speed should never be less than the likely legal speed limit.

*** Driving Lanes - AutoTURN will examine turn movement maneuvers and driving lanes along the roadway network.

Portland Cement Concrete Pavement: Portland Cement Concrete (PCC) shall be designed based on 700 psi flexural strength at 28 days. All pavements shall be reinforced with steel. Keyways will not be allowed.

Asphalt Pavement: Asphalt pavements shall be a heavy duty mix design to accommodate heavy freight truck traffic. Texas Department of Transportation (TxDOT) Specification No. 340 with approval. All joints between concrete and asphalt shall be sawed and sealed to retard moisture intrusion and vegetation growth.

The Engineer should specify the widest paving lane practicable in an effort to hold the number of longitudinal joints to a minimum. Additional job specific construction limitations may be added as necessary covering such items as echelon paving, hot joint construction, etc.

12.1.2 Design Vehicle

The WB-67 design vehicle should be the design vehicle considered for arterials and roads that carry moderate to high volumes of traffic or that provide local access for large trucks. The WB-62 may be used in design for turning movements when authorized and approved by the Authority Having Jurisdiction (AHJ).

12.1.3 Geometrics

Horizontal Alignment: The following design criteria shall be followed unless otherwise specified by the Airport:

1. Reverse Curves without an intervening tangent will not be permitted unless otherwise approved by DFW Airport.
2. Compound curves will not be permitted.

Vertical Alignment: The following design criteria shall be followed unless otherwise specified by the Airport:

1. Minimum grade for vertical profile shall not be less than 0.3 percent, unless approved by DFW Airport.
2. Maximum grade for vertical profile shall not be greater than 6% percent, unless approved by DFW Airport.
3. Longitudinal grade typically shall be 0.5 and at a minimum to be 0.3 percent. For maximum allowable designer shall follow roadway classification defined by TxDOT at Roadway Design Manual (current edition).
4. Rates of Grade shall be shown for all grades.

Driveways: Driveway width will need to be designed to allow an Interstate Semitrailer the ability to effectively make a turning movement into a driveway with a Privately-Owned Vehicle (POV) within the driveway throat.

The throat width and radii should be designed to incorporate AASHTO's standard commercial design vehicle for industrial areas, the WB-67. The WB-67 design vehicle is an appropriate design vehicle model for a DPS Fire Department emergency vehicle responding to an incident and to adequately turn into the driveway with an uninterrupted turn movement onto the driveway apron and into the facility.

Maximum vertical grade at driveways shall not exceed 8%.

For further design detail Refer to TxDOT Roadway Design Manual (current edition).

Auxiliary Lanes: Minimum width for a turning lane will be 12-feet to afford maximum safety and service to Airport emergency vehicles.

Cross Slopes: Preferred Cross slopes are 2 percent. Maximum allowable cross slope is 6 percent, unless approved by DFW Airport.

Clear Zone: Follow TxDOT Roadway Design Manual (latest edition) based on functional class to determine clear zone distances. Front slope from the edge of pavement shall have minimum 4:1 slope or flatter and maximum 3:1 slope.

Vertical Clearance: Reference the Roadway Design Manual (Current edition). Any deviation from the standard requires submission of Variance Form to be approved by DFW Airport (online at DFW Airport website).

The clearances must be verified in the early stages of design. Any deviation from the standards shall be approved by DFW Airport. Requester shall complete the Variance Request Form and obtain approval prior to implementation of variation.

Sight Distance: Sight distance is important in the design and operation of various types of road facilities. Poor sight distance can adversely impact safety. Reference TX DOT Roadway Design Manual (current edition) and AASHTO's Green Book (current edition) for guidelines and design criteria.

12.1.4 Pavement

12.1.4.1 General

Each project Designer shall prepare and recommend a pavement structure for DFW Airport to review and approve. A final geotechnical report shall be submitted with the final construction documents.

When organic content of soil exceeds 1%, Follow TxDOT "Treatment Guidelines for Soil and Base in Pavement Structure" (current edition).

For CRCP, longitudinal reinforcing bars shall be adequately lapped at splices, and that the splices be staggered in arrangement that will not cause localized strain in the pavement or nearby construction joint.

Dowels are recommended when positive load transfer is required. For dowel size and spacing with respect to pavement thickness, reference American Concrete Institute ACI 302.1R-96, Guide for Concrete Floor and Slab Construction Table 3.2.7. Dowel Size and Spacing. Dowel bars should have a diameter consistent with the slab thickness and be placed at the mid-depth of the existing slab.

12.1.4.2 Roadway Pavement Section

Surface Layer: Minimum 11 inch of Continuous Reinforced Concrete Pavement (CRCP).

Base Layer: Minimum 3 inch of Hot Mix Asphalt Concrete (HMAC).

Subbase Layer: Minimum 8 inch of flex base or a minimum 9 inch Cement Treated Base (CTB) (Stabilized Base).

Geosynthetic separation material for integrity between the surface layer and the base layers is required.

Subgrade Layer: Minimum 9 inch compacted soil. Establish minimum density requirements for cohesive soil type.

When organic content of soil exceeds 1%, Follow TxDOT "Treatment Guidelines for Soil and Base in Pavement Structure" (current edition).

12.1.4.3 Other Laneside Pavement Section

All Other laneside roadway pavements shall be rigid pavement.

Surface Layer: Minimum 8 inch concrete, with #4 rebar at 12 inch.

Base Layer: Minimum of 6 inch Flex Base. Geosynthetic separation material for integrity between the surface layer and the base layers is required.

Subgrade Layer: Minimum 6 inch compacted soil. Establish minimum density requirements for cohesive soil type.

Roadway Pavement Finishes

1. Carpet Drag - Provide a carpet drag mounted on a work bridge. Provide a single piece of carpet of sufficient transverse length to span the full width of the pavement being placed and adjustable so that a sufficient longitudinal length of carpet is in contact with the concrete being placed to produce the desired texture.
2. Texturing and Tining Equipment - Provide a self-propelled metal tine device equipped with steel tines with cross-section approximately 1/32" thick X 1/12" wide. Provide tines for transverse tining equipment spaced at approximately 1" center-to-center.
3. Placing, Consolidating, and Finishing Equipment -Provide approved self-propelled paving equipment that uniformly distributes the concrete with minimal segregation and provides a smooth machine finished consolidated concrete pavement conforming to the plan line and grade. Provide an approved automatic grade control system on slip-forming equipment. Provide approved mechanically operated finishing floats capable of producing a uniformly smooth pavement surface. Provide equipment capable of providing a fine, light water fog mist.
4. Finished Surface - Perform sufficient checks with long handled 10' and 15' straightedges on the plastic concrete to ensure the final surface is within the tolerances specified in Surface Test A in Item 585, "Ride Quality for Pavement Surfaces". Check with the straightedge parallel to the centerline.
5. Surface Texturing - Complete final texturing before the concrete has attained its initial set. Drag the carpet longitudinally along the pavement surface with the carpet contact surface area adjusted to provide a satisfactory coarsely textured surface. Prevent the carpet from getting plugged with grout. Do not perform carpet dragging operations while there is excessive water bleed. A metal-tine texture is required.
6. Provide transverse tining. Immediately following the carpet drag, apply a single coat of evaporation retardant, if needed, at the rate recommended by the manufacturer. Provide the metal tine finish immediately after the concrete surface has set enough for consistent tining. Operate the metal tine device to obtain grooves approximately 3/16" deep, with a minimum depth of 1/8" and approximately 1/12" wide. Do not overlap previously tined areas.

7. Ride Quality - Measure ride quality in accordance with Item 585, "Ride Quality for Pavement Surfaces"
 - a. Provide Surface Test Type A- Provide a 10' straightedge or where allowed, a high-speed or lightweight inertial profiler, certified at the Texas A&M Transportation Institute.
 - b. Surface Test Type B- Provide a high-speed or lightweight inertial profiler, certified at the Texas Transportation Institute. Provide equipment certification documentation.
 - c. Use diamond grinding method to correct surface areas that have more than 1/8" variation between any 2 contacts on a 10' straightedge. Use diamond grinding to correct any deficient 0.1-mi. section. Diamond Grinding Equipment. Provide self-propelled powered grinding equipment specifically designed to smooth texture pavements using circular diamond blades when grinding is required. Provide equipment with automatic grade control capable of grinding at least 3' of width longitudinally in each pass without damaging the pavement.
8. Maturity Method - Use the maturity method, Tex-426-A, to estimate concrete strength for early opening pavement to traffic unless otherwise shown on the plans. Install at least 2 maturity sensors for each day's placement in areas where the maturity method will be used for early opening. When used, Maturity sensors will be installed near the day's final placement for areas being evaluated for early opening. Use test specimens to verify the strength-maturity of the concrete before opening pavement to traffic.

Verify the strength-maturity relationship at least every 10 days of production after the first day. Establish a new strength-maturity relationship when the strength specimens deviate more than 10% from the maturity-estimated strengths. Suspend use of the maturity method for opening pavements to traffic when the strength-maturity relationship deviates by more than 10% until a new strength-maturity relationship is established.

The Engineer will determine the frequency of verification when the maturity method is used intermittently or for only specific areas.

9. Wet curing for concrete is preferred. If Poly-Alpha Methylstyrene curing compound is used it shall be approved by DFW Airport.

If wet curing method is implemented contractor shall provide "Continuous Watering Plan".

12.1.5 Bridges

Bridges and Bridge Class Box Culverts: Any new or rehabilitated bridge and box culvert structures shall follow TxDOT Bridge Design Manual (current edition), unless directed otherwise by DFW Airport.

Bridge Class Culvert: This provision shall apply to structures with spans greater than twenty (20) feet and whose function is to carry roadway traffic. This section does not apply to parking structures or ramp systems within them.

Live Loads: Selection of live load for bridges over arterial roads shall be designed for an HS20-44 live load plus impact, unless otherwise approved by DFW Airport.

Bridge Widths: Generally, bridge width from face of rail to face of rail shall be at least as wide as the approach roadway's travel lane and usable shoulder.

Concrete: Concrete materials, quality, classes of, and proportioning shall comply with the applicable sections of the TxDOT "Standard Specifications for Construction of Highways, Streets, and Bridges" (current edition).

Structural Steel: Structural steel, forgings, castings, anchor bolts, pipe, tubing, bolting of and welding of shall comply with the applicable sections of the TxDOT "Standard Specifications for Construction of Highways, Streets, and Bridges" (current edition).

Reinforcing Steel: Reinforcing steel material and bending shall comply with the applicable sections of the TxDOT "Standard Specifications for Construction of Highways, Streets and Bridges" (current edition).

Pre-stressing Steel: Pre-stressing steel, packing, storing, handling, working drawings and construction methods shall comply with the applicable sections of the TxDOT "Standard Specifications for Construction of Highways, Streets, and Bridges" (current edition).

12.1.6 Drainage

Refer to Chapter 5, Section 5.0.4 title "Stormwater Drainage Utilities" for drainage design requirements.

12.1.7 Pavement Marking and Signage

The design of signs for roadways shall be in accordance with the following standards:

1. Texas Manual of Uniform Traffic Control Devices (TMUTCD, current edition) and
2. DFW Airport Wayfinding Signage Standards & Guidelines

Designer shall submit "Summary of Small Signs and Large Signs" at 100 percent for Code/Design Review.

The retro-reflective sheeting for all signage will be 3M Diamond Grade DG3 Reflective Sheeting or approved equal.

All Post size and placement shall be according to Texas Manual of Uniform Traffic Control Devices, TMUTCD (current edition), and TxDOT Roadway Standards (current edition).

CLEARANCE signs (W12-2, W12-2a and W12-4T) need to reflect the vertical clearance between the roadway and the overhead obstruction. Consult the TMUTCD for more information on this topic.

Markings: Markings provide guidance and information to road users for both daytime and nighttime use. All pavement markings shall follow the Texas Manual of Uniform Traffic Control Devices (TMUTCD, current edition).

1. Stop Bar (Stop Lines) will be 24 inches.
2. Grinding is not an acceptable method of marking removal

Traffic Signals: DFW Airport utilizes an Advanced Traffic Management System to monitor and maintain traffic throughout the Airport. The system includes:

1. Controller Cabinet
2. Opticom System
3. Traffic Video Detection Camera

Any design package shall include this system to allow 100% compatibility and integration into the network.

12.1.8 Traffic Control Plan

All traffic control plans (TCP) shall follow Texas Manual of Uniform Traffic Control Devices Part 6, Temporary Traffic Control, (current edition).

Any deviation from the standards shall be approved by DFW Airport. Requester shall complete the Variance Request Form and obtain approval prior to implementation of variation.

DFW Airport "Work Zone Closure Requirements" form is online at DFW Airport website.

12.1.9 Variance from Design Criteria Manual

Any proposed variation from DCM must be submitted for review and approval by the DFW Airport prior to design submittal.

12.2 Land Development

12.2.1 General

For any new development in DFW Airport follow Design Guidelines located at:

DFW Airport website: sites.dfwairport.com/landhere/resources/index.php

12.2.2 Design Vehicle

The WB-67 design vehicle should be the design vehicle considered for arterials and roads that carry moderate to high volumes of traffic or that provide local access for large trucks, and for emergency vehicles for Fire and DPS Department.

The WB-62 may be used in design for turning movements when authorized and approved by the Airport.

12.2.3 Geometrics

Driveway width will need to be designed in order to allow an Interstate Semitrailer the ability to effectively negotiate a turning movement into a driveway with a POV within the driveway throat. The throat width and radii should be designed to incorporate AASHTO's standard commercial design vehicle for industrial areas, the WB-67.

12.2.4 Pavement

For CRCP, longitudinal reinforcing bars shall be adequately lapped at splices, and that the splices be staggered in arrangement that will not cause localized strain in the pavement or nearby construction joint.

Dowels are recommended when positive load transfer is required. For dowel size and spacing with respect to pavement thickness, reference American Concrete Institute ACI 302.1R-96, Guide for Concrete Floor and Slab Construction Table 3.2.7. Dowel Size and Spacing. Dowel bars should have a diameter consistent with the slab thickness and be placed at the mid-depth of the existing slab.

12.2.4.1 Pavement Fire Lane Section

Surface Layer: Minimum 8 inch of Continuous Reinforced Concrete Pavement (CRCP) (4400 psi).

Base Layer: Minimum 8 inch of Flex Base.

Subgrade Layer: Minimum 6 inch compacted soil. Establish minimum density requirements for cohesive soil type.

When organic content of soil exceeds 1%, Follow TxDOT “Treatment Guidelines for Soil and Base in Pavement Structure” (current edition).

Designer shall propose subgrade preparation method (Example: proof roll, ordinary compaction, or density control).

12.2.5 Drainage

Refer to Chapter 5, Section 5.0.4 title “Stormwater Drainage Utilities” for drainage design requirements.

12.2.6 Traffic Impact Analysis

A Traffic Impact Analysis (TIA) will be required for all landside development projects that will generate, distribute, and assign vehicular traffic to and from a proposed development onto the Airport’s roadway network. The TIA will determine what impact a development will have on the existing and proposed roadway network.

AutoTURN will be required to assess and design all turning movements from an adjacent road onto a facility’s driveway throat.

A Level of Service “C” or better is the desired objective for the airport’s roadways and intersections.

For the principles of design that are common to developments and roadway/driveway intersection, reference TxDOT Roadway Design Manual (current edition) and AASHTO’s Green Book (current edition) for guidelines and design criteria.

For proposed driveway locations, be cognizant of the “Functional Area of an Intersection.”

12.2.7 Variance from Design Criteria Manual

Any proposed variation from DCM must be submitted for review and approval by the DFW Airport prior to design submittal.

12.3 AOA Fencing

Security fencing shall be constructed at locations adjacent to the Air Operations Areas (AOA) with no gaps at grade (maximum of 2 inches below gates).

12.4 Fence and Gates

8 foot in height of number 10 gage, 2-inch galvanized steel chain link mesh with three strands of #12 gage, four-point barbed wire mounted above the fence with an outward extension of 45 degrees. In addition, Razor Wire shall be looped and attached on top of the barbed wire in 20-inch loops as per fence specifications. Any water crossings or outfalls shall have coverage equal in strength and durability to the actual fence. Fence posts shall be installed on 10-foot centers maximum. Post, rails, and braces shall be zinc-coated steel sized in accordance with the table below. Hinges and latches shall be zinc-coated. Concrete retaining walls 6 feet or higher will only require the barbed wire extensions and razor wire.

12.5 Genies and AOA Gate Actuators

Information regarding DFW gate actuators and genies has been declared SSI by the DFW DPS and the Transportation Security Administration. Specifications may be released only upon the approval of the Airport Security field spliced with special attachments provided by the manufacture, Polyolefin-coated frame and top track thermally fused.

12.6 Chain-Link Cantilever Slide Gates

12.6.1 Gate Frames

Fabricate chain link cantilever slide gates in accordance with ASTM F 1184, Type II, class 2, using 2-inch square aluminum members, ASTM B – 221. Alloy and temper 6063 – T6, weighing 0.94 pound per foot (lb/ft). Weld members together forming rigid one-piece frame integral with top track. (No substitution allowed.) Frame members are to be square and straight to within over a 40- foot span in an unstressed state. Provide two truck assemblies for each gate leaf, except for gates larger than 30 feet. Frame sizes over 27 feet in length shall be shipped in two parts and Coordinator/Chief of Police, or designee.

Within minimum 10-mil per ASTM 1043 (after fabrication). Coating before fabrication will not be allowed. If gate is not to be polyolefin coated, eliminate reference to polyolefin coating. For gate leaf size 23 to 30 feet, weld an additional 2-inch square lateral support rail adjacent to top horizontal rail. Bottom rail shall consist of 2-inch by 4-inch aluminum member weighing 1.71 lb/ft.

12.6.2 Bracing

Provide diagonal adjustable length truss rods of 3/8-inch galvanized steel in each panel of gate frames.

12.6.3 Top Track /Rail

Enclosed combined one-piece track and rail, aluminum extrusion with weight of 3.72 lb/ft. Track to withstand reaction load 2,000 pounds.

12.6.4 Truck Assembly

Swivel type, zinc die cast, with four seal lubricant ball bearing rollers, 2 inches in diameter by 9/16 inch in width, and two side rolling wheels to ensure truck alignment in track. (No substitution allowed.). Mount trucks on post brackets using 7/8-inch diameter ball bolts with 1/2-inch shank. Truck assembly to withstand same reaction load as track (2,000 pounds).

12.6.5 Miscellaneous

Gate hangers, latches, brackets, guide assemblies, and stops shall be malleable iron or steel, galvanized after fabrication. Provide positive latch with provisions for padlocking.

12.6.6 Bottom Guide Wheel Assembly

Each assembly shall consist of two 3"-inch diameter rubber wheels, straddling bottom horizontal gate rail, allowing adjustment to maintain gate frame plumb in proper alignment. Attach one assembly to each guidepost.

12.6.7 Rotating Beacon

A Red rotating beacon will be installed at a minimum height of 7 feet from grade and adjacent to electronically operated gates. The beacon will activate upon opening and closing of the gate and must be clearly visible to approaching drivers.

12.6.8 Gate Post

For gates under 31 feet: galvanized steel 4-inch OD Schedule 40 pipe, ASTM F – 1083, weighing 9.1 lb./ft. Provide one latch post and two support posts for single slide gates and four support posts for double slide gates.

For gates 31 feet or larger: two pairs of support posts for each leaf (dual) 4-inch OD Schedule 40 pipe, ASTM F – 1083 weighing 9.1 lb./ft each. Posts connected by welding 6-inch by 3/8-inch plate between posts as shown on drawings. Also one 4-inch latch post. Finish to match fence.

12.7 Signs**12.7.1 AOA Signs**

The Airport Board will furnish sign graphics in the form of a master suitable for photographic enlargement. Signs shall be constructed of screen-printed reflective vinyl film on a .080 inch anodized aluminum panel. Anchor signs to fence fabric with #6 galvanized wire ties.

12.7.2 Fence Signs

Warning signs shall be affixed to the fence on centers of 200 feet maximum and at each turn of the fence of 45 degrees or greater. Signs shall be 24 inches in height by 30 inches in width. Fence sign graphics will include the Airport brand and the following verbiage: "Air Operations Area – No Trespassing." In addition, the following will be added to the bottom of the sign:

"WARNING ~ RAZOR WIRE." Sign specifications approved by the DFW DPS can be obtained from the DFW Sign Shop.

12.7.3 Gate Signs

All AOA gates will be assigned a gate number by the DPS. Sign specifications approved by the DPS can be obtained from the DFW Sign Shop.

12.8 Concrete Mow Strip

A concrete mow strip 36 inches in width shall be installed at all permanent fencing. The concrete strip shall be 4 inches in thickness and reinforced with flat 6-inch by 6-inch, W2.9 X W2.9, welded wire fabric. Place the strip on a 2-inch sand cushion. Provide contraction joints spaced 4 feet on center and pre-molded 1/2-inch expansion joint material spaced at 32 feet on center.

12.9 Post and Cable System

A post and cable system shall be installed on the Air Side of AOA fences, 3 feet from and parallel to the fence, to separate International Parkway and the Service Roads. The posts must meet Standard DMS-7200 and the cable must meet Standard ASTM-A-475.

12.10 Bollards

Bollards will be used at locations where the Post & Cable System cannot be installed. Bollard shall include but are not limited to gaps where the Post and Cable System is anchored to the ground and changes directions, or where the terrain is not suitable for the system. Bollards shall be 6-inch galvanized pipe filled with concrete. The pipe shall be anchored in concrete 3 feet below grade. The exposed pipe shall be 3 feet 6 inches above grade. Bollards shall be spaced at 5 feet center to center. If a mow strip is being added to the AOA fence, the mow strip shall be expanded to incorporate the bollards.

12.11 Gate Barriers

Barriers are required for AOA gates as determined by the DPS. The system shall meet the Department of State standard for testing as called out in SD-STD-02.01, revised in March 2003, and must meet the K-12 impact rating of a 15,000-pound vehicle traveling at 50 mph.

Gate barriers must include sub-surface. Electrical components shall be located on the AOA side to avoid activation from the non- AOA side.

Barriers shall be automatically lowered/raised to allow passage of authorized vehicles by either a card swipe through the Airport Board's AACS, Intellikey, genie, or combination, and shall also be capable of being operated manually from inside a guard house.

12.12 Gate Locks

All unmanned AOA gates shall be equipped with an electronic lock and key system and shall comply with CFR 49, Section 1542.207. Provide Intellikey Stand-Alone System compatible with Intellikey ASC 4000 System or DPS-approved equal. The system shall utilize an audit trail and provide access control by user, time, date, and location. The system shall be constructed of weatherproof stainless steel and utilize a captive chain in the locking mechanism that automatically compensates for sags and misalignment.

12.13 AOA Screening

Screenings shall not be attached to AOA fencing or gates. No vehicles or property may be stored within 10 feet of an AOA fence or gate. Any construction design for the parking of vehicles will include the means to maintain the 10-foot separation such as striping, wheel stops, signs, or other approved architectural methods.

12.14 AOA Guard Stations

12.14.1 Guard Houses

Guard houses are required at AOA gates when determined by DFW DPS.

12.14.2 Lighting

Lighting requirements for guard stations shall provide area lighting with intensity at ground level of 10 foot candles within a 20-foot radius of the door of each guard station and 5-foot candles within 40 feet of each guard station. These requirements may change based on new technology or FAA rules that may prohibit same. Refer to IESNA Lighting Handbook, 9th Edition for additional technical information. LED Illumination is preferred.

12.14.3 Lightning Protection

Lightning protection for guard stations (see Division 26).

12.14.4 Building with AOA Interface

All building constructed at locations adjacent to the AOA shall be designed and equipped with Automatic Access Control on any door providing direct access to the AOA.

Chapter 13 Concrete

13.0 Concrete Reinforcing

13.0.1 Reinforcing Steel Clear Cover

For clear cover or protection, comply with the latest editions of TxDOT Standard Specifications, American Concrete Institute and Concrete Reinforcing Steel Institute guidance.

13.0.2 Stressing Tendons

Stressing tendons shall be per Post Tensioning Institute (PTI) standards.

13.0.3 Reinforcing Steel

Provide Grade 60 or above, deformed steel bar reinforcement unless otherwise indicated on the plans. Provide positioning and supporting devices (baskets and chairs) capable of securing and holding the reinforcing steel in proper position before and during concrete placement.

13.1 Structural Concrete

13.1.1 Non-Prestressed Concrete

Non prestressed concrete for structural elements shall be designed for a minimum 28-day compressive strength (f_c') of 4,000 pounds per square inch (psi), as dictated by design requirements.

Exception: Pier concrete may be 3,500 psi.

13.1.2 Concrete Pavement

Concrete pavement shall be designed for a minimum 28-day compressive strength (f_c') of 4,000 pounds per square inch (psi), as dictated by design requirements.

13.1.3 Cold/Hot Weather Concrete

Cold weather placement parameters shall be subject to the design of the Engineer of Record, concrete may not be placed when air temperature is below 40 degrees or over 95 degrees over a 24-hour period following concrete placement.

13.1.4 Precast Structural Concrete

Precast Structural Concrete and Prestressed Concrete for structural elements shall be designed for a minimum 28-day compressive strength (f_c') of 4,000 psi.

13.2 Structural Foundation Systems

13.2.1 Foundations

Allowable foundation capacities shall be determined from geotechnical investigations under the direction of a professional structural engineer registered in the State of Texas.

Foundations shall be designed to prevent uplift and differential settlement, as well as design loads meeting bearing requirements. Minimum ground cover footings shall be 12 inches. Obtain written approval if the concrete mix design exceeds 520 lb. per cubic yard of cementitious material.

13.2.2 Pier Shafts

Pier shafts shall be a straight shaft design. Bell-shaped piers will not be allowed.

13.2.3 Structural Piers

Structural piers used in buildings and bridges; Major structures should extend into the shale rock a minimum of 24 inches to minimize the potential for lateral movement.

13.2.4 Rebar Cages

The rebar cages for all piers shall extend the full length of the pier shaft.

13.3 Pavement Utilities**13.3.1 General**

All cast-in-place and precast concrete utility structures shall be reinforced.

13.3.2 Paving Block outs

Manhole and inlet-throat paving block outs shall be separated from the surrounding paving section with expansion material and connected to the concrete paving section with smooth dowels with expansion caps. The smooth dowel size, length, and spacing shall conform to ACI 302.1R-11 Code. The reinforcing rebar size shall match the longitudinal reinforcing of the surrounding pavement.

13.3.3 Grated Inlets

Grated inlets placed in pavement shall be separated from the surrounding pavement section with expansion material and will not be doweled.

13.3.4 Curb and Gutter Reinforcement

Curb and gutter reinforcing shall match the surrounding pavement. Provide curb forms for separately placed curbs that are not slip formed that conform to the requirements TxDOT Item 529, "Concrete Curb, Gutter and Combined Curb and Gutter". Ensure that changes in curb grade and alignment do not exceed $\frac{1}{4}$ inches between any 2 contacts on a 10 foot straightedge. Place curbs, gutters and combined curb and gutter in 50 foot maximum sections unless otherwise approved. For extruded or slip formed concrete, place concrete with approved self-propelled equipment. The forming tube of the extrusion machine or the form of the slipform machine must be easily adjustable vertically during the forward motion of the machine to provide variable heights necessary to conform to the established grade line.

13.3.5 Transverse Construction Joints

There shall be no laps within 10 feet on either side of a transverse direction continuously reinforced concrete pavement (CRCP) paving joint. All transverse construction joints across the paving lanes, between existing pavement and new pavement, shall be doweled with smooth dowels and shall conform to ACI 302.1R-11 Code. Transverse construction joints must line up and extend across all the paving lanes at the same location without staggers.

In the absence of tie-bars, all longitudinal construction joints between the paving lanes shall be connected with rebar dowels, 6 rebar, 3 feet long at 12 inch centers, with 12 feet embedded and epoxied into the existing paving lane.

13.3.6 Anchor Bolt Assemblies

All anchor bolt assemblies placed in the tops of piers and columns shall incorporate embedded anchor plate rings to maintain vertical alignment of the anchor bolts.

Chapter 14 Parking Structures

14.0 General Information

All criteria specified in Section 13.1; concrete shall also apply to parking structures except as amended in the following.

14.1 Design Guidelines

14.1.1 Material Selection

Structural steel shall not be considered for the vertical and horizontal framing system unless approved by the Building Official.

14.1.2 Steel

All exposed miscellaneous steel used for concrete supports and connections shall be galvanized and retouched after installation.

14.1.3 Corrosion Protection

Calcium chloride and admixtures containing chlorides should not be used in concrete for parking structures. Admixtures shall be used with care and compatibility shall be verified by testing laboratories.

Protection of embedded metals including concrete cover over reinforcement, post-tensioning tendons, pre-tensioned connections for precast systems, dissimilar metals, and embedded metal conduit should meet or exceed the minimum ACI 318 requirements.

ACI 362.1R, Guide to the Design of Durable Parking Structures should be consulted for pertinent information concerning corrosion inhibitors, cathodic protection and protection of concrete.

The guidelines for applied sealers or membrane treatments shall be followed.

14.1.4 Expansion Devices and Materials

Expansion joint seals and isolation joints shall be designed to prevent the following defects or failures:

1. Migration, bleeding into or staining abutting materials
2. Deformation sufficient to become unsightly or cause leakage
3. Chalking, picking up dust or excessive color change
4. Adhesive or cohesive failures

In addition to ACI 318, the Designer should refer to ACI 504.R for various joint treatments and to ACI 224.R and ACI 224.1R for crack controls.

14.1.5 Elastomeric Bearings

Bearings shall be designed as a plain pad with a 70- durometer elastomer or laminated pads with a (60-durometer elastomer.

14.1.6 Parapet Systems

Systems which are integral or monolithic with supporting structural systems shall be designed such that damage to the parapet will not adversely affect the supporting system. The use of isolation joints and membrane protection is important for roof connections.

14.1.7 Drainage Systems

Systems shall be designed and located such that structural elements (i.e., reinforcing steel, tendons, beam flanges, lighting column base plates, etc.) shall take precedence. Use the least amount of bends for unimpeded flow.

Chapter 15 Conveying Equipment

15.0 Baggage Handling System

15.0.1 General

The BHS system should have a recirculation point to reintroduce bags to the system in the event of a failure.

1. BHS ladders should be safety ladders with self-closing gates.
2. Baggage control rooms shall have independent and or redundant cooling systems.

15.0.2 Baggage Characteristics

The types of baggage which the systems may encounter are classified as follows:

1. Normal Baggage
 - a. Normal baggage is defined as items which can be transported by the Baggage Handling System (BHS) and processed by the project-specific EDS equipment without special handling as indicated in

Table 39 - Normal Baggage Characteristics

BHS	Length (inches)	Width (inches)	Height (inches)	Weight (pounds)
Minimum	12	12	4	5
Maximum	54	24	30	120

The BHS must handle items safely and securely while traveling through the system without causing interruption to the systems operation. The system must be able to process tubbed articles such as: car seats, garment bags, soft-sided bags, as well as items with loose straps, strings, ropes, or hooks.

2. Oversize (OS) Baggage
 - a. Items that do not fall within the normal characteristics described above or are of a nature that cannot be handled properly by the system (animal cages, skis, fishing poles, etc.) are considered oversize and will be handled according to the system-specific operating policy.

Table 40 - Oversize Baggage Characteristics

BHS	Length (inches)	Width (inches)	Height (inches)	Weight (pounds)
Minimum	12	9	4	5
Maximum	120	36	33	150

15.0.3 Clearances

1. The BHSC must conduct a field survey of all conveyor paths prior to the submission of installation drawings.
2. Bag Clearances
 - a. A minimum of 36 inches of vertical clearance for normal bags from the conveying surface.
 - b. If conditions exist where this clearance cannot be maintained, BHSC must notify the Owner or their designated representative in writing as soon as possible for any clearance deviations.
3. Conveyor Clearances
 - a. Where conveyors must cross over baggage cart ways or workstations, a design goal clearance of 8 foot – 0 inches to a minimum clearance of 7 foot - 6 inches (only when larger is not achievable) must be maintained to the lowest portion of the conveyor or its support structure, unless otherwise noted in the drawing package.
 - b. A minimum of 12 inch clearance must be maintained from motors, drive units and bearing to fixed obstructions such as walls, beams, and columns.
 - c. A minimum of 24 inch clearance between conveyor guards for adjacent conveyors.
4. EDS Maintenance Access Clearance
 - a. BHSC must maintain the manufacturer's recommended clearance for access and maintenance for the machine as outlined in their respective integration guidelines.
5. Install belly/catch pans underneath conveyors above common work areas to minimize risk to workers.

15.0.4 Baggage Transport

1. The BHSC must provide a system to transport the baggage smoothly and effectively through the system.
2. Consideration and care must be used, particularly in the fabrication of all projections, welds, surfaces, and transfer points between conveyors to eliminate damage to various types of bundles, handbags, suitcases, and trunks encountered.
3. Bottom glides on cases, strings, tags, straps, bag handles, destination tags, etc. must be protected against damage from sideguards, transfer points and all other surfaces against which baggage may come into contact during baggage handling.
4. Maximum Incline/Decline Angles
 - a. The maximum angle on inclines and declines must not exceed 20° from the horizontal; 15° or less must be maintained as a design objective for non-tracked conveyor lines.
 - b. Incline and decline angles must be 12° or less in tracked zones.
 - c. The BHSC must promptly notify the Owner or their designated representative if conditions exist that exceeds these requirements.

15.0.5 Equipment Loads

1. Input Load conveyors must be sized for a live load of 60 lbs./ft.
2. All other conveyors must be sized for live loads based on the following formula and Table 35.

$$\text{Live Load} = 40 \text{ lbs./ft.} \times (90 \text{ fpm/actual fpm}) \times (\text{Throughput (BPM)}/20 \text{ BPM})$$

Table 41 - System Throughput Rates

Subsystem	Maximum (BPM)
Inbound Conveyor Lines	20
Inbound Oversize Conveyor Line	15
Outbound Oversize Conveyor Line	15

15.0.6 Conveyor Design Speed

1. Individual component speeds with and without VFDs
 - a. Any component with a VFD must operate within 2 percent of design speed.
 - b. Any component without a VFD must operate within 5 percent of design speed.
 - c. If conditions exist where this design speed cannot be obtained, BHSC must notify the Owner or their designated representative in writing as soon as possible for any design deviations.

15.0.7 Performance and Dependability

1. General Requirements
 - a. The BHS must be capable of continuous operation, 20 hours per day, 365 days per year during normal operating hours, with provisions for routine preventive maintenance as specified in the maintenance manual.
 - b. The system must operate reliably and be reasonably free of breakdowns.
 - c. Major components of the system must be pre-tested so that life expectancy under the design operating conditions is known prior to installation.
 - d. Design of the components comprising the system must provide for ease of maintenance, replacement accessibility, and service requirements.
 - e. Ensure that no pipes containing any type of liquid other than life safety systems are installed above the control room or motor control panels.
 - f. Size of Conveyors
 - i. The size of standard beds should be a minimum width of 39 inches.
 - (1) The size of oversized beds should be a minimum width of 46 inches.
 - (2) Minimize the use of 90 degree turns.
2. Breakdowns
 - a. The system breakdowns must be limited to the following repair times:

Table 42 - Repair Time Requirements

Breakdowns	Repair Times (Minutes)
90.00%	< 15
5.00%	< 30
4.90%	< 120
0.10%	< 24 hrs

3. Availability

- a. The weekly system availability as a whole (excluding EDSs) must be 99%.
The system availability must be calculated as follows:

$$SA = (OH - RT) / OH$$

Where:

SA = System Availability

OH = Operational Hours

RT = Repair Time

15.0.8 Operating Conditions

1. Normal operations will be considered to be 20 hours per day, 365 days per year.
2. All mechanical and electrical components must operate satisfactorily within a temperature range of 0°F to +140°F, with a relative humidity of 0 to 100 percent (non-condensing on optical surfaces).

15.0.9 Accessibility

1. Equipment components requiring inspection and servicing must be readily accessible.
2. Suitable doors or removable enclosures must be provided.
3. Access holes in frames or guards are acceptable but must be held to a minimum number and size and must not create protrusions or discontinuities detrimental to the baggage being conveyed.
4. All components and assemblies must be easily disconnected and removed from the equipment without requiring extensive disassembly.

15.0.10 Vibration Control

1. All parts of the conveyor system, including catwalk, supported from structural elements or hangers must be on vibration isolators to eliminate such vibrations from being transmitted to the building. Vibration must not be noticeable in public or office spaces.
2. Surface-mounted conveyors and other associated components, including catwalk, using header steel connected to building elements and mezzanines must not transmit any vibration to these structures and must be fitted with vibration isolators.

15.0.11 Seismic Bracing

1. The BHSC must comply with all applicable codes regarding seismic bracing for the project.

15.0.12 Noise Levels

1. Noise measurements will be made under free field conditions according to DIN 45635/ISO 3744, no closer than three feet from the sound source. The dBA values do not include the noise due to baggage.
2. BHSC must measure noise levels with a calibrated A-weighted sound level meter at slow response or a calibrated 1/3 band octave analyzer and converted to A-weighted sound pressure levels.

3. BHSC must conduct baseline noise level readings during peak and non-peak operating hours prior to any work done on the system. The BHSC must record the time, location and the noise level for each reading.
4. BHSC must conduct noise level readings after work is complete on the system at the same locations for the same peak operating period during test.
5. Noise must be limited throughout the system to the following levels:

Table 43 - Noise Level Requirements

	Public Areas	Equipment Areas
Maximum	70 dBA	80 dBA

15.0.13 Materials

1. General
 - a. Steel used in the fabrication of the equipment must be free from rust, rust pits, kinks, and sharp bends.
 - b. Methods used to form the material must not fracture or otherwise damage the metal.
 - c. Burrs, sharp edges, and sharp corners must be removed. All parts must be formed and cut properly to assure uniformity.
 - d. All joints must be smooth, and all corners rounded.
 - e. Holes for photo eye beams must be punched into metal sideguards and tapered from the inside (wide) to the outside (narrow). Burning will not be allowed.
 - f. Joints between sections of conveyor bed must be chamfered to ensure that there is no step-up condition between bed sections.
2. Plate
 - a. Structural steel plates and shapes must conform to ASTM A36/A36M-12, Standard Specification for Carbon Structural Steel.
 - b. Hot-rolled sheets and coils must conform to ASTM A-569.
3. Welding
 - a. General procedure and qualification of welders must be in accordance with AWS D.1.1/D1.1M:2020, Structural Welding Code - Steel.
 - b. All welding electrodes must conform to AWS A.5.1/A5.1M:2012, Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding.
4. Fasteners
 - a. All fasteners at a minimum must conform to ASTM A325, Type 1, or SAE Grade 5, Class 2A thread fit for bolts and Class 2B thread fit for nuts.
 - b. All fasteners must be zinc-plated, cadmium plated or stainless steel.
 - c. Jam nuts must not be used for structural connections.

15.1 Electrical

All equipment must be the current version of manufacture at time of bid. When any product has available factory upgrades or modifications, such upgrades and modifications must be fully implemented at the factory and receive full factory testing and certification.

15.1.1 Electrical Clearances

1. Electrical clearance for safety and maintenance must meet the requirements of NFPA 70 National Electric Code (NEC) article 110.26, state, and local Electrical Codes for Personnel and Other Protective Equipment.
2. If conditions exist where this clearance cannot be maintained, BHSC must notify the Owner or their designated representative in writing as soon as possible for any clearance deviations.

15.1.2 Radio Frequency Interference

1. The equipment provided must not cause interference with communications within the airport or between the airport and aircraft or ground vehicles.
2. All electrical and electronic equipment, including interconnecting wires and cables, must be designed to operate without malfunction in the presence of normal electromagnetic emissions generated by other equipment installed or used at the airport.
 - a. The normal airport environment includes the aircraft communications bands and high-power radar systems. It also includes various electrical motors and controls, power tools (including welders), automotive vehicles, etc.

15.1.3 System Power Provisions

1. Source for Systems
 - a. Existing building panel boards or MDPs feed the newly installed Motor Control Panels (MCPs) for each inbound and outbound subsystem. Each MCP will have an adjacent PDP that feeds power to the conveyors. The MCPs will not have any 480v power inside.
 - b. Distributed SEW PMM and Movimot drives will be mounted on the conveying system and powered from the associated MCPs.
 - c. Each MCP will contain a main breaker for power distribution to branch circuits, control circuits and other BHS related components.
 - d. PLCs and controls can be installed integral within MCPs or in separate panels.
 - e. 480 Volts Alternating Current (VAC), three-phase, 60 Hz power will be furnished by the General Contractor (GC) or Construction Manager (CM) from existing PDPs to a disconnect located within the newly installed MCPs for each inbound subsystem.
2. Distribution for Systems
 - a. BHSC must provide all power distribution equipment from the building PDP to all BHS/BHS equipment. This includes the following:
 - i. Approximate Locations of the MCPs are as shown on the plans. The BHSC is responsible for coordinating the final location of the MCPs with the Owner or their designated representative.
 - ii. Coordinate the location and distance between the PDPs or MCPs and the field devices so that the voltage drop does not exceed two percent.

- iii. Furnish and install all panel boxes, wire ways, conduits, conductors, transformers, fuses, equipment, and materials required to complete the electrical power distribution for the operation of the system.
- iv. Indicate the full load amperage of each MCP fuse or breaker size on shop drawings.
- b. BHSC must show on its shop drawings the electrical power requirements (amperage) of each MCP main circuit breaker.
- c. BHSC must validate harmonic distortion induced by the BHS and correct if it does not meet the acceptable limits. The acceptable limits on total harmonic distortion (THD) is at 5% and any single harmonic is at 3% as recommended by IEEE std 519.

15.2 Controls

15.2.1 General Controls Requirements

The control system must interface and/or control, at a minimum, the following equipment and all functionalities relevant to the equipment:

1. All transport, queue conveyor belts and claim devices
2. Security/Fire Doors
 - a. Fire input signals
3. BHSC must program each system to function as specified herein.
4. During startup and testing, the functionality of the conveying system, devices, control, and reporting must be demonstrated through the integrated BHS system.
5. Erroneous code must be removed prior to commissioning and final testing.
6. The BHSC is responsible for providing Controls system which is compatible with the existing system.
7. The BHS control system must interface with the fire/security system.
8. The hardware, software, and firmware must comply with state-of-the-art technologies and represent current industrial standards. The components must be compatible and tested product families to provide continuity. The components must be readily available.
9. All field components used within the system must be suitable for use under harsh environmental conditions (e.g., temperature, soiling, electrically contaminated environments, vibrations, shocks, high humidity, and various climate temperatures).
10. All field network components must meet the environment class specified in the MICE table (ISO/IEC/TR 29106).
11. The system components must permit easy and rapid servicing and replacement.
12. Proprietary hardware, software, or firmware is prohibited within this project. Software must be off-the-shelf or open source and be the most recent version.
13. The computer system hardware must be comprised of standard, off-the-shelf components. No custom-built hardware or interface devices are to be used.
14. The computers used must be open-standard non-proprietary computers and the latest technology at the time of bid submittal.
15. All new industrial computer equipment must be available for two years.
16. All new PLC components must be available for three years.
17. The system controls must be capable of functioning for a minimum of 10 years.
18. If the BHSC cannot comply with the availabilities of equipment, they must indicate alternative source(s) of supply.

19. The functionality and compatibility of spare parts from alternative source(s) of supply must be identified by the BHSC.
20. The control equipment for the system must be designed to control and monitor the baggage from the time the bag is inducted into the system until it reaches its destination.
21. The control system must be designed to meet the specific functional requirements of the BHS.
22. The control system must include the ability for an operator to take any diverter out of automatic and manually control diverters for desired routing scheme from the Control Room.
23. The loss of a single power circuit or controls must not disable more than one conveyor line.

15.3 TSA – PGDS

1. Compliance with the mandatory requirements of the TSA PGDS. If conflicts arise, the PGDS will govern until an approved Request for Variance (RFV) is received from TSA.
2. Designer will coordinate with the DFW Project Representative the submittal requirements of the TSA PGDS, including a Basis of Design Report and Construction Work Estimates
3. Produce drawings and specifications that describe in detail the requirements for a Baggage Handling System Contractor (BHSC) to design, develop, manufacture, install, test and commission a complete BHS. The system must include a Checked Baggage Inspection System (CBIS) that complies with DFW requirements and the latest version of the Transportation Security Administration (TSA) Planning Guidelines and Design Standards (PGDS).

ACRONYMS AND ABBREVIATIONS

AACS	Automatic Access Control System
AASHTO	American Association of State Highway and Transportation Officials
AC	Advisory Circular
AC	Alternating Current
ACI	American Concrete Institute
ACM	Asbestos Containing Material
ACS	Announcement Control System
ADA	Americans with Disabilities Act
ADG	Airplane Design Group
AED	Automated External Defibrillator
AHJ	Authorities Having Jurisdiction
AISC	American Institute of Steel Construction
ALP	Airport Layout Plan
ANS	Ambient Noise Sensors
ANSI	American National Standards Institute
AOA	Aircraft Operation Area
ARFF	Aircraft Rescue and Fire Fighting
ARV	Air Release Valve
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigeration, and Air-Conditioning Engineers
ASME	American Society of Mechanical Engineers
ASTM	International (formerly American Society for Testing and Materials)
ATC	Air Traffic Control
AWG	American wire gauge
AWWA	American Water Works Association
BAS	Building Automation System
BHS	Baggage Handling System
BIM	Building Information Management
BL-ETELs	Blue Light Emergency Telephones
BTU	British thermal unit
CADD	Computer Aided Design and Drafting
CATV	Cable Television
CBP	Customs and Border Patrol

CCTV	Closed Circuit Television
CCU	Cluster Control Unit
CDA	Critical Design Aircraft
CFAS	Central Fire Alarm System
CFM	Cubic Feet Per Minute
CFR	Code of Federal Regulations
CIP	Capital Improvement Project
CPS	Cathodic Protection System
CPS	Corrosion Protection System
CPVC	Chlorinated Polyvinyl Chloride
CRCP	Continuously Reinforced Concrete Pavement
CRSI	Concrete Reinforcing Steel Institute
CSI	Construction Specifications Institute
CTA	Central Terminal Area
CTS	Corrosion Test Station
DAR	Design Analysis Report
DC	Detector Check
DDC	Direct Digital Control
DIP	Ductile Iron Pipe
DFW	Dallas Fort Worth International Airport
DPS	Department of Public Safety
DREFs	Data Reference Files
DWV	Drain-Waste-Vent System
EAD	Environmental Affairs Department
EMT	Electrical Metallic Tubing
EOR	Engineer of Record
EP	Energy Plaza
EPA	U.S. Environmental Protection Agency
ESP	East Side Plant
ETAM	Airport Energy, Transportation and Asset Management
EVAC/MN/ PA	Emergency Voice Alarm Communications/ Mass Notification/ Public Address
FAA	Federal Aviation Administration
FARs	Federal Aviation Regulations
FAS	Fire Alarm System
FCU	Fan Coil Unit
FDC	Fire Department Connection
FFETs	Fire Fighter Emergency Telephones
FFSW	First Flush Stormwater System
FHWA	Federal Highway Administration
fps	Feet per second
GBR	Geotechnical Baseline Report
GDM	Geotechnical Design Memorandum
GDR	Geotechnical Data Report

GFI	Ground Fault Interrupter
GI/LID	Green Infrastructure/Low Impact Development
GIS	Geographical Information System
GP	Galvanic Protection
gpm	Gallons per minute
GPS	Global Positioning System
HARP	Hardware Architecture Reference Platform
HDPE	High Density Polyethylene Pipe
HID	High Intensity Discharge
HMAC	Hot Mix Asphalt Concrete
HOA	Hand Off Auto
HSLA	High Strength Low Alloy
IBC	International Building Code
ICC	International Code Council
ICCP	Impressed Current Cathodic Protection
IEC	International Electrotechnical Commission
IECC	International Energy Conservation Code
IEEE	Institute of Electrical and Electronics Engineers
IES	Illumination Engineering Society
IESNA	Illuminating Engineering Society of North America
IFC	International Fire Code
IMC	Intermediate Metal Conduit
ISO	International Organization for Standardization
iSWM	Integrated Stormwater Management
ITS	Information Technology System
ITE	Institute of Transportation Engineers
KPI	Key Performance Indicators
kV	Kilivolts
kVA	Kilivolt amps
LCD	Liquid Crystal Display
LED	Light-emitting Diode
LEED	Leadership in Energy and Environmental Design
LLDPE	Liner Low Density Polyethylene
LP	Liquid Petroleum
LPI	Lightning Protection Institute
LRFD	Load and Resistance Factor Design
MBI	Modular Building Institute
MC	Metal Clad
MDF	Medium-density Fiberboard
MEP	Mechanical/Electrical/Plumbing
Mph	Miles Per Hour
MSS	Manufactures Standardization Society
MUFIDS	Multi-User Flight Information Systems
NACE	National Association of Corrosion Engineers

NAVAIDS	Navigational Aids
NEC	National Electrical Code
NELAP	National Environmental Laboratory Accreditation Program
NEMA	National Electrical Manufacturers Association
NEPA	National Environmental Policy Act
NFPA	National Fire Protection Association
NPSH	Net Positive Suction Head
NRCA	National Roofing Contractors Association
O&M	Operation and Maintenance
O.D.	Outside Diameter
OSHA	Occupational Safety and Health Administration
PAVE	Public Address and Voice Evacuation System
PCB	Polychlorinated Biphenyl
PC	Point of Curvature
PCC	Portland Cement Concrete
PDWF	Peak Dry Weather Flow
PI	Point of Intersection
POV	Privately-Owned Vehicle
PRV	Pressure Reducing Valves
PSF	Pounds Per Square Foot
PSI	Pounds Per Square Inch
PSIG	Pounds Per Square Inch Gauge
PTI	Post Tensioning Institute
PTWS	Pre-Treatment Waste System
PVC	Polyvinyl Chloride
PVI	Point of Vertical Intersection
PWS	Potable Water System
PWWF	Peak Wet Weather Flow
QA/QC	Quality Assurance/Quality Control
RAM	Random Access Memory
RCCP	Reinforced Concrete Cylinder Pipe
RFI	Request for Information
RMC	Rigid Metal Conduit
RPM	Rotations Per Minute
RVI	Remote Virtual Inspections
SCADA	Supervisory Control and Data Acquisition
SDMP	Stormwater Drainage Master Plan
SEER	Seasonal Energy Efficiency Ratio
SIDA	Security Identification Display Area
SMACNA	Sheet Metal and Air Conditioning Contractors National Association
SOW	Scope of Work
SPDT	Single Pole-Double Throw
SPRI	Single Ply Roofing Industry
SPT	Standard Penetration Test

SPRI	Single Ply Roofing Industry
SPv	Streambank Protection Volume
SRD	Signal Receiving Device
SRS	Supplemental Radiating System
SSI	Security Sensitive Information
SSS	Sanitary Sewer System
STC	Soild Transmission Class
SUE	Subsurface Utility Engineering
SWMM	Storm Water Management Model
SWMP	Storm Water Management Plan
TAS	Texas Accessibility Standard
TCAS	Courtesy Announcement System
TC	Termination of Curve
TCEQ	Texas Commission on Environmental Quality
TCP	Tree Conservation Plan
TD	Trench Depth
TIA	Traffic Impact Analysis
TMUTCD	Texas Manual on Uniform Traffic Control Devices
TRACON	Terminal Radar Approach Control
TRE	Trinity Railway Express
TSA	Transportation Safety Administration
TxDOT	Texas Department of Transportation
UL	Underwriters Laboratory
UPS	Uninterruptible Power Supply
USACE	U.S. Army Corps of Engineers
UST	Underground Storage Tank
VAV	Variable Air Volume
VFD	Variable Frequency
WQv	Water Quality Volume